

GenAI for .NET: Build LLM Apps with OpenAI and Ollama

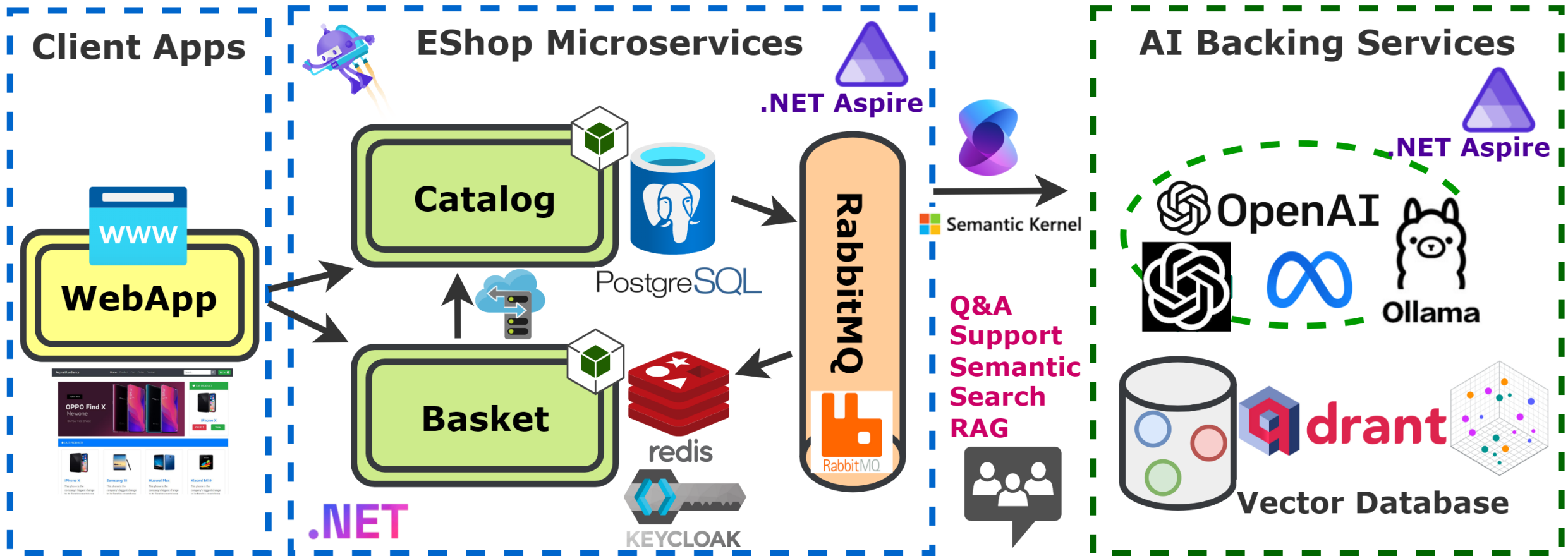
Developing Chat, Vector Search, VectorDB, RAG Apps using OpenAI, Ollama and Microsoft.Extensions.AI (MEAI)

Build Eshop Vector Search App w/ .NET Aspire, gpt-5-mini and Qdrant Vector DB

Mehmet Ozkaya



EShop Vector Search App with .NET Aspire, gpt-5-mini and Qdrant Vector DB



GenAI for .NET: LLM Apps with OpenAI & Ollama

.NET + AI Ecosystem

Chat, Text Completions, Function Calling

.NET AI Vector Search App

Retrieval Augmented Generation (RAG) App

Image Analysis with .NET AI

Eshop Vector Search App



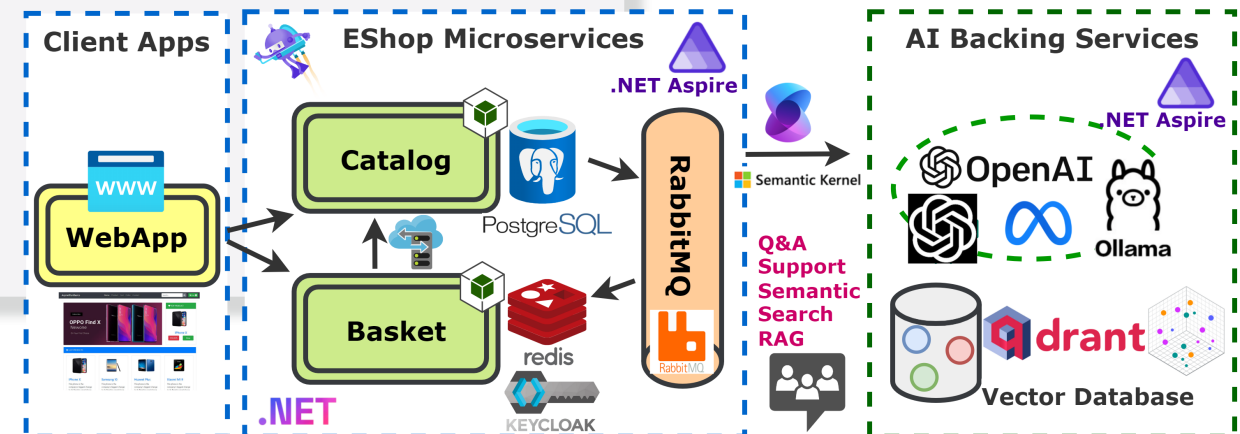
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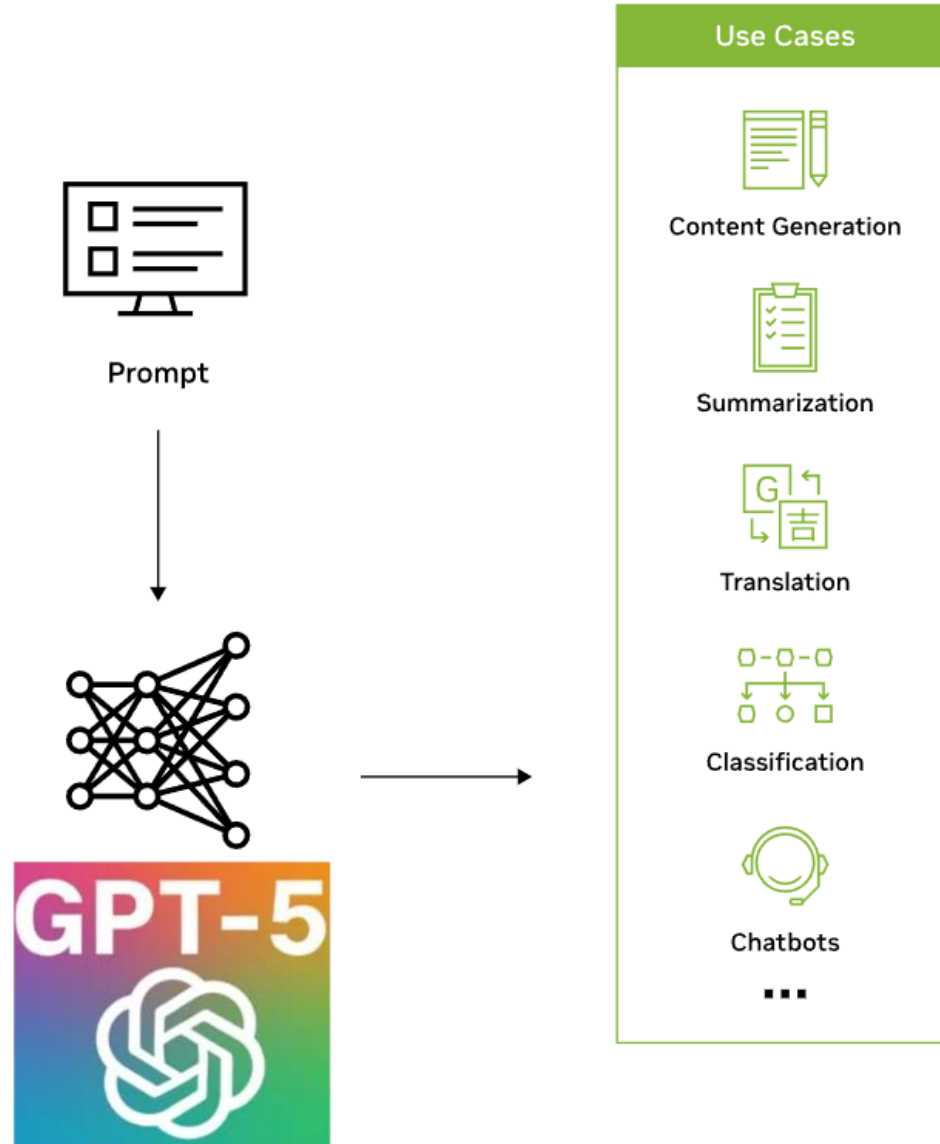
OpenAI



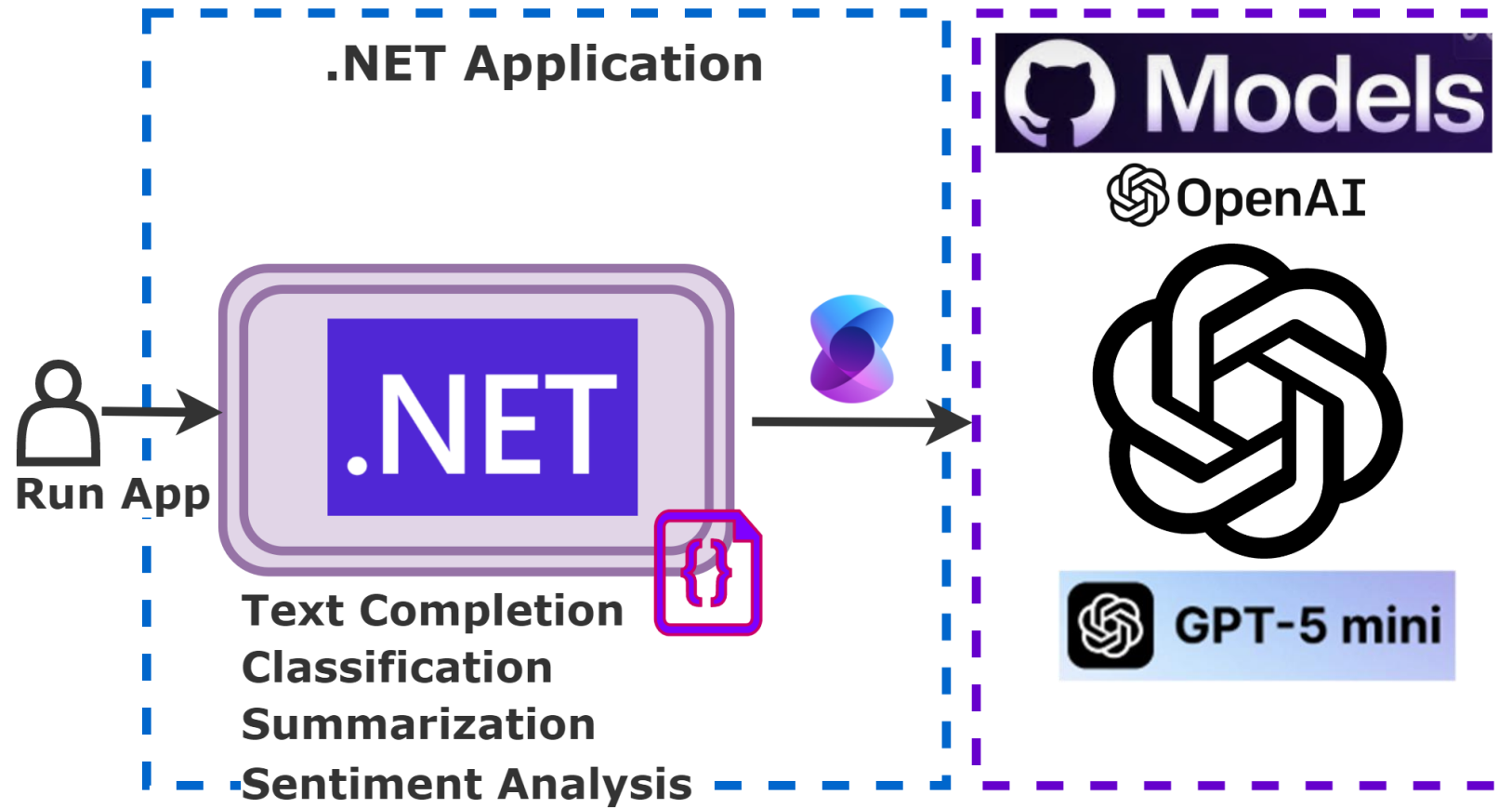
Ollama



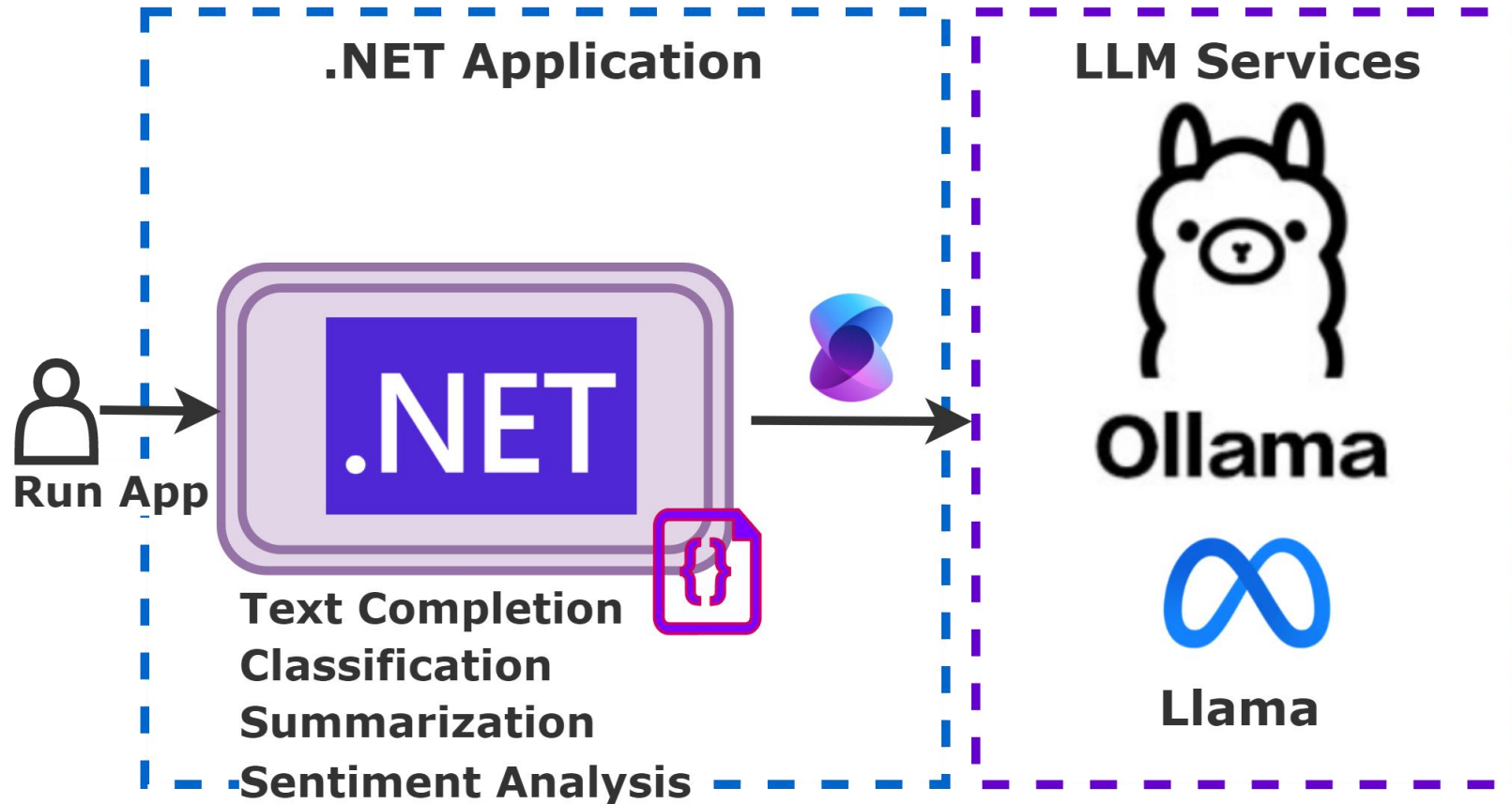
Generate Text with OpenAI Models: Understand Capabilities and Limitations of LLMs



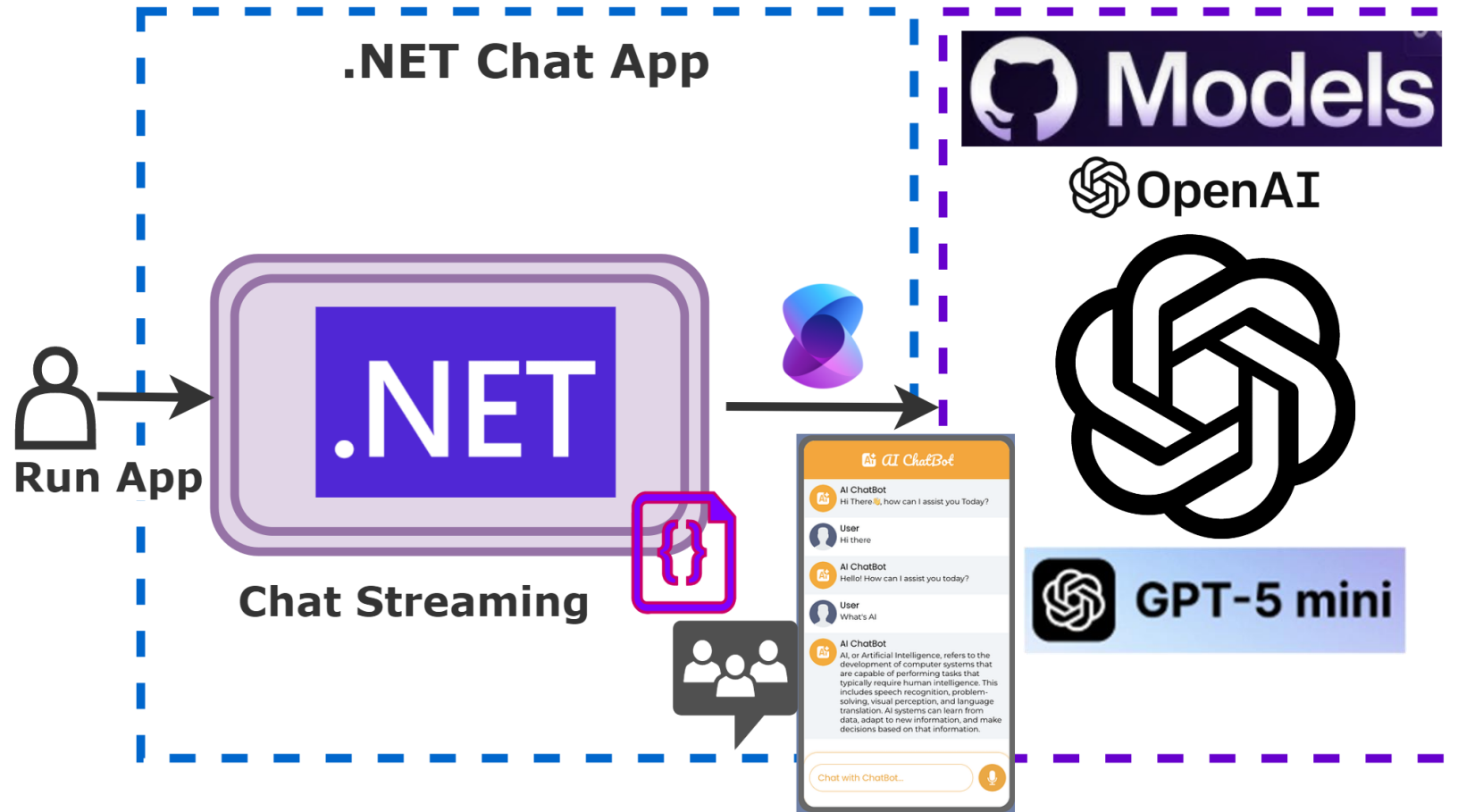
Text Completion LLM w/ GitHub Models OpenAI gpt-5-mini



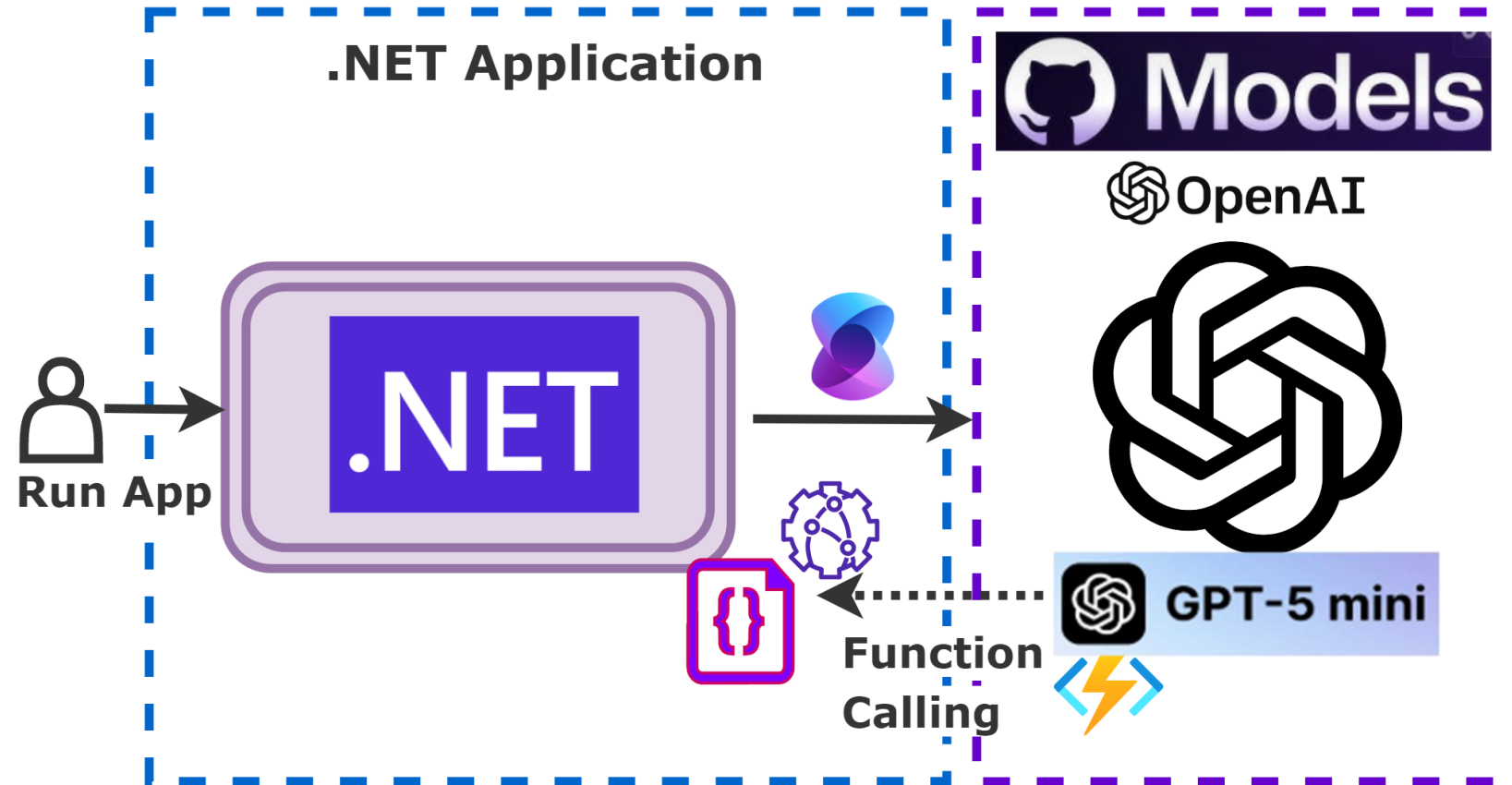
Text Completion LLM with Local Ollama llama3.2 Model



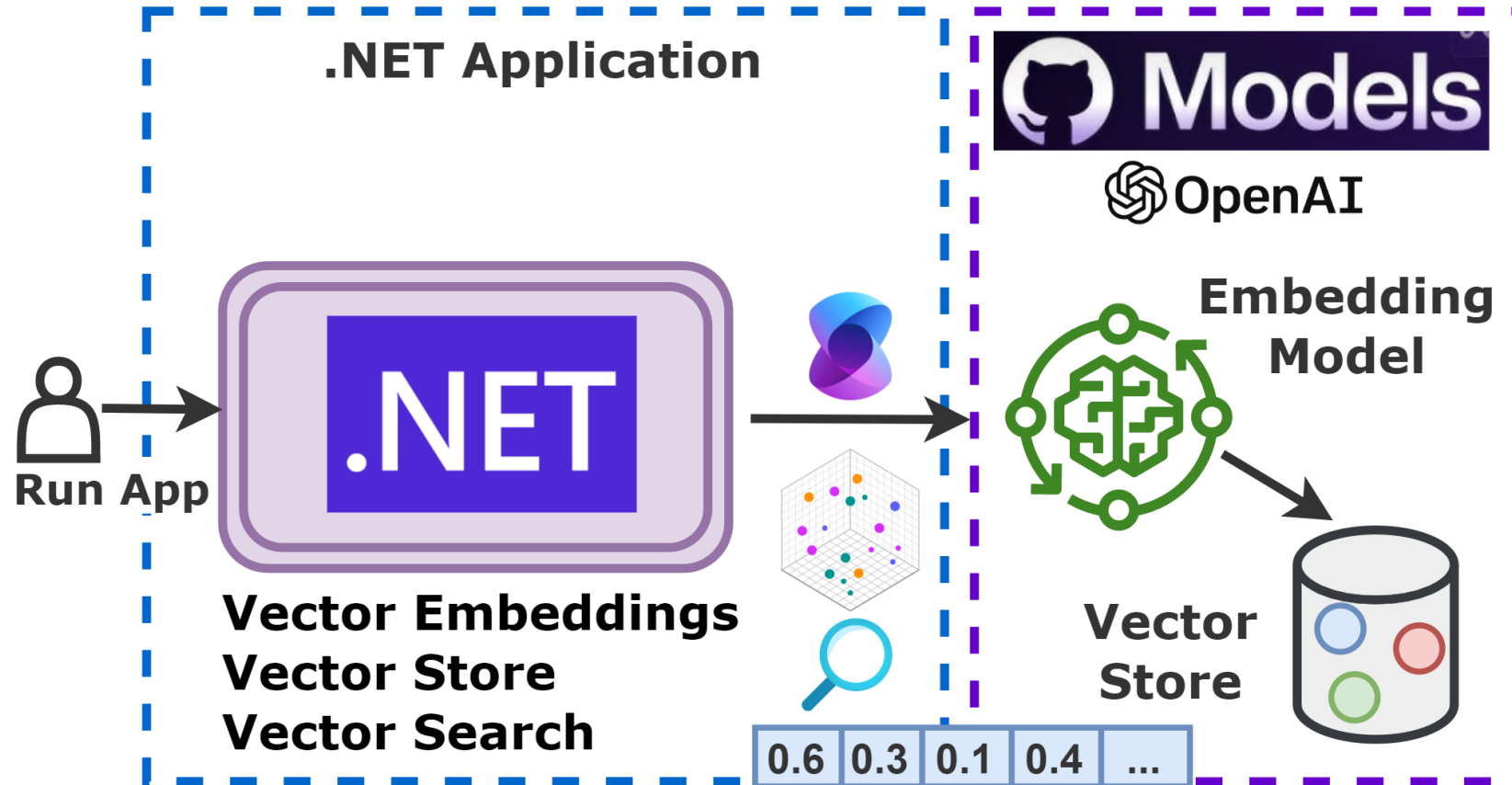
Build AI Chat App with .NET and gpt-5-mini model



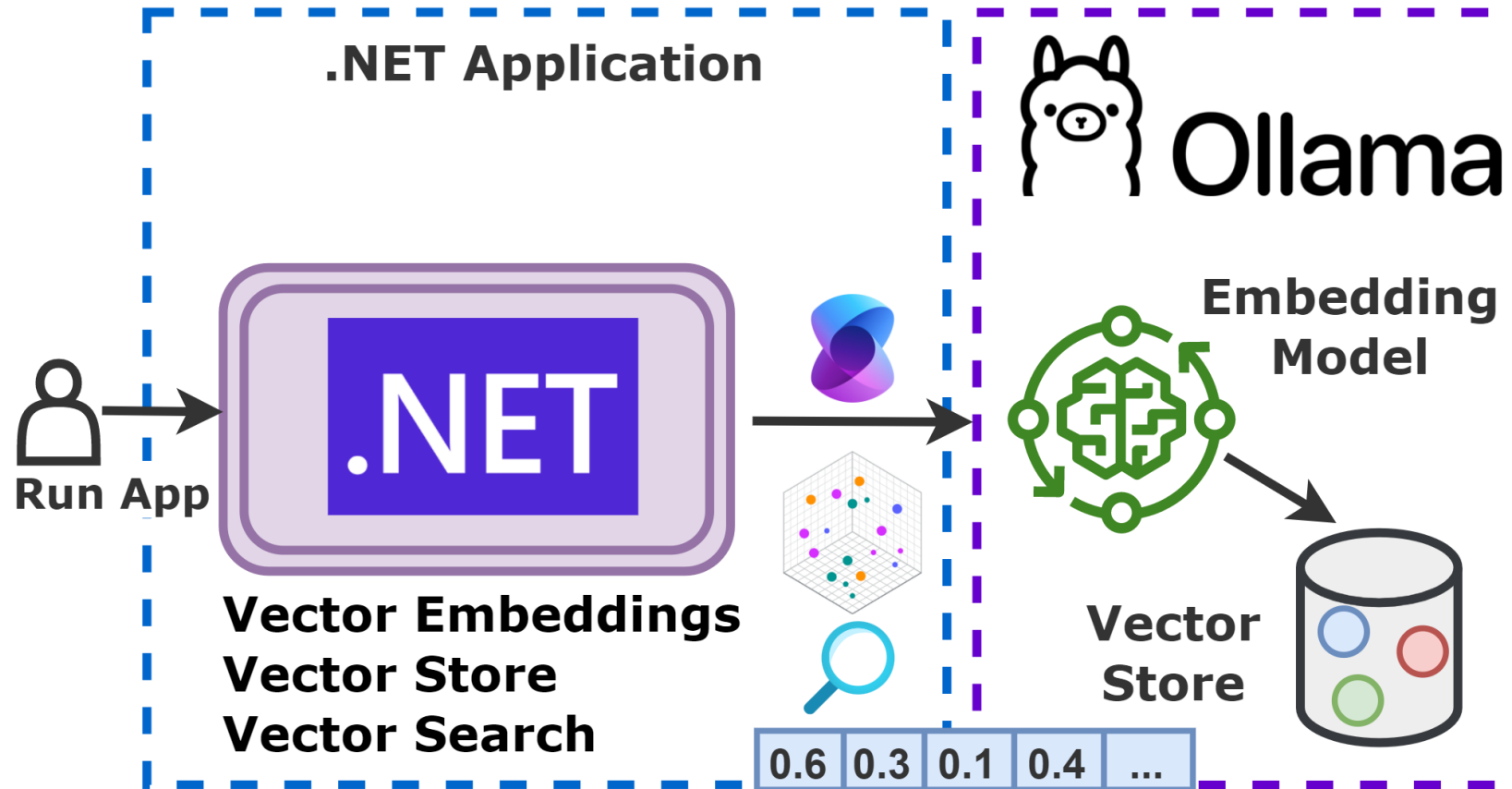
Function Calling in Large Language Models (LLMs)



.NET AI Vector Search using Vector Embeddings and Vector Store



.NET AI Vector Search using Vector Embeddings and Vector Store - Ollama



Retrieval Augmented Generation (RAG) Application w/ .NET

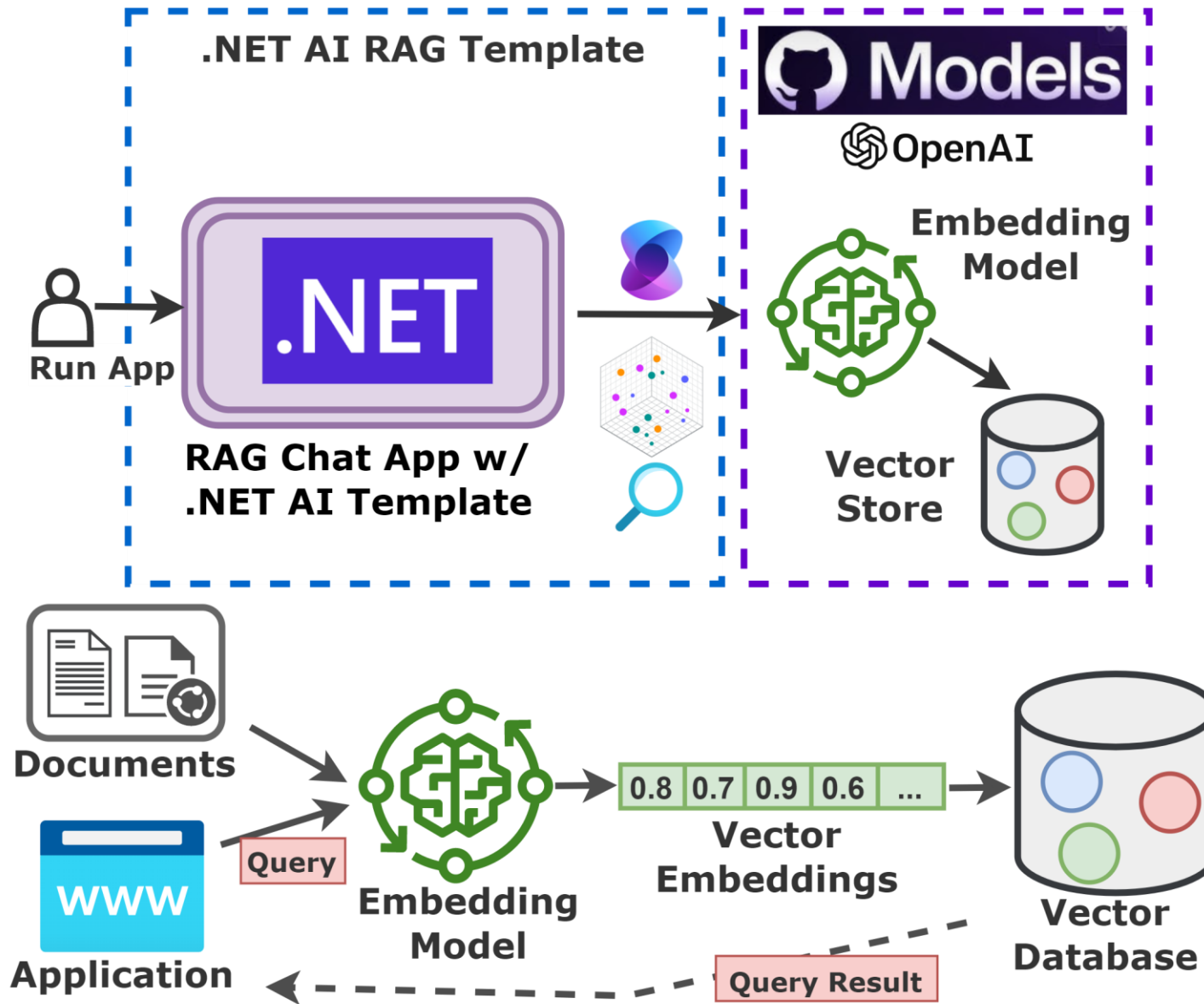


Image Analysis Apps with .NET AI

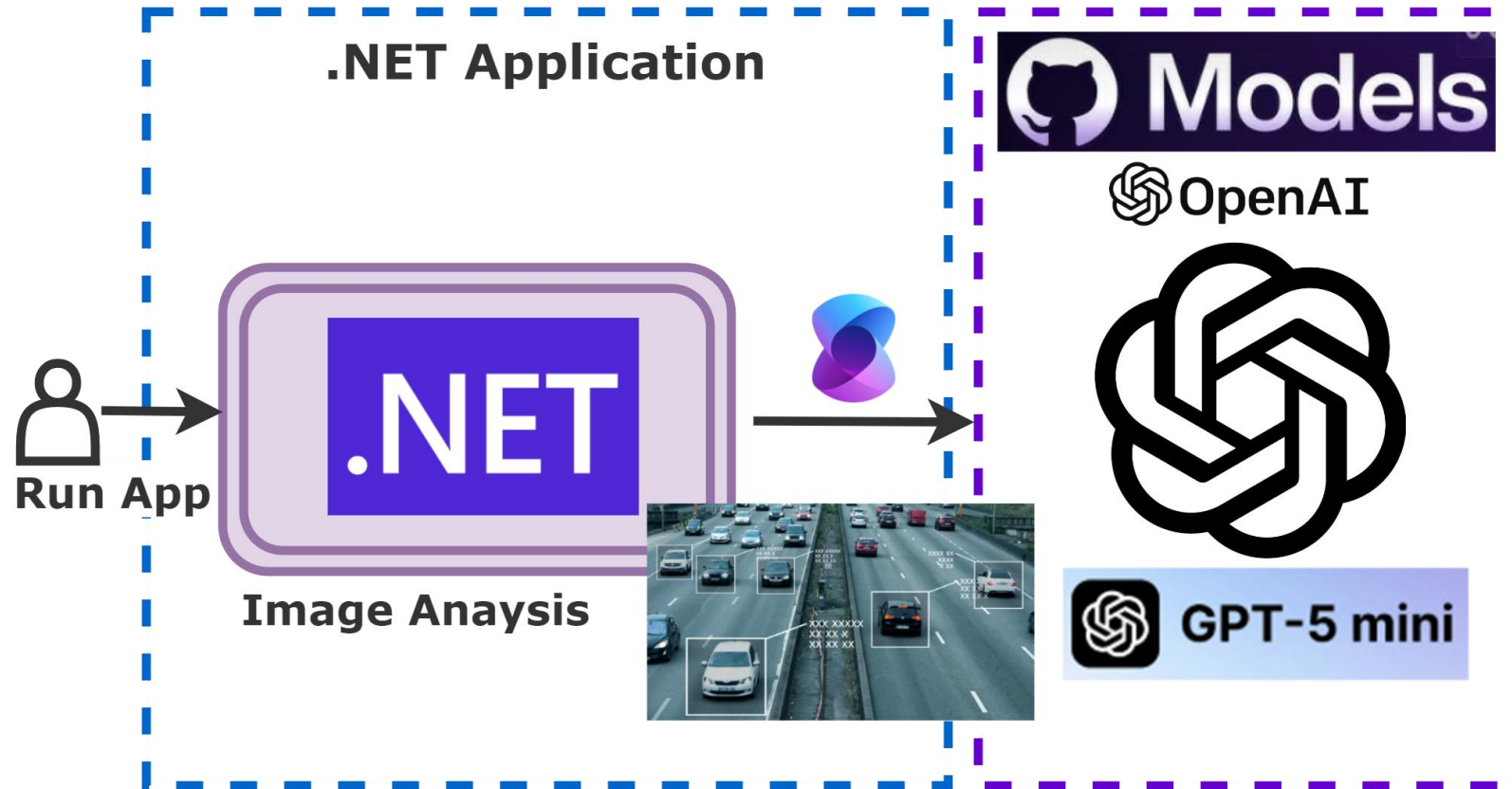
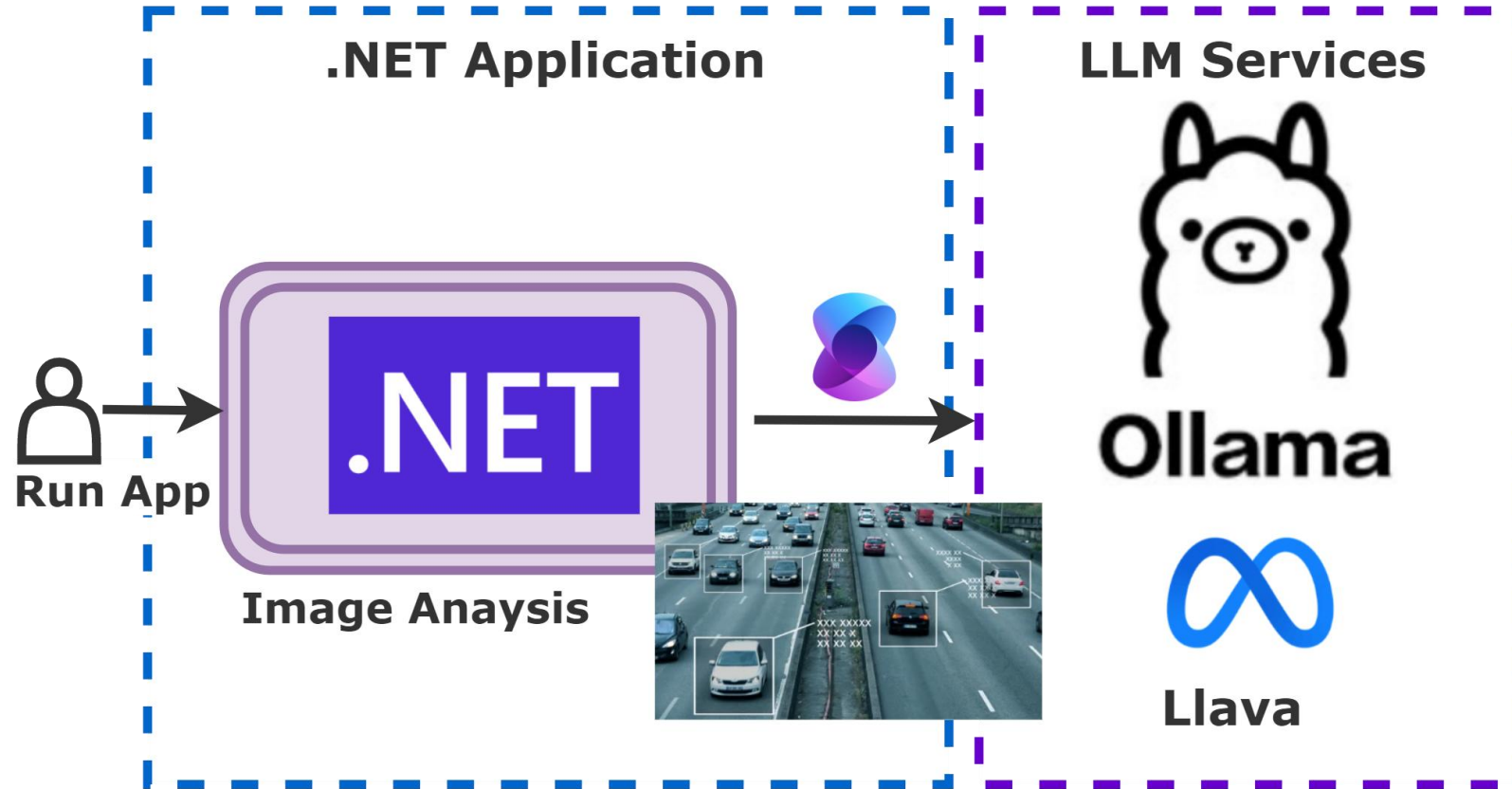
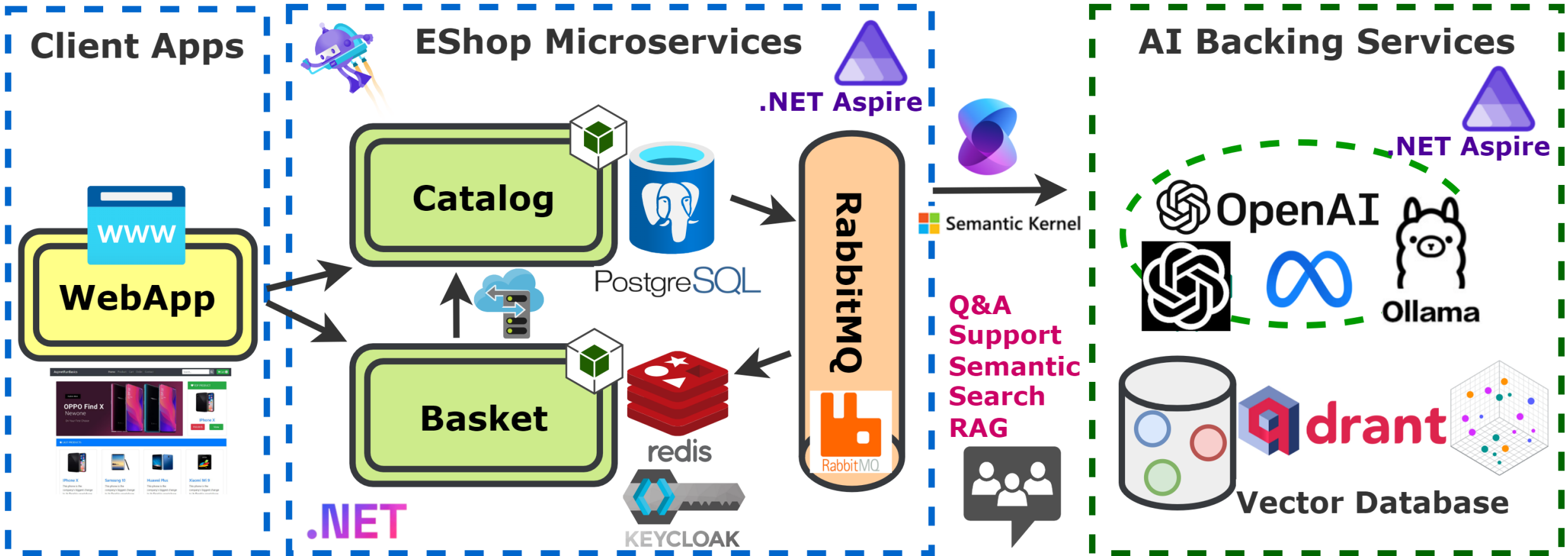


Image Analysis Apps with .NET AI - Ollama



EShop Vector Search App with .NET Aspire, gpt-5-mini and Qdrant Vector DB



Who is this course for ?

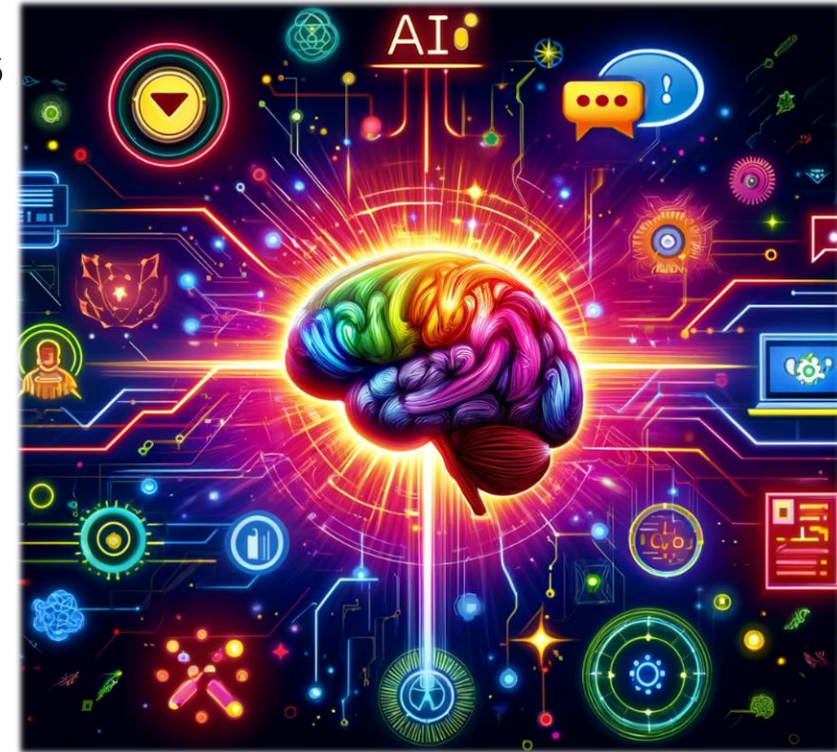
.NET developers wanting to break into AI

Software architects exploring Generative AI use cases

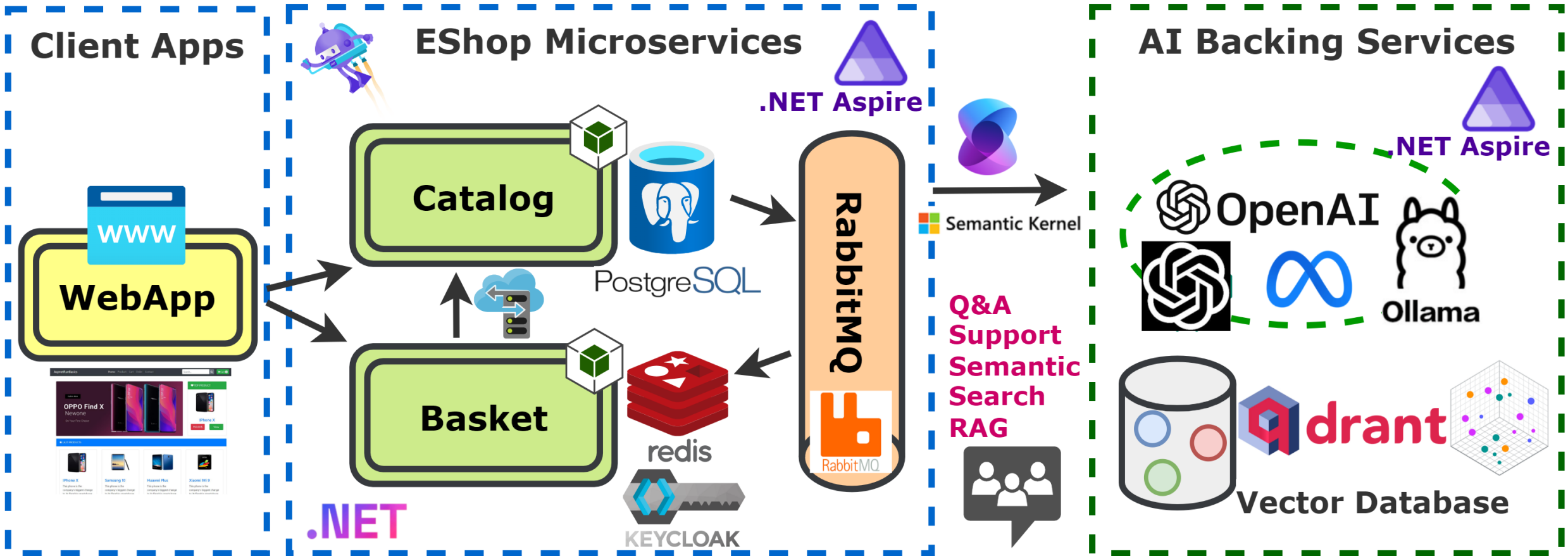
Engineers looking to integrate GPT features, RAG or image analysis into enterprise .NET apps

No prior AI experience is needed—just a solid foundation in .NET

The course is very practical, about 95%+ of the lessons will involve you coding



EShop Vector Search App with .NET Aspire, gpt-5-mini and Qdrant Vector DB



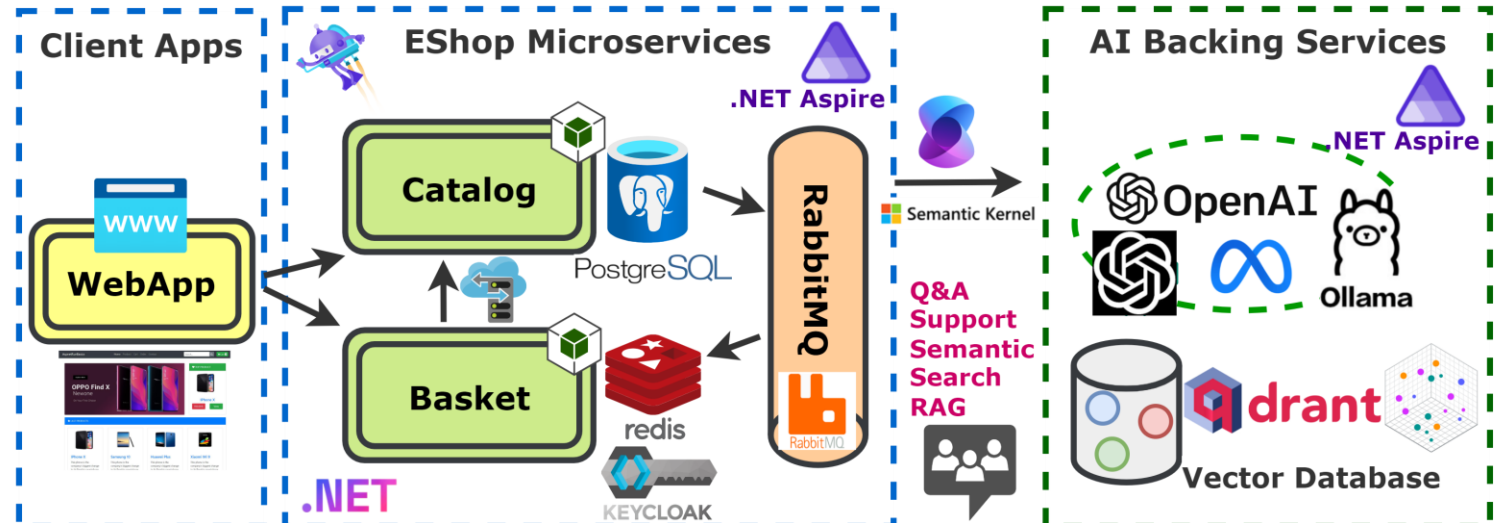
Prerequisites – What You Should Know

[.NET 9 \(Includes .NET Aspire by default\)](#)

[Visual Studio 2022 or Later](#)

[Docker Desktop](#)

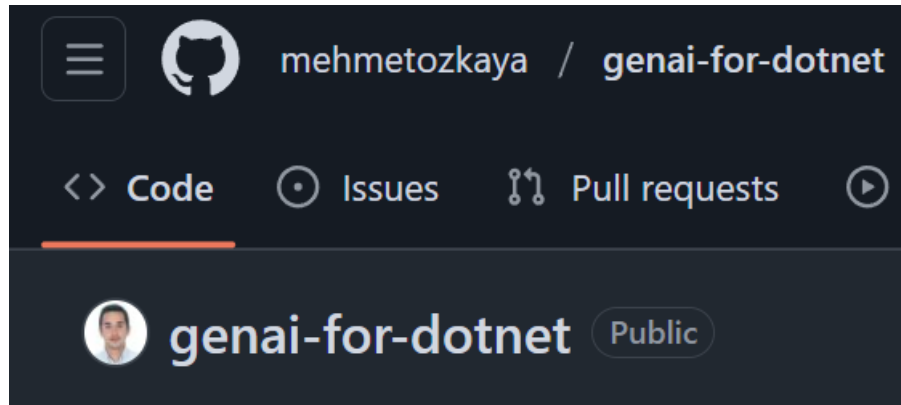
[GitHub Account](#)



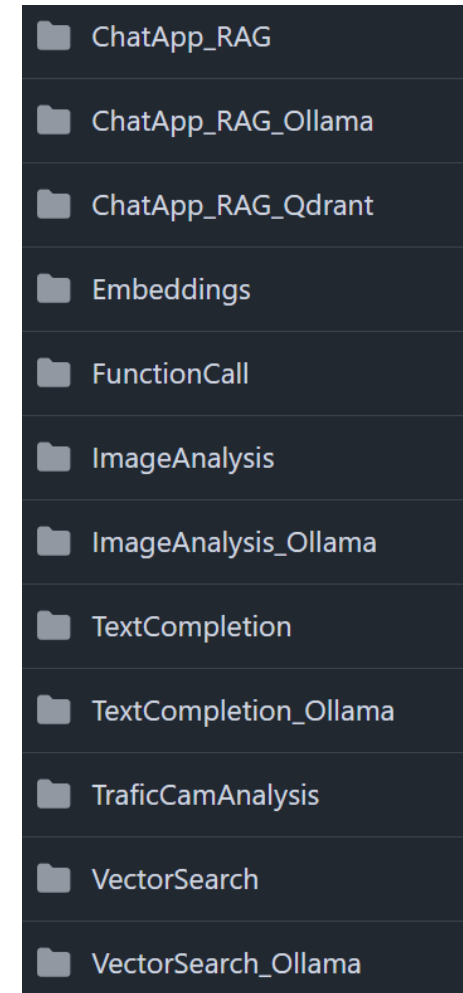
Slide Source Code on GitHub

All the code written and demonstrated in this course is available on GitHub

- <https://github.com/mehmetozkaya/genai-for-dotnet>



- Clone and download the source codes



Course Slides and All Resources

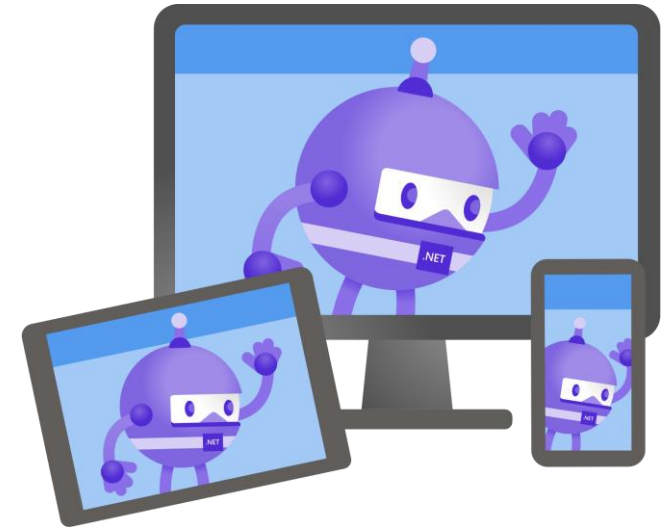
Download Course Slides and All Resources from This Lecture Resources

Section 1: Introduction
1 / 9 | 39min

- ☒ 1. Introduction
7min
Resources ▾
- ☐ 2. Pre...
Microservices Architecture on...

Course content ✕

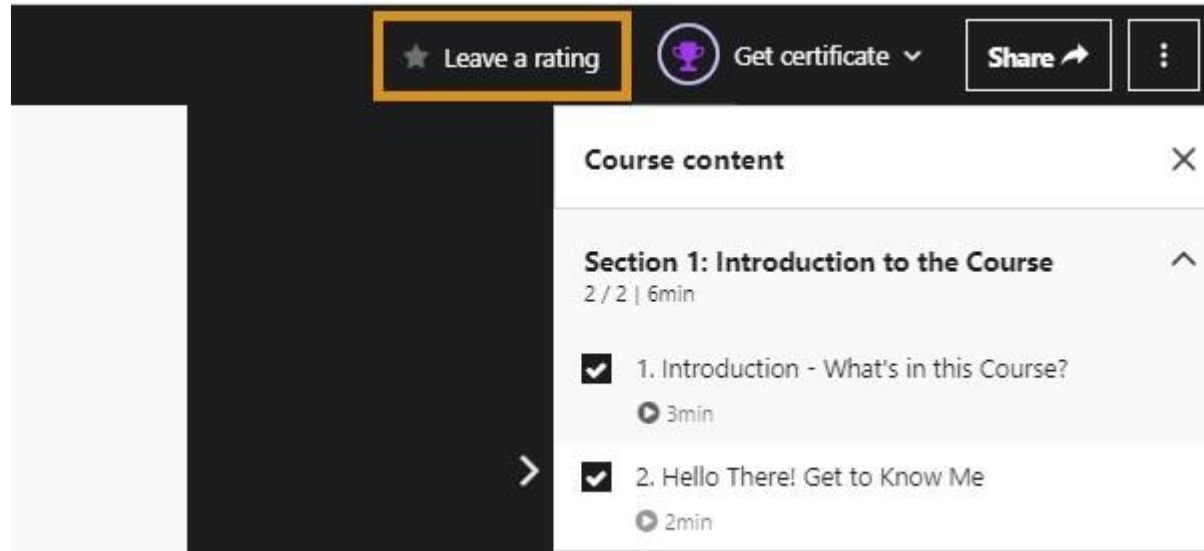
- ☐ 23. Variables
9min
Resources ▾
- ☐ 24. Starting out with Expressions
7min
Resources ▾
- ☐ 25. Primitive Types
12min
Resources ▾
- ☐ 26. byte, short, long and width
10min
Resources ▾



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Featured review



Fender McIntosh

8 courses

3 reviews

★★★★★ 4 months ago

I am taking this course for both personal and professional improvement. I'm usually not good with video courses because I get bored and distracted, but I was able to focus on this course. The video pacing is great and so is the teaching style. Each topic/section is broken down into very manageable and thankfully short videos. I started this course as a beginner and I feel like I will be given the tools and a solid foundation to use Tableau effectively and efficiently.

Course Projects on Code

01-TextCompletionSentiment

02-ChatApp

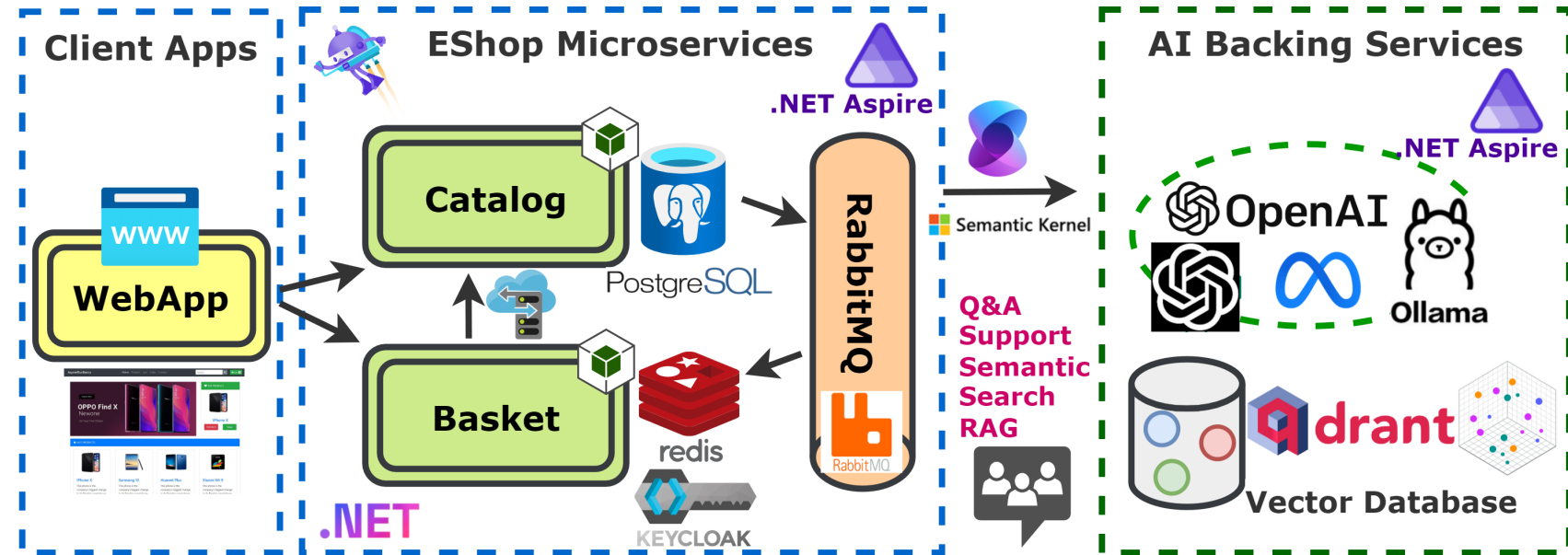
03-FunctionCalling

04-VectorSearch

05-RAGApplication

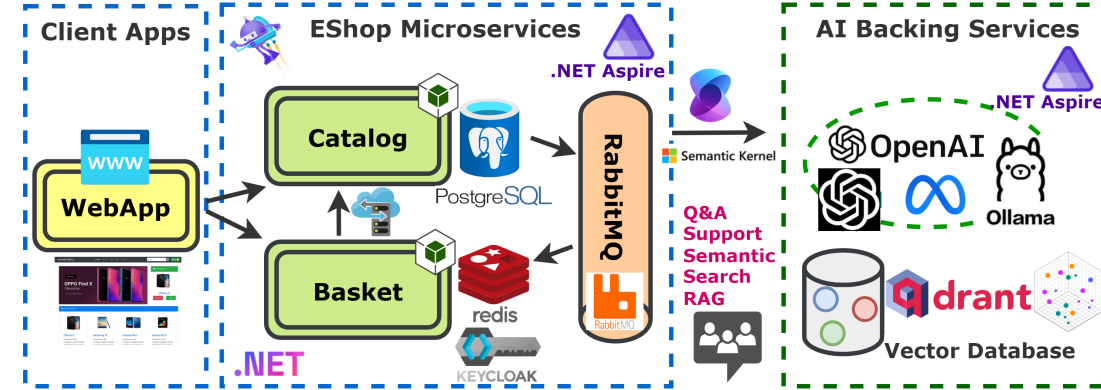
06-ImageAnalysis

07-EShopVectorSearch

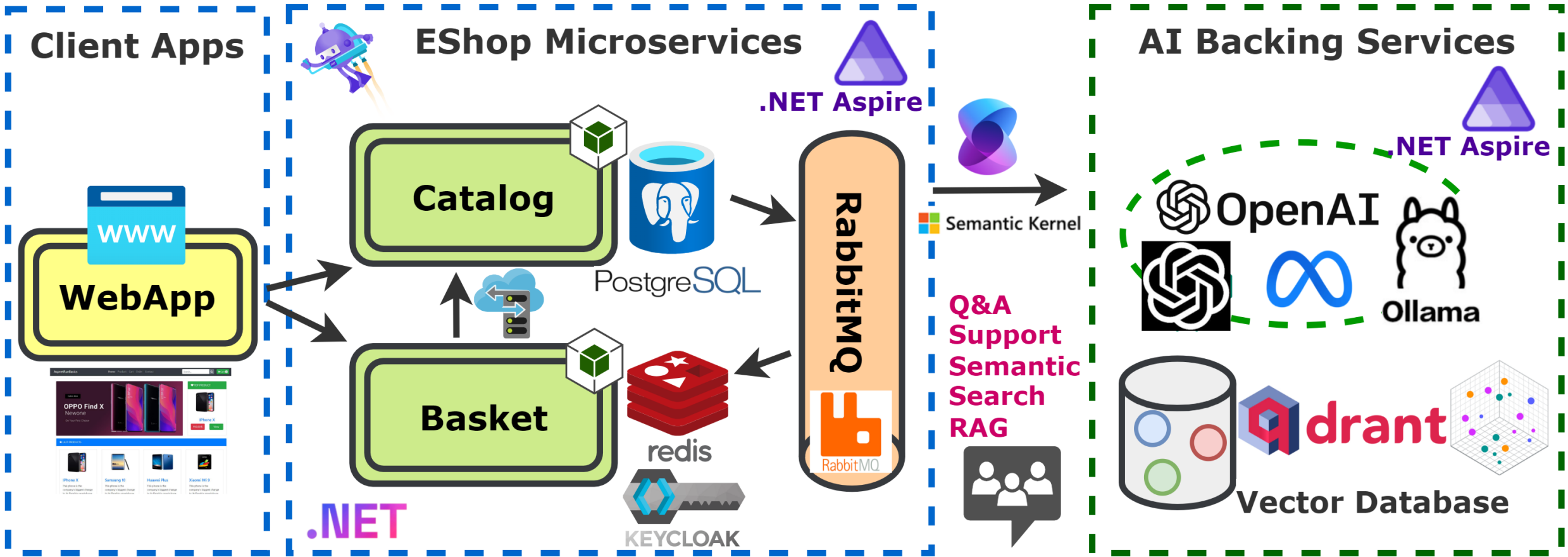


Course Project: EShop Microservices with AI-Powered Capabilities using LLMs

1. Clone the Repository
2. Start Docker Desktop
3. Setting Up in Visual Studio
4. Review .NET Aspire Project and EShop Microservices
5. Review AppHost Project – Cloud-Native Backing Services
6. Run the AppHost Project
7. Review Aspire Dashboard and Resources
8. Blazor Web App Test
9. Perform Vector Search operation on Search Page



EShop Vector Search App with .NET Aspire, gpt-5-mini and Qdrant Vector DB



GenAI Concepts: LLM, Tokens, Prompt Engineering

What are Large Language Models (LLMs) ?

How Large Language Models (LLMs) works ?

What is Token And Tokenization ?

What are Small Language Models (SLMs) ?

What is Prompt and Prompt Engineering ?

Mehmet Ozkaya



What are Large Language Models (LLMs) ?

LLMs are **advanced AI models** trained on **large datasets** - massive amounts of **text data** from **books, websites, research papers**

They can **understand, generate, and process natural language** that mimics natural human communication

Heart of **AI chatbots**, content **creation tools**, **translation services**



Large Language Models (LLMs)

LLMs are **machine learning models** specialized in **natural language**

They operate on a **vast number of parameters**—**billions** or **even trillions**

Trained to perform **tasks** such as **text generation**, **summarization**, **Q&A**, **translation**, and more

Building **intelligent language applications**



Why Are LLMs Considered Large ?

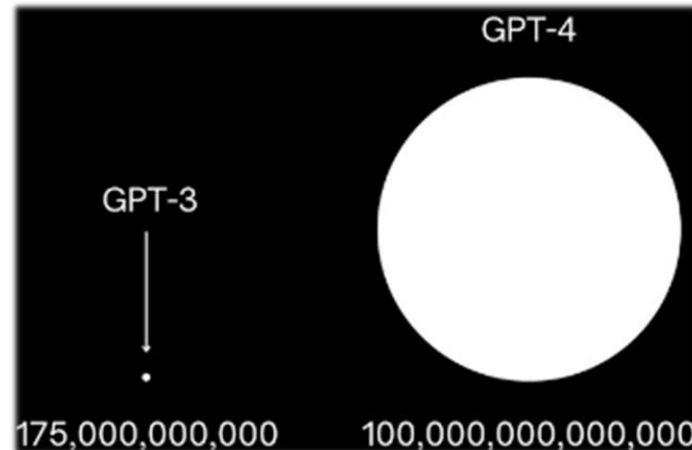
'Large' in LLM refers to both the size of the model: **the number of parameters** and **the training data**

LLMs contain **billions of parameters** (GPT-3 has 175 billion)

Training data: Trained on diverse text datasets (books, articles, web pages)

Larger models provide **better accuracy** and **context understanding**

More parameters = better performance but **higher computational costs**



How Large Language Models (LLMs) works ?

Transformer Architecture – The Core of LLMs

Introduced in "**Attention is All You Need**" (2017)

Uses **self-attention** to process text

Processes words **simultaneously** (unlike older models like RNNs)

arXiv > cs > arXiv:1706.03762

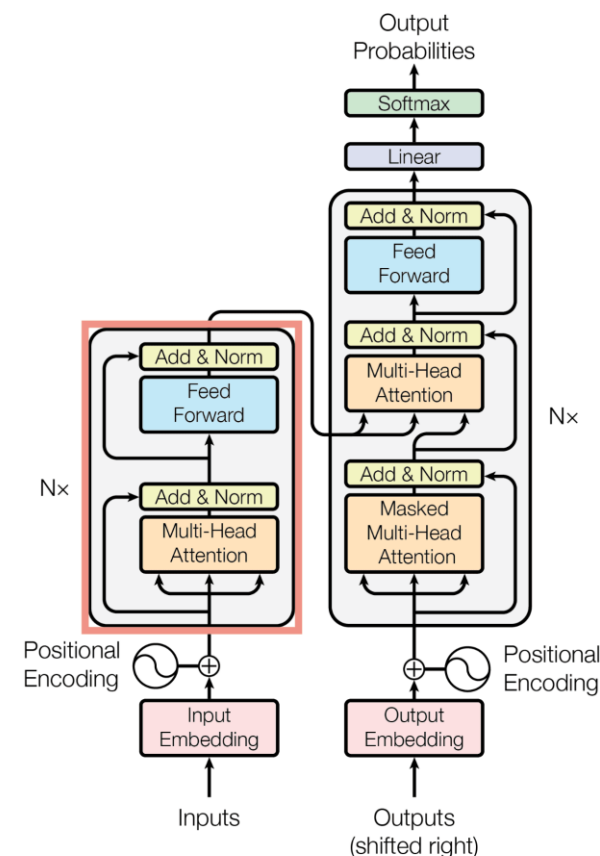
Computer Science > Computation and Language

[Submitted on 12 Jun 2017 (v1), last revised 2 Aug 2023 (this version, v7)]

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration and process the entire input sequence at once. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, without relying on positions or convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and efficient. On the French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a new model for the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing

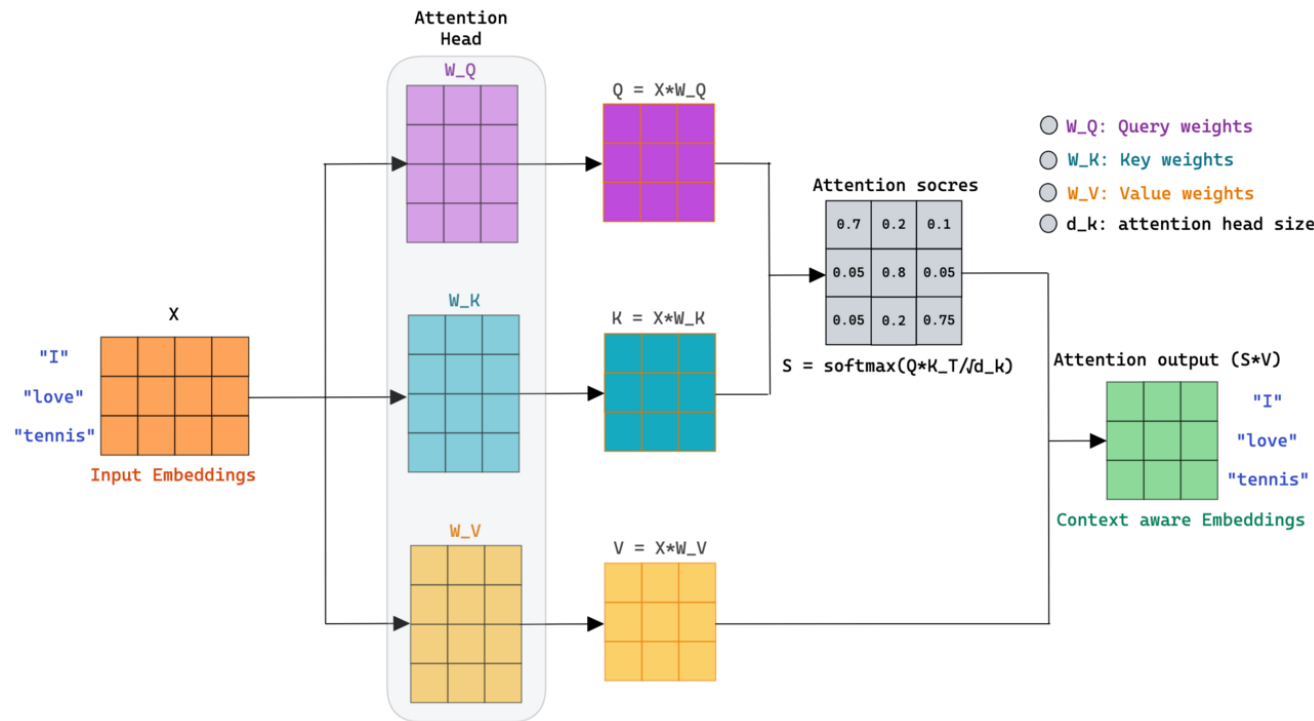


Self-Attention – Understanding Context

Models don't just read words individually; they understand context

Self-attention helps **capture relationships** between words in a sentence

The dog chased the ball, and **it** was fast



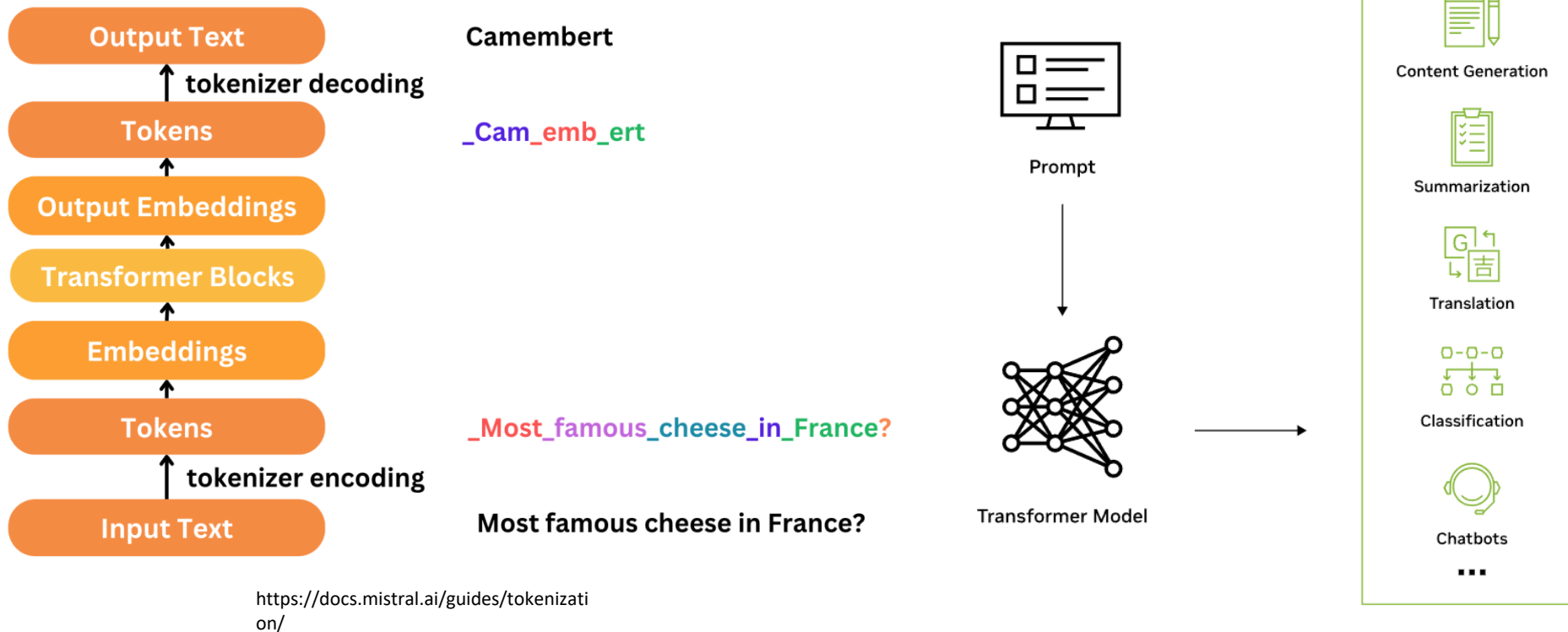
<https://mlspring.beehiiv.com/p/attention-need-self-attention-clearly-explained>

Tokenization

Text is broken into **smaller, manageable pieces** called **tokens**

Tokens are **words, subwords, or characters**

'I love AI' would be broken into **three tokens**: 'I', 'love', and 'AI'



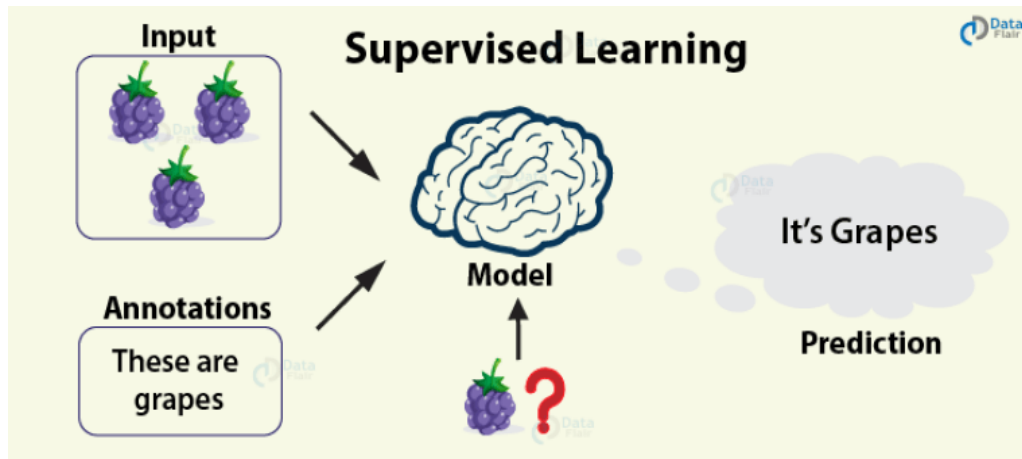
Self-Supervised Learning – Learning from Predictions

LLMs use training method called **self-supervised learning**

The model **learns by predicting the next word** in a sentence

No need for labeled data – text itself provides the supervision

Constantly **predicting what comes next**, model learns to **understand grammar, syntax** and even **deeper relationships**



<https://data-flair.training/blogs/types-of-machine-learning-algorithms/>

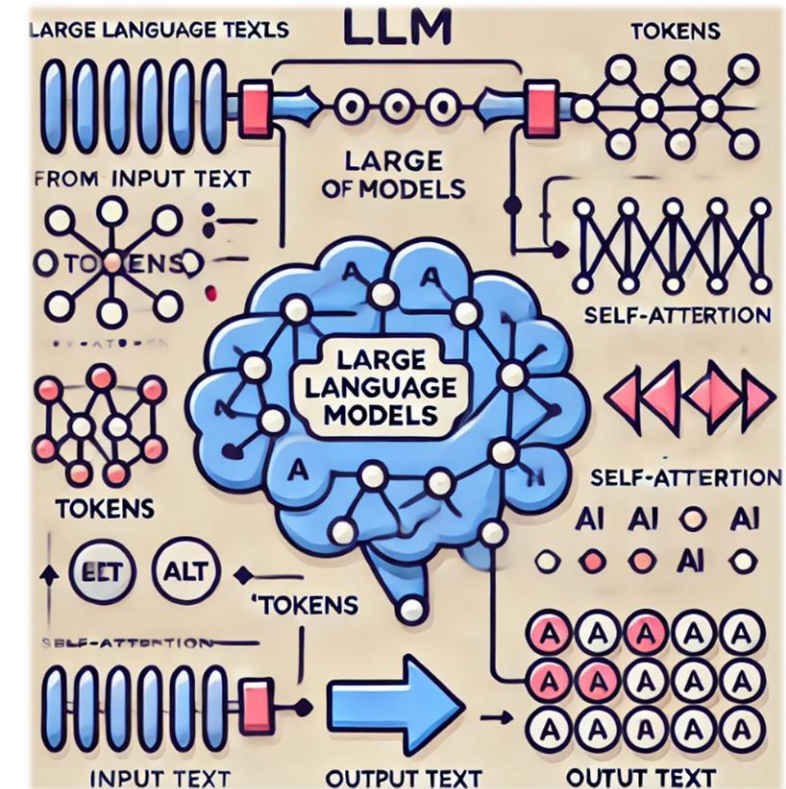
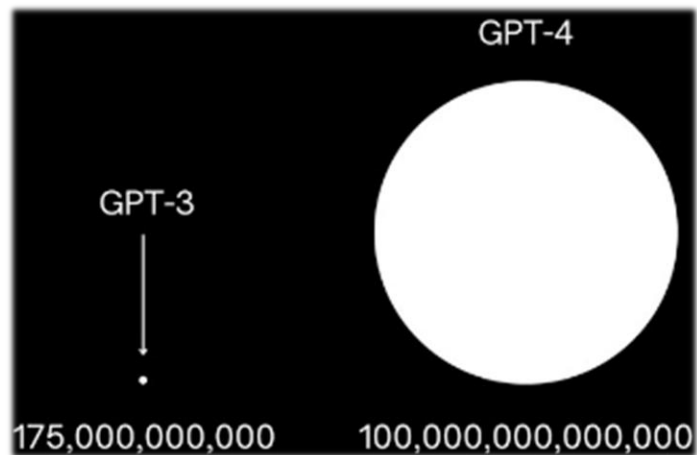
Bigger Models Perform Better

Larger models with **more parameters** perform better

Handle **more complex tasks** and generate more accurate, human-like responses

More parameters = more computational resources needed

GPT-4 require **significant hardware resources**, making them expensive to run, more **resource-intensive**



What is Token ?

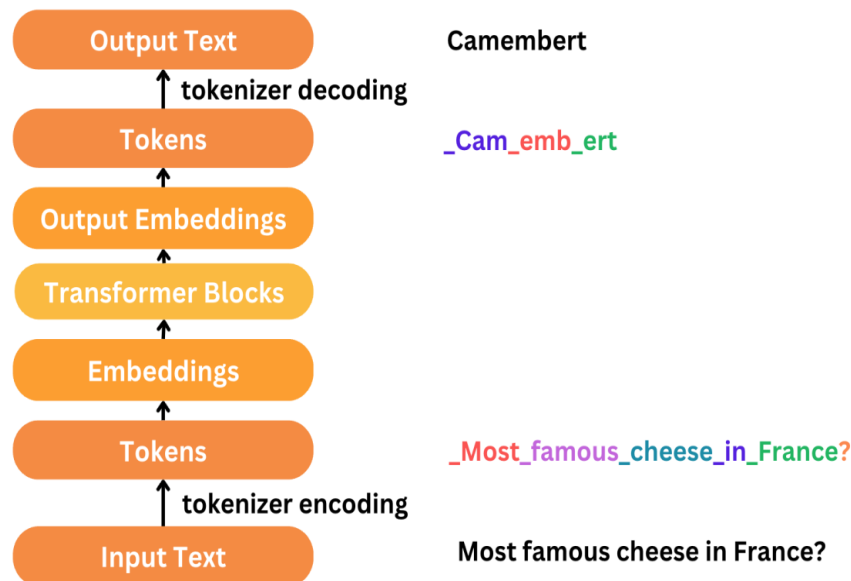
A **token** is a **small unit of text** that the model can understand

Token can be an **entire word**, a **punctuation mark**, or a **piece of a word**

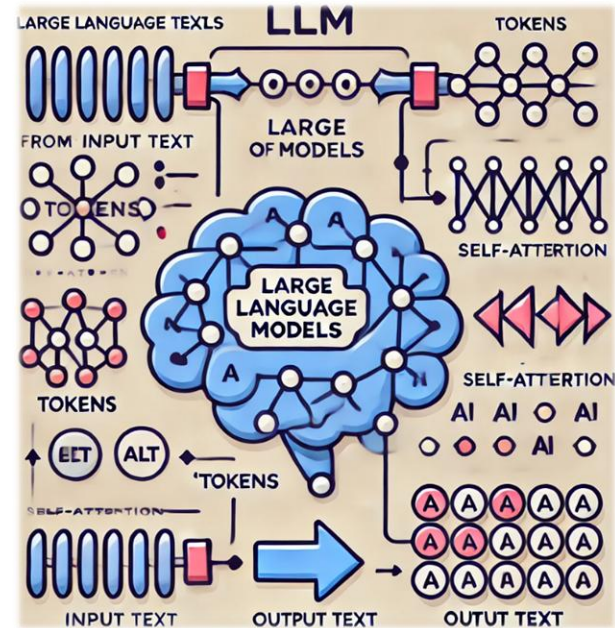
'**playing**' might be broken into **two tokens**: '**play**' and '**ing**'

'**Hello, how are you?**' might be split into tokens like '**Hello**', ',', '**how**', '**are**', and '**you**'

LLMs **don't read sentences** the way humans do; instead, they **rely on tokens** to process information



<https://docs.mistral.ai/guides/tokenization/>



Tokenization

Tokenization is the **process** of **splitting text into tokens**.

LLMs use **tokens** to **process** and **generate** responses.

Like chopping up a **long paragraph** into **bite-sized pieces** that the model can analyze

Tokens	Characters
63	278

The screenshot shows a tokenization tool interface. It displays four lines of text, each with colored segments representing individual tokens. Below the text, there are two tabs: 'TEXT' and 'TOKEN IDS'. The 'TEXT' tab is currently selected, showing the original text. The 'TOKEN IDS' tab would show the numerical IDs for each token.

What the Heck are Tokens?

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 000000

Sequences of characters commonly found next to each other may be grouped together: 1234567890

TEXT TOKEN IDS

Example of Tokenization

Input sentence: "Artificial intelligence is amazing. "

Tokenized into: ["Artificial", "intelligence", "is", "amazing", "."]

- 'Artificial'
- 'intelligence'
- 'is'
- 'amazing'
- '.'

OpenAI Tokenizer

<https://platform.openai.com/tokenizer>

You can use the tool below to understand how a piece of text would be tokenized by the API, and the total count of tokens in that piece of text.

GPT-3 Codex

Finding the right balance is crucial to ensure that the tokenizer effectively represents the text while keeping computational efficiency in check. GPT uses a custom tokenizer, and you can even try out how it works.

Clear Show example

Tokens	Characters
41	214

Finding the right balance is crucial to ensure that the tokenizer effectively represents the text while keeping computational efficiency in check. GPT uses a custom tokenizer, and you can even try out how it works.

TEXT TOKEN IDS

How LLMs Use Tokens

Tokens are processed **one by one**

The model understands patterns and relationships between tokens

Tokens are used to **generate new text**: 'Artificial' -> 'intelligence'

Predicts the next token based on the **sequence of tokens**

Tokens	Characters
63	278
What the Heck are Tokens? Many words map to one token, but some don't: indivisible. Unicode characters like emojis may be split into many tokens containing the underlying bytes: 000000 Sequences of characters commonly found next to each other may be grouped together: 1234567890	
TEXT	TOKEN IDS



As an AI language model, I have knowledge of various commonly used by journalists. Here are some of them:

1. Inverted pyramid: This technique involves starting with the beginning of the article and gradually moving to the end. The idea is to make sure that the reader gets the most important information first, so they don't have time to read the entire article.
2. Active voice: Using the active voice is a technique that makes writing more direct and engaging. It also helps to make the article more concise and easier to read.
3. Short sentences and paragraphs: Journalists often use short sentences and paragraphs to make the article easier to read and digest. This also helps to maintain the reader's interest and excitement.

How Tokenization Affects Model Usage

LLMs have a **token limit** for input and output

Large inputs need to be **broken down** into **smaller chunks** of **tokens**

If a model has a **limit** of **4,000 tokens**, it can **only process** a combination of **4,000 tokens** from the **input** and the **output**

Understanding **token limits** helps **avoid truncation**

```
color_fix_type, linear_CFG, linear_s_stage2, spt_linear_CFG, spt_linear_s_stage2))
submit_button.click(fn=submit_feedback, inputs=[event_id, fb_score, fb_text], outputs=
[fb_text])
if args.port is not None: # Check if the --port argument is provided
    block.launch(server_name=server_ip, server_port=args.port, share=args.share, inbrowser=True)
else:
    block.launch(server_name=server_ip, share=args.share, inbrowser=True)
```

 ChatGPT

The message you submitted was too long, please reload the conversation and submit something shorter.

LLM	TOKEN LIMIT
GPT-4 Turbo	128,000
GPT-4	8,192
GPT Vision	128,000
GPT 3.5 Turbo	16,385
Llama3	8,000
Claude3	200,000

Free

Tier 1

Tier 2

Tier 3

Tier 4

Tier 5

Tier 1 rate limits

This is a high level summary and there are per-model exceptions (models with larger context windows have different rate limits). For more information, visit the [limits](#) section of your account settings.

MODEL	RPM	RPD	TPM
gpt-4	500	10,000	10,000
gpt-4-1106-preview*	500	10,000	150,000
gpt-4-vision-preview*	80	500	10,000
gpt-3.5-turbo	3,500	10,000	60,000

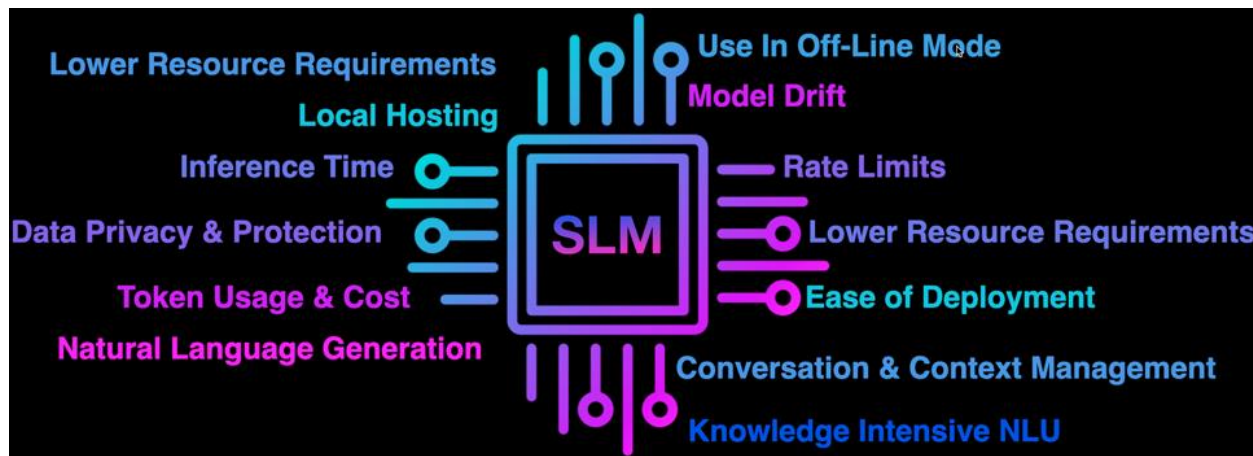
Small Language Models (SLMs)

SLMs are smaller, more efficient and scaled-down versions of **LLMs**

Retain key functionalities like text generation, classification, language understanding but with **significantly fewer parameters**

This makes them **faster, cheaper, and more efficient** to run

SLMs operate with **millions of parameters**, ideal for **scenarios** where **resources are limited** or where we need **faster, real-time performance**



Why Use Small Language Models (SLMs) ?

Efficient with **lower computational requirements**, optimized for **speed** and **lower resource consumption**

Suitable for **real-time apps**, **mobile**, **edge devices**, and specific tasks

ChatGPT 4o-170 trillion, **Gemini-540** billion, **Llama-70** billion parameters but SMLs are **a few billion** parameters

Cost-effective compared to LLMs, don't need the full capabilities of an LLM, require some level capabilities

Category	Benchmark	Phi-3-Small		Gemma-7b	Mixtral-8x7b	Llama-3-8B-Instruct	GPT3.5-Turbo-1106
		Phi-3-Small-8K-In	Phi-3-Small-128K-In				
Popular Aggregate Benchmarks	MMLU (5-shot)	75.7	75.5	63.6	70.5	66.5	71.4
Language Understanding	HellaSwag (5-shot)	77.0	79.6	49.8	70.4	71.1	78.8
	WinoGrande (5-shot)	81.5	80.1	55.6	62	65	68.8
Reasoning	Social IQA (5-shot)	79.2	78.7	65.5	75.9	73.9	68.3
	TruthfulQA (MC2) (10-shot)	70.2	69.6	52.1	60.1	63.2	67.7
	MedQA (2-shot)	65.4	46.3	49.6	62.2	60.5	63.4
Factual Knowledge	TriviaQA (5-shot)	58.1	66.0	72.3	82.2	67.7	85.8
Math	GSM8K CoT (8-shot)	89.6	87.3	59.8	64.7	77.4	78.1
Code generation	HumanEval (0-shot)	61.0	59.1	34.1	37.8	60.4	62.2
	MBPP (3-shot)	71.7	70.3	51.5	60.2	67.7	77.8



<https://datasciencedojo.com/blog/small-language-models-phi-3/>

Use Cases for Small Language Models (SLMs)

Edge Computing and IoT Devices

SLMs enough to run on smart devices, embedded systems, or other hardware with limited processing power

Mobile Applications

For real-time interactions, like virtual assistants on smartphones or voice-based commands

Customer Support Chatbots

Chatbots handle predefined queries where an LLM might be overkill, SLMs handle without the extra cost or resource consumption

Offline applications for personalization

Offline personal assistant run on a smartphone can still provide personalized recommendations without needing to connect to the cloud

Ensures data privacy since the processing happens locally



What is a Prompt ?

The **input** or **instruction** you give to an **LLM** to **guide** its **response**

It could be a **question**, a **command**, or even **just a phrase**

The way you **phrase your prompt** plays a **huge role** in **how the model responds**

Prompt is like a **set of directions** that you give the **LLM**

Answer a question, **summarize a text**, **generate a story** or even providing **code snippets**



Understanding a Prompt

We can think of **prompts** as **programs** for LLMs

How a **traditional program** gives **instructions** to a computer to perform a specific task

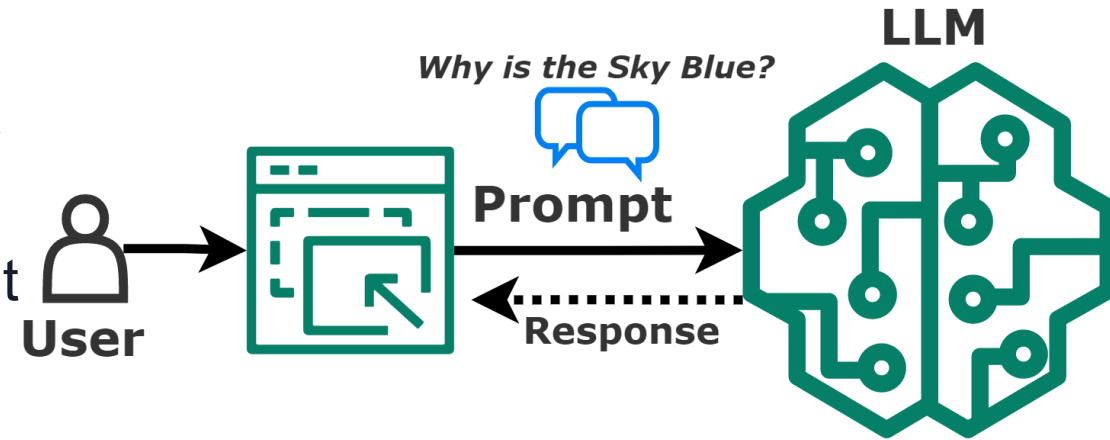
Tells the LLM how to respond based on your input

Designed to get the model to **produce responses** that **meet** your **specific requirements**

The **better your prompt**, the **better the model's response**

The **foundation** of the **interaction** with the LLM

Its **structure** and **clarity** directly **influence** the **quality** of the **response**



Example of a Simple Prompt

Prompt: "Explain the process of photosynthesis"

Clear and direct prompts lead to **better answers**

Different types of prompts we can use, **depending on** what we want the model

Questions

You ask the model for information, like 'What is the capital of Japan?'

Commands

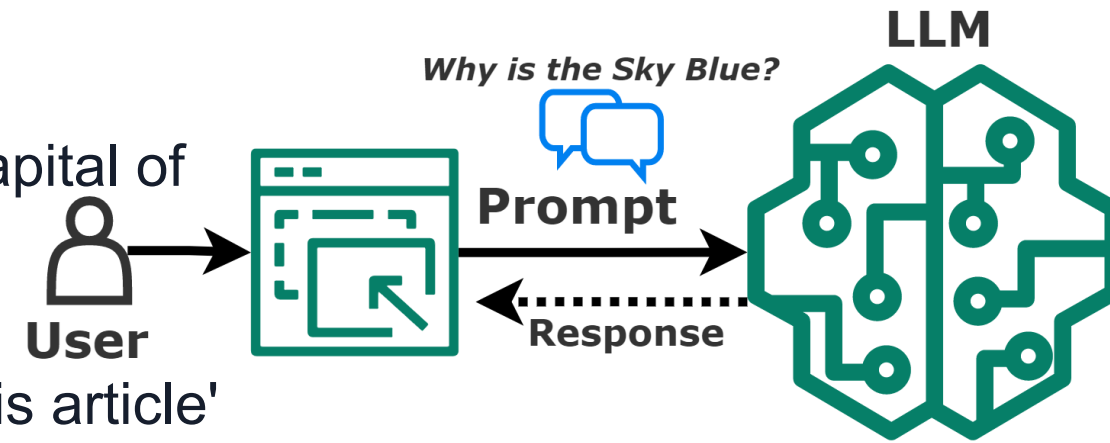
You give the model an instruction, like 'Summarize this article'

Fill-in-the-blank

You start a sentence and leave space for the model to complete it, such as 'The weather today is ____.'

OpenAI Prompt Examples:

<https://platform.openai.com/docs/examples>



Elements of a Prompt

Instruction or Task

The core request for the model, a specific task or instruction you want the model to perform

Context

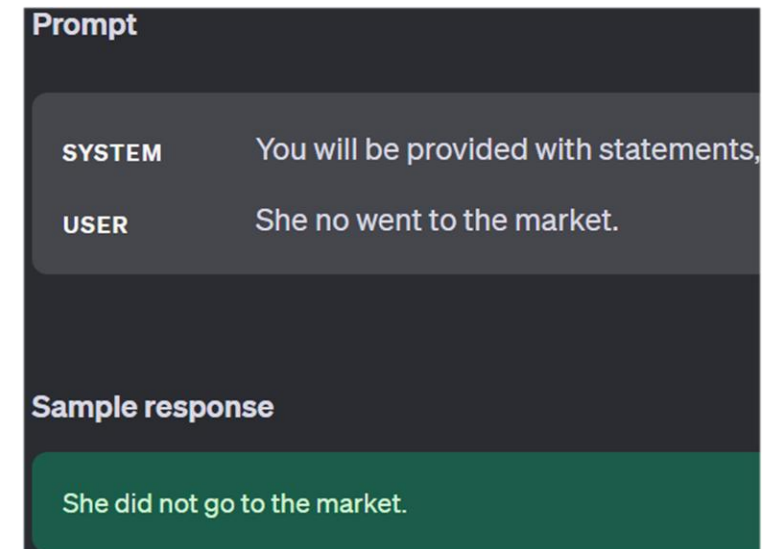
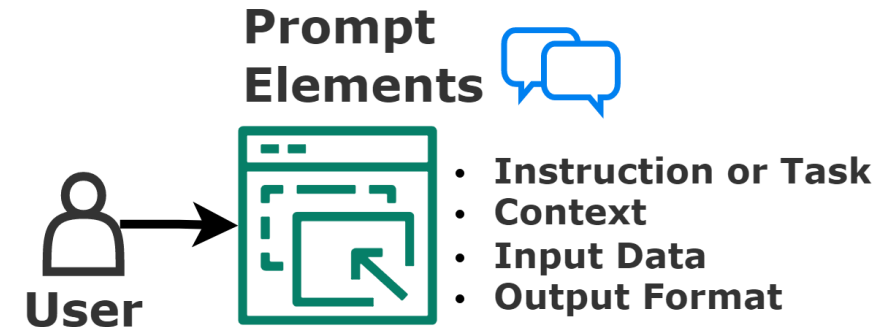
Background information to guide the response, external information or additional context

Input Data

The input or question that we are interested to find a response for

Output Format

Defines how the response should be structured, the type or format of the output



Examples - Elements of a Prompt

Instruction or Task

You will be provided with statements, and your task is to convert them to standard English

Context

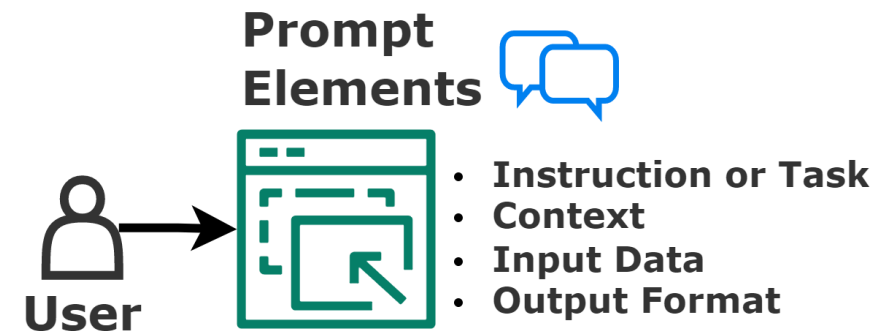
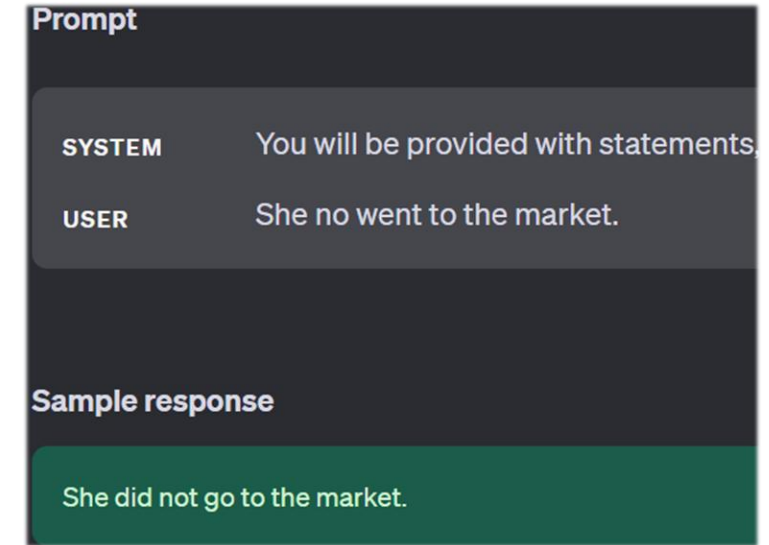
You are a customer service representative. Explain the return policy to a customer who is unhappy with their purchase

Input Data

The customer purchased a smart speaker but reported that it is not connecting to their Wi-Fi. They have tried restarting both the router and the speaker, but the issue persists

Output Format

Provide a list of the top 5 features of this product
Write a 2-paragraph summary of the article
Create a table comparing solar energy



Roles of Prompts – System and User Roles

System Role

Sets up the model's behavior and environment, defines guidelines for the entire conversation

*You are an assistant with expert knowledge in cybersecurity.
Provide detailed explanations and advice*

User Role

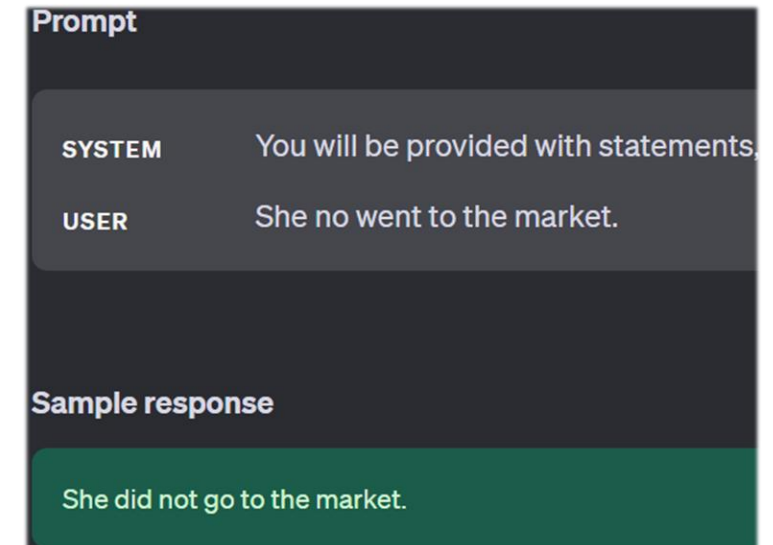
The specific instructions or questions you give the model

What are the best practices for protecting personal data online?

Both roles **influence how the model interprets and generates responses**

Open AI Prompt Examples

<https://platform.openai.com/docs/examples>



What is Prompt Engineering ?

Crafting and **refining prompts** to improve model responses

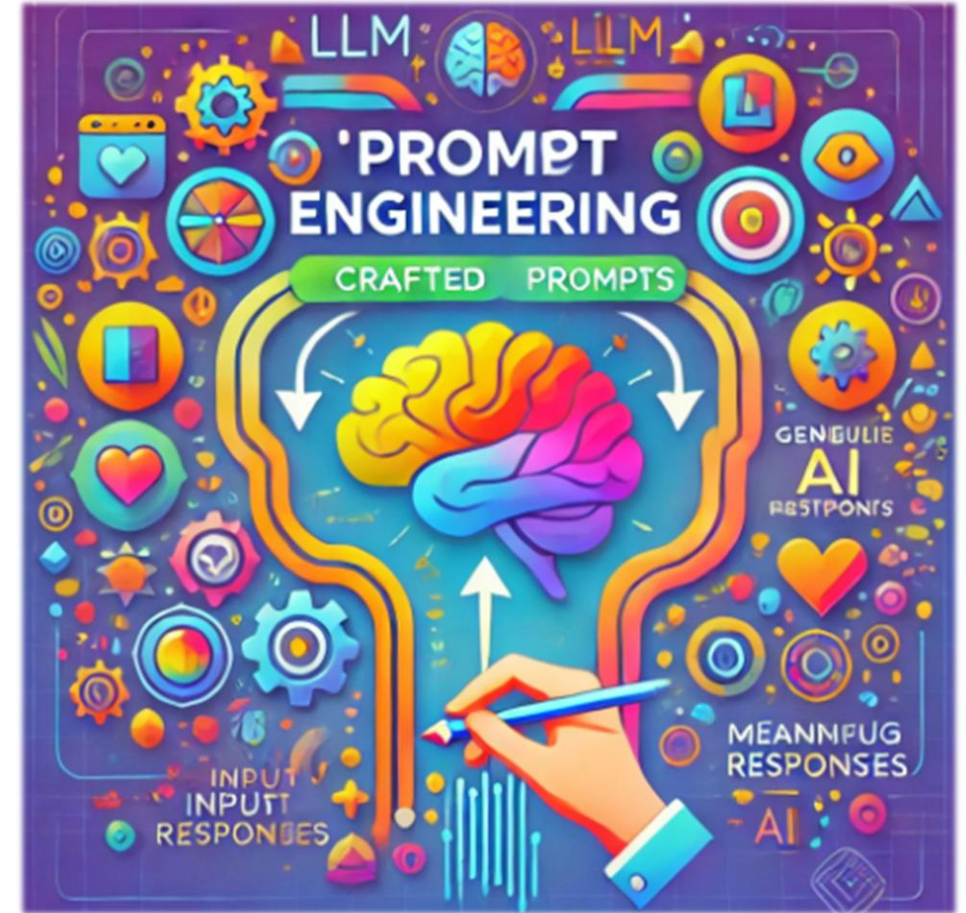
Thinking strategically to **guide** the **model** in the **right direction**

Involves **experimenting** and **iterating** with **different wording, structure and styles**

The **first prompt** you use **won't give** you the **best result**

Refine it through **iteration** to **optimize** the response

Maximize the **performance** of the model and get answers that **meet your specific needs**



Why is Prompt Engineering Important ?

Helps you get more **accurate** and **relevant** responses

Allows you to **shape** the **model's** behavior

You can **design prompts** to respond in a **specific tone**, such as **formal**, **friendly** or **technical**

Reduces vague or **incorrect outputs** by clarifying your instructions

There are **different types** of **prompts**: Zero-shot, Few-shot and One-shot



Example of Prompt Engineering in Action

Initial Prompt

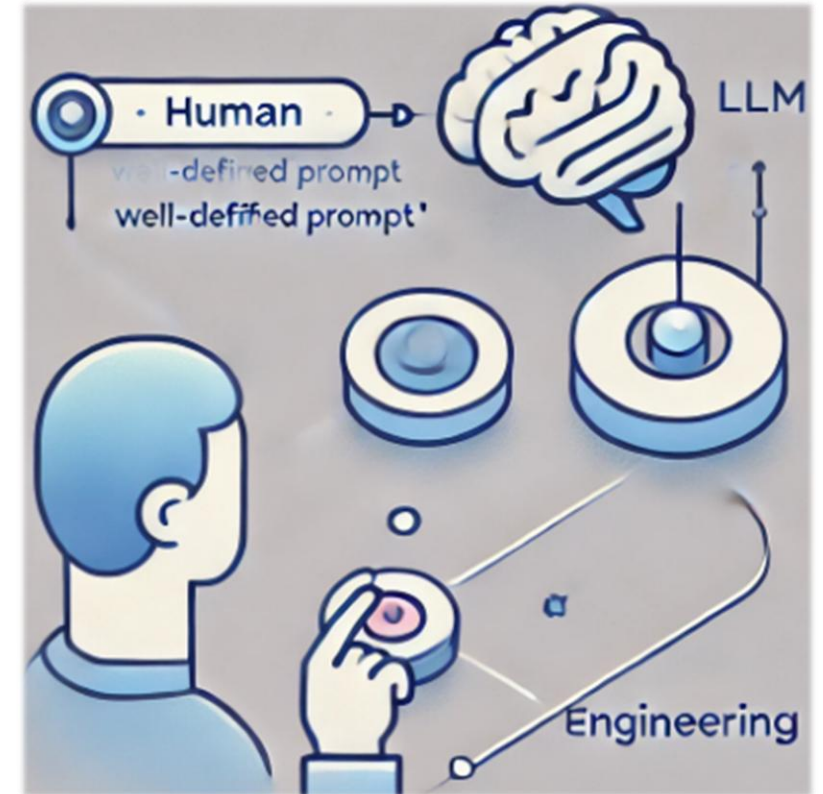
Tell me about space exploration

Improved Prompt

Summarize the history of space exploration, focusing on the key milestones from the 1960s to today in 100 words

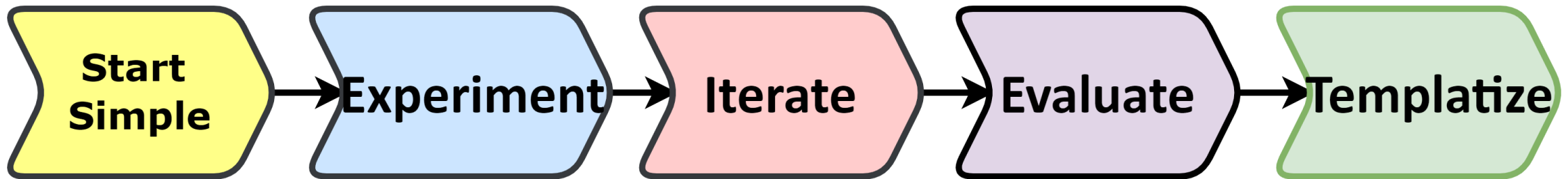
Prompt Engineering Actions:

- Narrowed the focus to key milestones
- Added a timeframe (from the 1960s to today)
- Given a word limit (100 words)



Steps to Effective Prompt Engineering

1. Start simple
2. Experiment
3. Iterate
4. Evaluate
5. Templatize



1. Start Simple

Begin with a **simple prompt** to see the model's response

Avoid overcomplicating at the start, test the model's baseline performance

Start simple prompt like **'Summarize this article'**

Once you see the **model's response**, you can then **refine** or **add complexity** as needed

Understand **how the model interprets** your instructions and gives you a **foundation**

Next, **be specific** and **direct** with your **prompts**

'Tell me about space' → 'Explain the history of space exploration in 200 words'

**Start
Simple**



2. Experiment

Try different **structures** and **wording** to see how the model's output changes

'**Explain**' vs '**Summarize**' or '**Compare**'

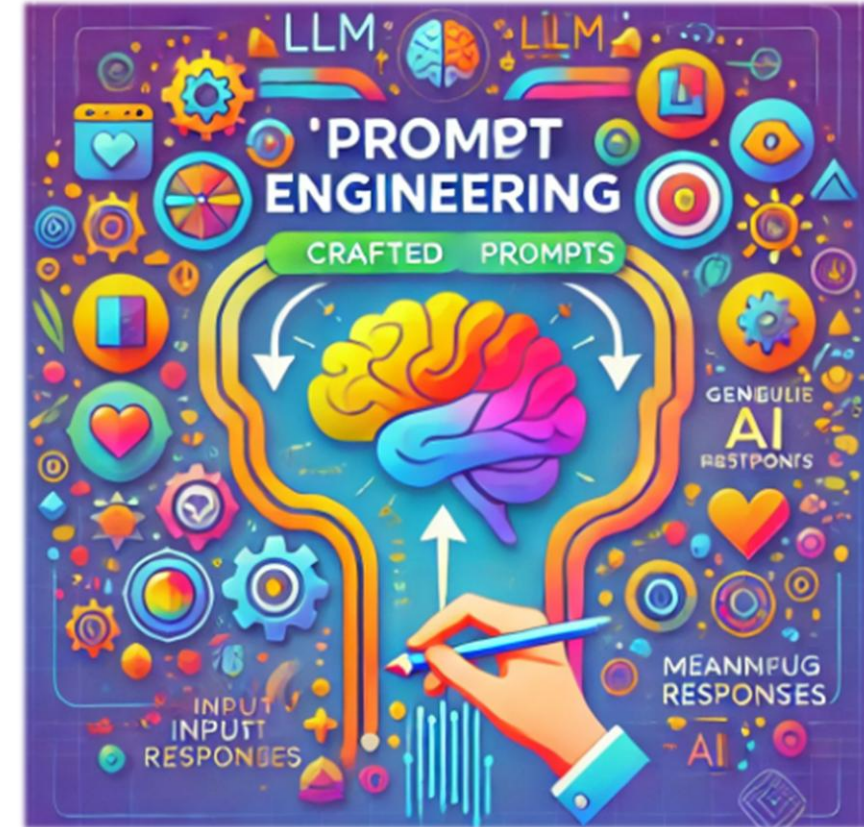
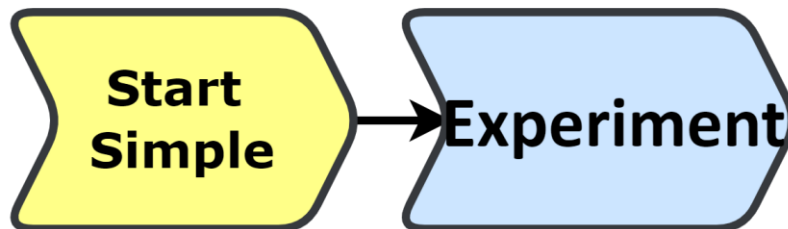
Few-shot prompting **gives examples** for the model to follow

Include one or two examples of what a good product description looks like

Use **Constraints** and **Parameters**, **Constraints** guide the **length, tone** or **format** of the response

Word Limit: Summarize this article in **150 words**

Tone: Write an email response in a **professional tone**



3. Iterate

One of the most important strategies is to **test** and **iterate** with **different inputs**

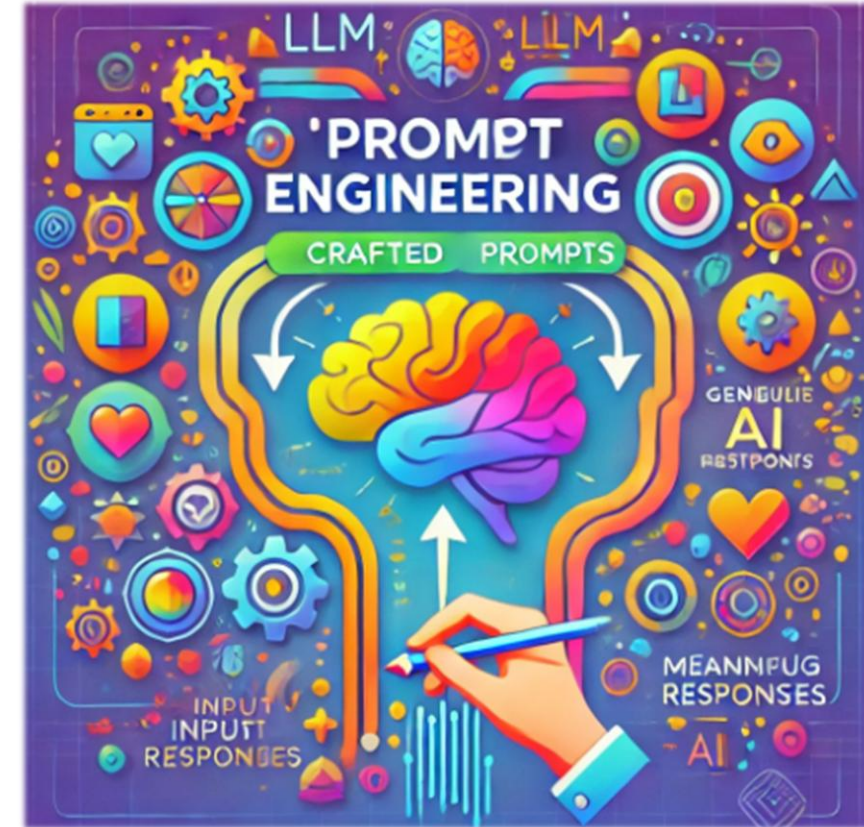
Review and **adjust** your **prompt** as needed

If the **response** is **too broad**, make your prompt **more specific**

Iteration is key to **improving** the **quality** of **responses**

First Prompt: Write a blog post about renewable energy

Refined Prompt: Write a 300-word blog post about the benefits of renewable energy, focusing on solar and wind power



4. Evaluate

Determine if the prompt has **successfully guided** the model

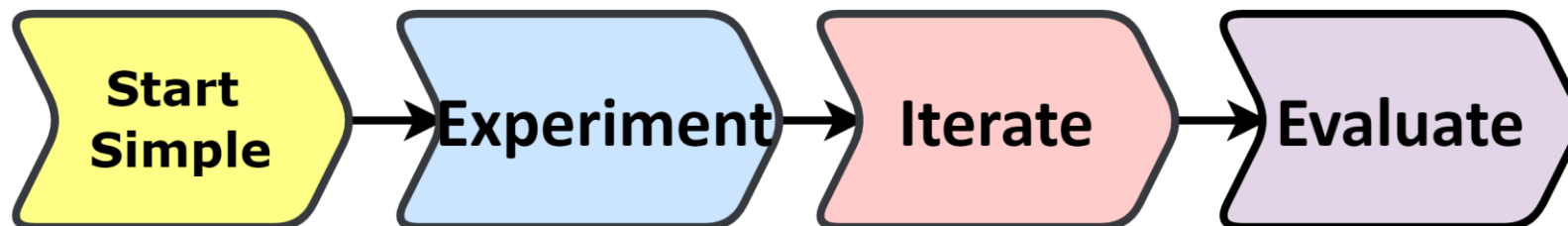
Does the **response meet the requirements** of the task?

Is it **accurate, relevant, and clear**?

Are there **any points** where the **model strayed** from the topic or didn't provide **enough detail**?

You may **need** to **further adjust** the **prompt**

Maybe the **instructions weren't clear enough** or you need to add **more constraints** or **examples**



5. Templatize

Create a **reusable structure for prompts** that can be applied to **similar tasks**

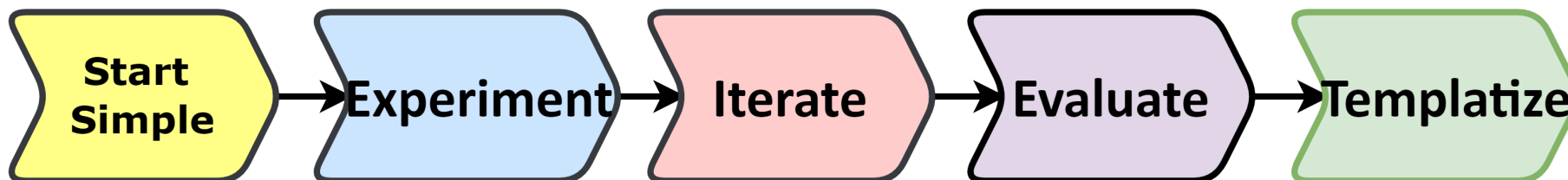
Useful when doing **repetitive tasks** like summarize articles, generate reports, answer questions

Templates save time and **ensure consistent outputs**

Summarize the **[topic]** in **[word count]**, focusing on **[key points]**

'Summarize **climate change** in **150 words**, focusing on **sea level rise**'

'Summarize **artificial intelligence** in **200 words**, focusing on **ethical concerns**'



.NET + AI ecosystem: AI Development Tools and Libraries for .NET

Why choose .NET to build AI applications ?

AI frameworks and SDKs for .NET

Unified AI building blocks: Microsoft.Extensions.AI (MEAI)

Semantic Kernel for .NET

Mehmet Ozkaya

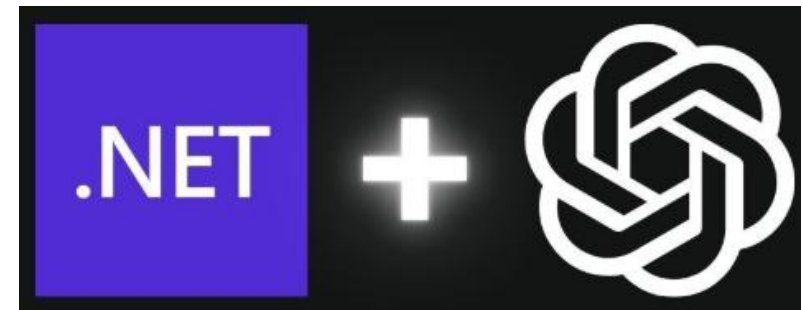


Introduction to AI with .NET

With **.NET**, you can **integrate AI** into your apps using **tools**, **libraries** and **services** you're **already comfortable with**

Devs **rely on .NET** to **build performant apps** that run on the **web**, **mobile devices**, **desktops** or in the **cloud**

Using **AI** will **supercharge these apps** with capabilities like **automation**, **intelligent decision-making** and **interactive experiences**



Benefits of Building AI Apps with .NET

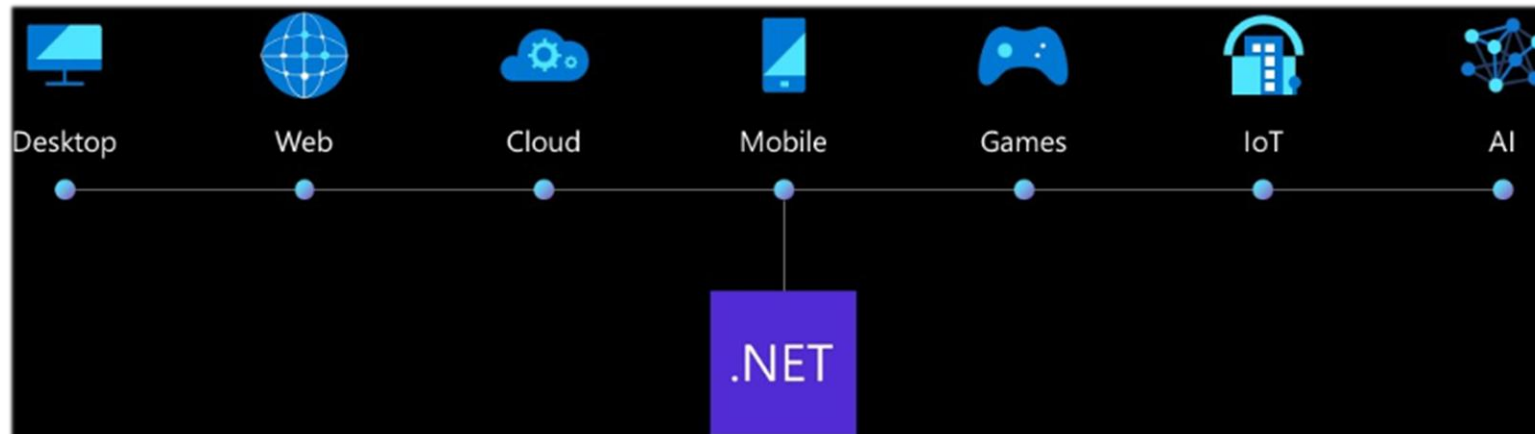
Unified dev environment to create **web front-ends**, **REST APIs** and **entire apps** under one umbrella

Build, deploy and run AI apps on **Cross-Platform** (Windows, macOS or Linux)

Ecosystem is open-source, supported by an **active community**, Many **AI libraries** are **developed** collaboratively

Integration with popular **Cloud Platforms**, **deploy LLMs** in **containers**

Visual Studio and **VS Code** provide robust **debugging**, **IntelliSense** for AI libraries



What Can You Build with AI and .NET ?

Language Processing

Build chatbots that can understand and respond to user queries, or create assistants that generate new content

Computer Vision

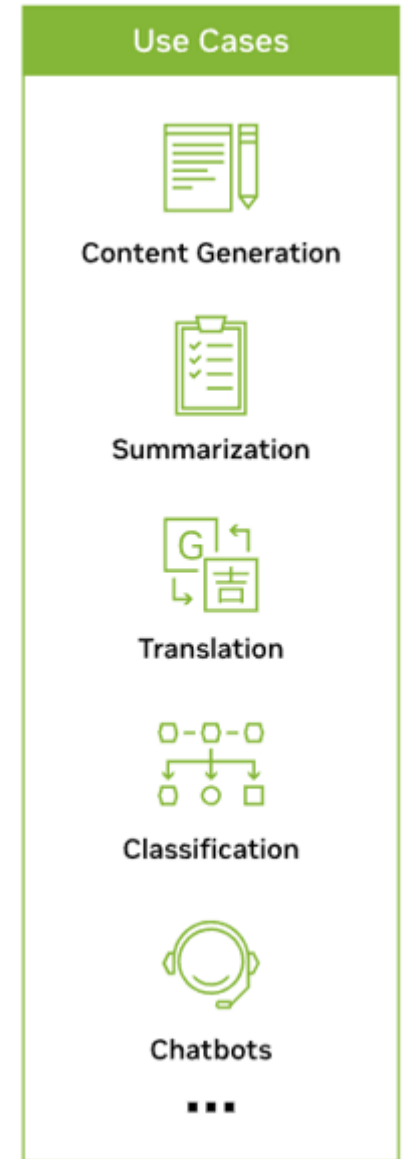
Integrate models to identify objects in images or videos—useful for surveillance, inventory management

Audio Generation

Synthesized voices to interact with customers in a more natural way, creating voice-based assistants or audio notifications

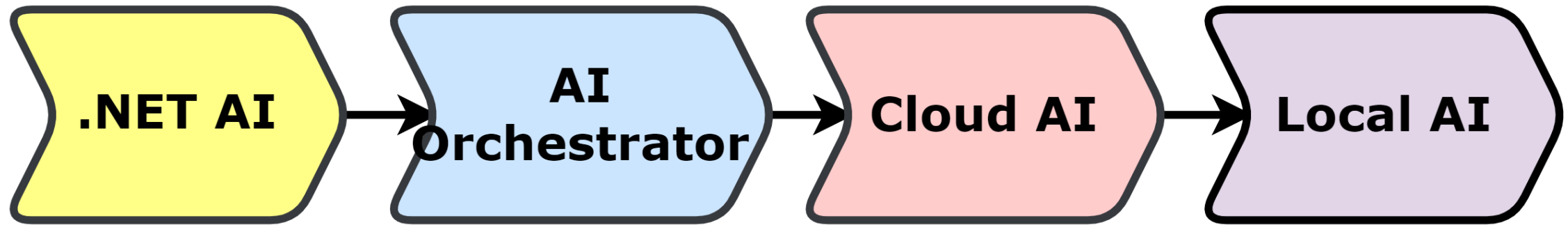
Classification & Prediction

Predict the severity of a customer-reported issue or categorize product information



AI frameworks and SDKs for .NET

.NET offers a rich set of libraries and SDKs for integrating AI into your apps



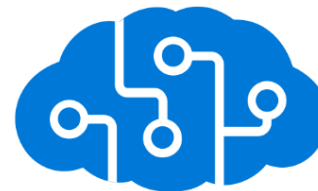
**Microsoft
.Extensions.AI**

**Semantic Kernel
Autogen**

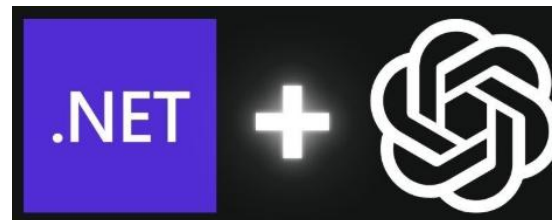
**Azure AI
OpenAI Platform
AWS Bedrock**

Ollama

**Introducing
Microsoft.Extensions.AI**



Ollama



AI frameworks and SDKs for .NET

Microsoft.Extensions.AI Libraries (MEAI)

Collection of libraries is designed to simplify AI integration, consistent way to interact with different AI services

Semantic Kernel for .NET

Structure complex AI workflows, combine multiple AI services and data sources w/ Plugins and Extensions

.NET SDKs for OpenAI Models

Official packages that make requests to OpenAI's GPT and embedding endpoints

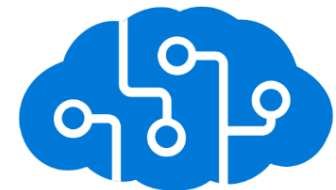
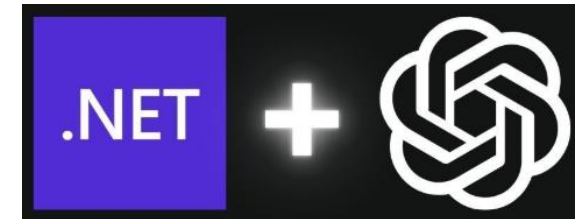
.NET SDKs for Azure AI Services

Includes vision, speech, language understanding, leverage pre-trained models

Develop with Local AI Models

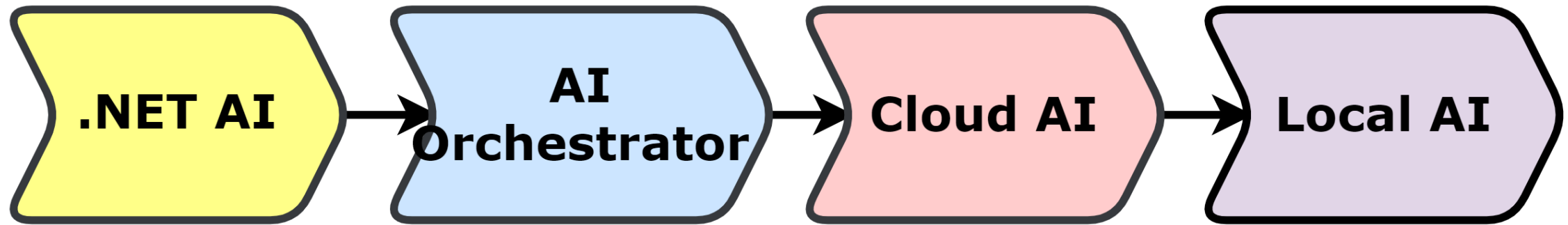


Introducing
Microsoft.Extensions.AI



AI frameworks and SDKs for .NET

.NET offers a rich set of libraries and SDKs for integrating AI into your apps



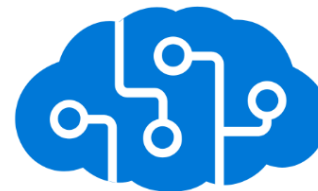
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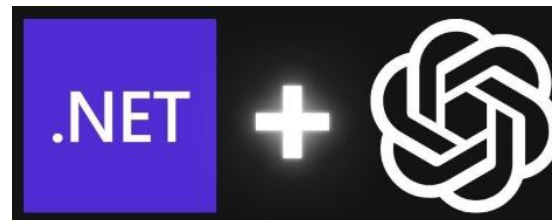
Azure AI
OpenAI Platform
AWS Bedrock

Ollama

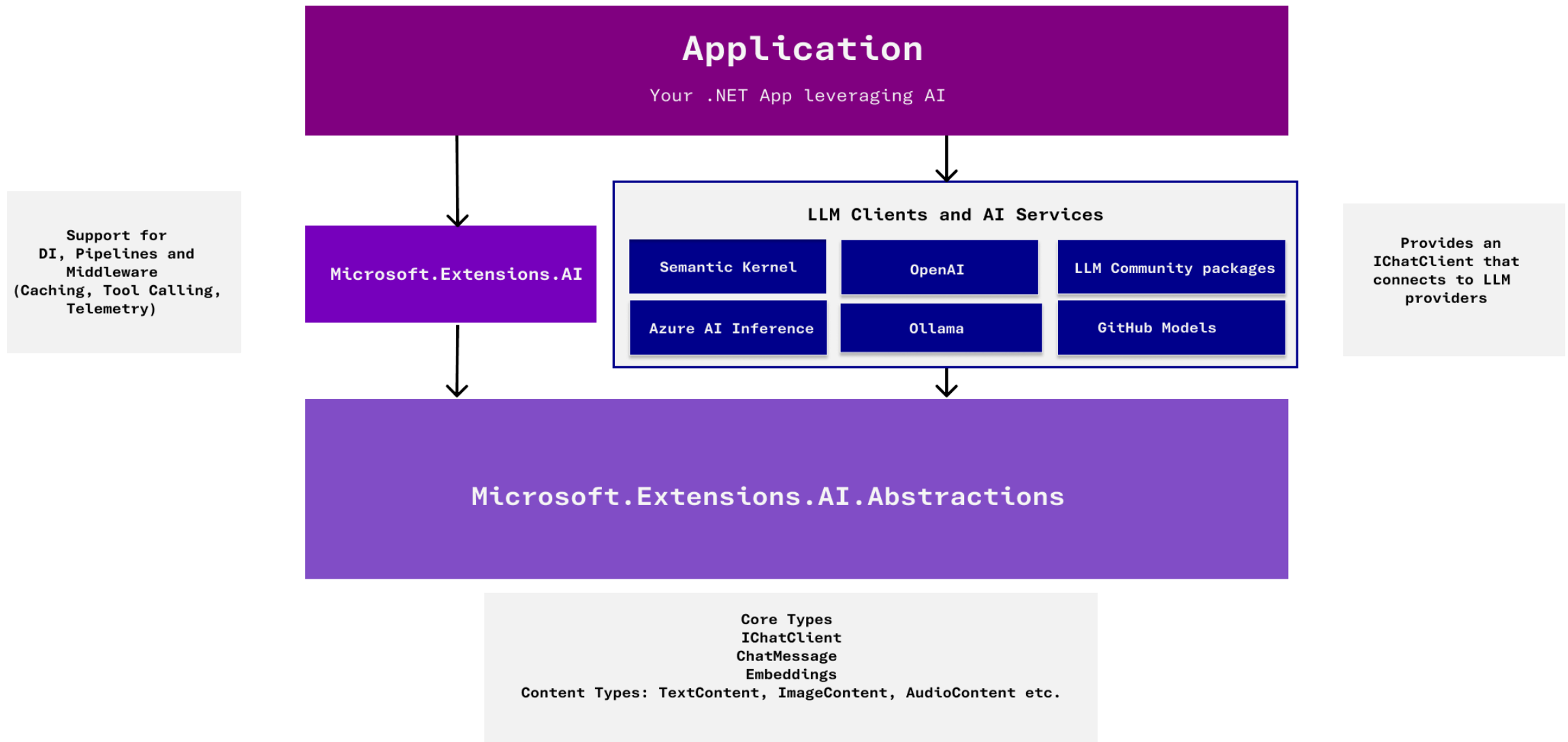
Introducing
Microsoft.Extensions.AI



Ollama



Microsoft.Extensions.AI Structure

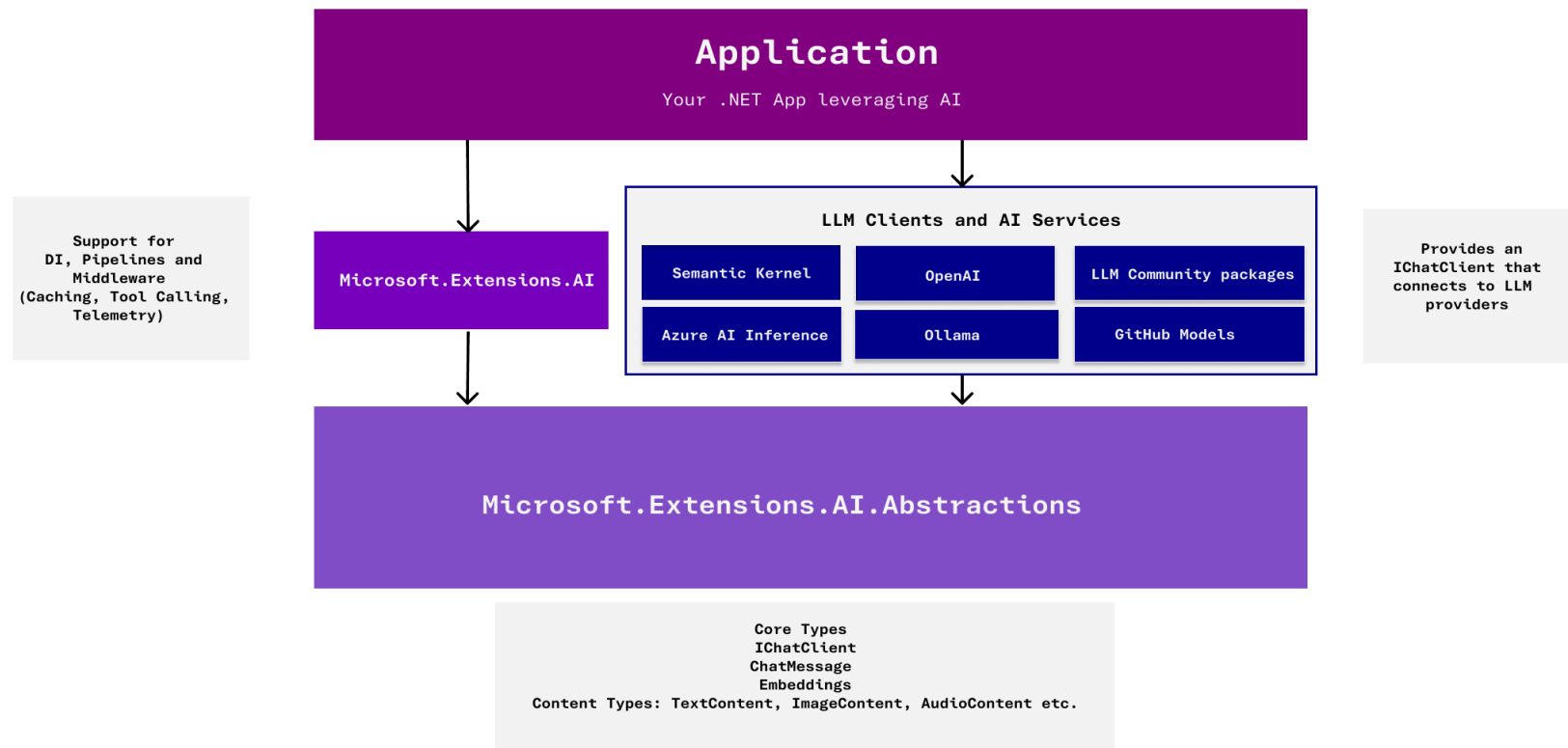


Work with Abstractions for Common AI Services

Chat: Conversational AI or ChatGPT-like interactions

Embeddings: Vector search & semantic understanding

Tool calling: Orchestrate AI requests with external services

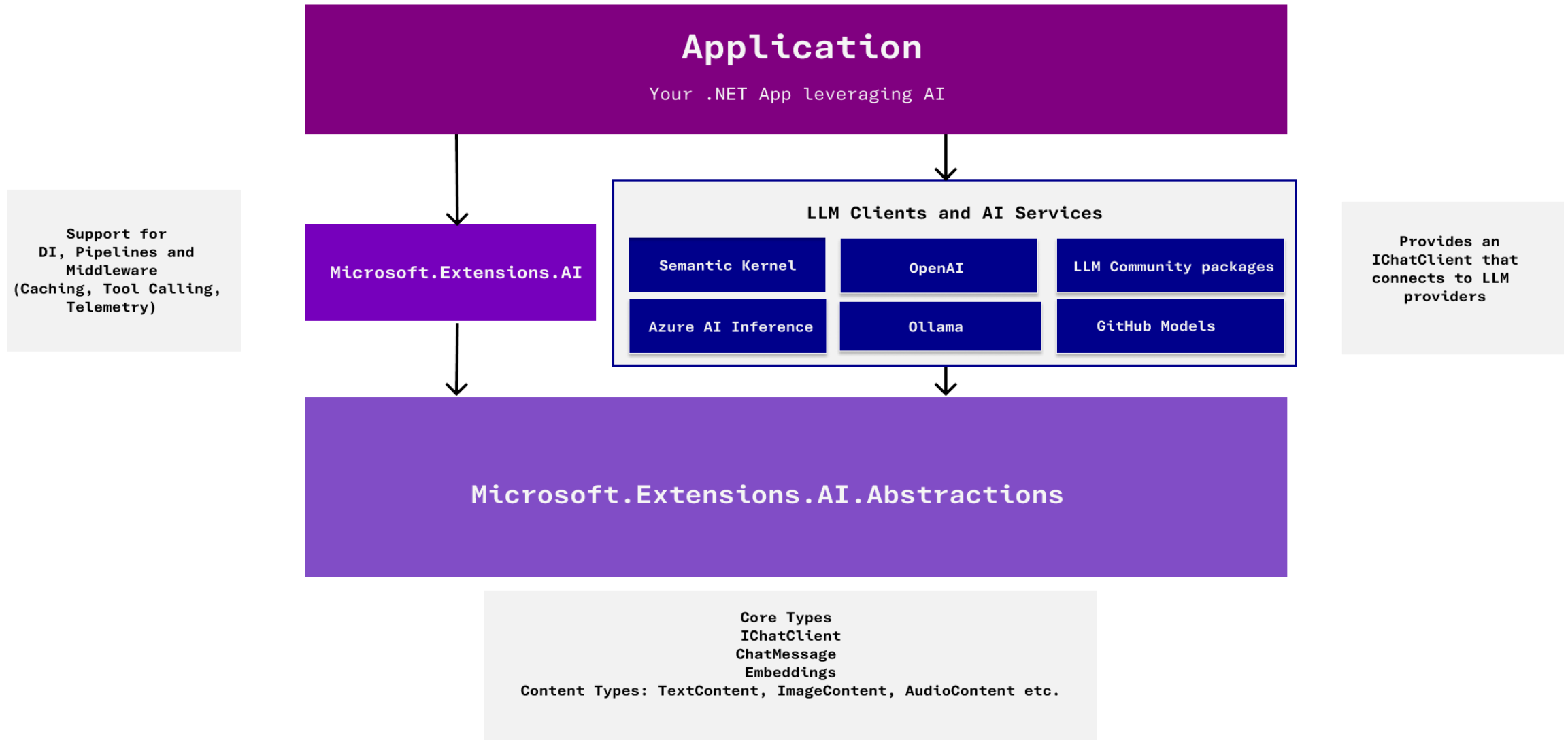


Example: IChatClient



```
IChatClient client =  
    environment.IsDevelopment  
        ? new OllamaChatClient(...) : new AzureAIInferenceChatClient(...);  
  
var response = await client.GetResponseAsync(messages, options, cancellationToken);
```

Microsoft.Extensions.AI Structure



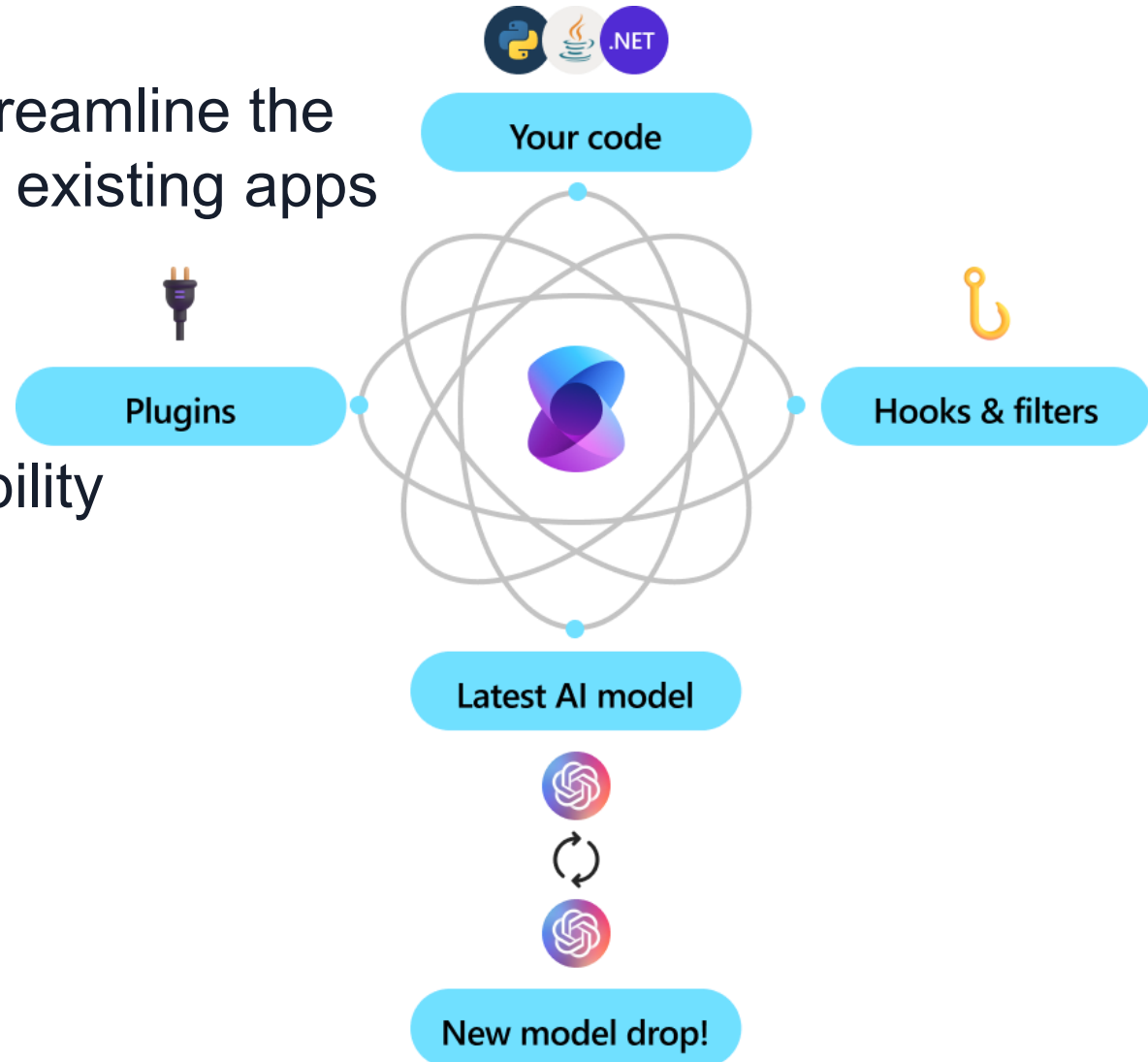
Semantic Kernel (AI Orchestration Framework)

Open-source SDK designed to streamline the **integration of AI capabilities** into existing apps

Streamlining AI Integration

Minimizing the Learning Curve

Improving Reliability and Predictability



Core Concepts in Semantic Kernel

Connections

Bridges between your application code and external AI services

Plugins

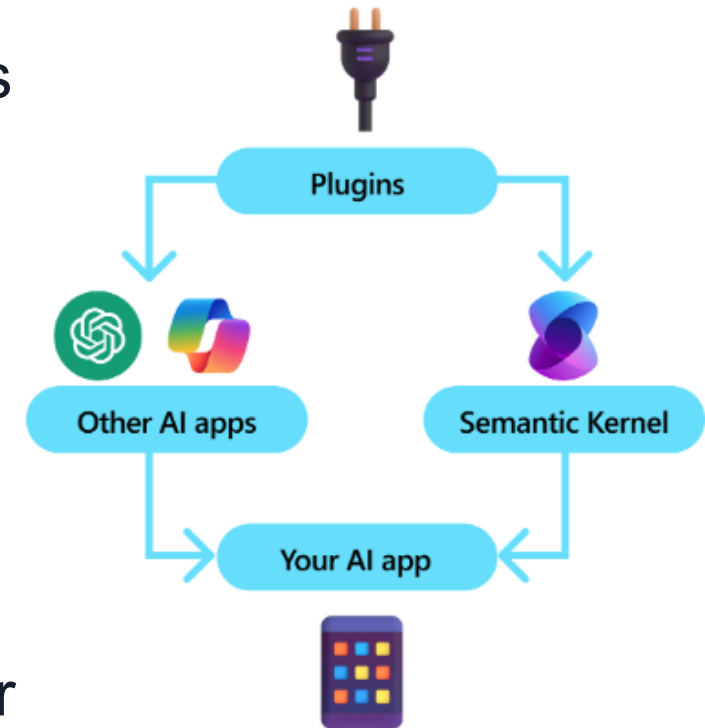
Encapsulate functionalities that your AI application might need

Planner

Orchestrates user requests by dynamically calling the right plugins and AI models

Memory

Manages context and stored data for AI apps, Leverages vector databases or caches



Setup LLM Providers: GitHub Models, Ollama, Azure AI Foundry

Azure OpenAI from Azure AI Foundry and OpenAI platform

GitHub Models

Create Access Token for GitHub Models

Ollama – Run LLMs Locally

Download Ollama and llama3.2, all-minilm models with Docker

Mehmet Ozkaya



Which AI provider should I use ?

Paid Services



Free Services



Factors to Consider When Choosing an AI Provider

Cost and Subscription

If budget aren't an issue choose Azure AI services. Or start with GH Models and easily shift to Azure AI services for production. With changing API keys only

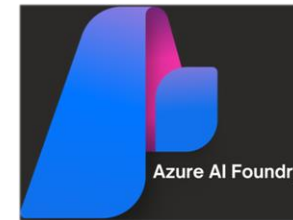
Performance and Scalability

Azure OpenAI typically provides high availability and can scale easily. Ollama relies on your local hardware

Ease of Setup

GitHub Models can be integrated quickly. Ollama setup might be more involved initially

Paid Services



OpenAI

Free Services

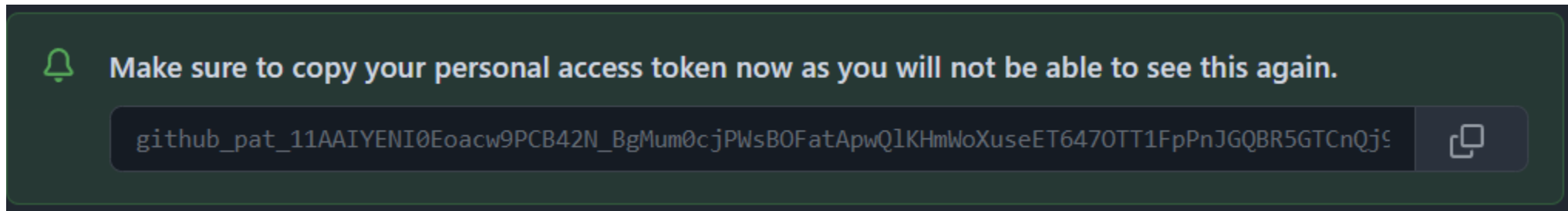


Recommended Options for This Course

Free Services



GitHub Access Token – Save Securely



Alternatively run the following commands to configure your OpenAI API key as a secret for the sample app:

```
dotnet user-secrets init
```

```
dotnet user-secrets set GitHubModels:Token <your-openai-key>
```

Ollama – Run LLMs Locally

Ollama is a platform to **run large language models (LLMs)** or **small language models (SLMs) locally**

Offers **private, secure** AI solutions **without requiring the cloud**

Ideal for **developers** and **businesses** seeking **offline AI** capabilities w/ **privacy, low latency**, and **control** over AI models

Excellent choice for those looking to **integrate AI without depending** on constant **internet access**



Key Features of Ollama

Local Execution

Run LLMs and Small Language Models (SLMs) directly on your device

Pre-built Models

Includes optimized models for coding, chat, creative tasks, and more

Privacy-First

All data remains on your machine, protecting sensitive information

Customization

Allows model fine-tuning and adaptation for specific needs

Low Latency

Responses are quick without network dependency



Available Models on Ollama

Variety of Models: **llama**, **gemma**, **qwen**, **phi**, **mistral**

Code Generation Models-codegemma,codellama

AI assistants specialized in code generation

Creative Models-llava

Text-to-image, story generation, and poetry models

Domain-Specific Models-medllama

Finance, healthcare, and other industry-specific LLMs



Models

Filter by name...

llama3.2

Meta's Llama 3.2 goes small with 1B and 3B models.

[Tools](#) [1B](#) [3B](#)

↓ 1.5M Pulls 🔖 63 Tags ⌚ Updated 3 weeks ago

llama3.1

Llama 3.1 is a new state-of-the-art model from Meta available in 8B, 70B and 405B parameter sizes.

[Tools](#) [8B](#) [70B](#) [405B](#)

↓ 7M Pulls 🔖 94 Tags ⌚ Updated 5 weeks ago

gemma2

Google Gemma 2 is a high-performing and efficient model available in three sizes: 2B, 9B, and 27B.

[2B](#) [9B](#) [27B](#)

↓ 1.6M Pulls 🔖 94 Tags ⌚ Updated 5 weeks ago

Benefits and Use Cases for Ollama

Benefits

Offline Capability, Data Privacy, Cost Savings, Low Latency

Use Cases

Software Development, Customer Support, Education and Creative work



Models

Filter by name...

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[Tools](#) [1B](#) [3B](#)

↓ 1.5M Pulls 🔖 63 Tags ⌚ Updated 3 weeks ago

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↓ 1.6M Pulls 🔖 94 Tags ⌚ Updated 5 weeks ago

Chat, Text Completions, Analysis and Function Calling w/ .NET

Text Completion LLM with GitHub Models OpenAI gpt-5-mini model
Classification, Summarization, Sentiment Analysis LLM Use Cases
Structured Output in LLM for Data Extraction Use Case
Build AI Chat App with .NET and gpt-5-mini model
Function Calling in Large Language Models (LLMs)

Mehmet Ozkaya



Introduction to This Section

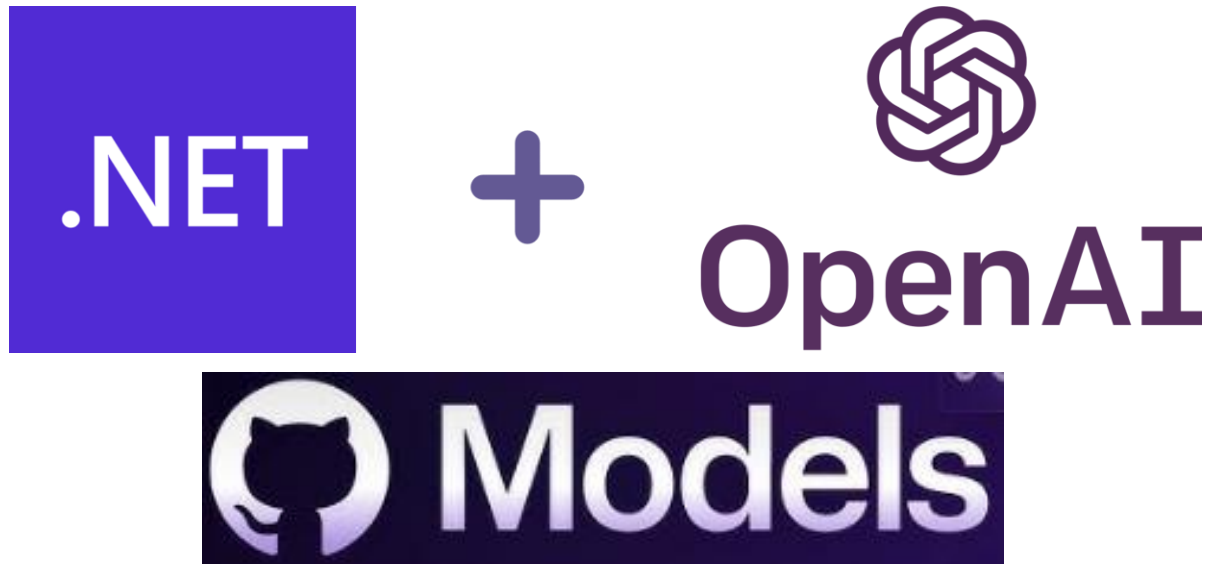
Text Completion LLM with GitHub Models OpenAI gpt-5-mini model

Classification, Summarization, Sentiment Analysis LLM Other Use Cases

Structured Output in LLM for Data Extraction Use Case

Build AI Chat App with .NET and gpt-5-mini model

Function Calling in Large Language Models (LLMs)

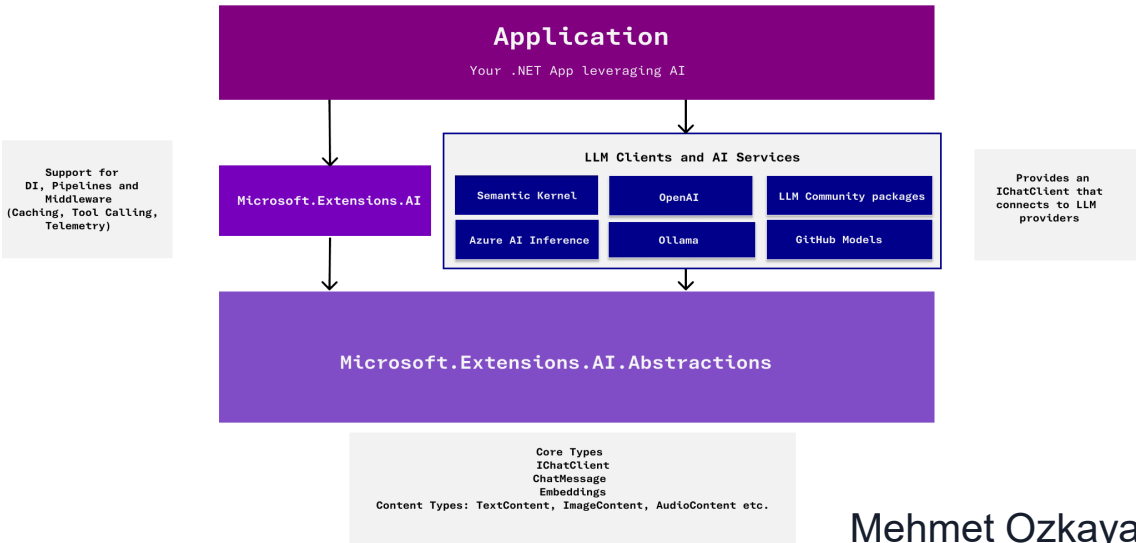
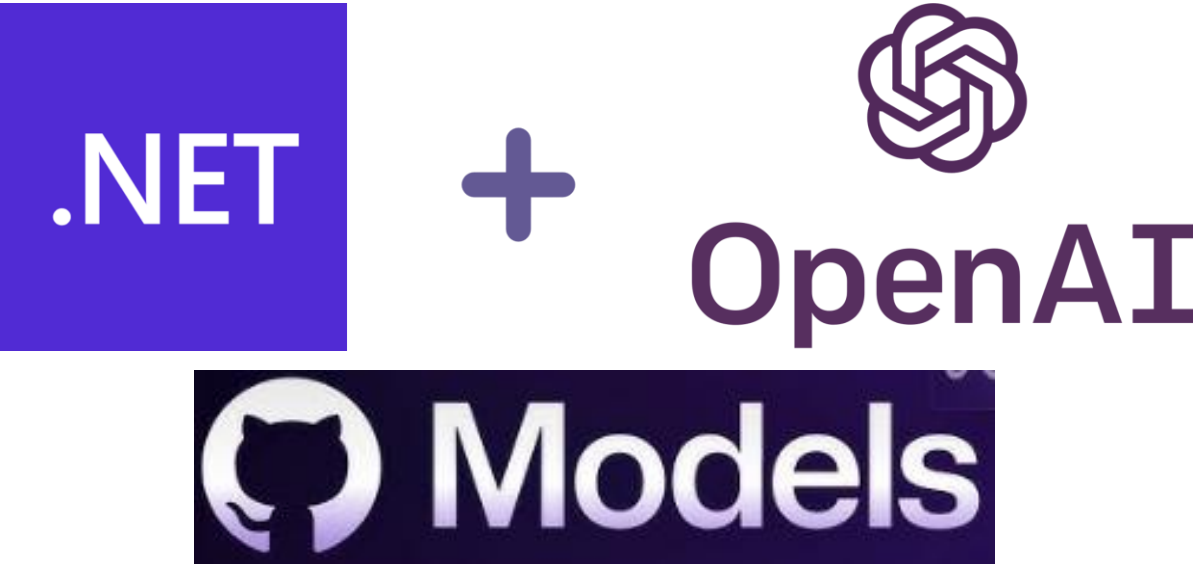


Development Process

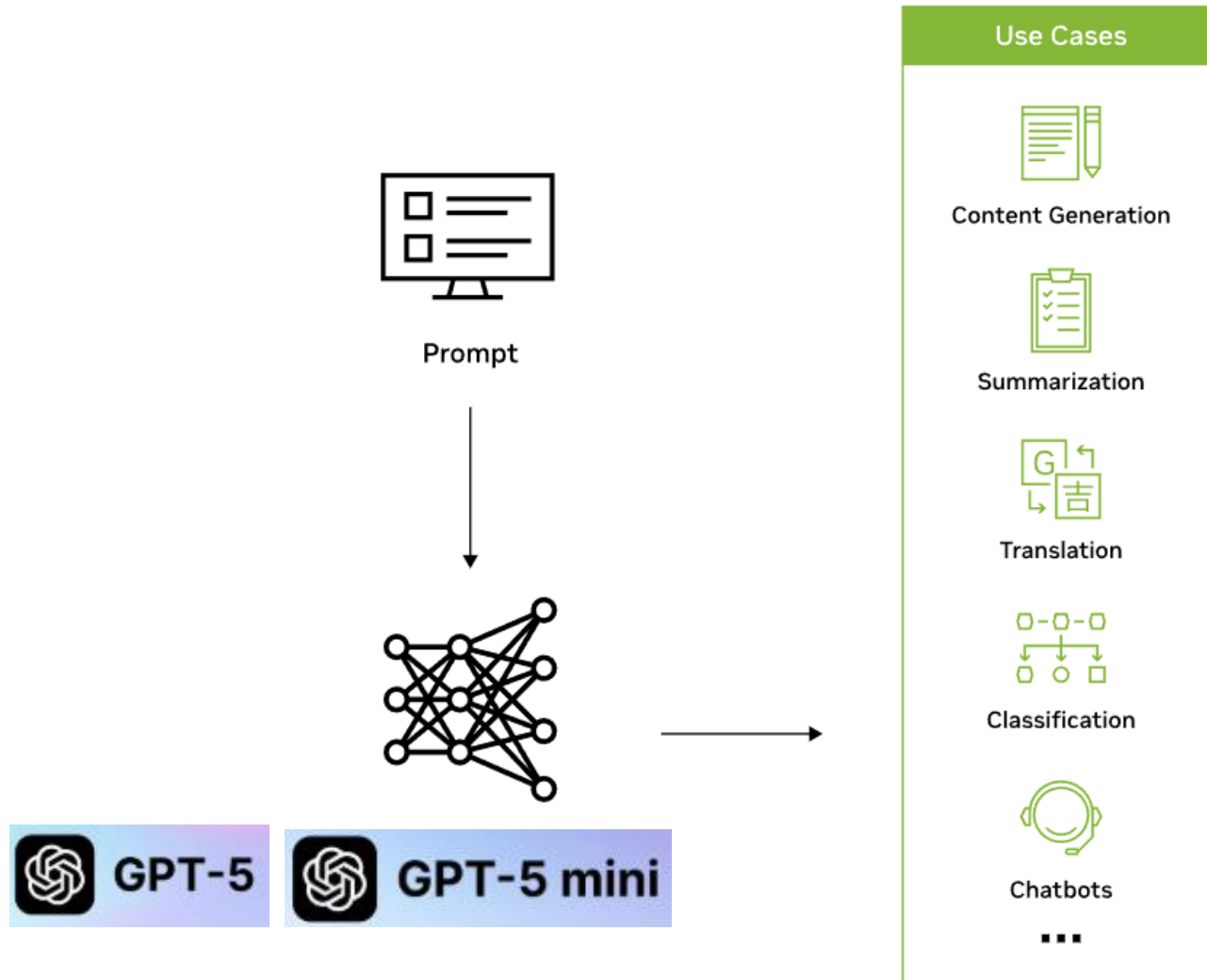
Using GitHub Models with OpenAI gpt-5-mini

Switching to Ollama's Local llama3.2

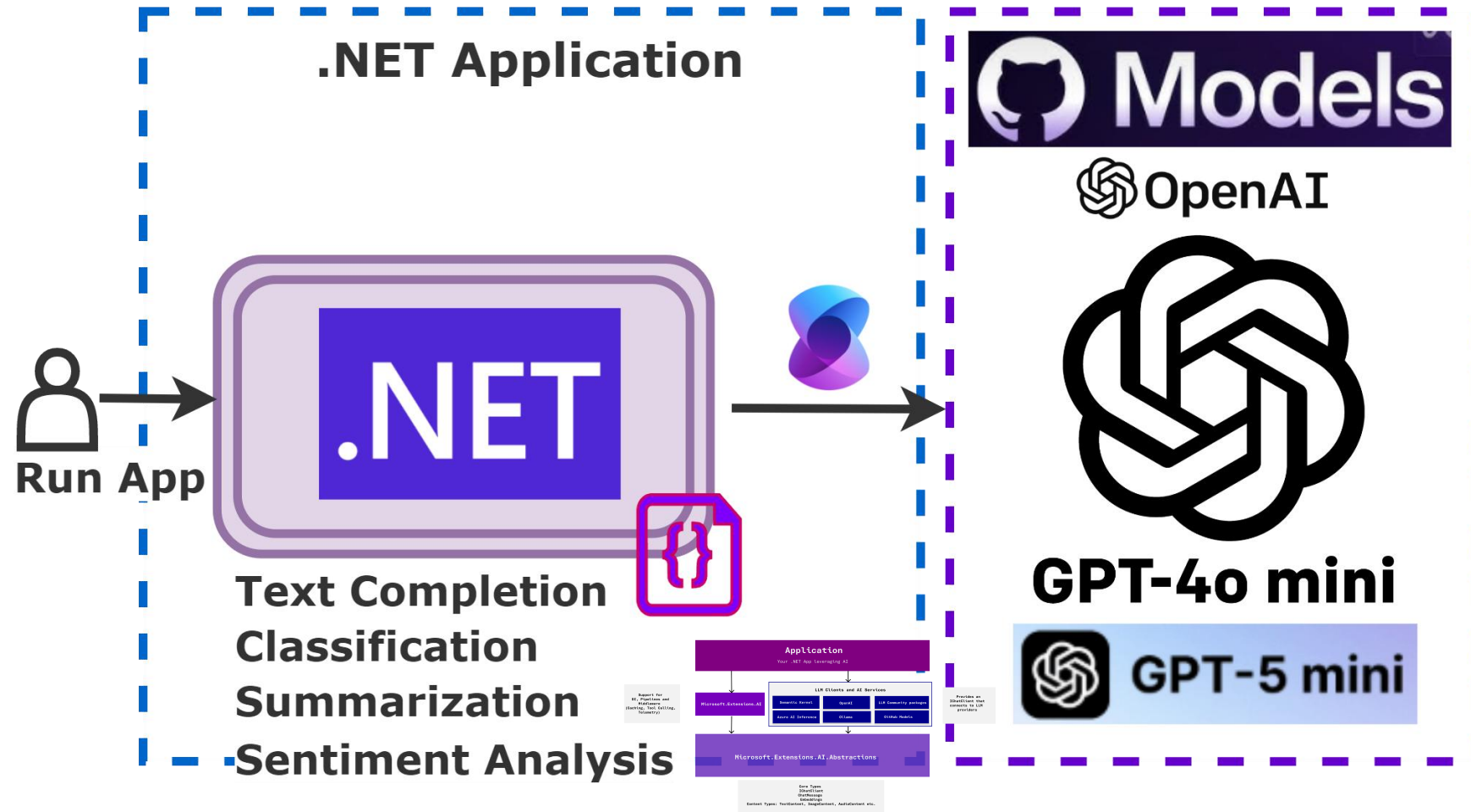
Core Packages: Microsoft.Extensions.AI



Generate Text with OpenAI GPT Models: Understand Capabilities and Limitations of LLMs



Text Completion LLM w/ GitHub Models OpenAI gpt-5-mini



Microsoft.Extensions.AI

C# Regions: Collapsible Block of Code

```
3  - public class SomeClass
4  {
5  +   Public Fields
8
9  +   Private Fields
12
13 +   Public Properties
29
30 +   Private Properties
33
34 +   Public Methods
40
41 +   Private Methods
332
333 }
```

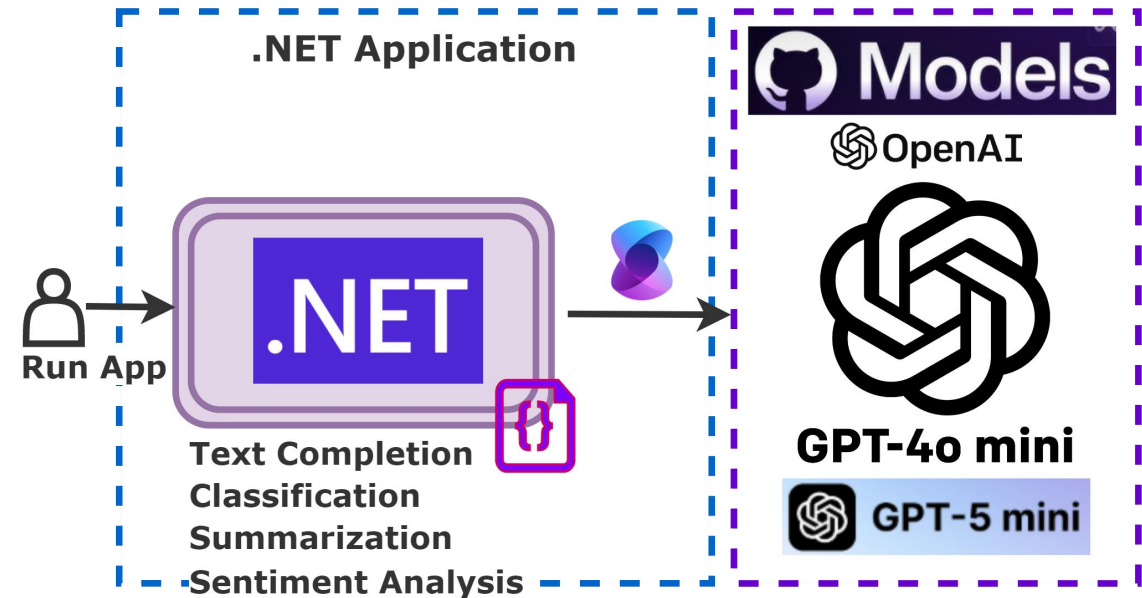
Required Packages

Microsoft.Extensions.AI

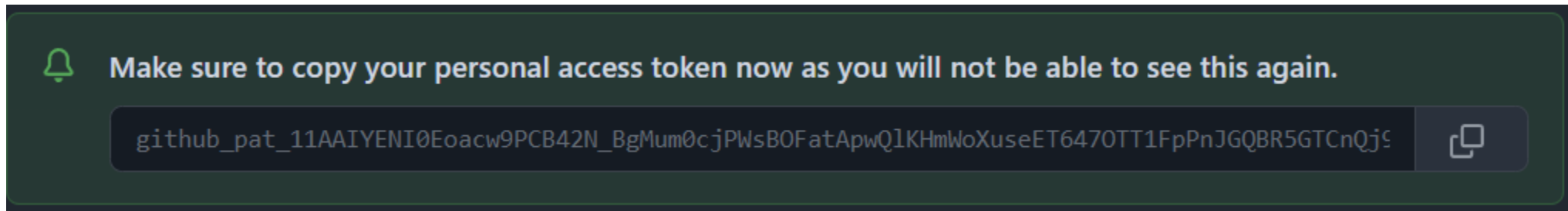
Microsoft.Extensions.AI.OpenAI

Microsoft.Extensions.Configuration

Microsoft.Extensions.Configuration.UserSecrets



GitHub Access Token – Save Securely

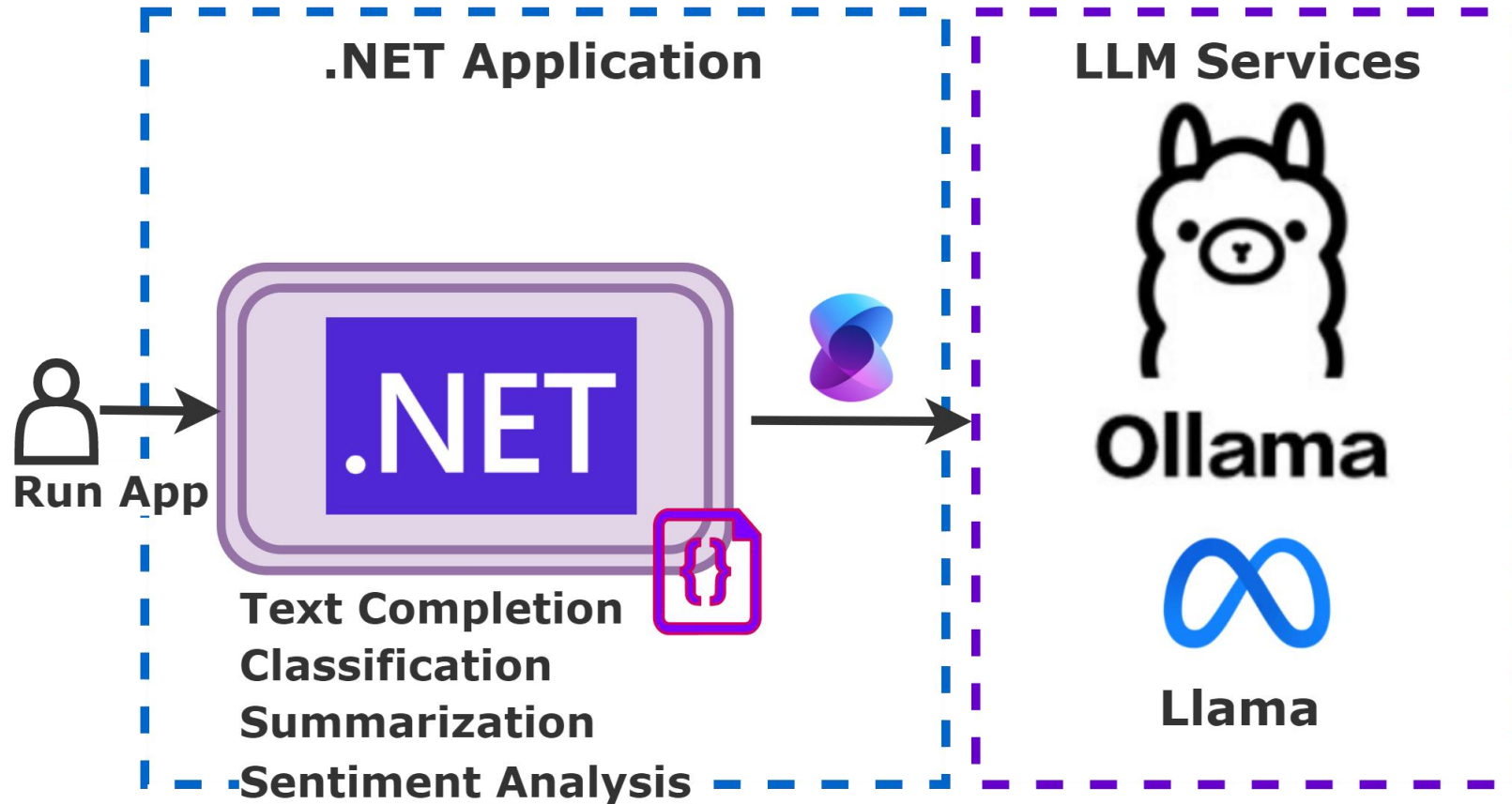


Alternatively run the following commands to configure your OpenAI API key as a secret for the sample app:

```
dotnet user-secrets init
```

```
dotnet user-secrets set GitHubModels:Token <your-openai-key>
```

Text Completion LLM with Local Ollama llama3.2 Model



Required .NET AI Packages

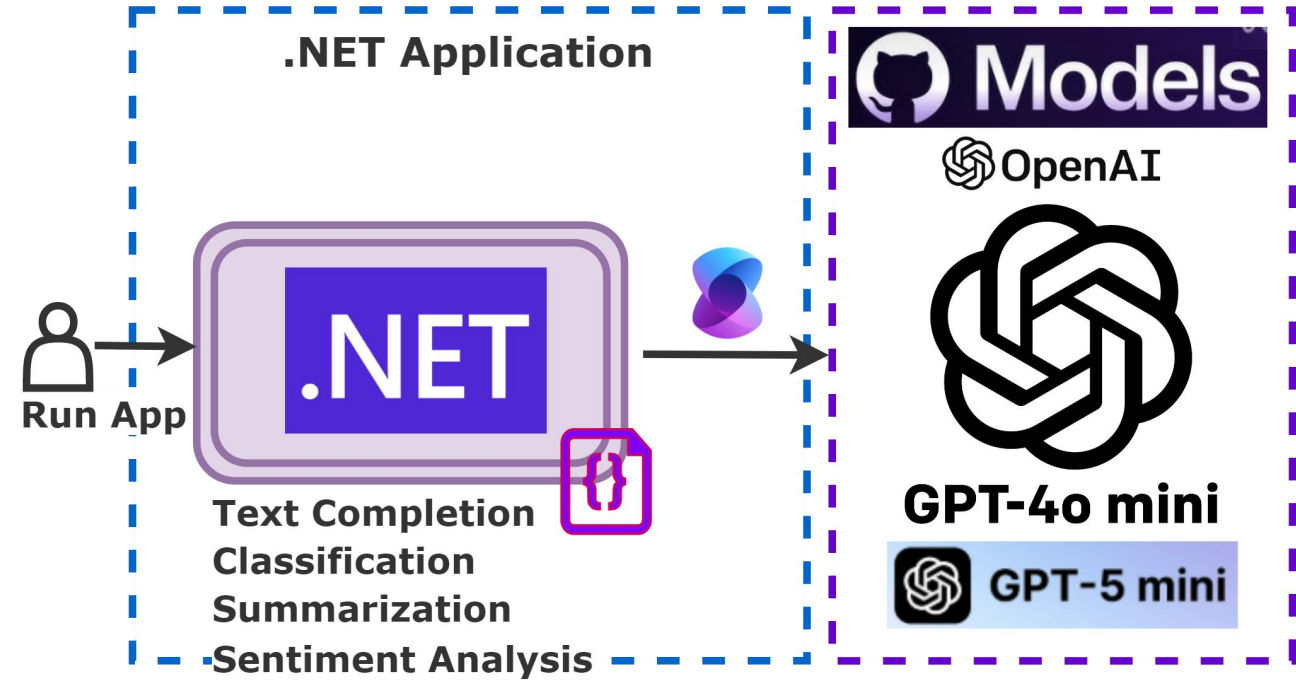
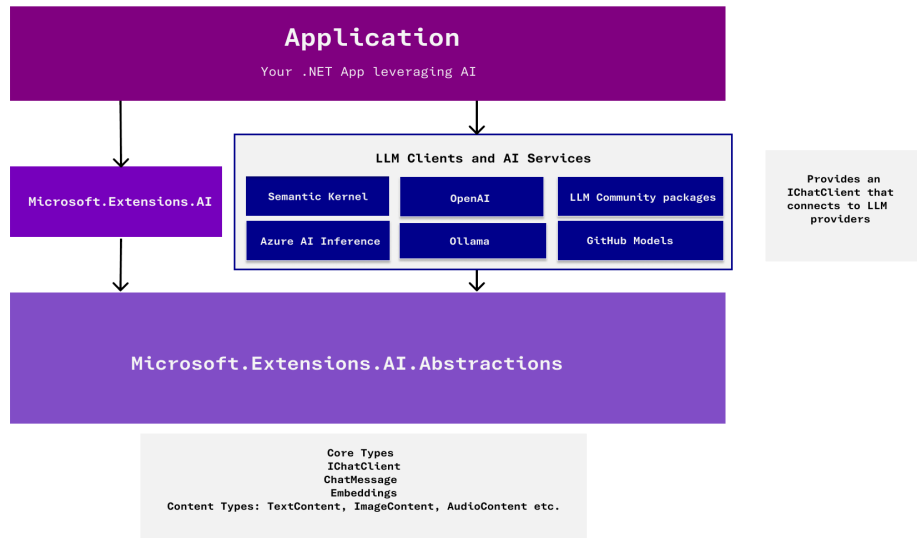
Microsoft.Extensions.AI

Microsoft.Extensions.AI.OpenAI

Microsoft.Extensions.Configuration

Microsoft.Extensions.Configuration

.UserSecrets



Structured Output in Large Language Models (LLMs)

LLMs can generate responses in a **well-defined format like JSON, XML or key-value pairs**

When the **output needs to be machine-readable and integrated into downstream systems**

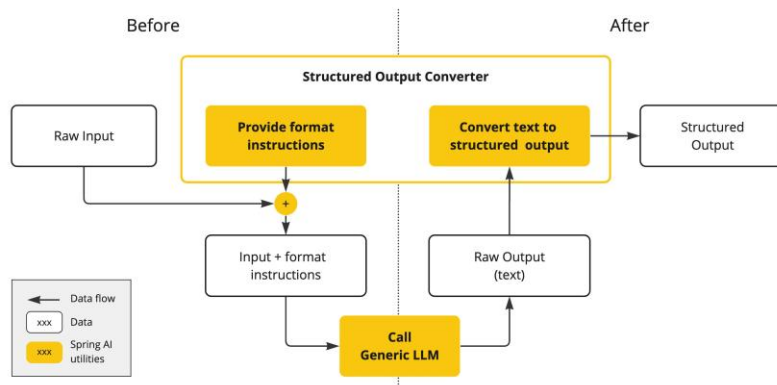
Difference between **writing a paragraph** and **filling out a form**

Why Do We Need Structured Output ?

LLMs are stochastic and vary their output, Ensures predictability and consistency

Examples of When Structured Outputs Are Useful

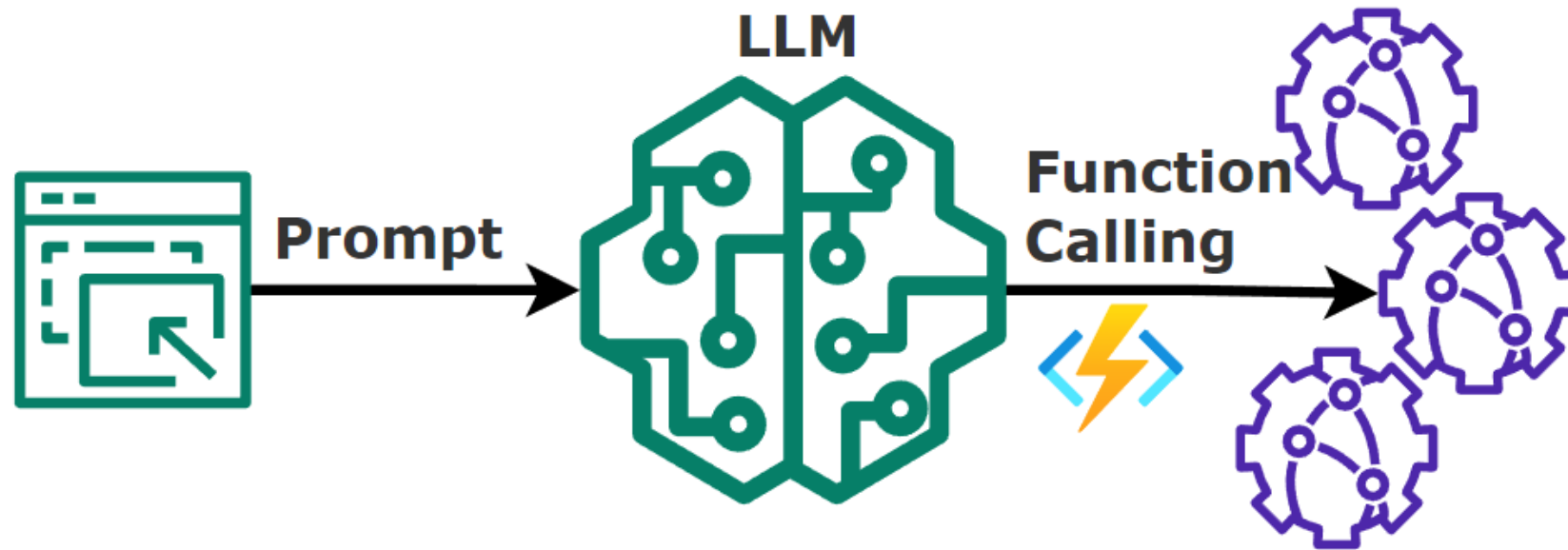
API calls and function responses, Automating workflows with predictable outputs, Data extraction and storage in databases



```
weather: [
  - {
    id: 800,
    main: "Clear",
    description: "clear sky",
    icon: "01n"
  },
]
```

How Does Structured Output Work ?

Prompt: Please provide the weather information for New York City in the following **JSON** format: { "city": "city name", "temperature": "value", "conditions": "description" }



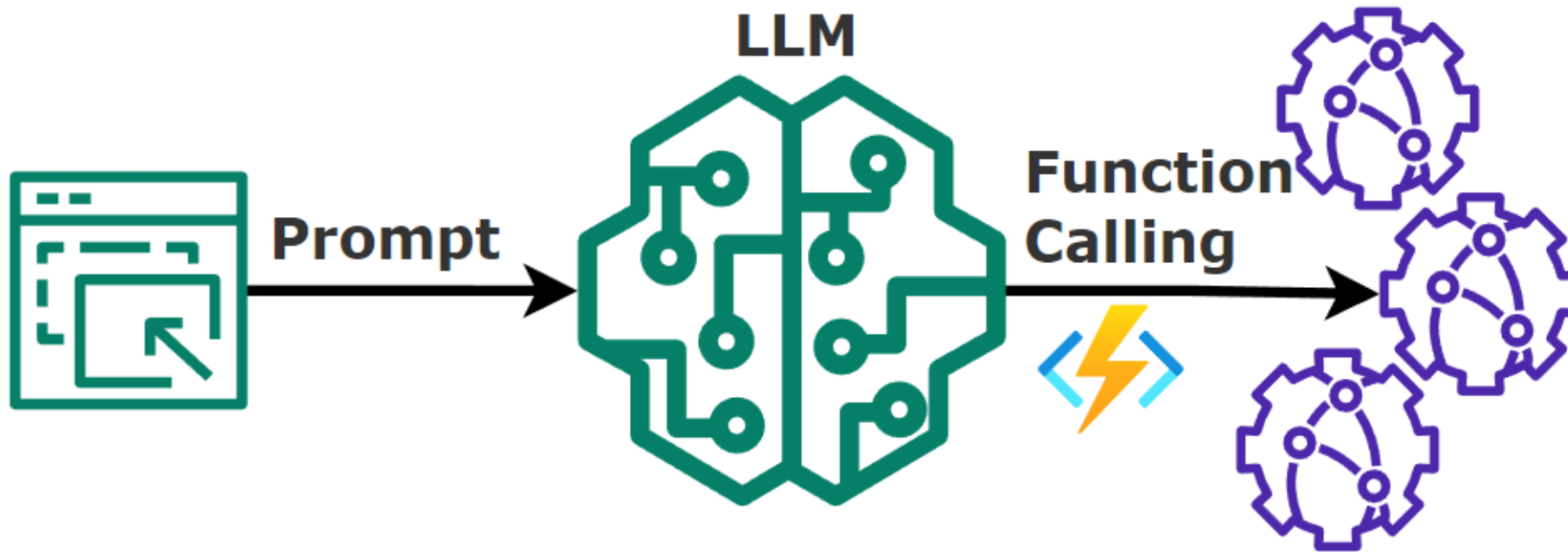
```
weather: [  
  - {  
    id: 800,  
    main: "Clear",  
    description: "clear sky",  
    icon: "01n"  
  }  
],
```

Car Listing Case

Comparing car prices

Filtering by make/model or condition

Alerting potential buyers or matching them to relevant listings



Car Listing Case

"Check out this stylish 2019 Toyota Camry. It has a clean title, only 40,000 miles on the odometer, and a well-maintained interior. The car offers great fuel efficiency, a spacious trunk, and modern safety features like lane departure alert. Minimum offer price: \$18,000. Contact Metro Auto at (555) 111-2222 to schedule a test drive."

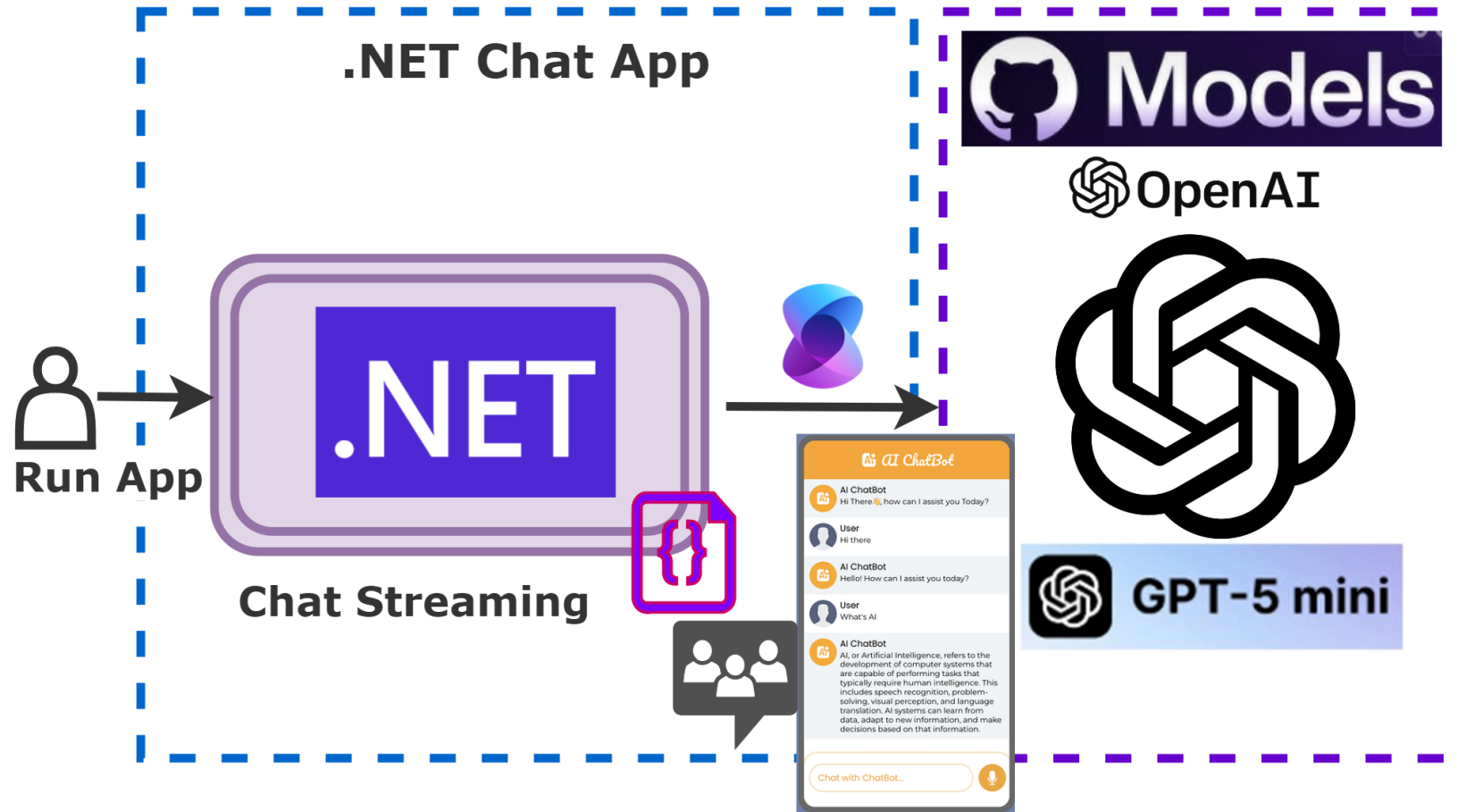
"Lease this sporty 2021 Honda Civic! With only 10,000 miles, it includes a sunroof, premium sound system, and backup camera. Perfect for city driving with its compact size and great fuel mileage. Located in Uptown Motors, monthly lease starts at \$250 (excl. taxes). Call (555) 333-4444 for more info."

"A classic 1968 Ford Mustang, perfect for enthusiasts. The vehicle needs some interior restoration, but the engine runs smoothly. V8 engine, manual transmission, around 80,000 miles. This vintage gem is priced at \$25,000. Contact Retro Wheels at (555) 777-8888 if you're interested."

"Brand new 2023 Tesla Model 3 for lease. Zero miles, fully electric, autopilot capabilities, and a sleek design. Monthly lease starts at \$450. Clean lines, minimalist interior, top-notch performance. For more details, call EVolution Cars at (555) 999-0000."

"Selling a 2015 Subaru Outback in good condition. 60,000 miles on it, includes all-wheel drive, heated seats, and ample cargo space for family getaways. Minimum offer price: \$14,000. Contact Forrest Autos at (555) 222-1212 if you want a reliable adventure companion."

Build AI Chat App with .NET and gpt-5-mini model



Understanding Chat Applications

System messages

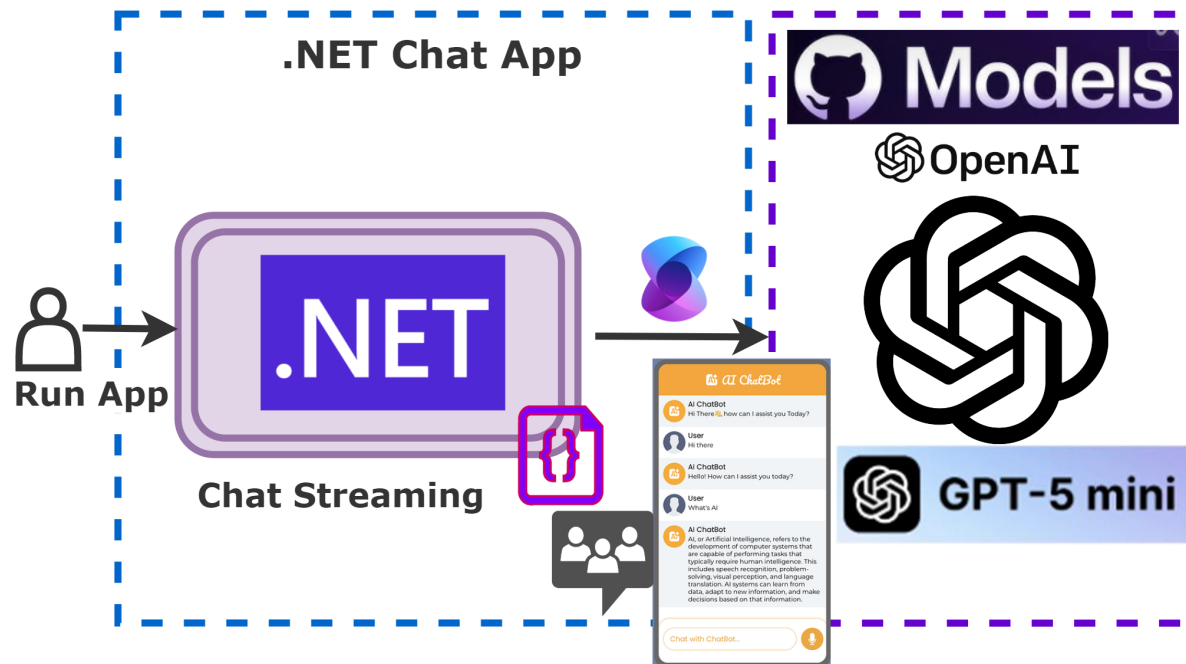
The “invisible” instructions guiding the model’s personality, style, or domain expertise

User messages

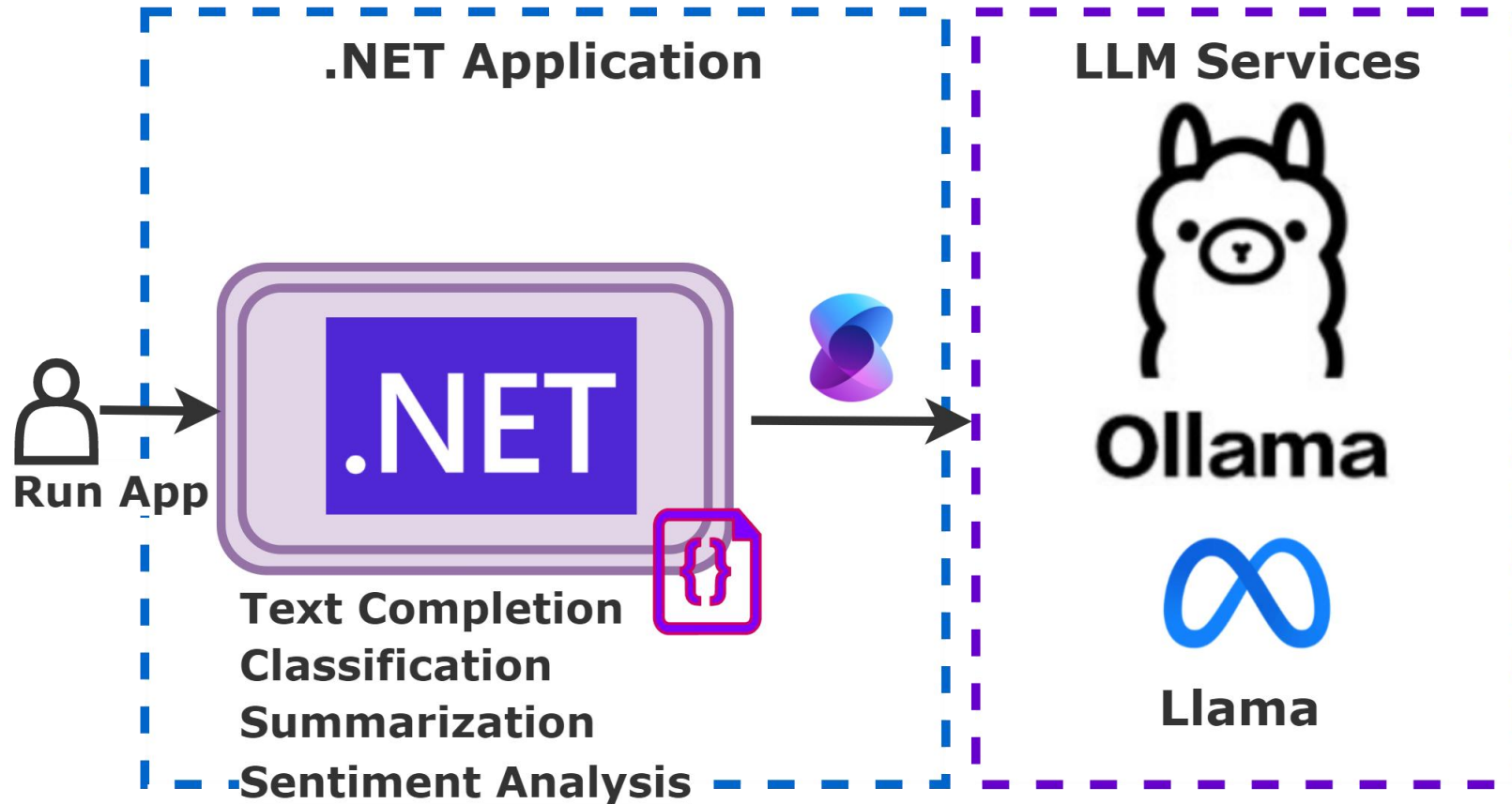
The prompts or questions from the actual end user

Assistant messages

The model’s replies, which incorporate context from both the system and user messages



Text Completion LLM with Local Ollama llama3.2 Model

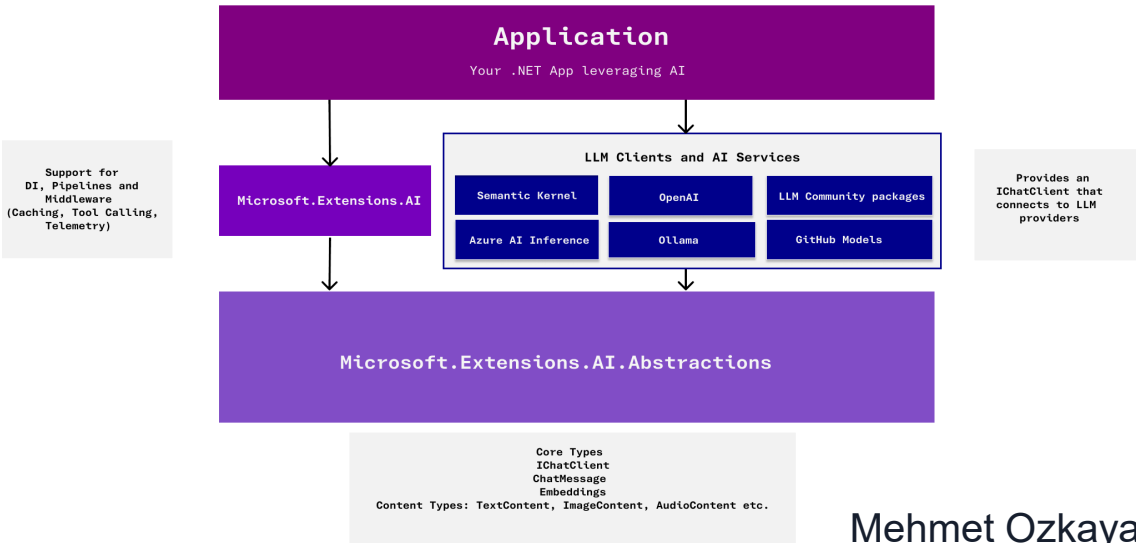
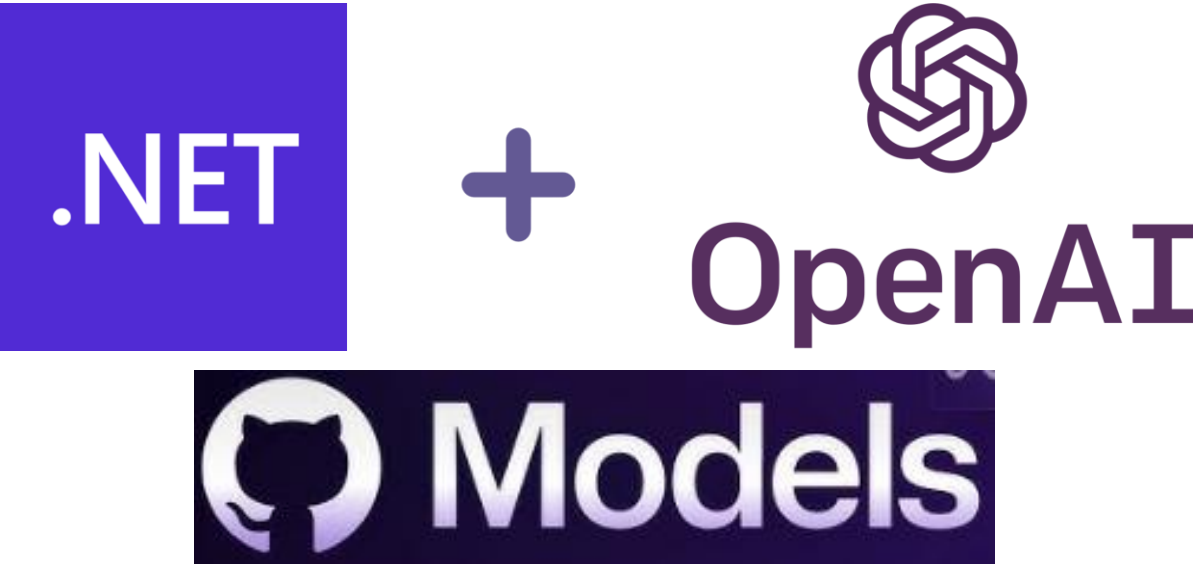


Development Process

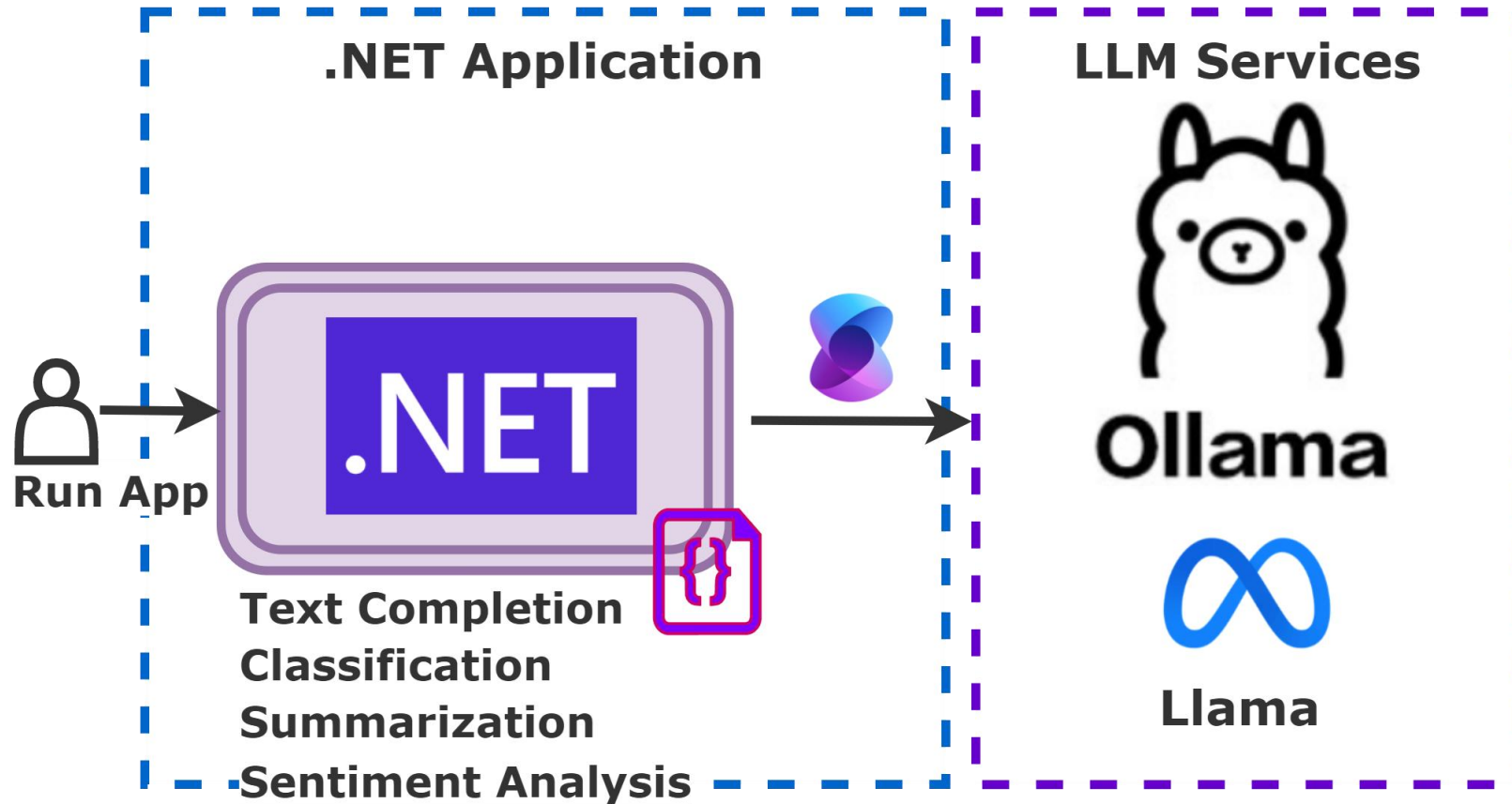
Using GitHub Models with OpenAI gpt-5-mini

Switching to Ollama's Local llama3.2

Core Packages: Microsoft.Extensions.AI



Text Completion LLM with Local Ollama llama3.2 Model



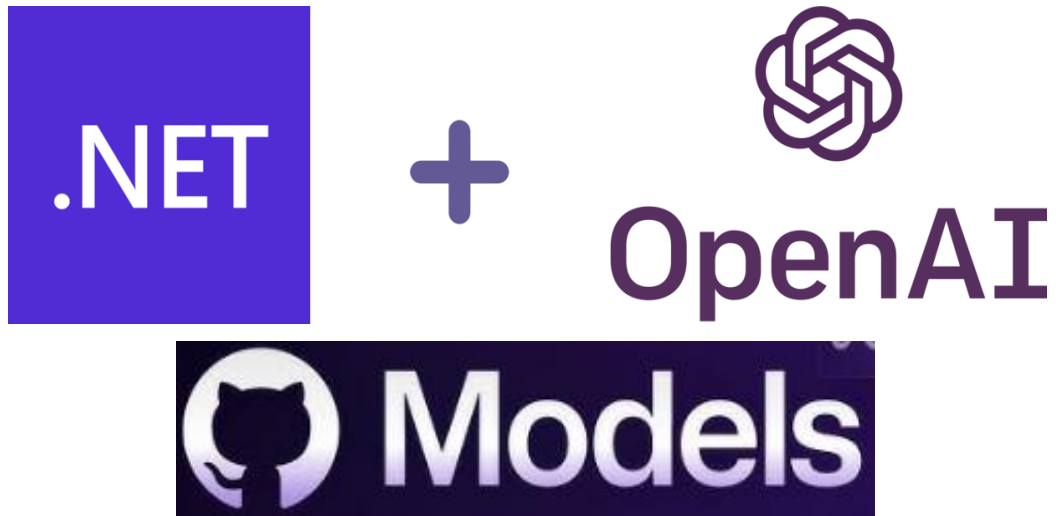
The Local vs. Cloud Trade-Off: A Performance Debrief

Ollama

Total data privacy and offline capability. Zero API costs but, it uses your own computer's resources

GitHub Models

Incredible speed and access to the most powerful hardware. Zero local resource usage. But it requires an internet connection, has potential costs



Benefits and Use Cases for Ollama

Benefits

Offline Capability, Data Privacy, Cost Savings, Low Latency

Use Cases

Software Development, Customer Support, Education and Creative work



Models

Filter by name...

llama3.2

Meta's Llama 3.2 goes small with 1B and 3B models.

[Tools](#) [1B](#) [3B](#)

↓ 1.5M Pulls 🔖 63 Tags ⌚ Updated 3 weeks ago

llama3.1

Llama 3.1 is a new state-of-the-art model from Meta available in 8B, 70B and 405B parameter sizes.

[Tools](#) [8B](#) [70B](#) [405B](#)

↓ 7M Pulls 🔖 94 Tags ⌚ Updated 5 weeks ago

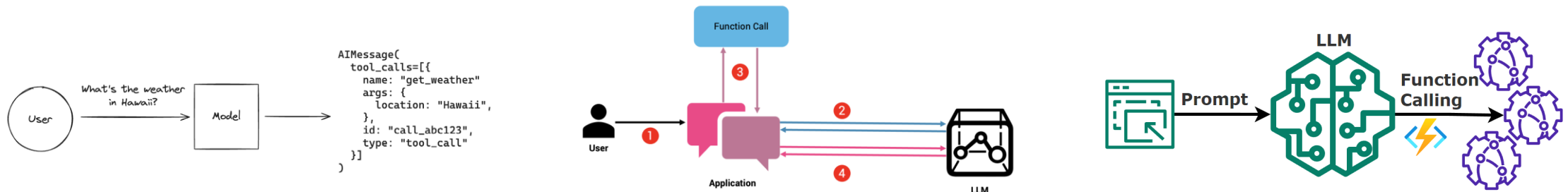
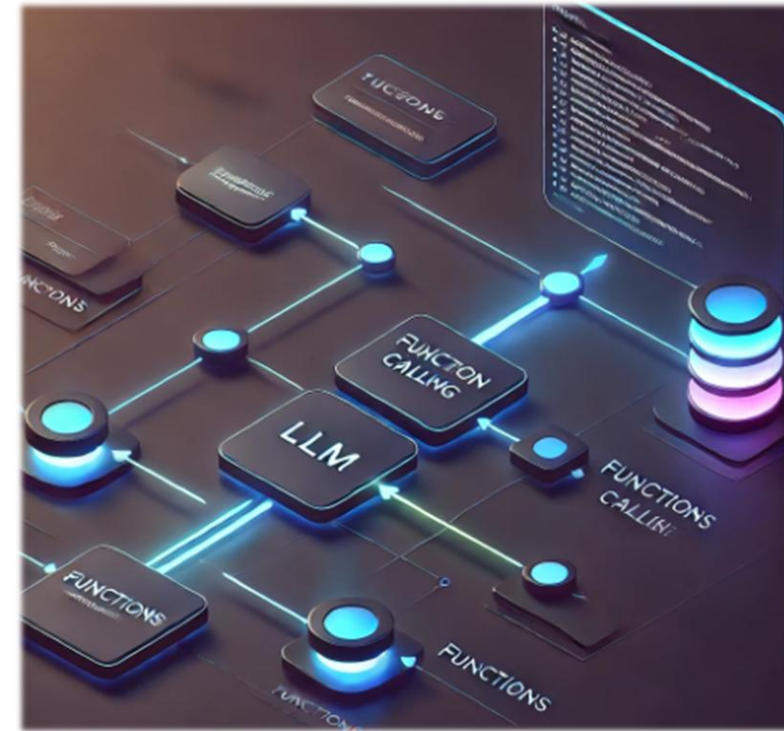
gemma2

Google Gemma 2 is a high-performing and efficient model available in three sizes: 2B, 9B, and 27B.

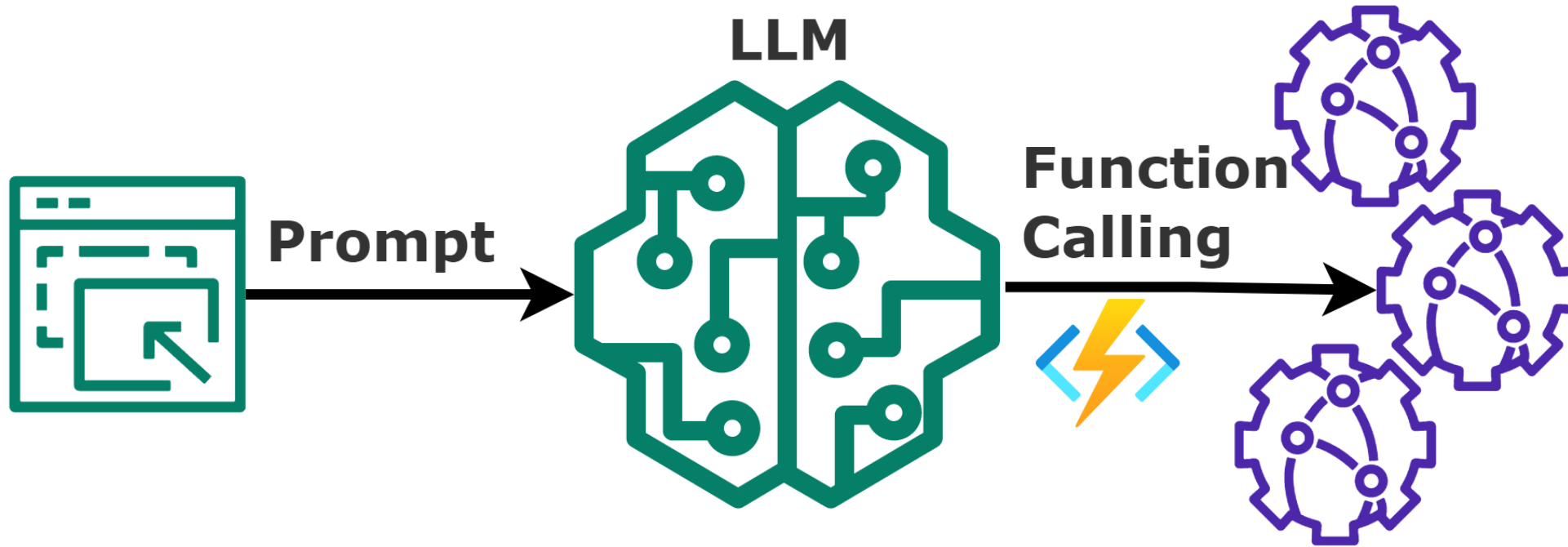
[2B](#) [9B](#) [27B](#)

↓ 1.6M Pulls 🔖 94 Tags ⌚ Updated 5 weeks ago

A function to retrieve the current weather in a given location might require parameters like the **city** and **unit of measurement**



How Does Function Calling Work ?



Examples of Function Calling

Retrieving Information

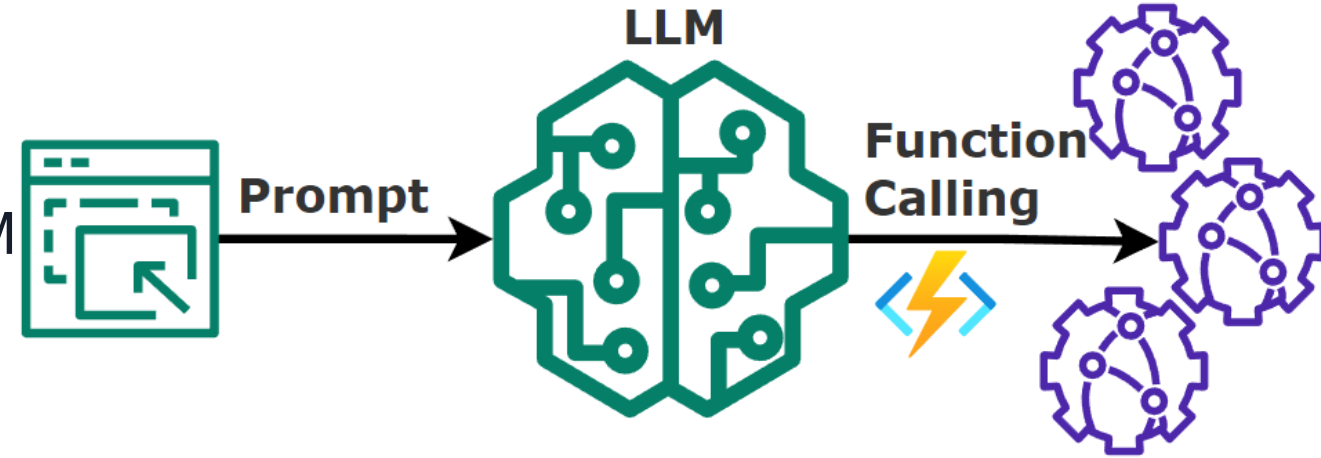
What's the weather in New York City?

Booking an Appointment

Schedule a meeting for next Monday at 2 PM

Running Calculations

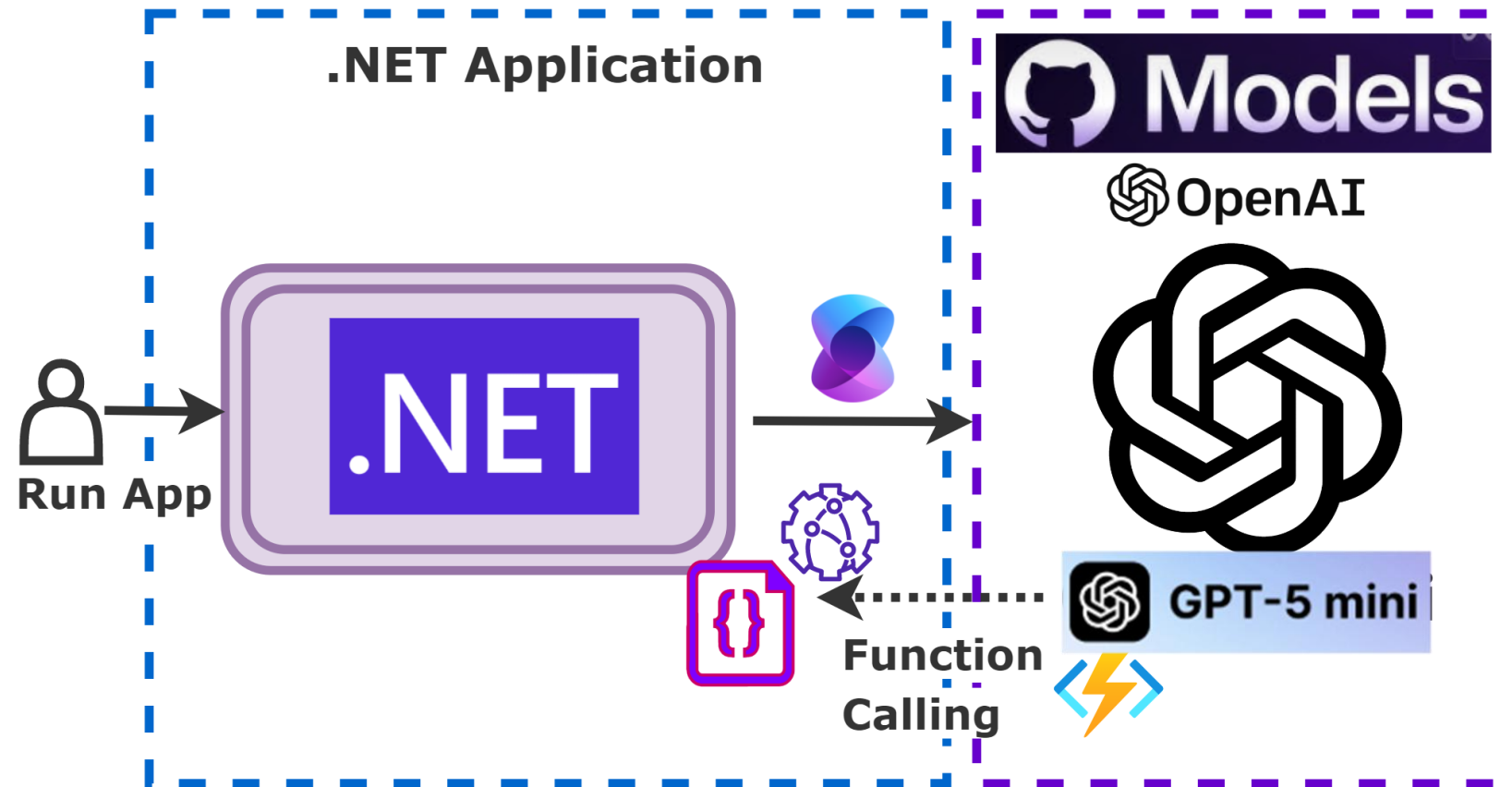
Calculate the square root of 256



Simulate Function Calling with OpenAI Playground

<https://platform.openai.com/playground>

Function Calling in Large Language Models (LLMs)



.NET AI Vector Search using Vector Embeddings and Vector Store

What are Vectors, Vector Embeddings and Vector Database ?

Generate Embeddings and Calculate Similarity w/ CosineSimilarity

.NET AI Vector Search w/ Extensions.VectorData.Abstractions

Create Query Embedding and Perform Vector Search



Introduction to This Section

What are Vectors and Vector Embeddings ?

Generate Embeddings and Calculate Similarity w/ CosineSimilarity

What is a Vector Database ?

.NET AI Vector Search w/ Extensions.VectorData.Abstractions

Create VectorSearch App and Install Packages

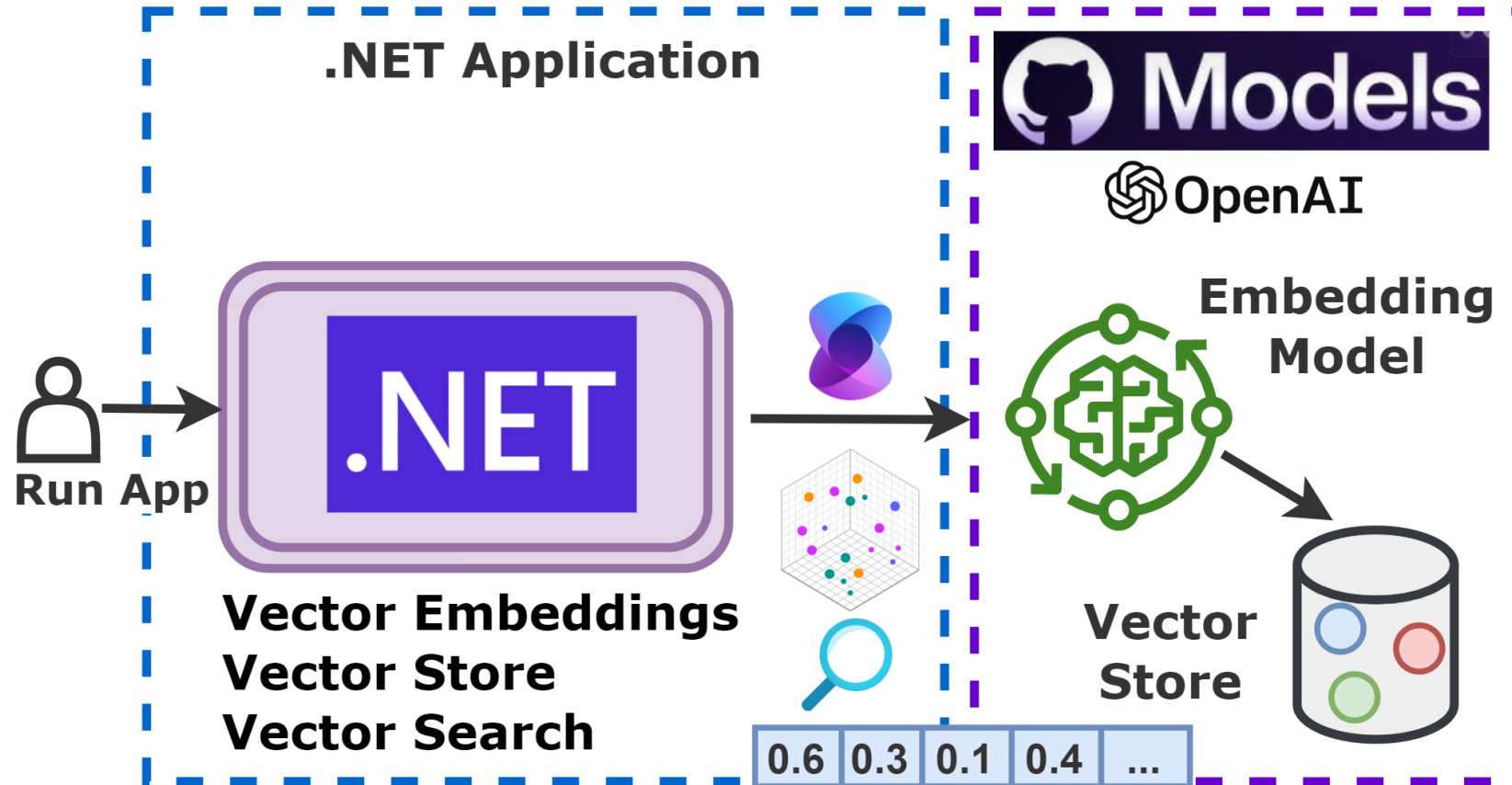
Develop Vector Data Entity and Populate Data

Generate Embeddings and Store in In-memory Vector Store

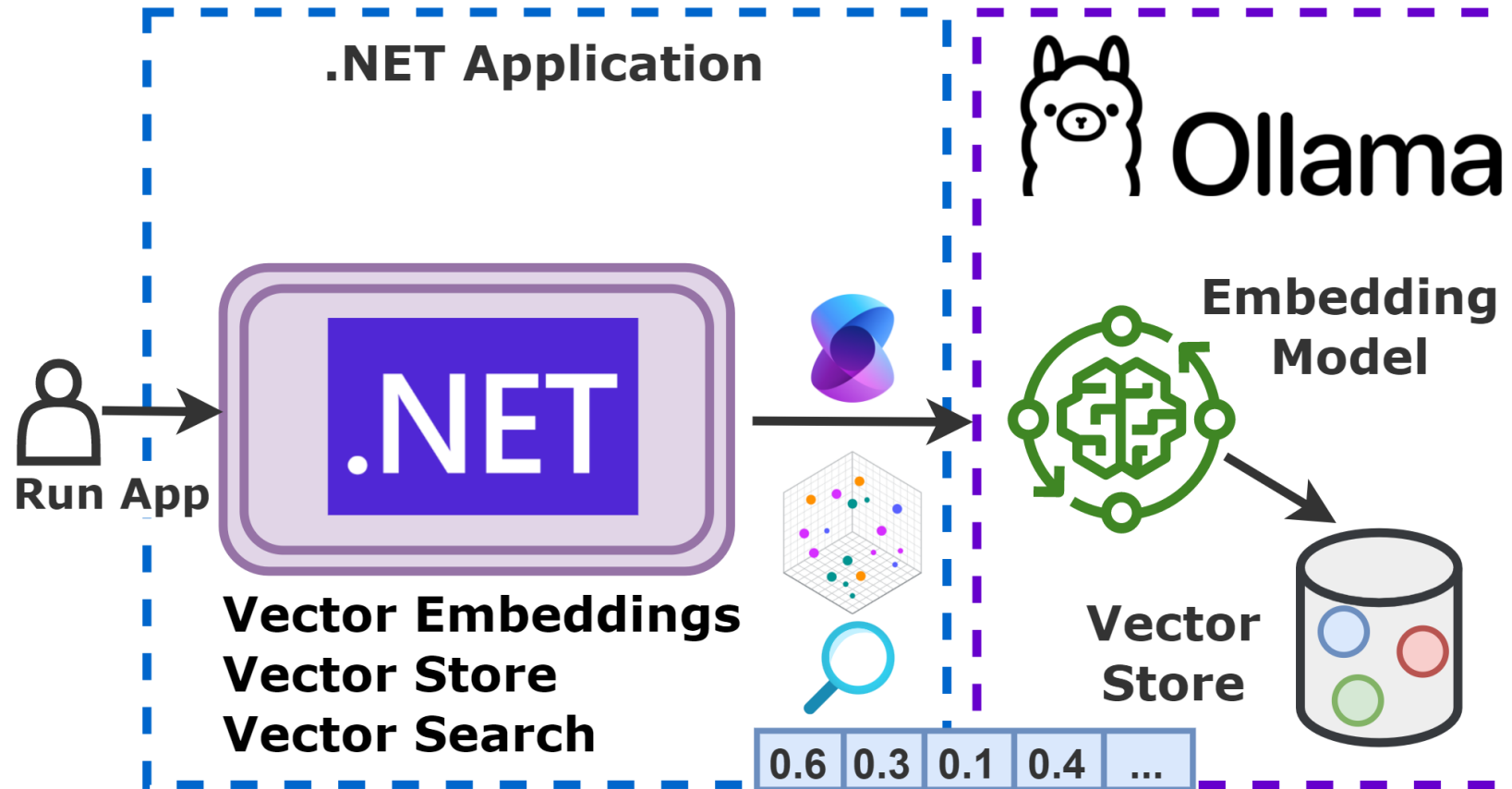
Create Query Embedding and Perform Vector Search

Develop .NET AI Vector Search App w/ Ollama and all-minilm embedding model

.NET AI Vector Search using Vector Embeddings and Vector Store



.NET AI Vector Search using Vector Embeddings and Vector Store - Ollama

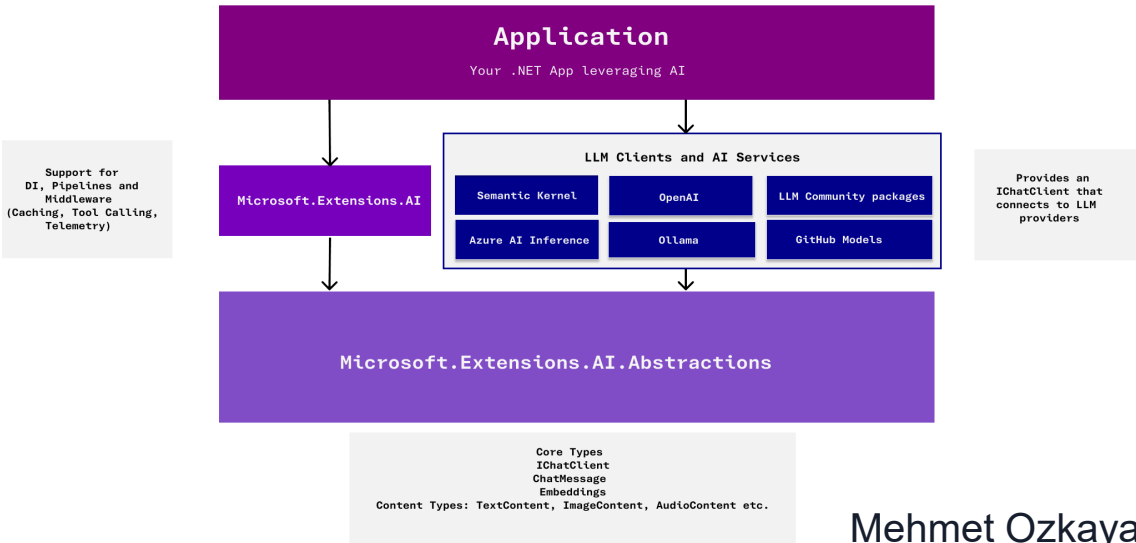
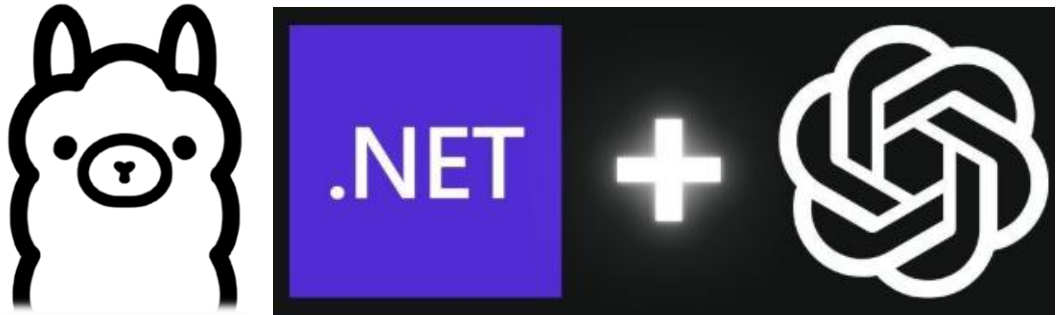
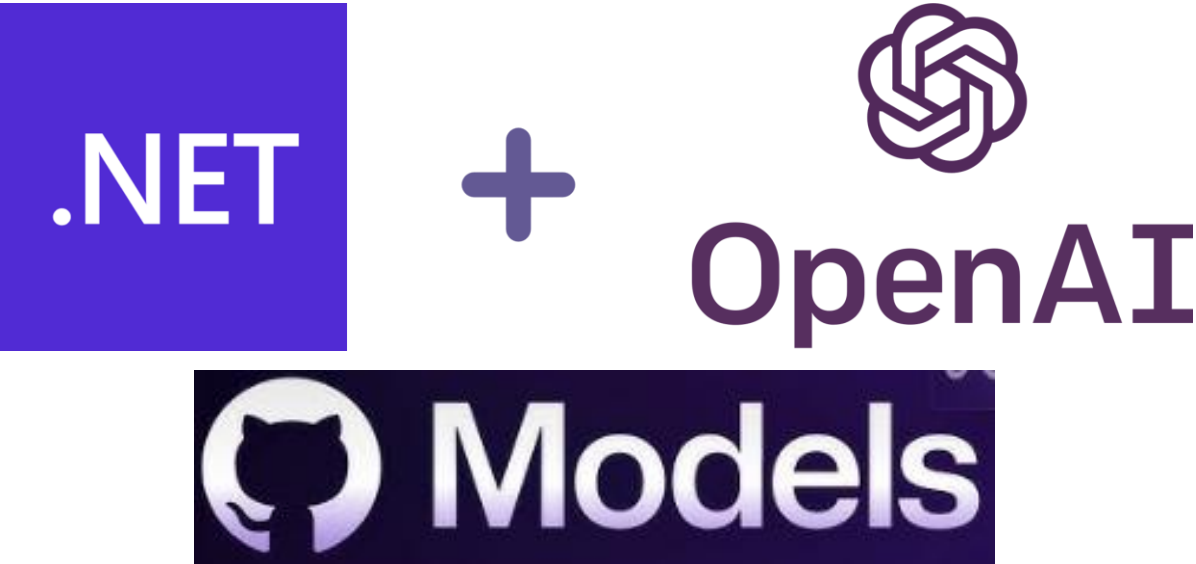


Development Process

Using GitHub Models with OpenAI gpt-5-mini

Switching to Ollama's Local llama3.2

Core Packages: Microsoft.Extensions.AI



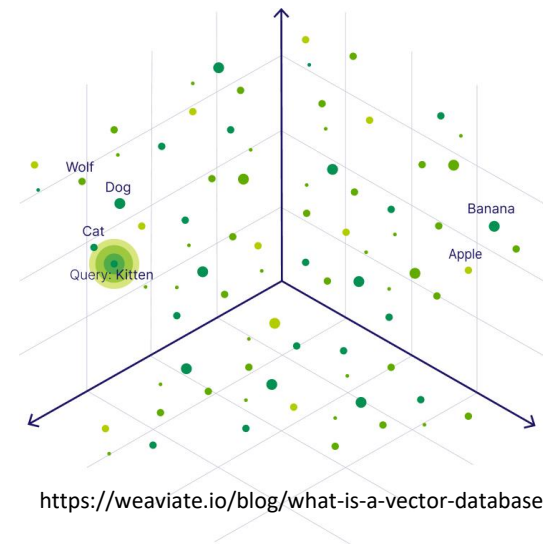
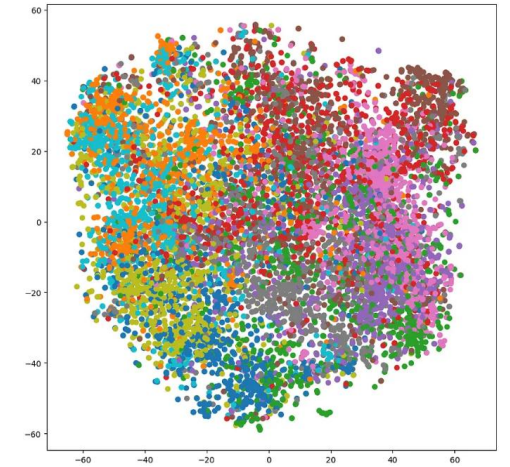
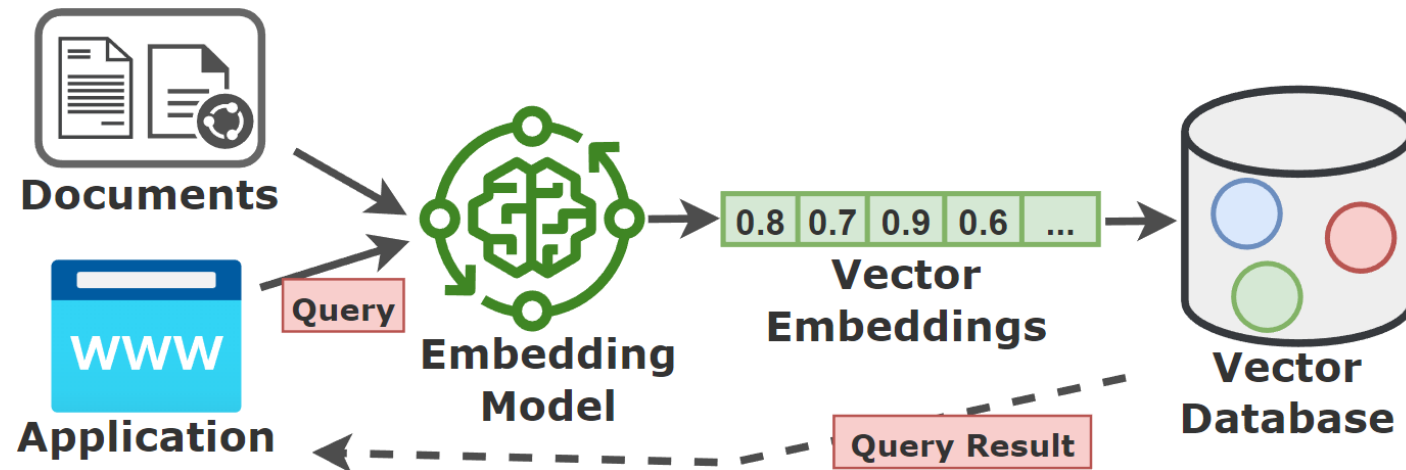
What is a Vector Database?

Vector Database is a **specialized database** designed to store, manage, and query **high-dimensional vectors**

Vectors are **numerical representations** that capture the **semantic meaning** of data

Indexes and **stores vector embeddings** for **fast retrieval** and **similarity search**

Enable **contextual searches** based on the **meaning** behind the data



The Evolution from Traditional to Vector Databases

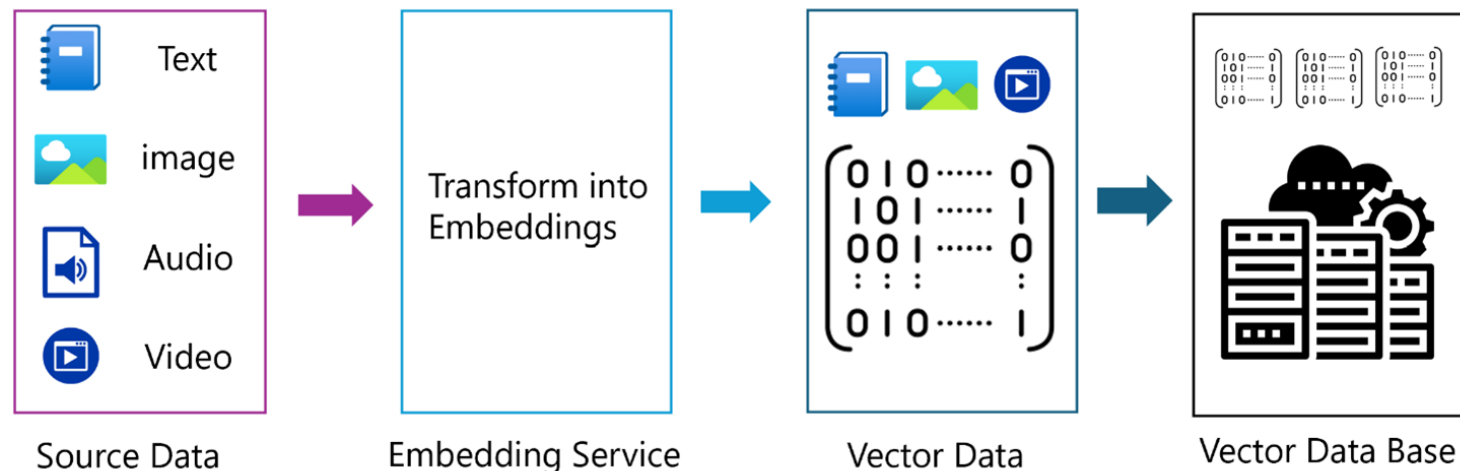
Traditional databases store data in **structured formats** like **rows** and **columns**

Fall short when we **need to perform semantic searches**—
searches **based on the meaning of data** rather exact words

Search for “**laptop**”: **Traditional Db** retrieve **exact word**, **Vector Db** understand that “**notebook computer**”

Vector Db store **unstructured data** like documents, images, audio, video, social media posts

Customers			
CustomerId	FirstName	LastName	DateCreated
1	Homer	Simpson	13/06/2014 3:33:37 PM
2	Peter	Griffin	13/06/2014 9:09:56 PM
3	Stewie	Griffin	13/06/2014 9:16:07 PM
4	Brian	Griffin	13/06/2014 9:16:36 PM
5	Cosmo	Kramer	13/06/2014 9:16:41 PM
6	Philip	Fry	13/06/2014 9:17:02 PM
7	Amy	Wong	13/06/2014 9:22:05 PM
8	Hubert J.	Farnsworth	13/06/2014 9:22:19 PM
9	Marge	Simpson	13/06/2014 9:22:37 PM
10	Bender	Rodriguez	13/06/2014 9:22:52 PM
11	Turanga	Leela	13/06/2014 9:23:37 PM
*	(New)		15/06/2014 9:00:01 PM



Why Are Vectors Important?

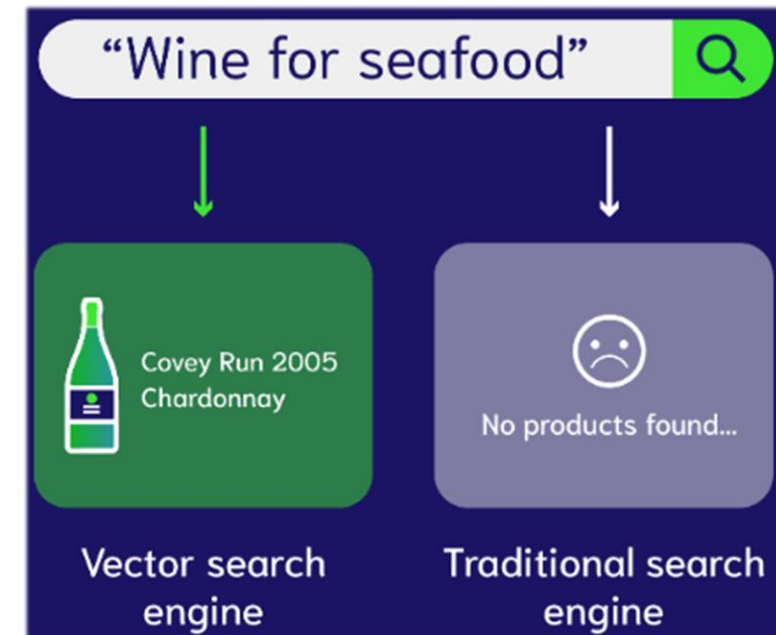
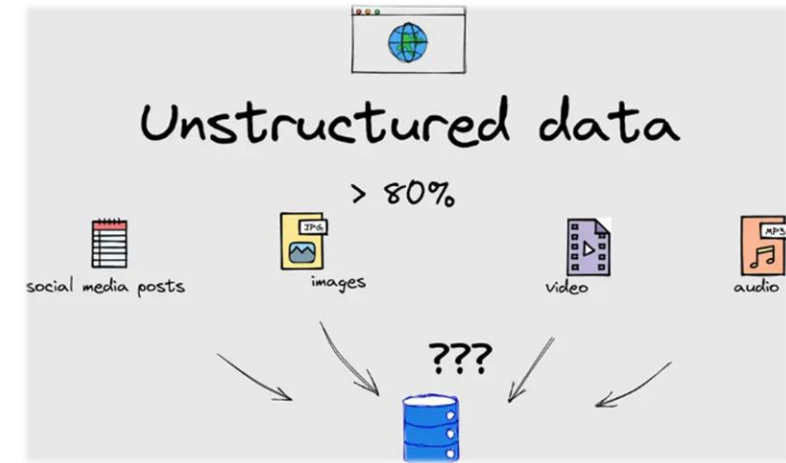
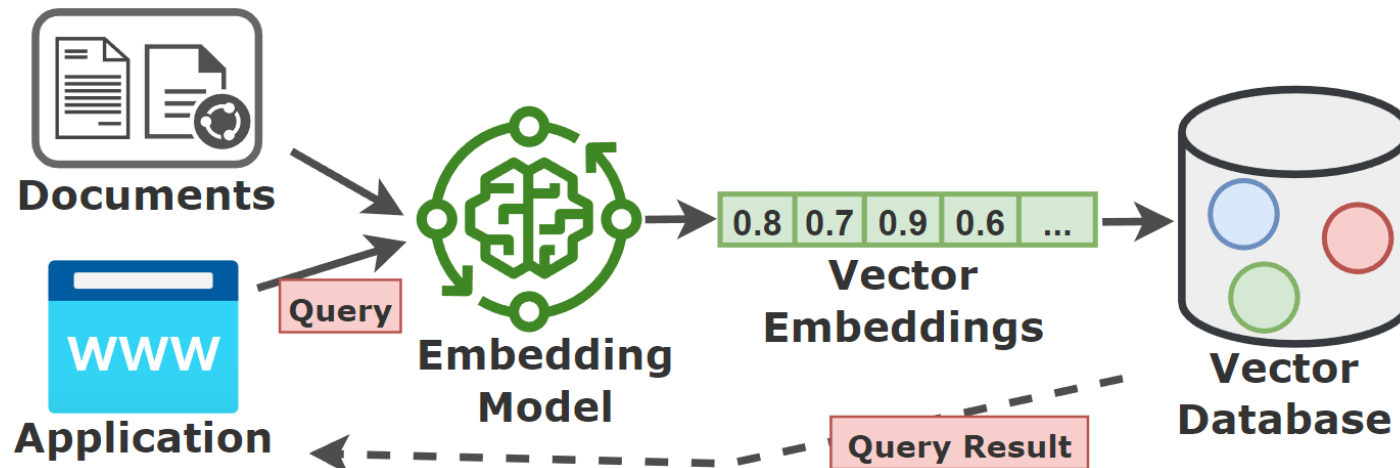
%80 of data on the internet is **Unstructured Data** (like documents, images, audio, video, social media posts)

Vectors **capture the meaning behind data**, not just its surface

Understand synonyms, paraphrases, and even **nuanced relationships** between data points

Search for “**mobile phone**” retrieve results for “**smartphone**”

Semantic search, personalized **recommendations**



<https://weaviate.io/blog/vector-embeddings-explained>

Core Features of Vector Databases

High-Dimensional Data Handling

Efficiently stores and queries data represented as vectors with hundreds or thousands of dimensions

Fast Similarity Search

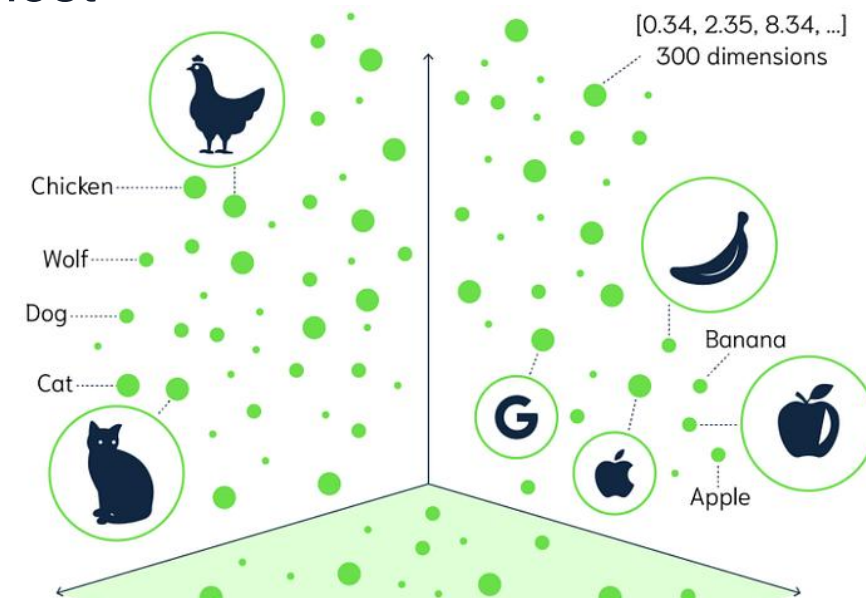
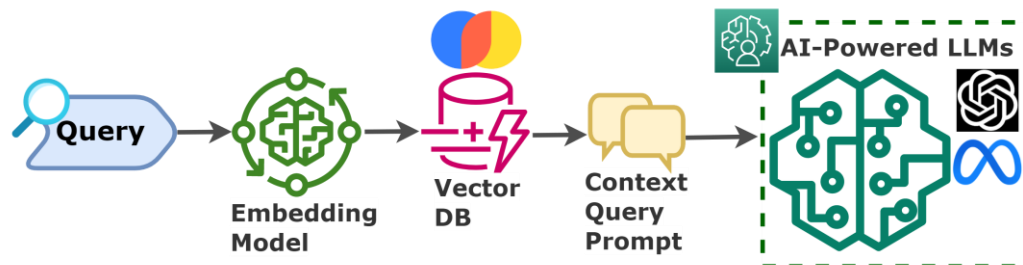
Approximate Nearest Neighbor (ANN) to quickly retrieve the most relevant vectors, even from large datasets

Scalability

Supports millions or even billions of vectors, suitable for enterprise-scale applications

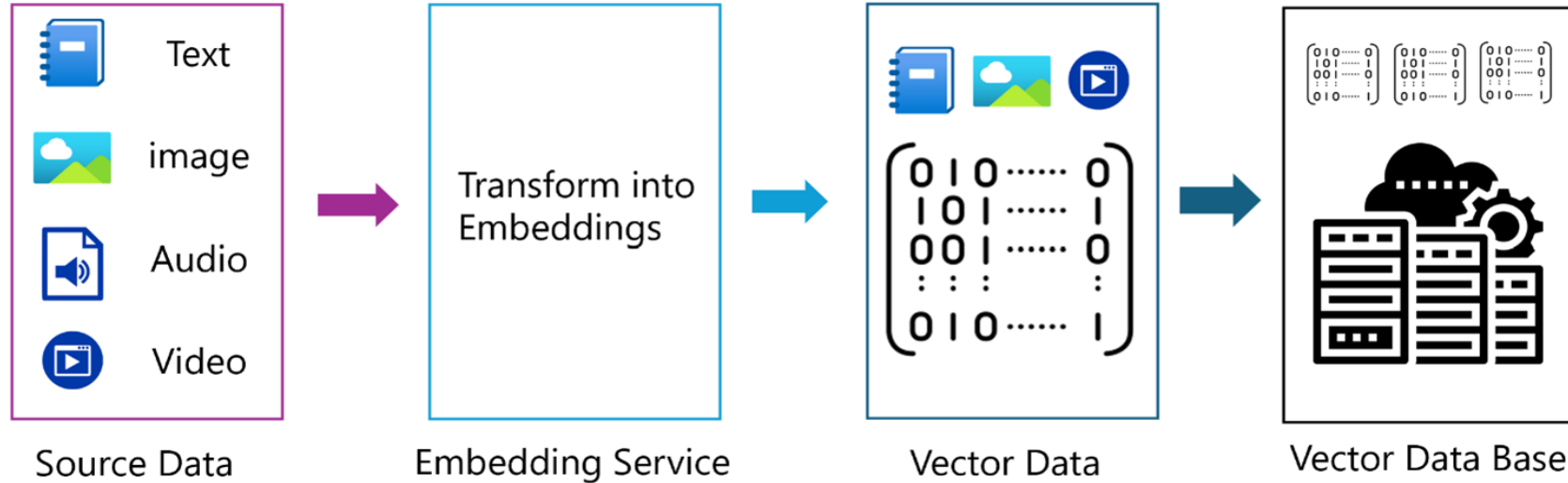
Integration with AI Models

Seamlessly works with embedding models from platforms like OpenAI, Hugging Face, or custom-trained models

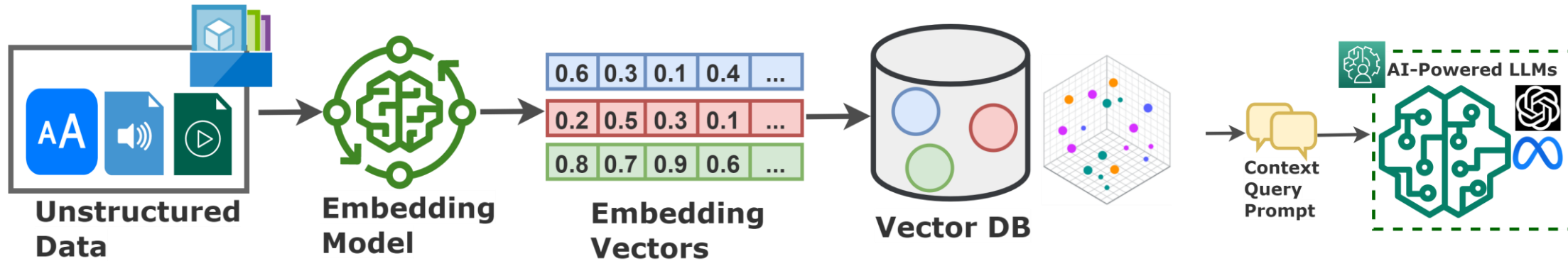


<https://www.truefoundry.com/blog/similarity-search>

What is a Vector Database? - Summary



<https://learn.microsoft.com/en-us/data-engineering/playbook/solutions/vector-database/>



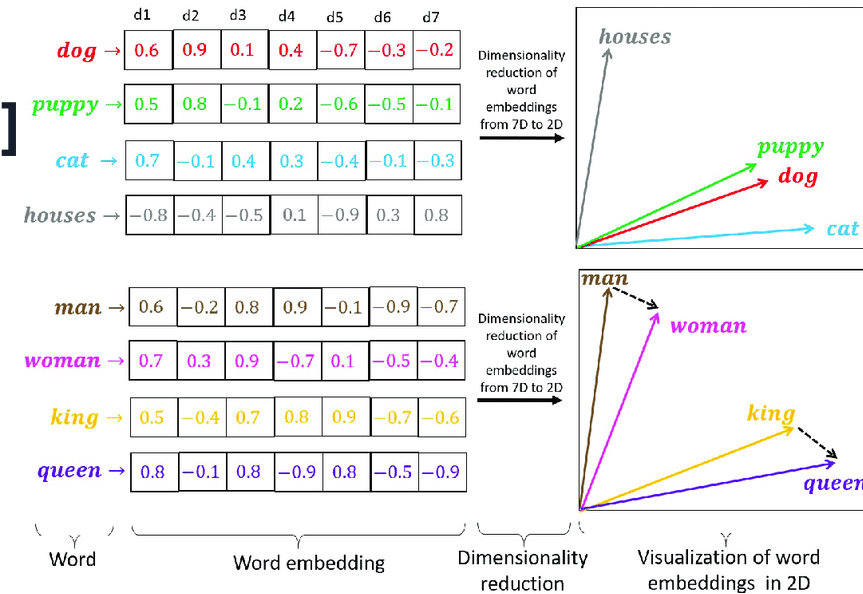
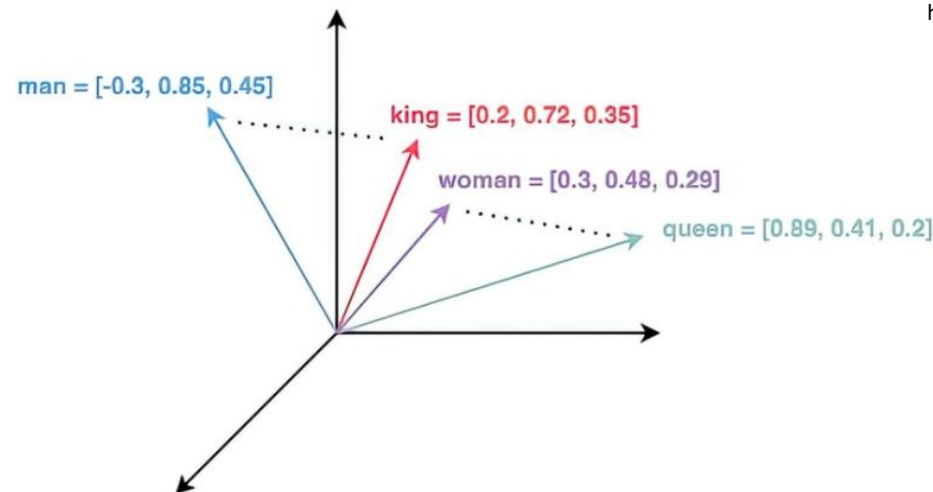
What is a Vector?

A **vector** is a mathematical object that has **both magnitude and direction**, represented as a list of numbers: **[1.2, 3.4, -0.8]**

Vectors are used to **represent complex data** in a way that **AI models** can **process**, numerical summaries of information

Vectors exist in **high-dimensional space**, which is **harder to visualize** but incredibly **powerful for computations**

Instead of saying “it’s round, orange, and sweet”, encode these features **numerically** in a **vector**: **[roundness: 0.9, color: 0.8, sweetness: 0.7]**



Rozado, David (2020). Word embeddings map words in a corpus of text to vector space.. PLOS ONE. Figure.
<https://doi.org/10.1371/journal.pone.0231189.g008>

What are Vector Embeddings?

Dense numerical representations of data, capture the semantic meaning of text, images, audio or other data types

Step 1: Input Data

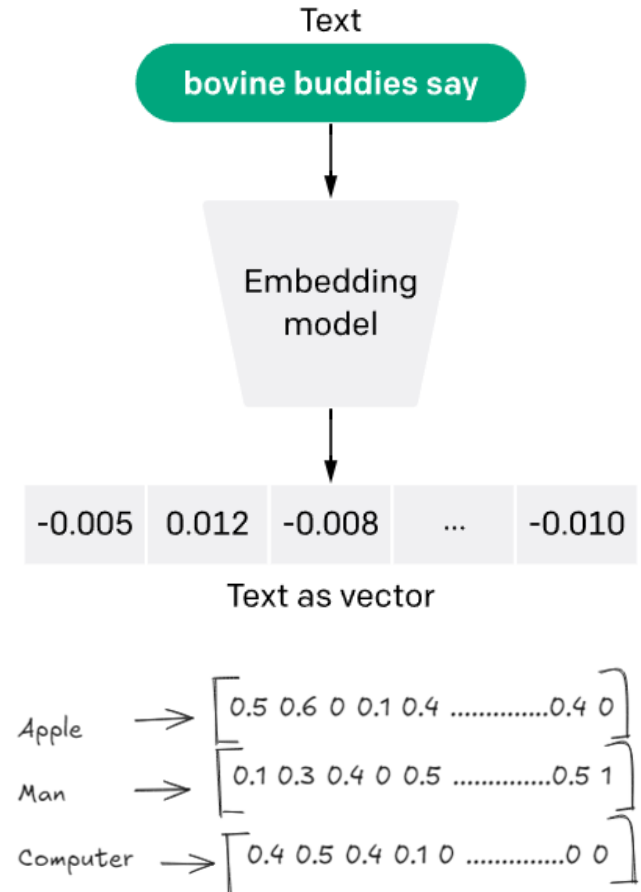
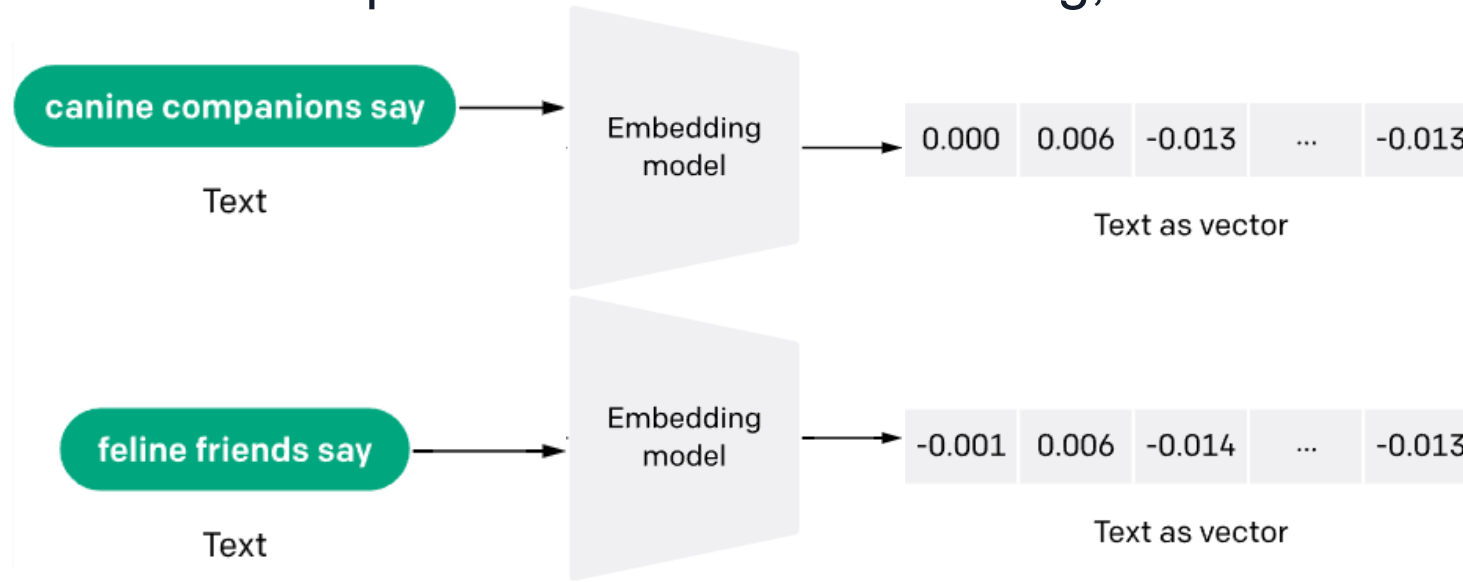
Start with raw data, like a sentence, an image, or a sound file

Step 2: Use an AI Embedding Model

Pass the data through an AI model, like a Transformer model

Step 3: Output Embedding

Transforms the input into a vector embedding, list of numbers



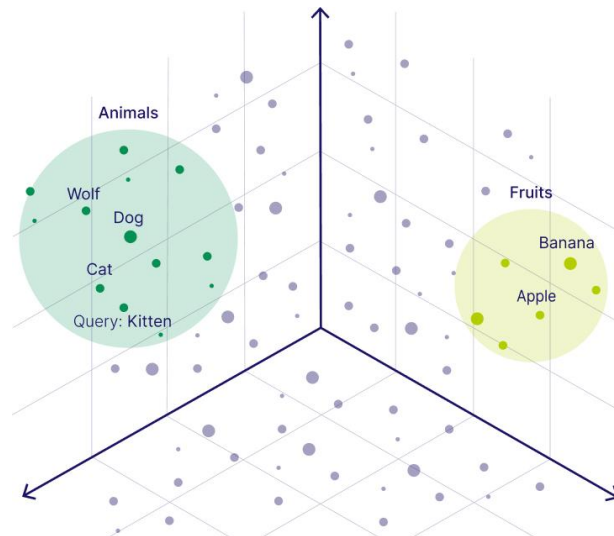
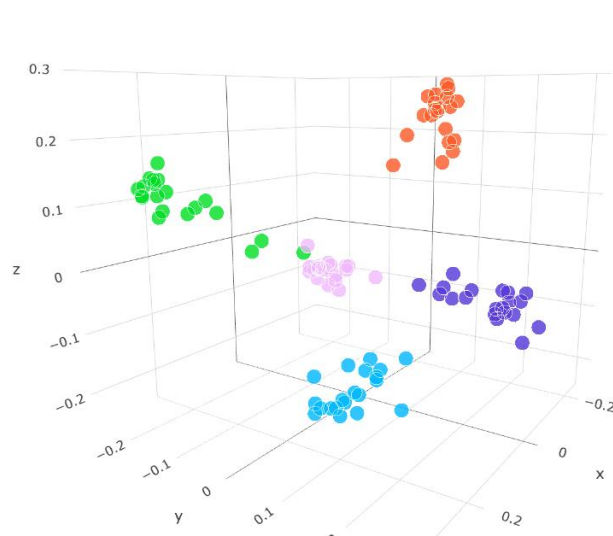
Why Are Embeddings Important?

Compare and analyze data based on semantic meaning rather than surface-level features

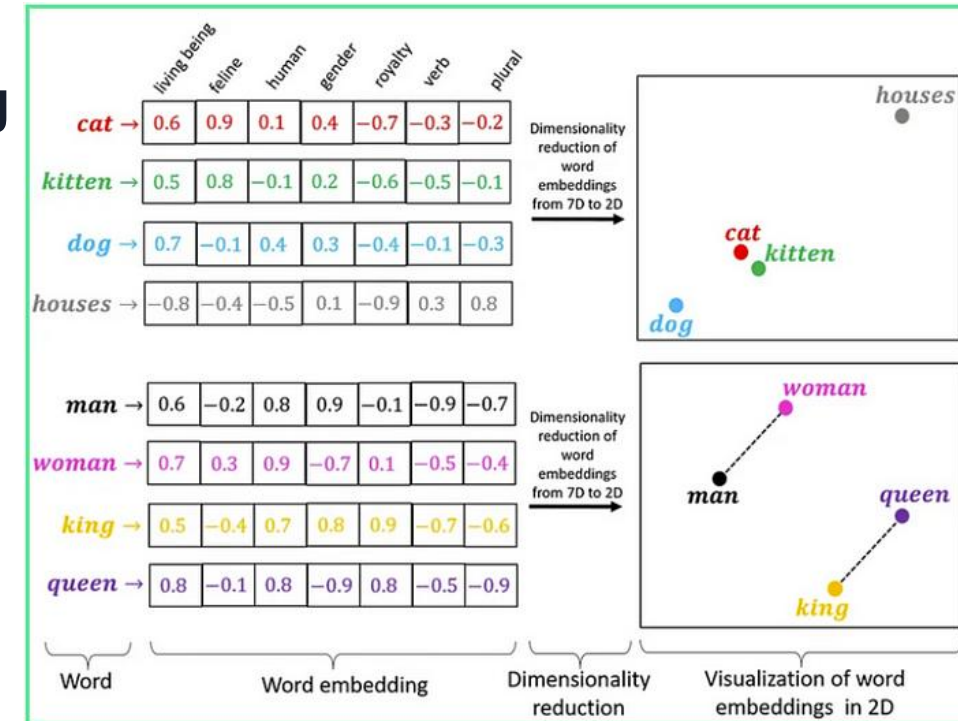
If two sentences have similar meanings, their embeddings will be close in high-dimensional space

Enable systems to retrieve relevant information quickly

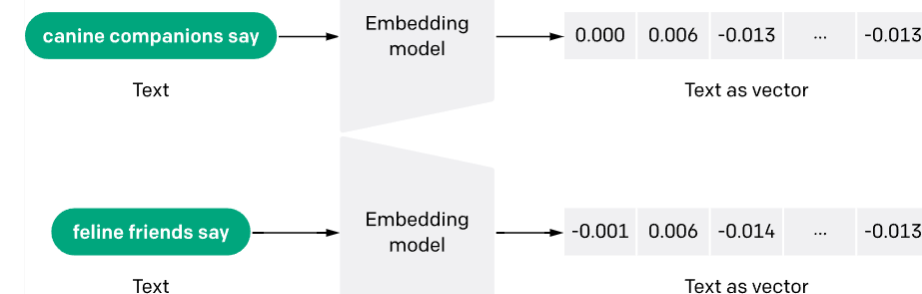
Same embedding techniques can be applied to text, images, and audio



<https://weaviate.io/blog/what-is-a-vector-database>



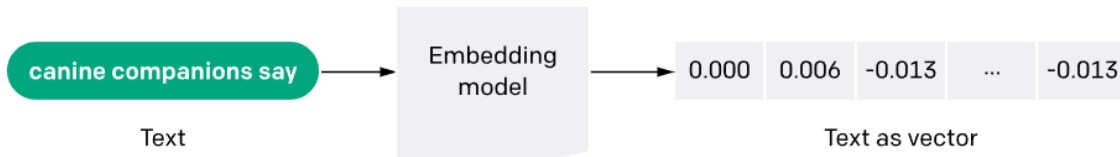
Rozado, David (2020). Word embeddings map words in a corpus of text to vector space.. PLOS ONE. Figure.
<https://doi.org/10.1371/journal.pone.0231189.g008>



Explore Vector Embedding Models: OpenAI and Ollama



OpenAI Embedding



OpenAI: text-embedding-3-small, text-embedding-ada

<https://platform.openai.com/docs/guides/embeddings/embedding-models>



"Ollama is the easiest way to get up and running with large language models."



```
[ -0.15521588921546936,  
  -0.3130679428577423,  
  -0.2622824013233185,  
  -0.10730823874473572,  
  ...  
  0.26006409525871277,  
  0.14494779706001282,  
  -0.01514953002333641,  
  0.04403747618198395]
```

Ollama: all-minilm, mxbai-embed-large, nomic-embed-text

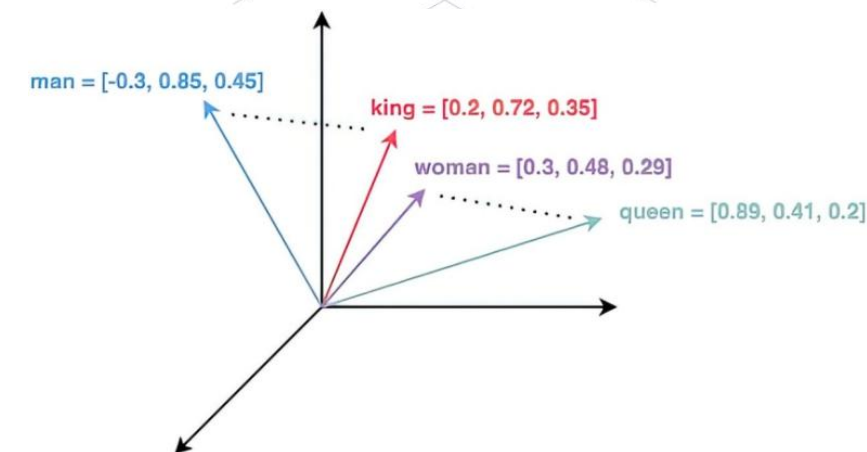
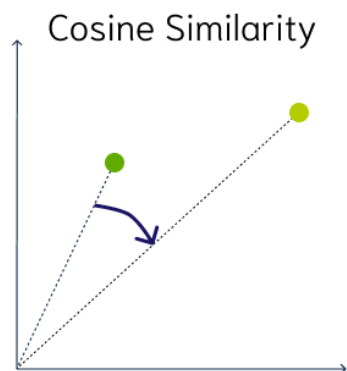
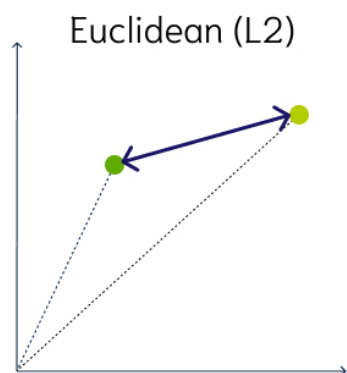
<https://ollama.com/blog/embedding-models>

Semantic Meaning and Similarity Search

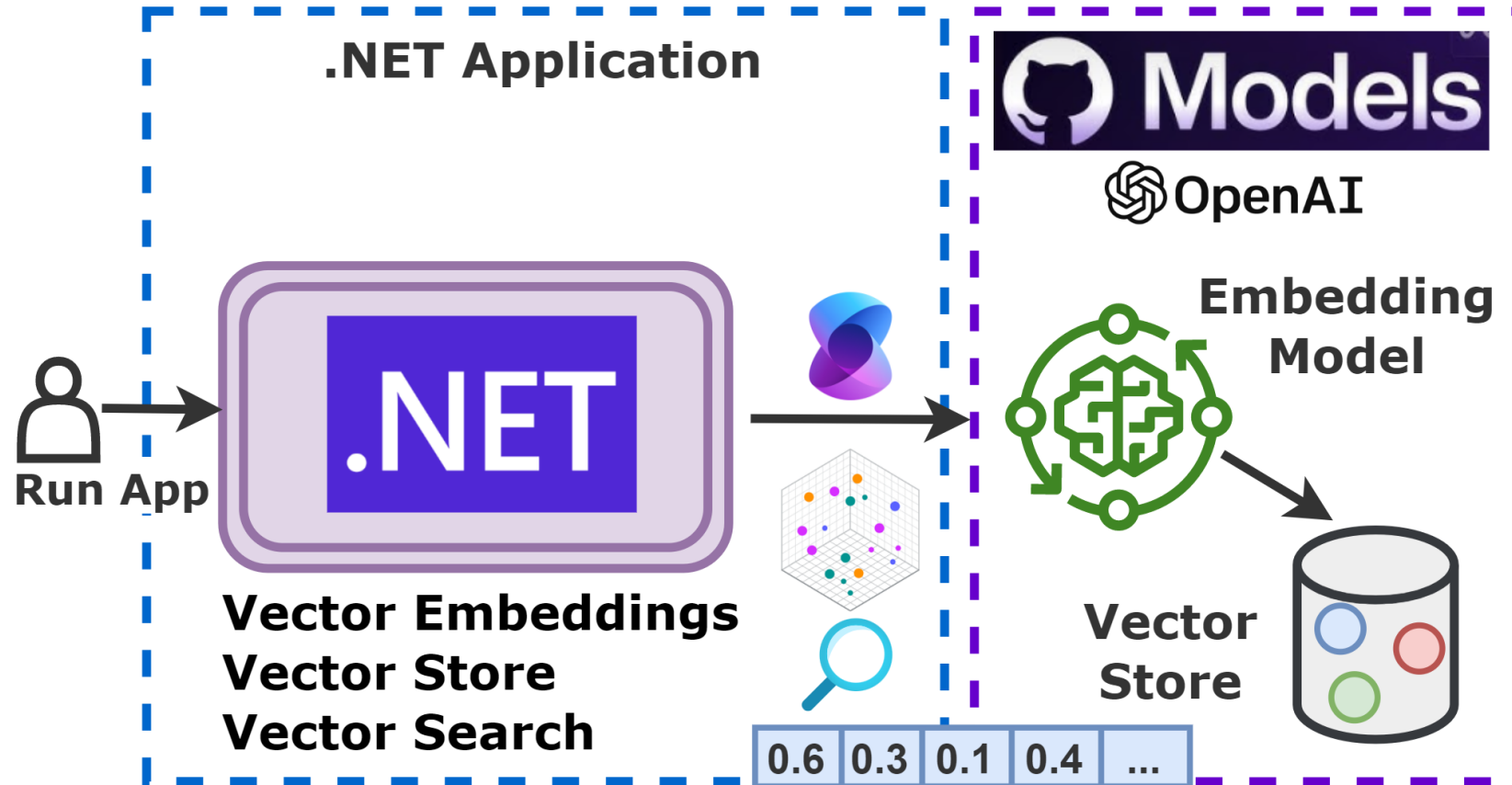
Understanding semantic meaning in **Vector Embeddings**

Exploring **Similarity Search** techniques

Applications of **Cosine Similarity** and **Euclidean Distance**



.NET AI Vector Search using Vector Embeddings and Vector Store



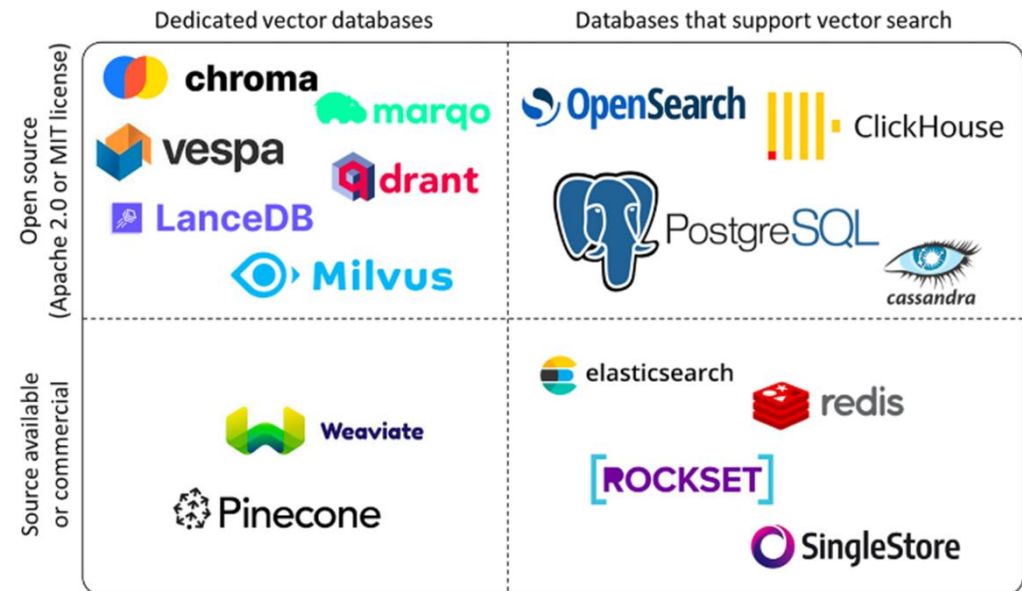
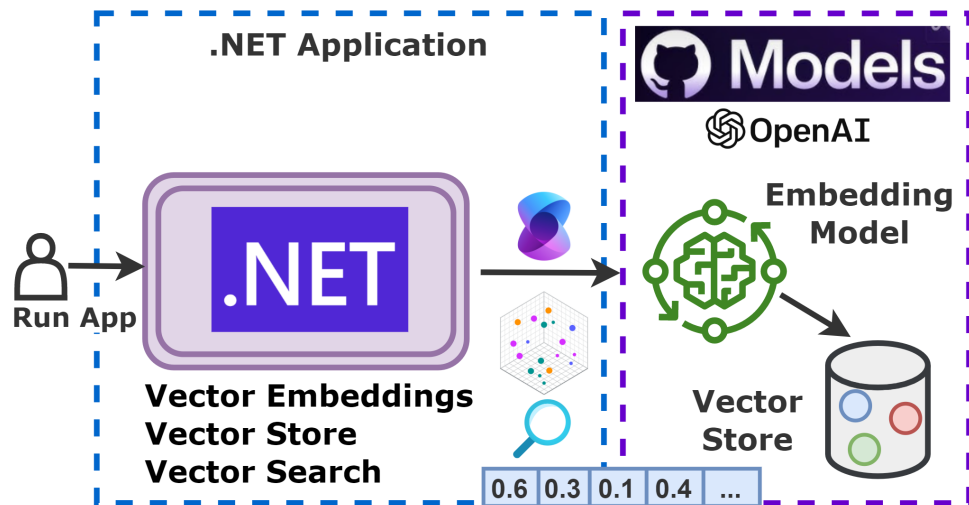
Connect to Various Vector Databases

The AI ecosystem is exploding with options for vector databases—Chroma, Qdrant, Pinecone, Weaviate

Each have their own unique SDK, their own specific methods

Write all your app code directly particular DB SDK makes locked in

Decide to switch to Qdrant for better performance, you're facing a significant rewrite of your data access layer



App code remains flexible, Swap underlying vector store implementations (local or cloud-based) with minimal changes



Core Concepts and APIs of VectorData.Abstractions

Nuget Packages

Microsoft.Extensions.VectorData.Abstractions, Microsoft.Extensions.AI

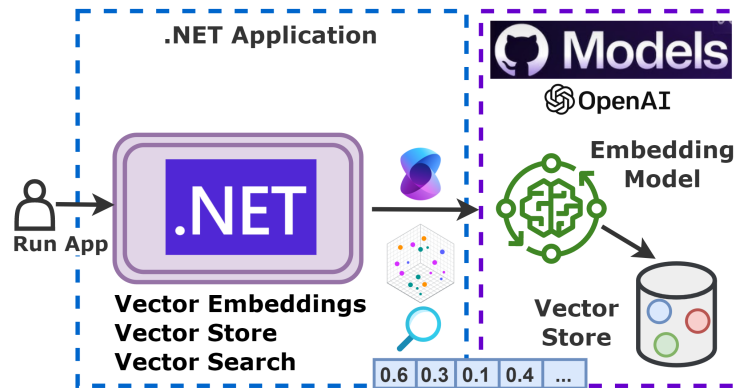
VectorData Stores

Configure VectorData Store in your app, VectorData.Abstractions provides interfaces to Add, Update, Delete, and Search vector records

Vector and Text Search

Vector Search: Query a vector store by specifying an embedding vector and similarity metric

Text Search: Convert text into embeddings, then query the store using the same semantic approach



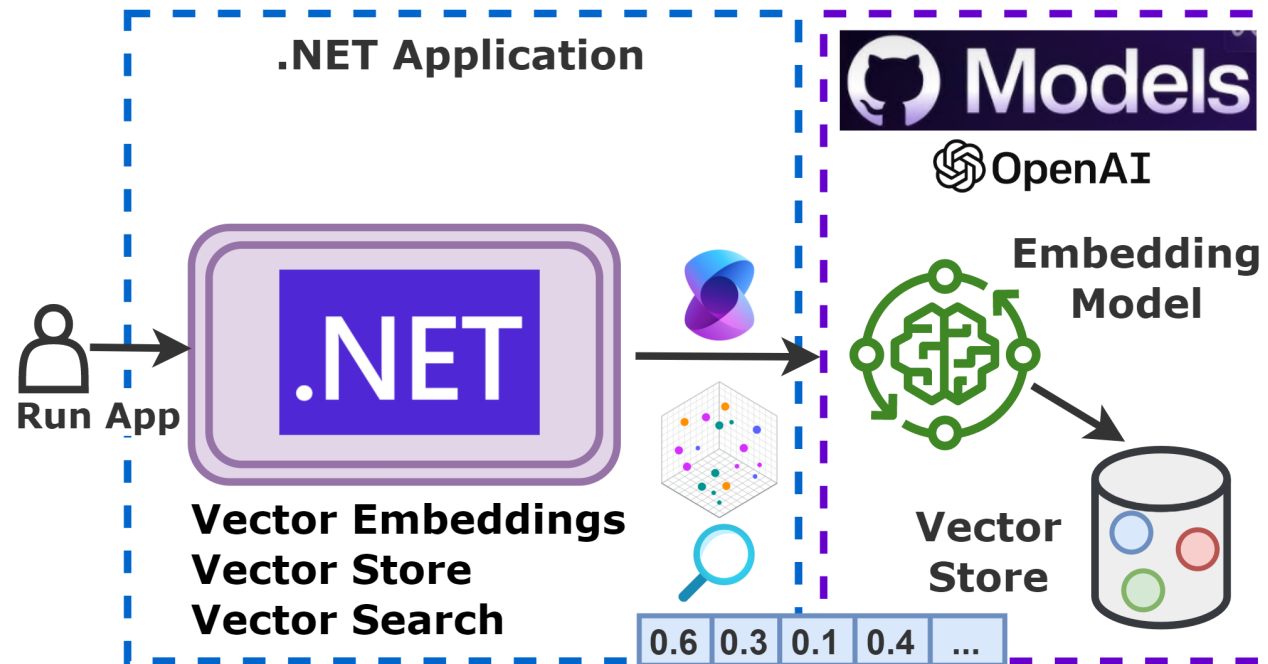
Create VectorSearch App

Create and populate a vector store by generating embeddings for a data set

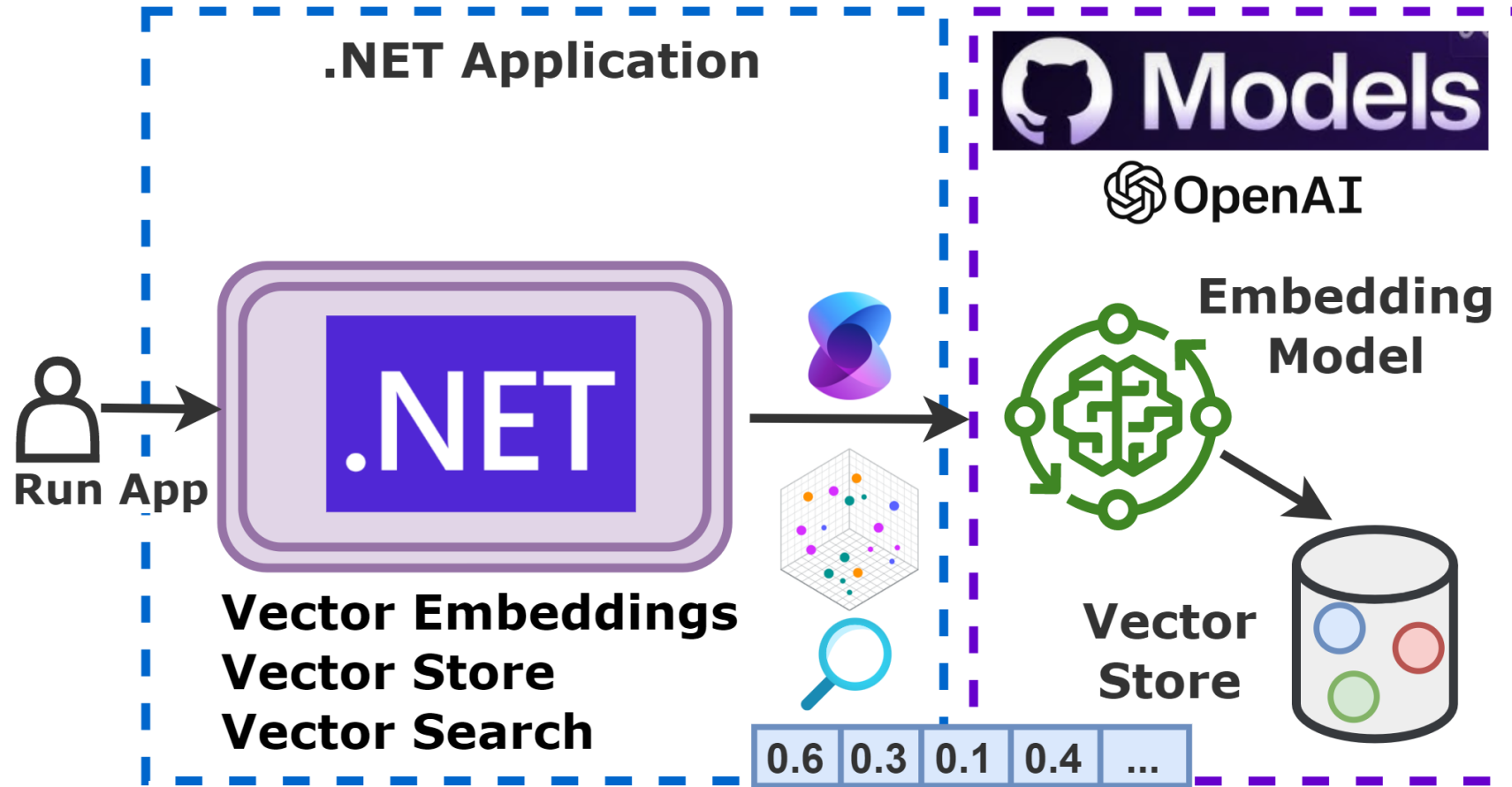
Generate an embedding for the user prompt

Query the vector store using the user prompt embedding

Display relevant results from the vector search



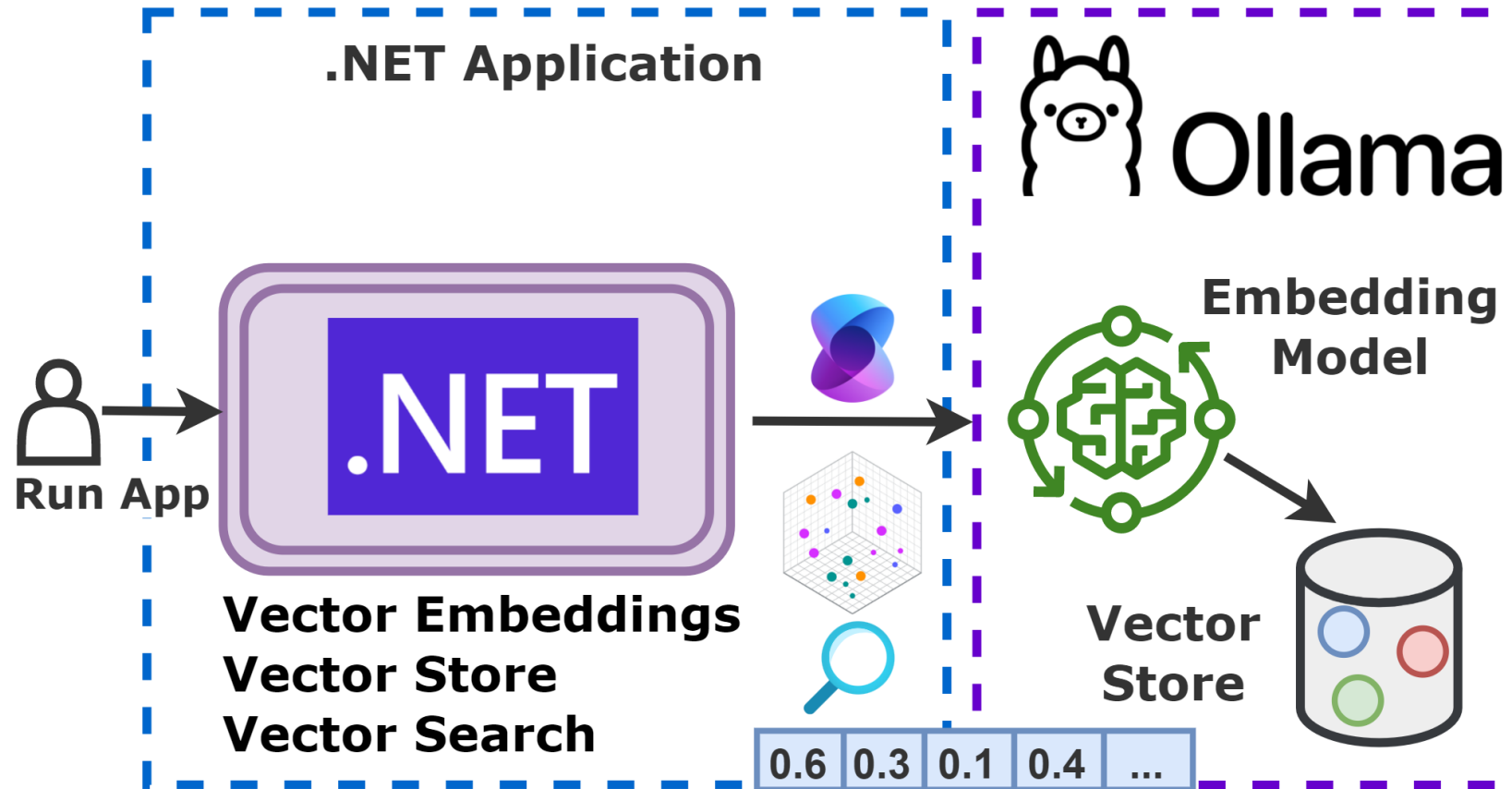
.NET AI Vector Search using Vector Embeddings and Vector Store



Nuget Packages

Microsoft.Extensions.VectorData.Abstractions,
Microsoft.Extensions.AI

.NET AI Vector Search using Vector Embeddings and Vector Store - Ollama



Retrieval Augmented Generation (RAG) Application w/ .NET AI

What is Retrieval-Augmented Generation (RAG) ? How RAG Work ?

Build & Explore .NET Chat App w/ RAG Template

Extend RAG Chat App with Your Own Documents (Ingest and Embeddings Custom Pdf files)

Extend RAG Chat App Functionality with Function Calling Tool

Mehmet Ozkaya



Introduction to This Section

What is Retrieval-Augmented Generation (RAG) ?

Why Need Retrieval-Augmented Generation (RAG) ? Why is RAG Important ?

How Does Retrieval-Augmented Generation (RAG) Work ?

Build .NET Chat App w/ RAG Template

Explore .NET Chat App RAG Solution

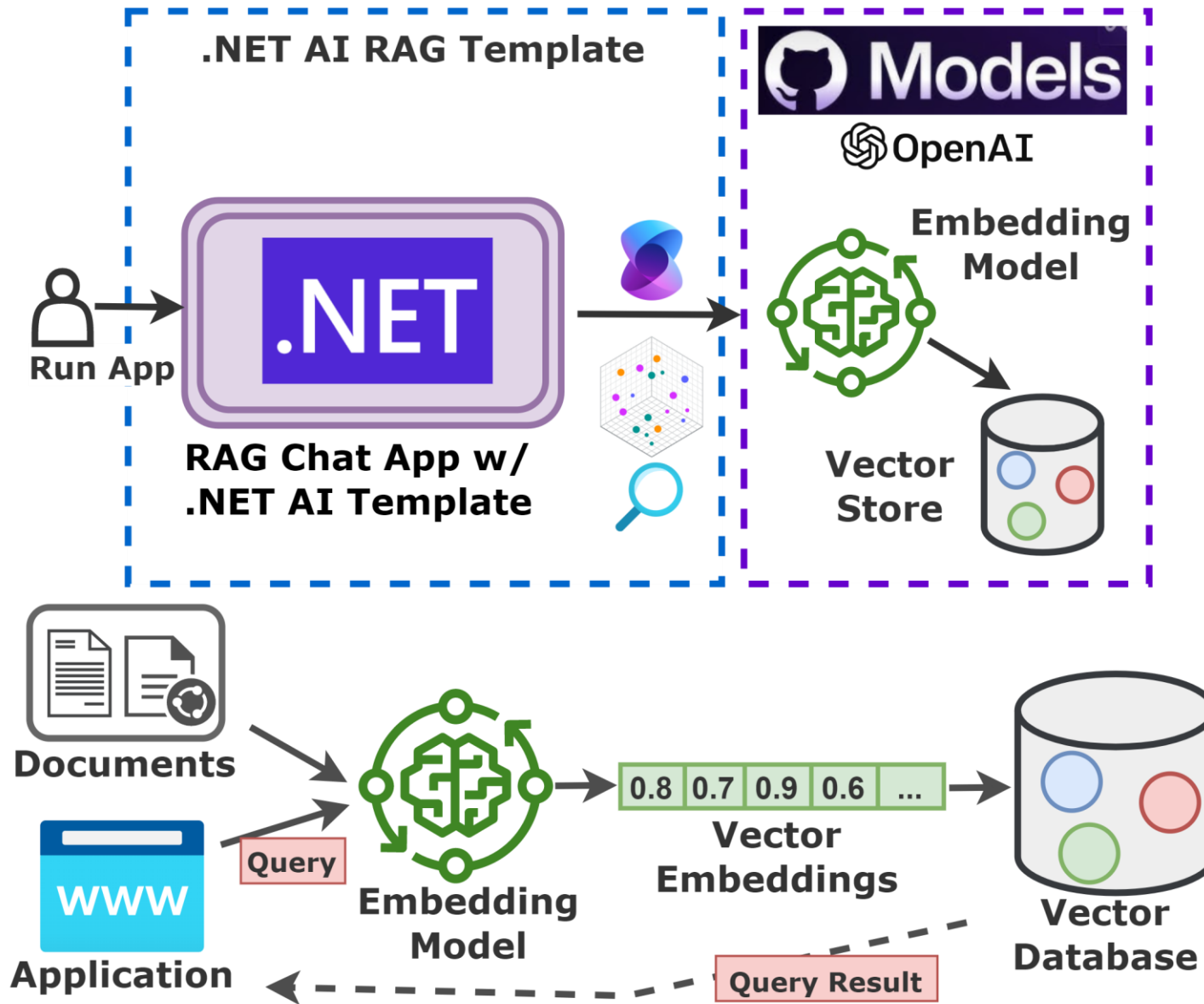
Run & Test .NET Chat App RAG Solution

Extend RAG Chat App with Your Own Documents (Ingest and Embeddings Custom Pdf files)

Extend RAG Chat App Functionality with Function Calling Tool

Build .NET Chat App w/ RAG Template using Ollama and all-minilm

Retrieval Augmented Generation (RAG) Application w/ .NET



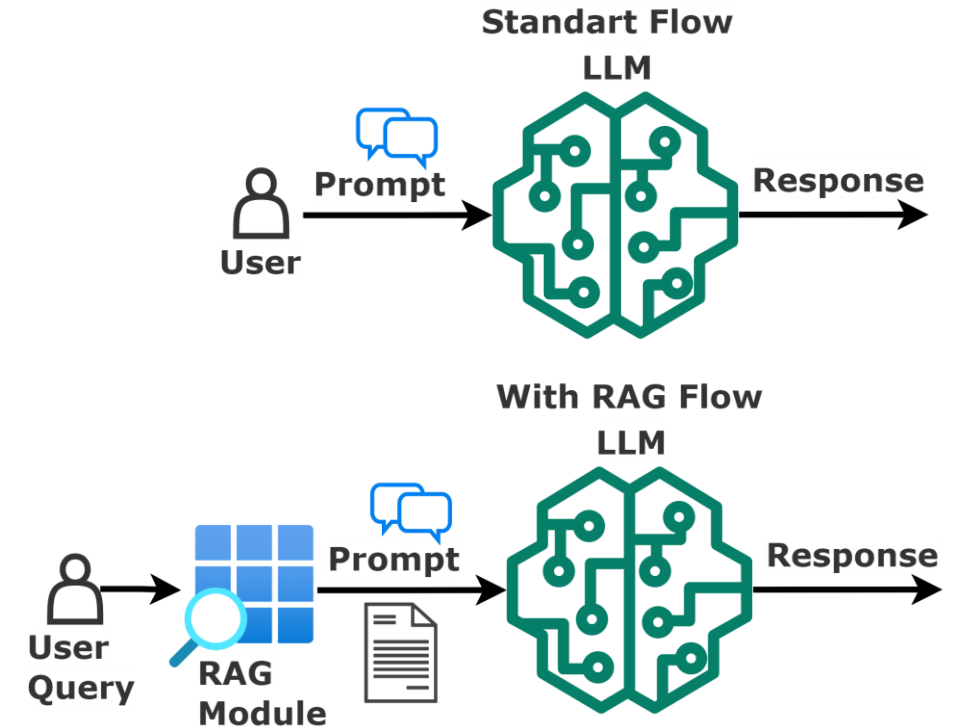
What is Retrieval-Augmented Generation (RAG) ?

Enhances the model's **ability** to **generate more accurate and relevant information**

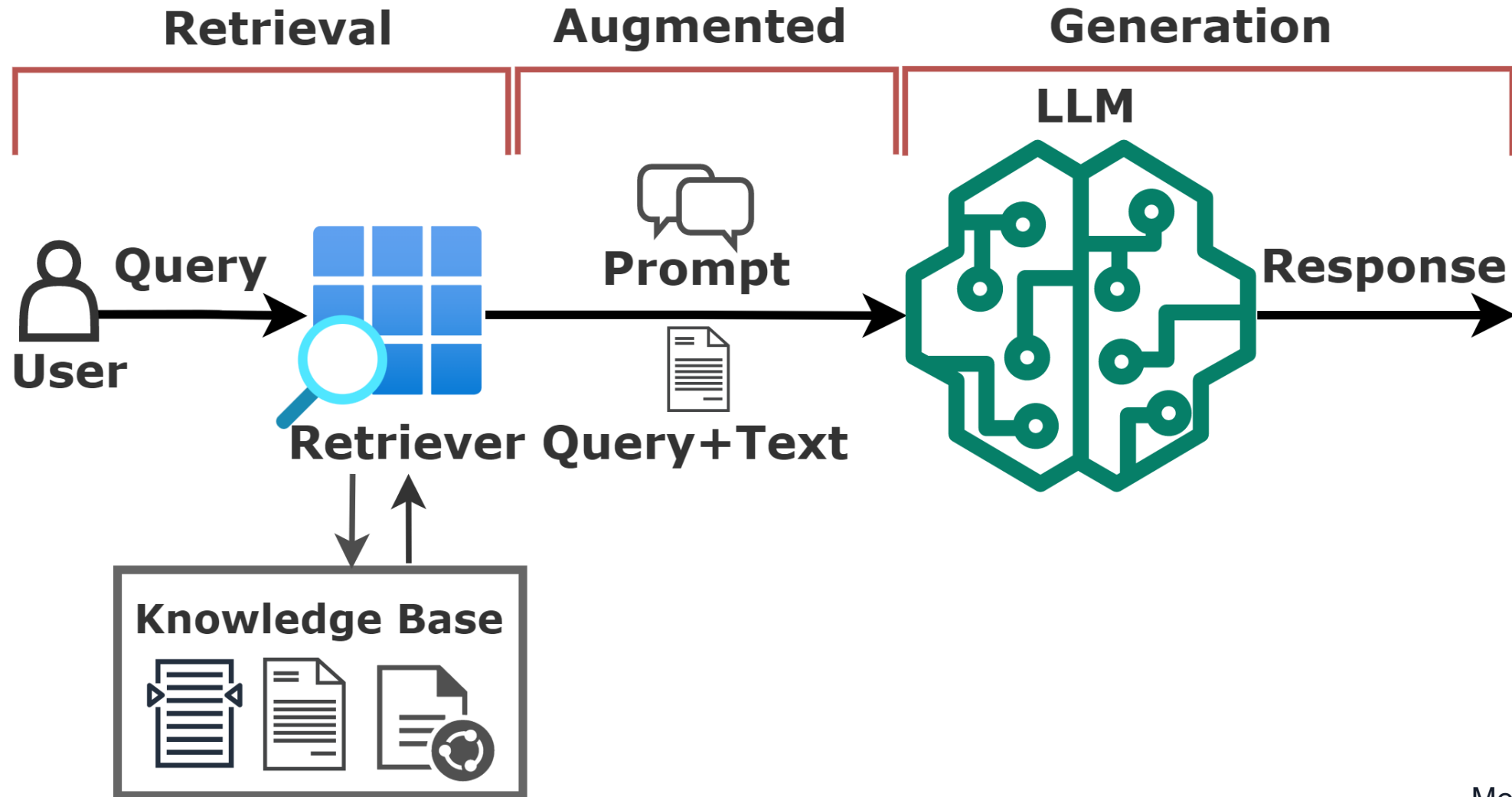
By **integrating external knowledge** into the response

Bridges the gap with **pre-trained knowledge** and the **real-time information** that is not be part of its training data

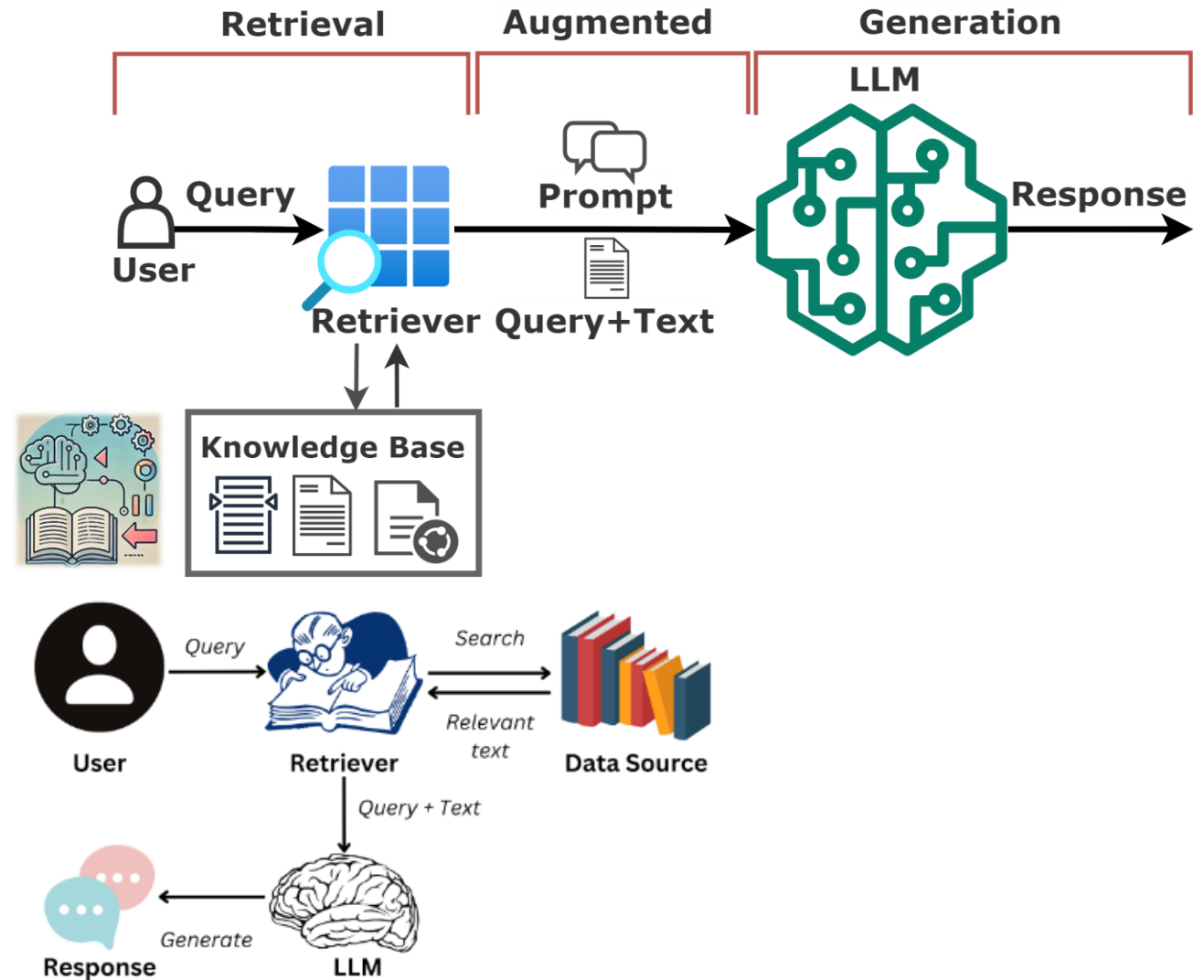
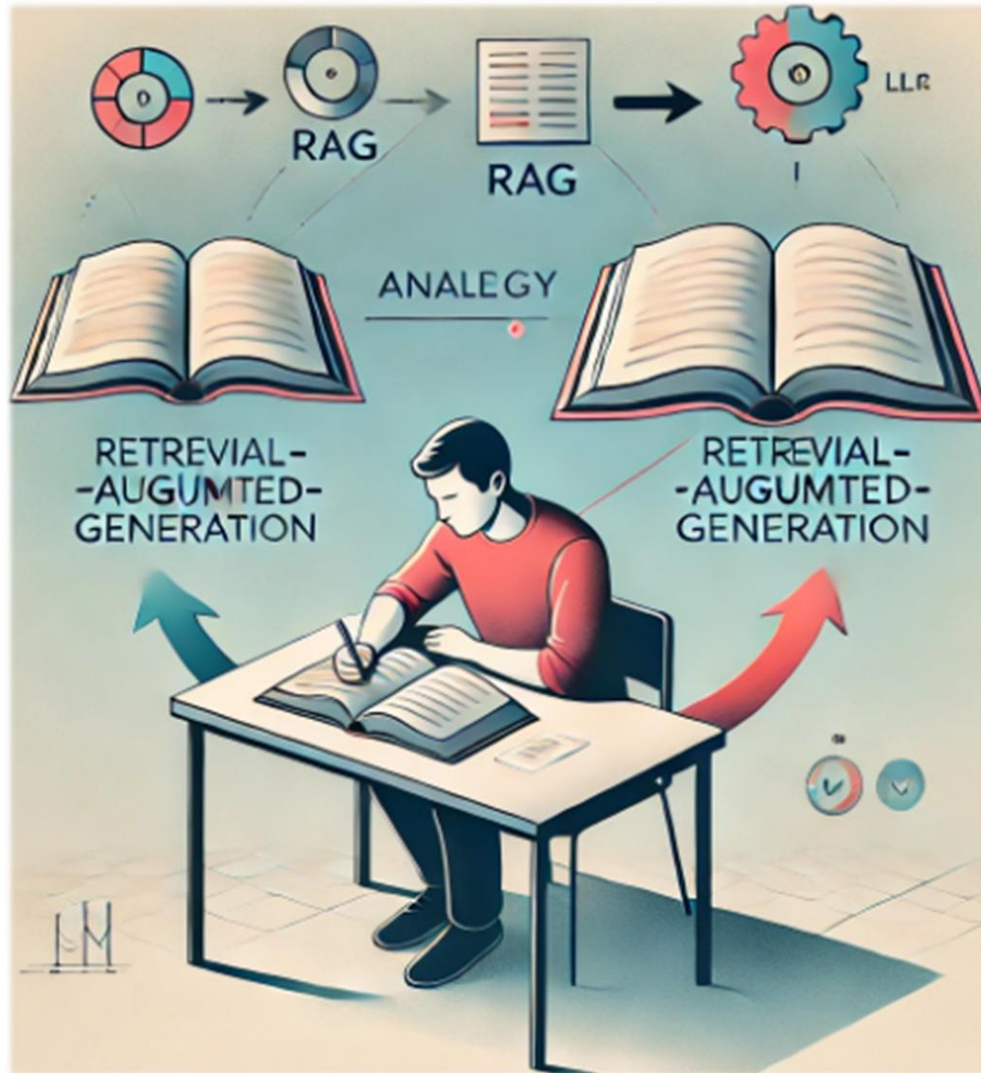
Pull in information from **real-time** or **external knowledge** sources, make responses **more accurate and relevant to specific queries**



Basics of RAG = Retrieval + Augmented + Generation



RAG is Like an Open-Book Exam



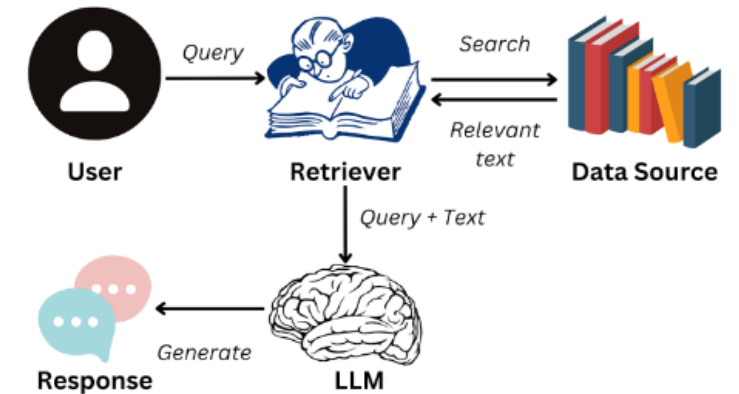
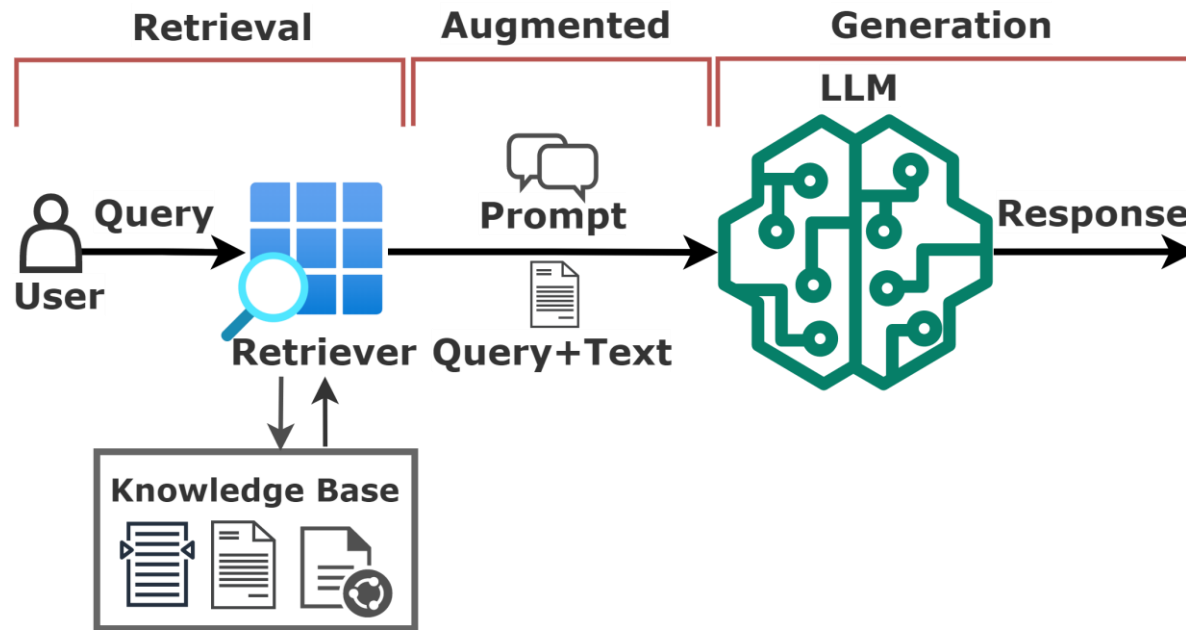
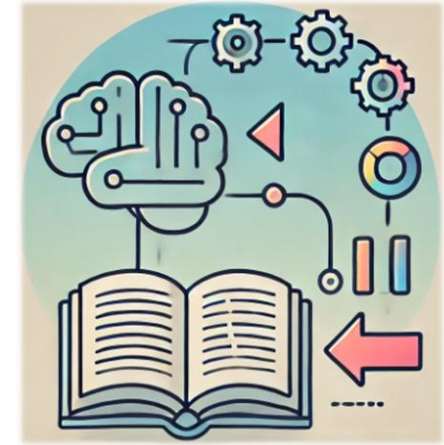
Why Do We Need RAG ?

LLMs are **powerful**, but they are **limited** to the **data** they were **trained on**

Their **knowledge becomes static** the moment **training stops**

They **lack access** to **real-time** or **specialized information**

RAG combines a retrieval mechanism with the **generative power of LLMs**



Limitations of Traditional Language Models

Outdated Information

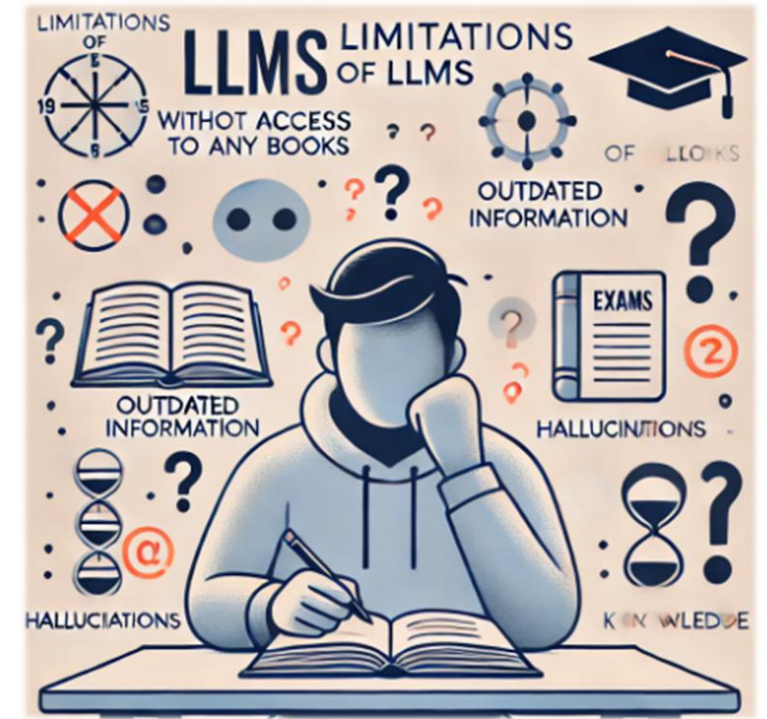
LLMs are only as current as their training data. GPT was trained in 2021, won't have any information from 2022 or beyond

Hallucinations

LLMs sometimes generate incorrect or misleading information, predict the next word without understanding the facts

Lack of Contextual Knowledge

LLMs cannot access specific or proprietary data. Can't provide information from specific product manuals or private databases

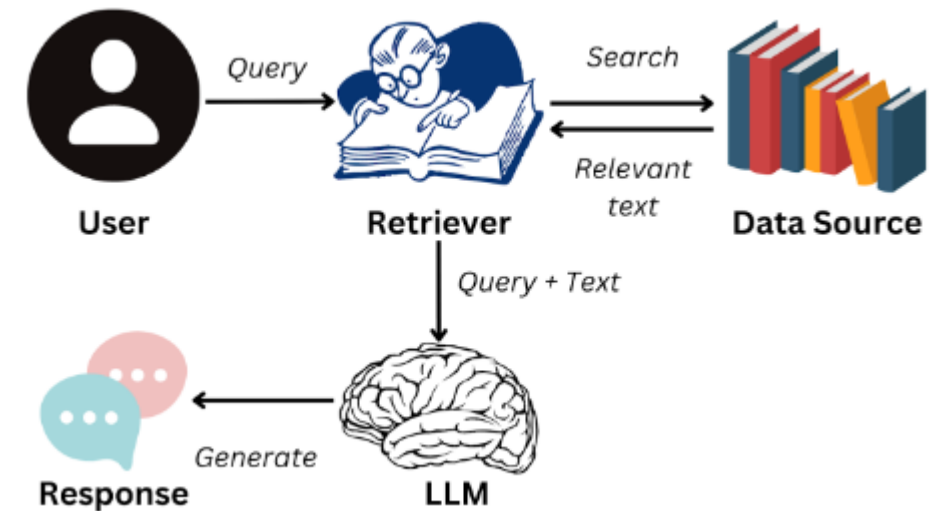
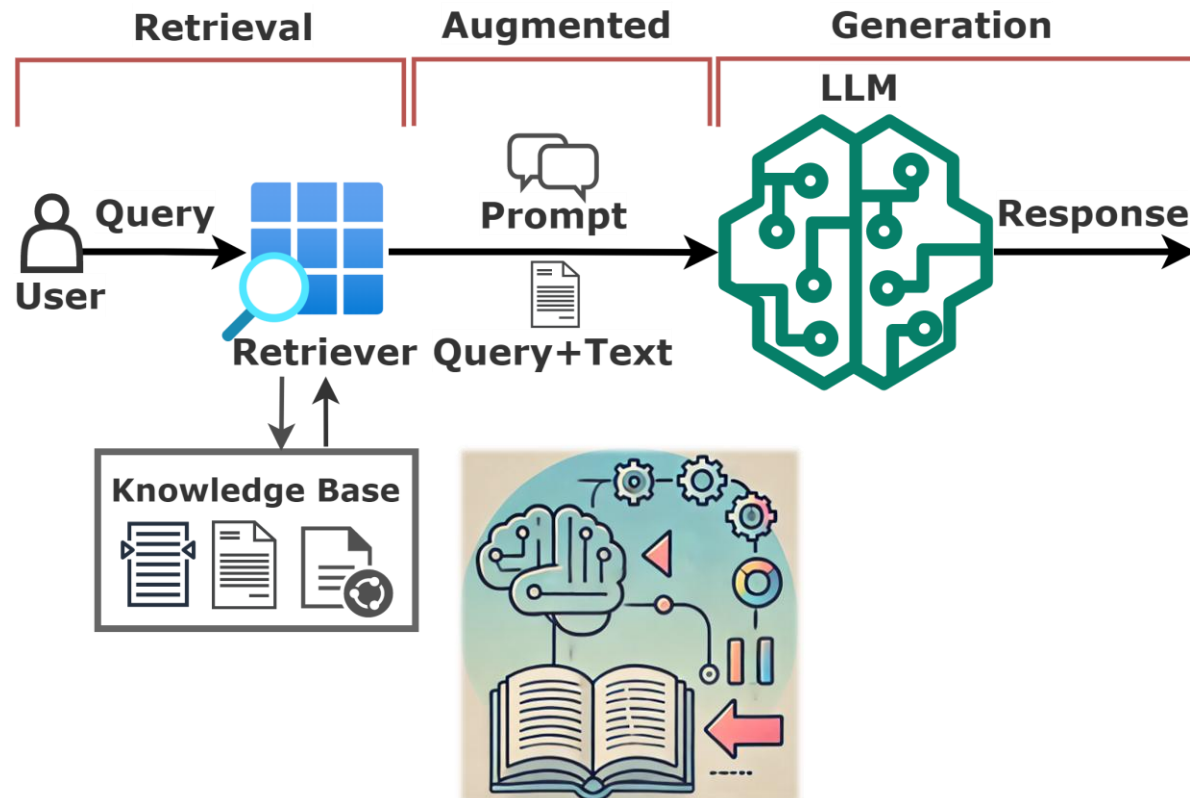


Why Do We Need RAG ? - Continue

RAG Solves the Problem of Static Knowledge: It retrieves real-time information

Access to External Sources: RAG pulls from documents, databases, and APIs

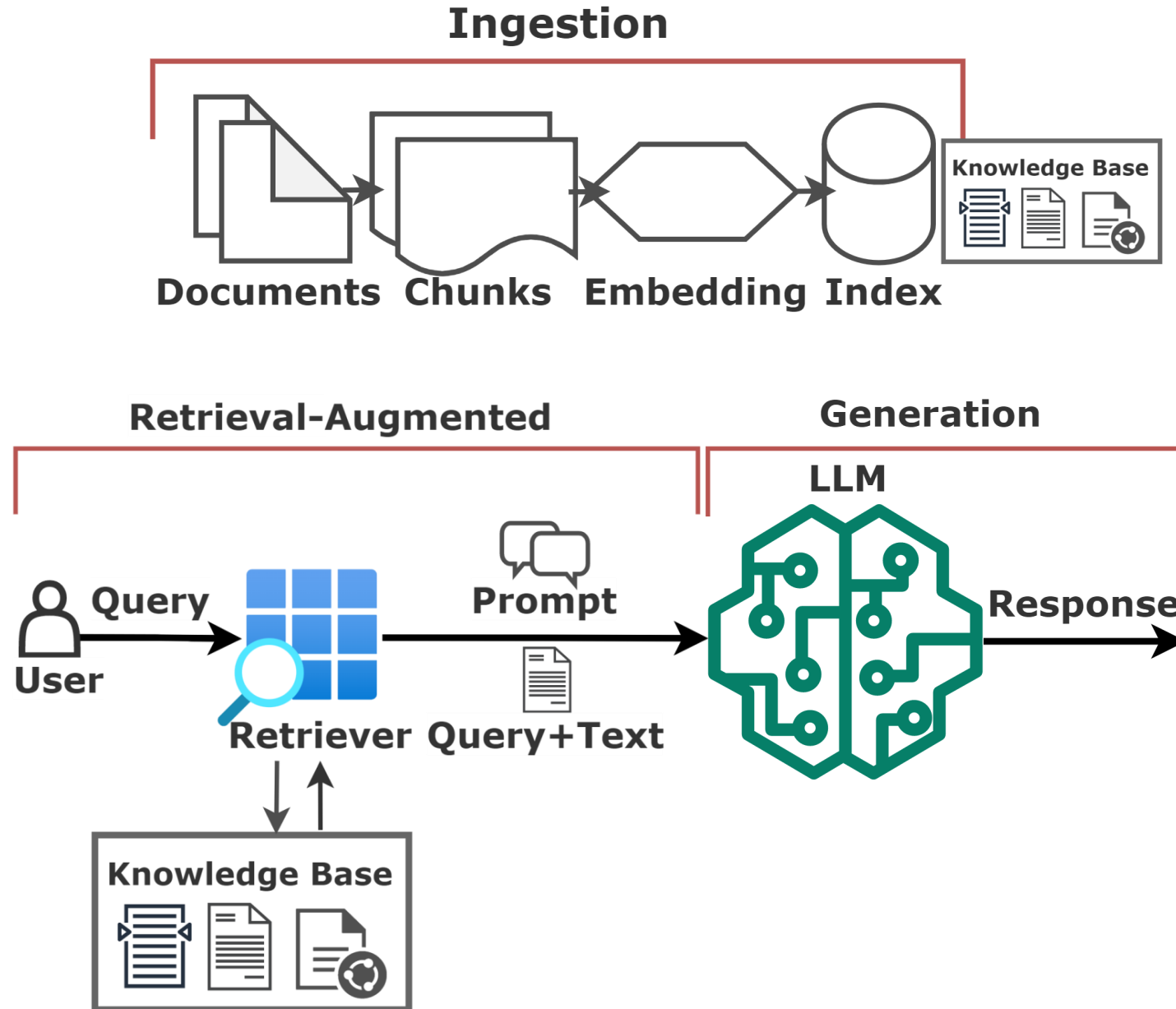
Combines Knowledge with Context: Bridges the gap between general knowledge and specific, up-to-date information



How Does RAG Work ?

3 Steps:

1. Ingestion
2. Retrieval
3. Generation



1. Ingestion – Building a Knowledge Base

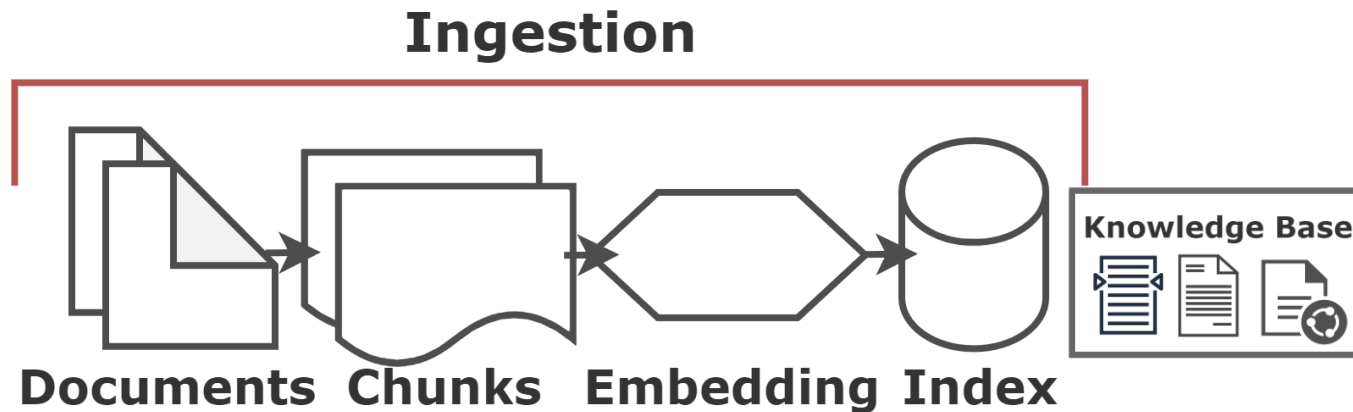
Collecting Data

Gather information from a wide variety of sources—from databases, documents, real-time information from APIs

Organizing Information

Needs to be organized for easily accessible in the retrieval step. Separate chunks, insert embeddings and create indexes

Creating **numerical representation** of documents into **vector databases**



2. Retrieval – Search Relevant Data

Searching the Knowledge Base

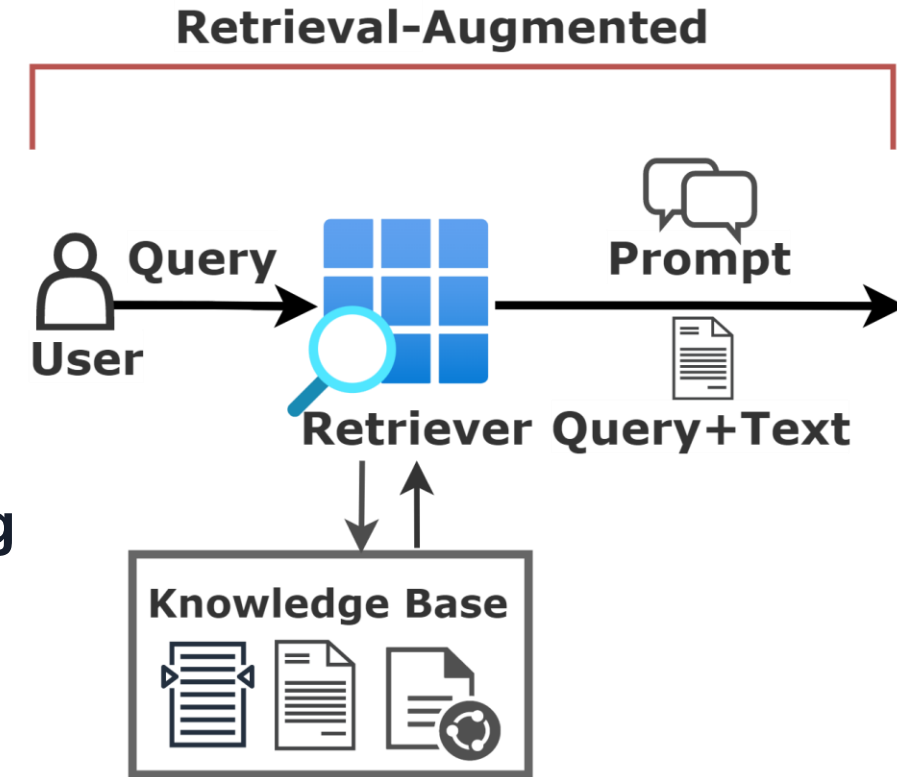
Search through the knowledge base: internal databases, documents, or external sources

Identifying Relevant Data

Identify and pulls the most relevant pieces of information related to the user's query

AI isn't just **guessing based** on its **training data**—it's **pulling** from **real-time** and **proprietary data sources**

Retrieve **specific troubleshooting** steps from a **product support document**



3. Generation – Create Coherent Response

Using Retrieved Data

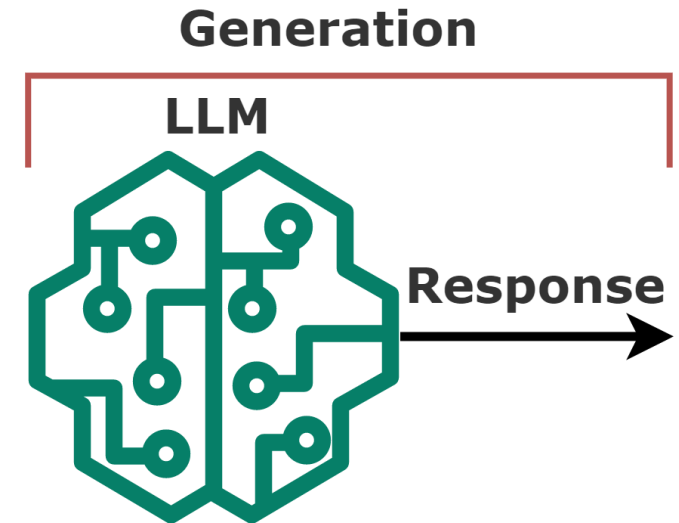
Uses retrieved information to generate a coherent, relevant response by synthesizes the retrieved information

Combining Knowledge

Retrieved data + pre-trained knowledge; fill in the gaps and create a response both accurate and contextual

Natural Language Output

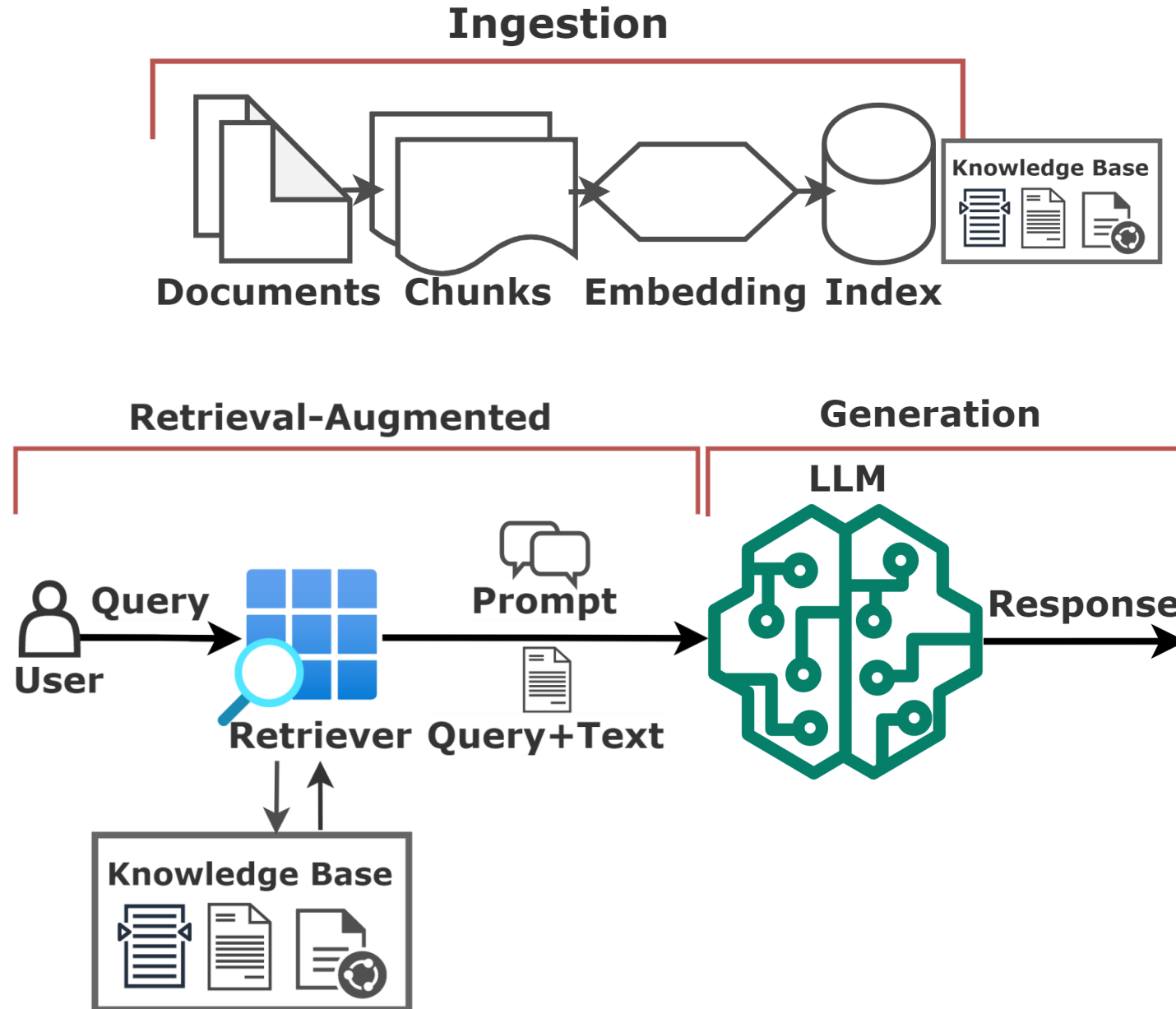
Reads naturally and is grounded in real data—whether that data comes from product manual, database or external API



How Does RAG Work ?

3 Steps:

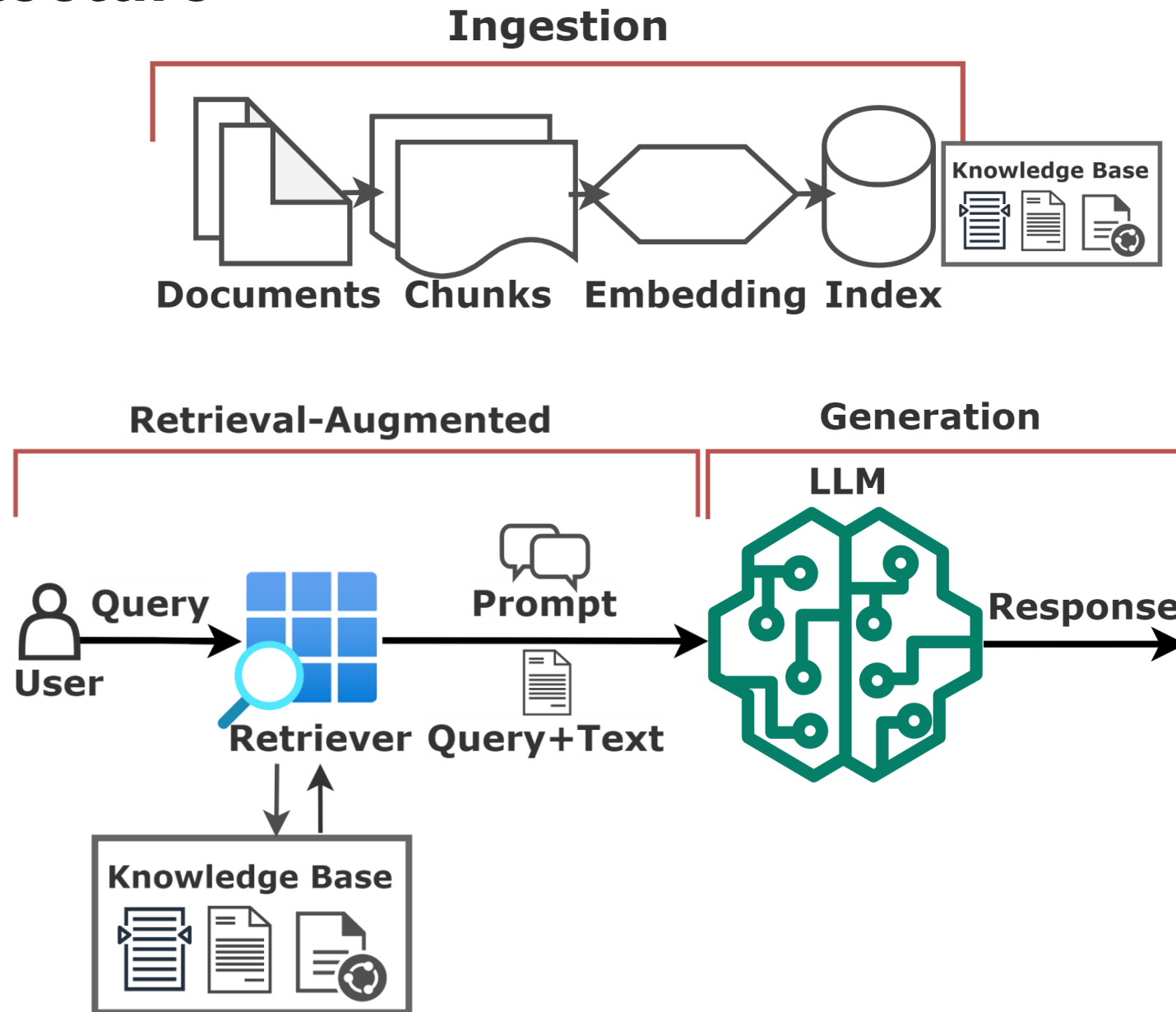
1. Ingestion
2. Retrieval
3. Generation



RAG Architecture

3 Steps:

1. Ingestion
2. Retrieval
3. Generation



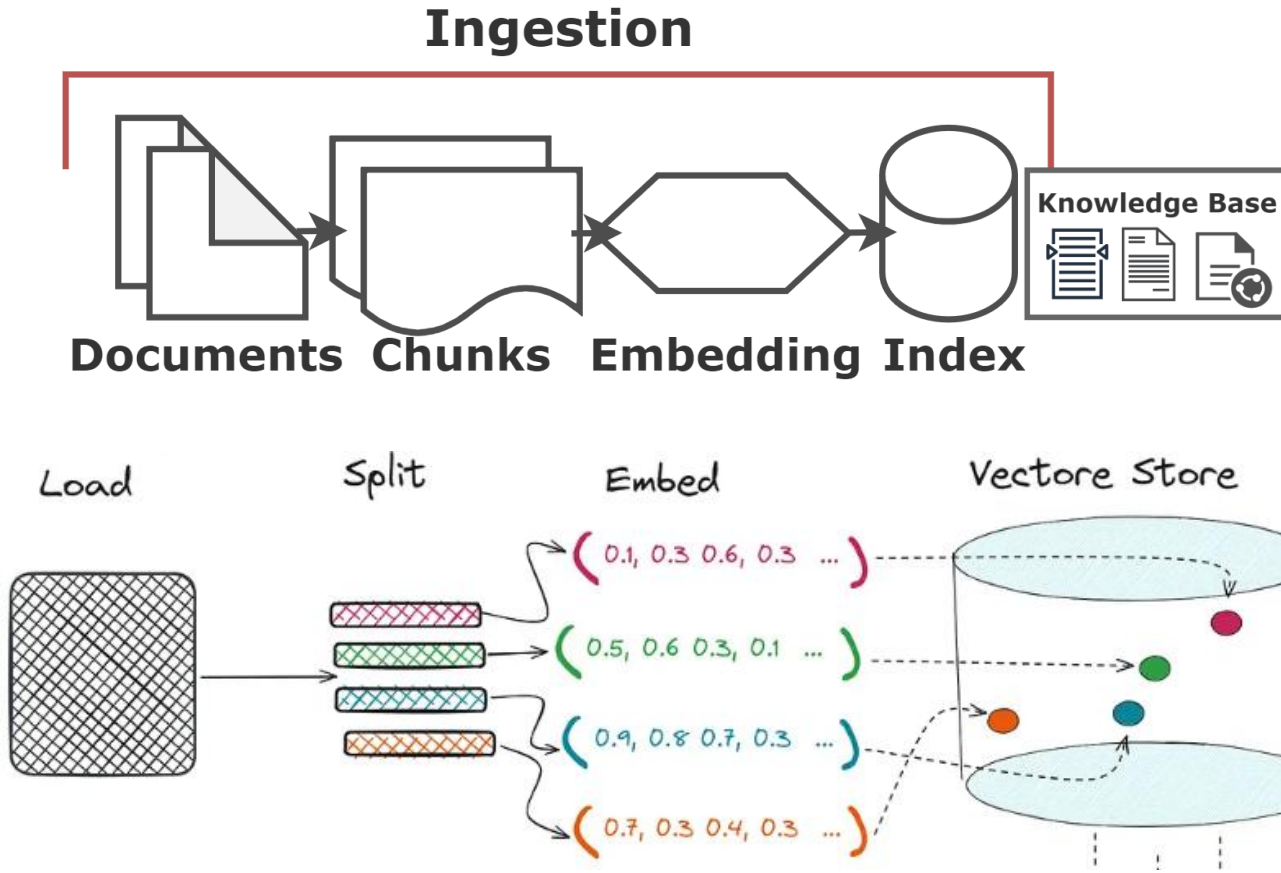
Ingestion

Retrieval

Generation

RAG Components : Focus on Ingestion

Documents, Embedding, Embeddings Model, Vector Search, Vector Store



<https://towardsdatascience.com/advanced-retriever-techniques-to-improve-your-rags-1fac2b86dd61>

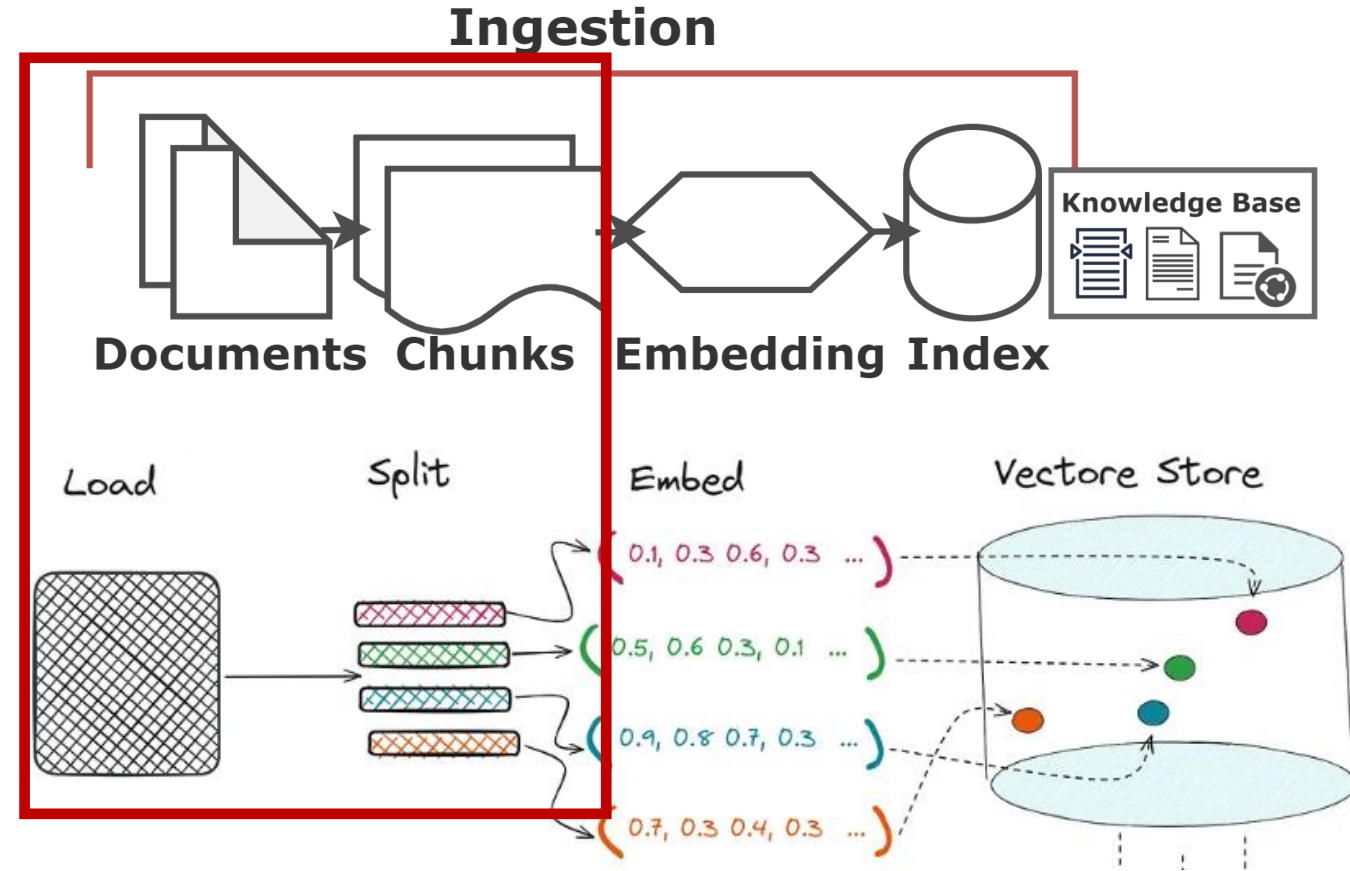
Documents – Building the Knowledge Base

Collection of documents forms the **core knowledge base**

Documents are the **sources of information** that the system **draws** from to **generate answers**

Can include **manuals**, **research papers**, or **legal documents**

The system **ingests** this **data** and **indexes** it for **retrieval**



<https://towardsdatascience.com/advanced-retriever-techniques-to-improve-your-rags-1fac2b86dd61>

Embeddings – Converting Text into Vectors

Embeddings are numerical representations of text that capture the semantic meaning of words and phrases

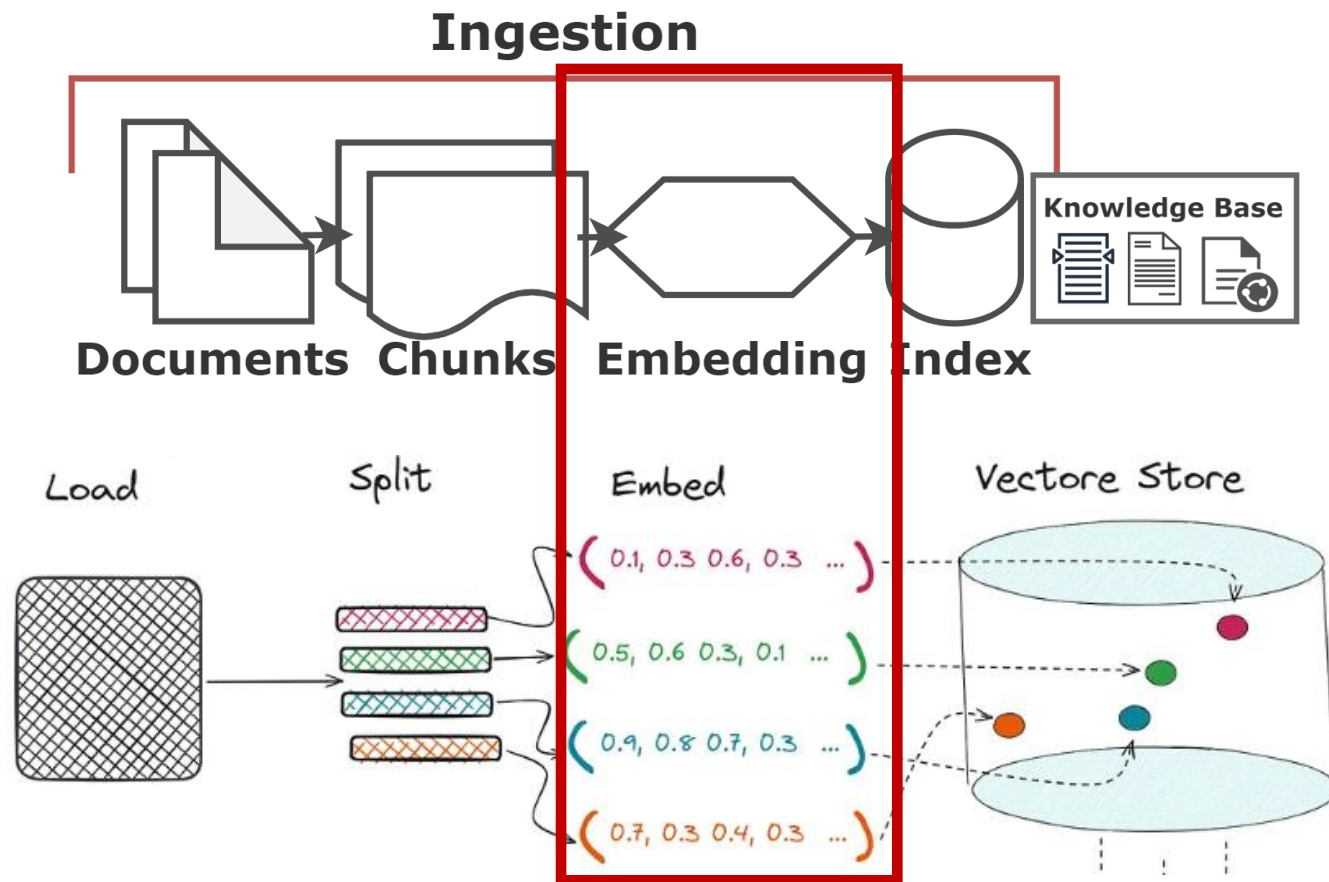
Embedding models for Transforms text into numerical vectors (embeddings)

Encodes semantic meaning of the text

Enables to perform similarity-based retrieval

'car' and 'automobile' have similar embeddings

Find documents that semantically relate to the user's query



<https://towardsdatascience.com/advanced-retriever-techniques-to-improve-your-rags-1fac2b86dd61>

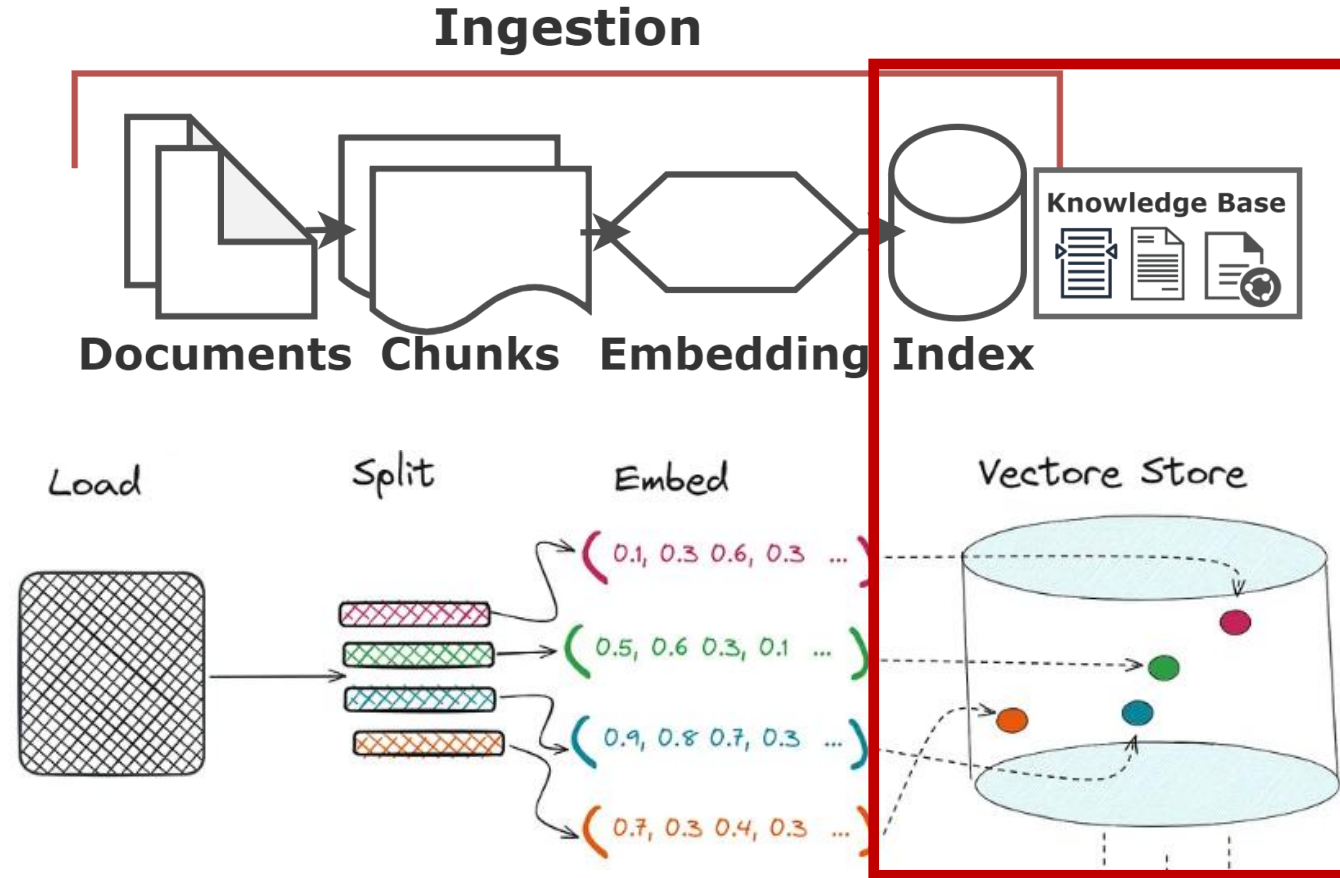
Vector Store – Storing and Managing Vectors

A specialized database for storing high-dimensional vectors

Stores millions of **vectors** and **access data efficiently** during retrieval

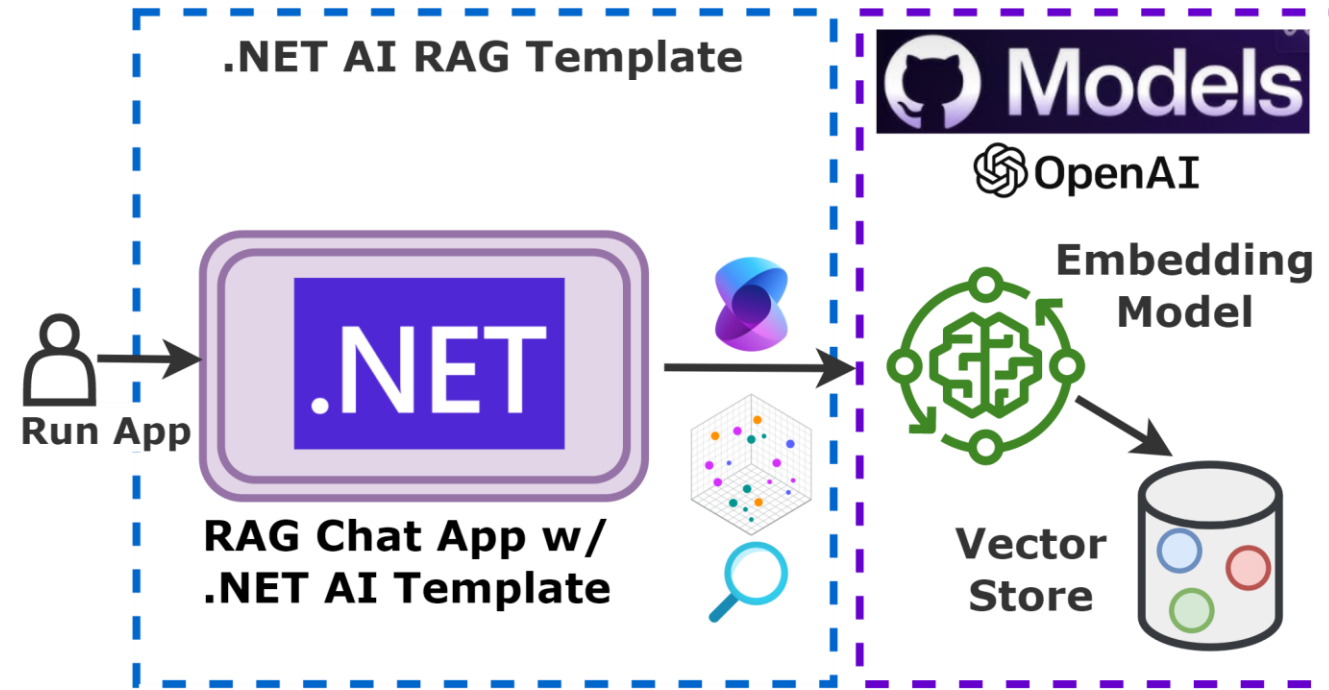
Optimized for **similarity-based searches**, quickly find the **closest vectors**

Vector Search to find documents that are **semantically similar** to the **query**



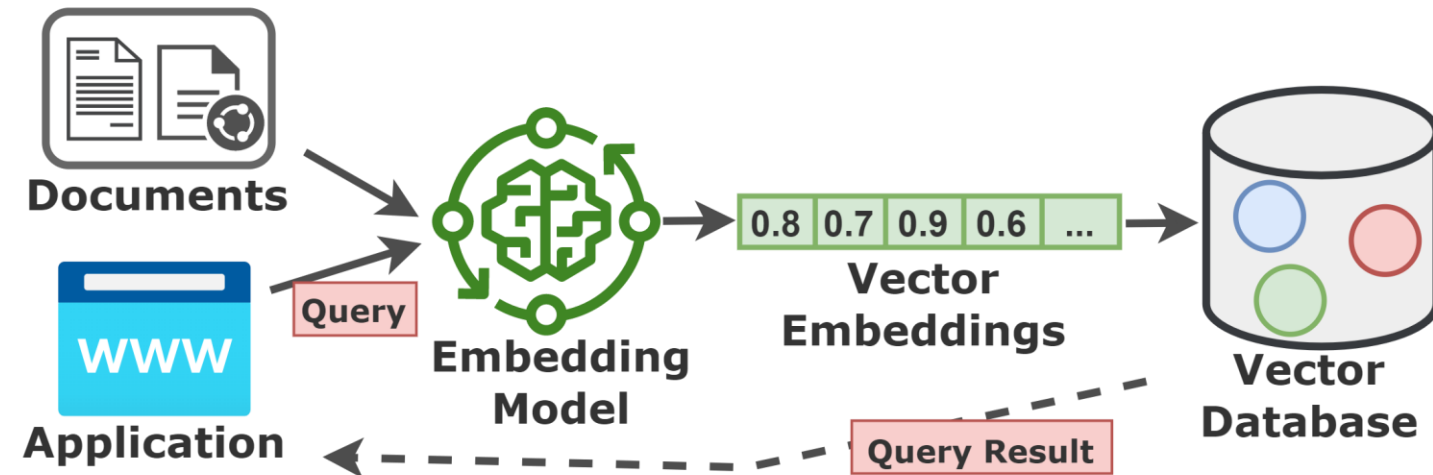
<https://towardsdatascience.com/advanced-retriever-techniques-to-improve-your-rags-1fac2b86dd61>

Retrieval Augmented Generation (RAG) Application w/ .NET



<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview1/>

<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview2/>



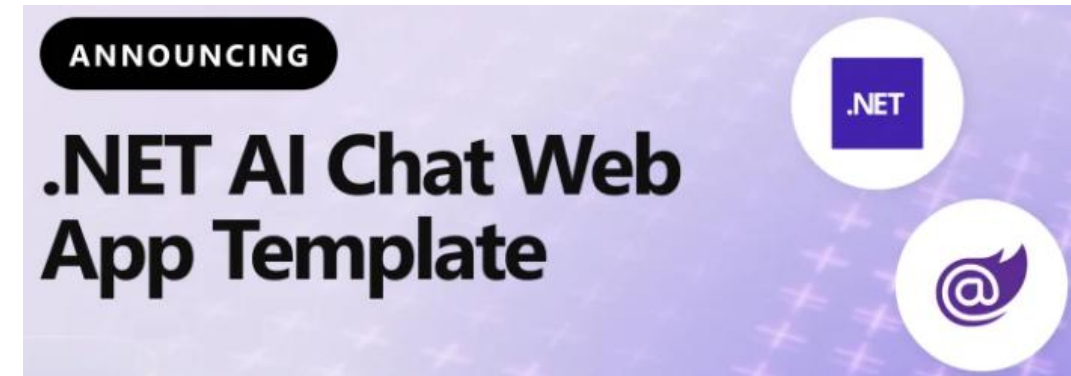
.NET AI App Template

Create a **.NET AI app** to chat with custom data using the .NET AI app template

Building **AI** apps with .NET by handling **common setup tasks** and **configurations**

Simplifies the **initial setup** for building **AI-powered** apps: **configuring** your **AI provider**, **setting up best practices** and adding **vector store support**

RAG ground AI responses by **retrieving relevant context** from a **knowledge store** before generating final answers



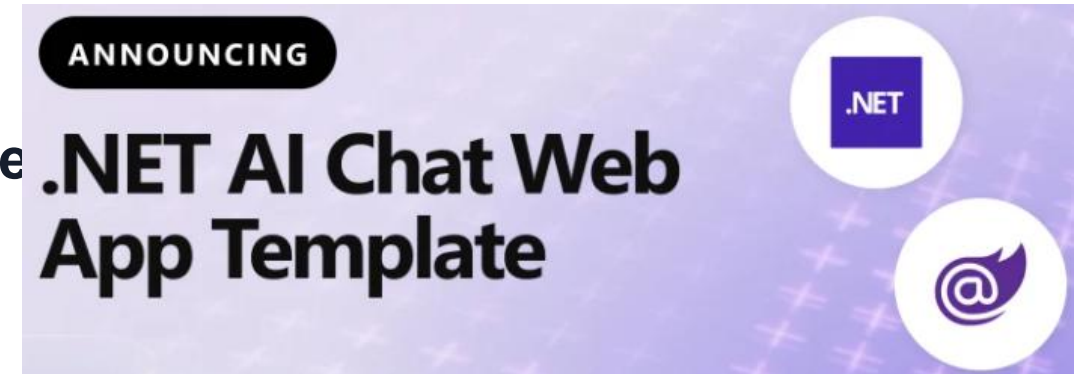
<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview1/>

<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview2/>

Install the .NET AI App Template

RUN: dotnet new install Microsoft.Extensions.AI.Templates

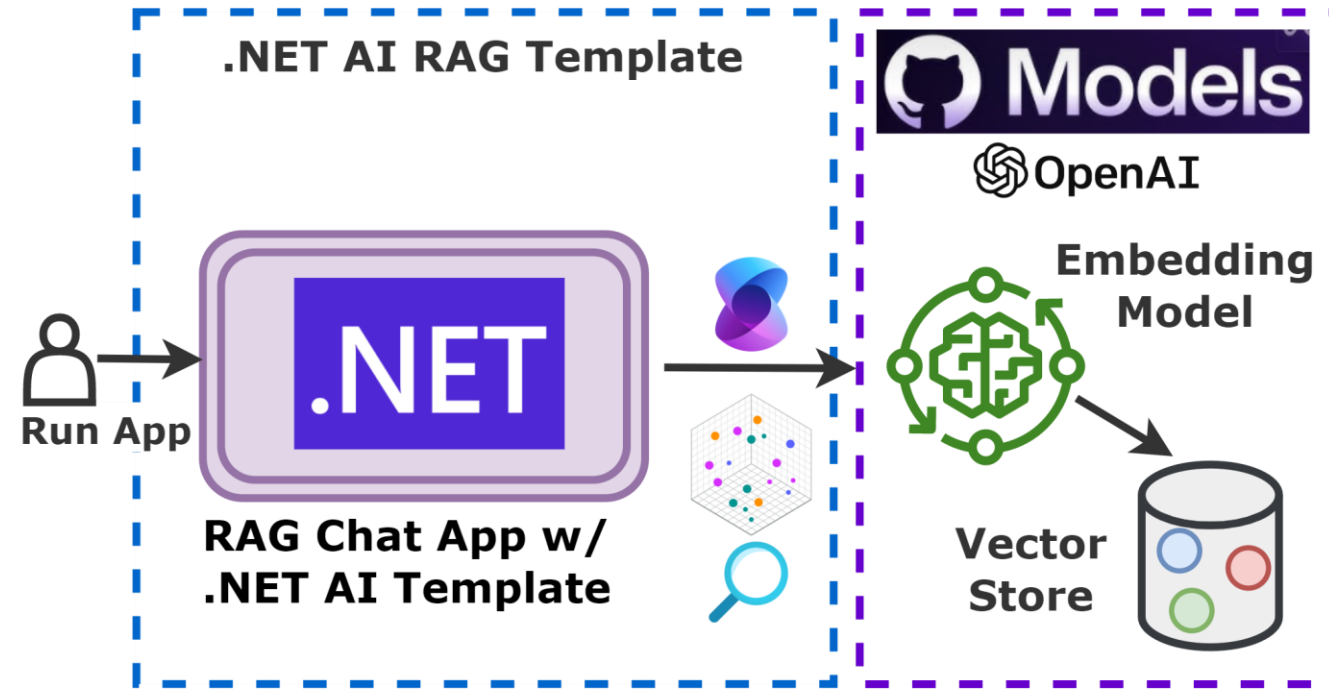
Fetches the Microsoft.Extensions.AI.Templates package, giving you access to new project templates in your .NET environment



<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview1/>

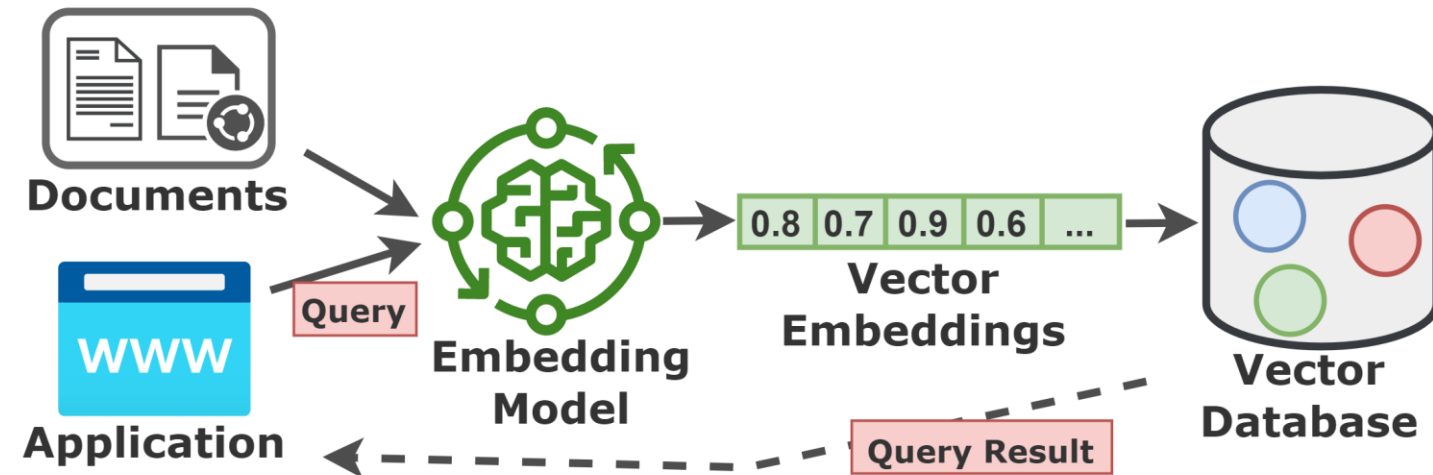
<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview2/>

Retrieval Augmented Generation (RAG) Application w/ .NET



<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview1/>

<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview2/>



Why Extend the Chat App with Your Documents?

Domain-Specific Knowledge

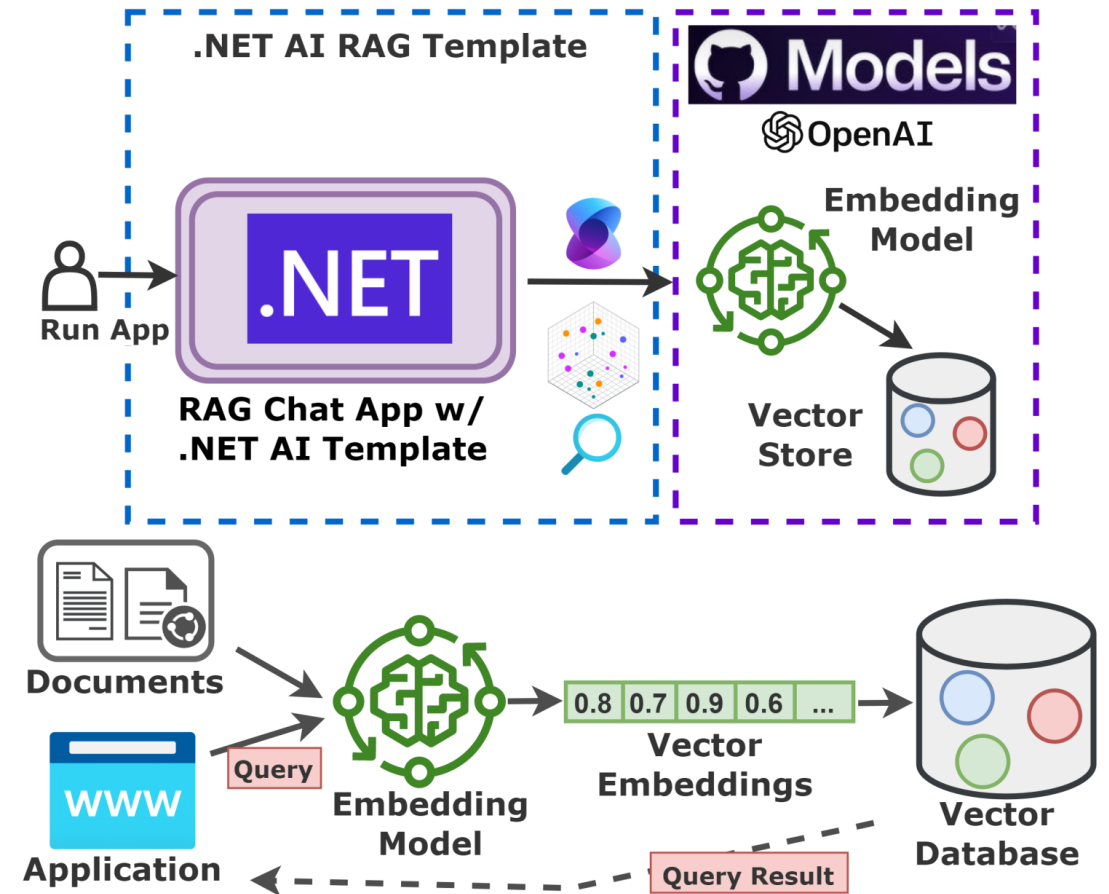
Accurate and Credible Responses

Automatic Citations

How Document Ingestion Works in RAG

1. Upload Documents
2. Chunking
3. Embedding Generation
4. Vector Store

Example: Adding a New Solar Charger PDF



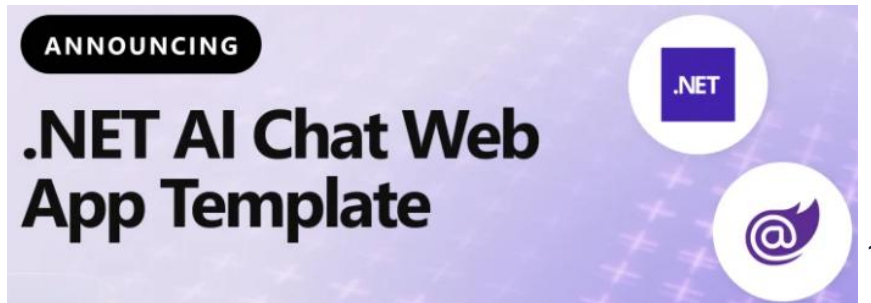
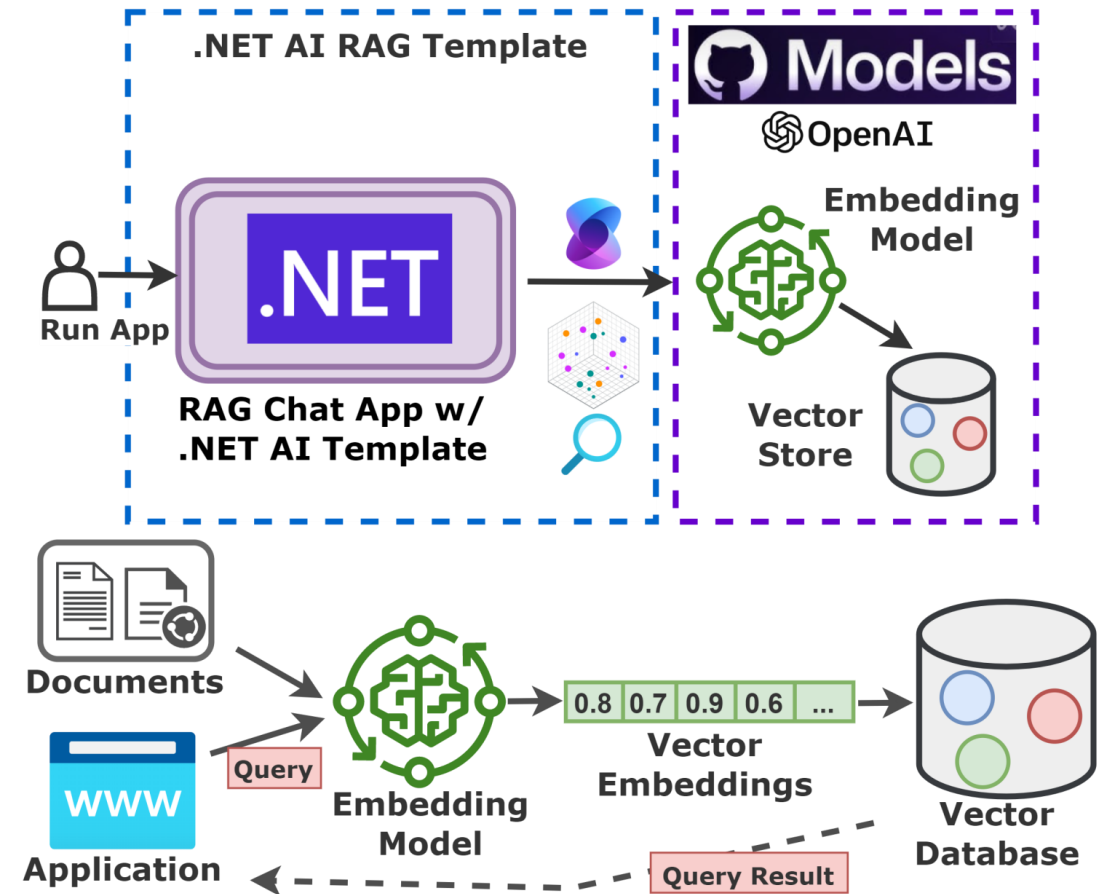
Why Extend a RAG Chat App with Function Calling?

Beyond Document Data

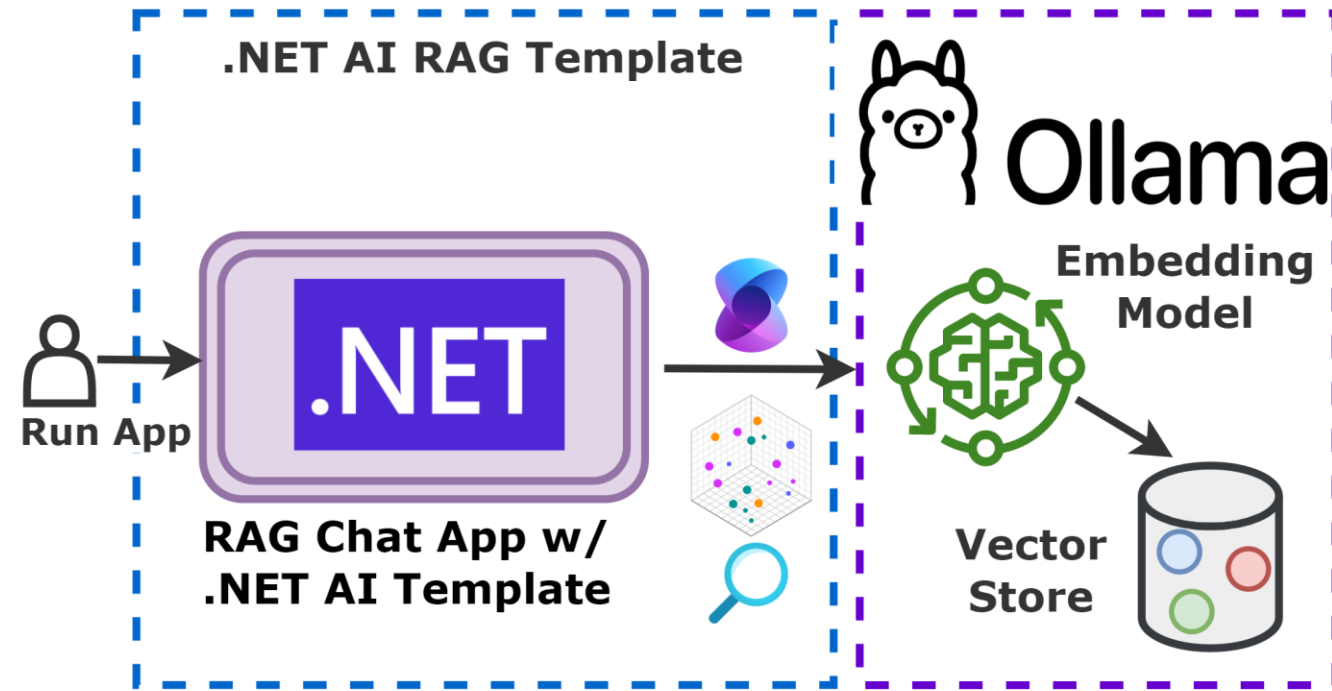
LLM as an Orchestrator

Enhanced Automation

Example: Chat App Asks for Current Temperature

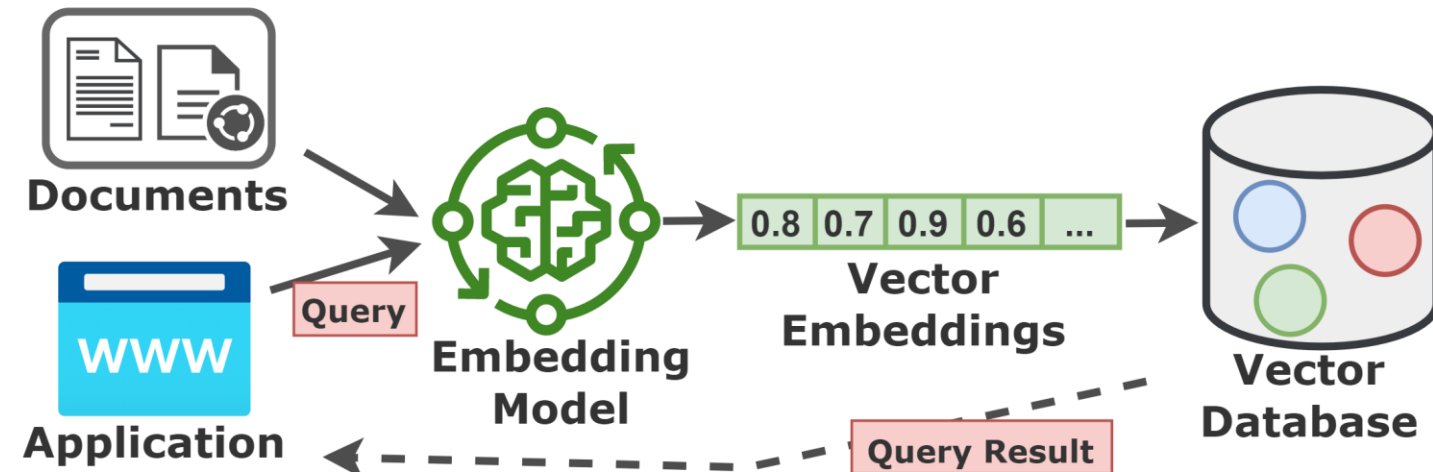


RAG App w/ .NET – Ollama using all-minilm embeddings



<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview1/>

<https://devblogs.microsoft.com/dotnet/announcing-dotnet-ai-template-preview2/>



Add Qdrant Vector Database into our RAG Chat App

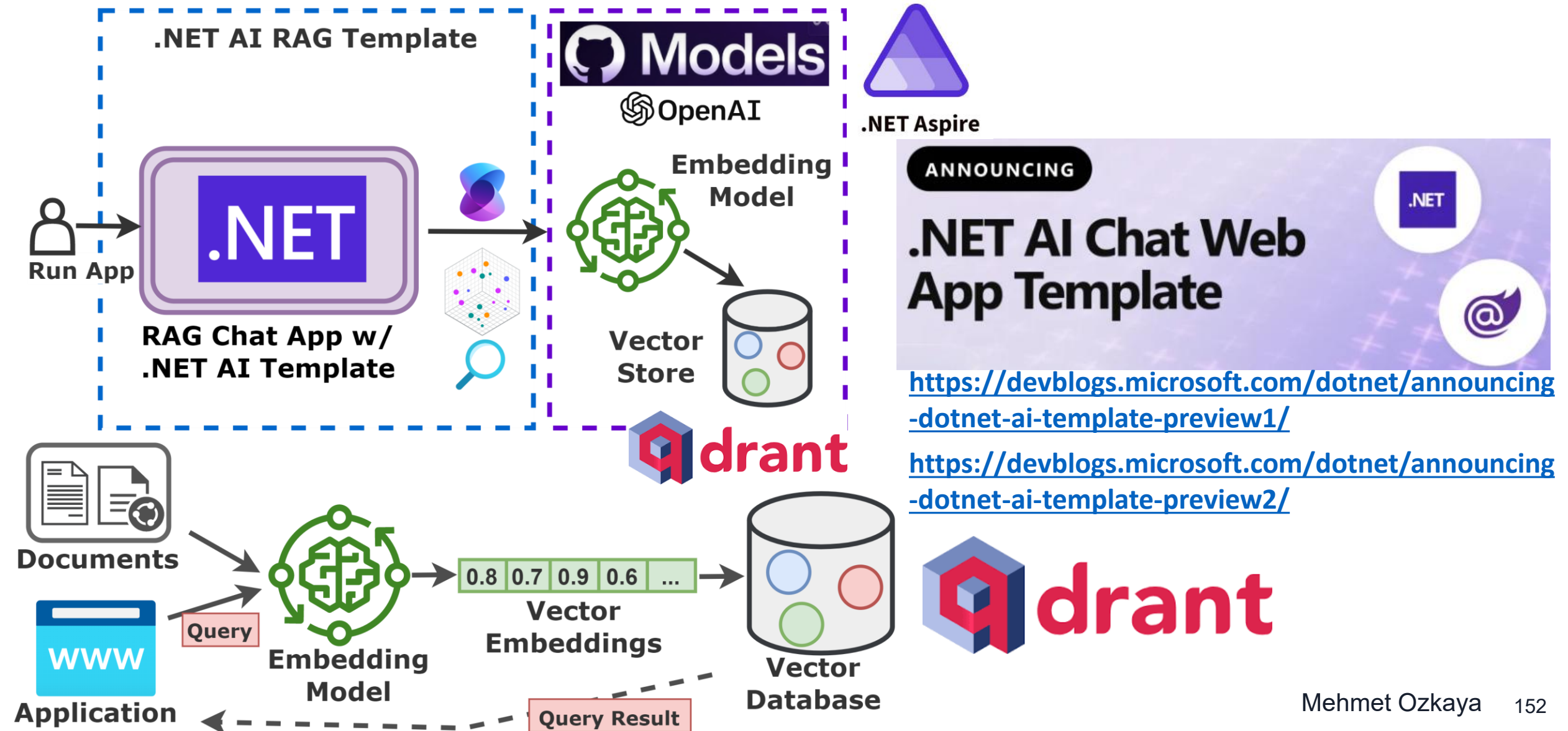


Image Analysis Apps with .NET AI

Image Recognition with .NET and MEAI

Image Analysis App w/ .NET and GH Models - OpenAI gpt-5-mini

Analysis Traffic Cams with Structured Output

Build Image Analysis App w/ .NET and Ollama llava

Mehmet Ozkaya



Introduction to This Section

Image Recognition with .NET and MEAI

Build Image Analysis App w/ .NET and GH Models - OpenAI gpt-5-mini

Analysis Traffic Cams with Structured Output

Download llava vision model for Ollama

Build Image Analysis App w/ .NET and Ollama llava



Why Image Recognition Matters

Current Models are multimodal

Understand and process a variety of inputs—text, images, audio, and even video

ask complex, natural language questions and get back nuanced, insightful answers

Estimate the number of calories in this dish

What can I make for dinner with these ingredients?

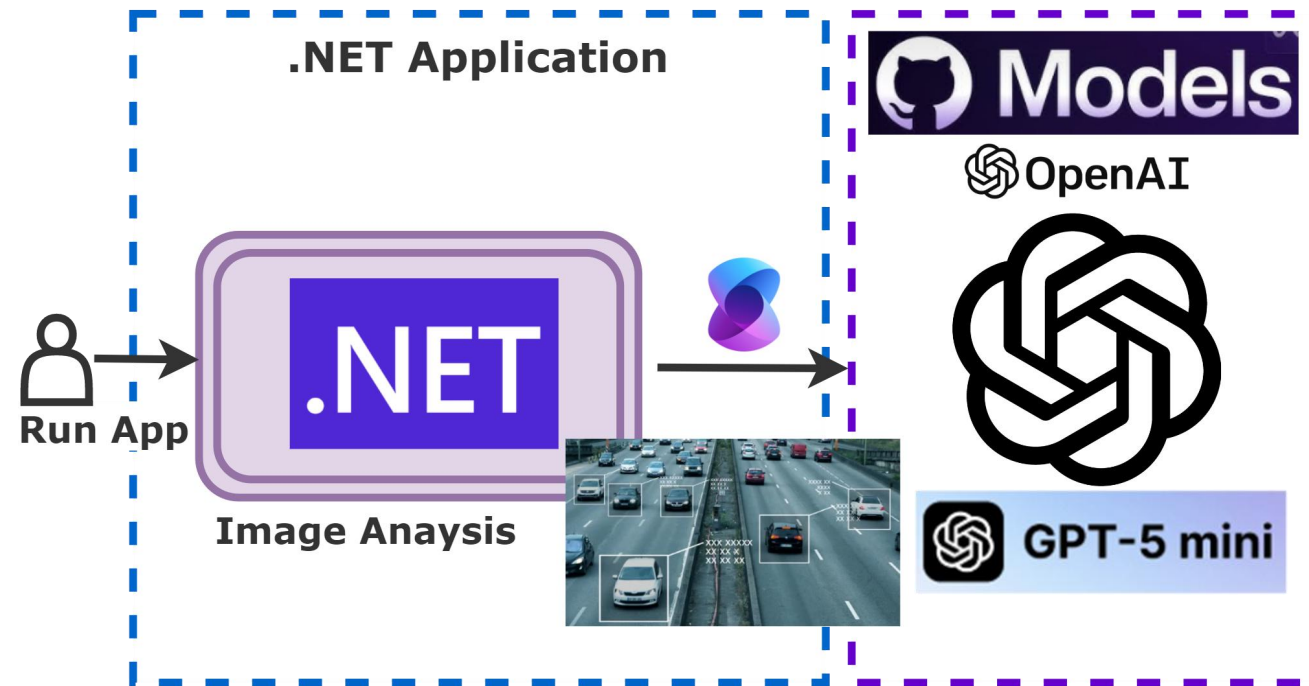


Basic Flow for Image Recognition

Obtain an Image

Send Image to the Model

Ask Questions or Request Descriptions



Why Image Recognition Matters

From **object detection** to **scene understanding** and **contextual Q&A**

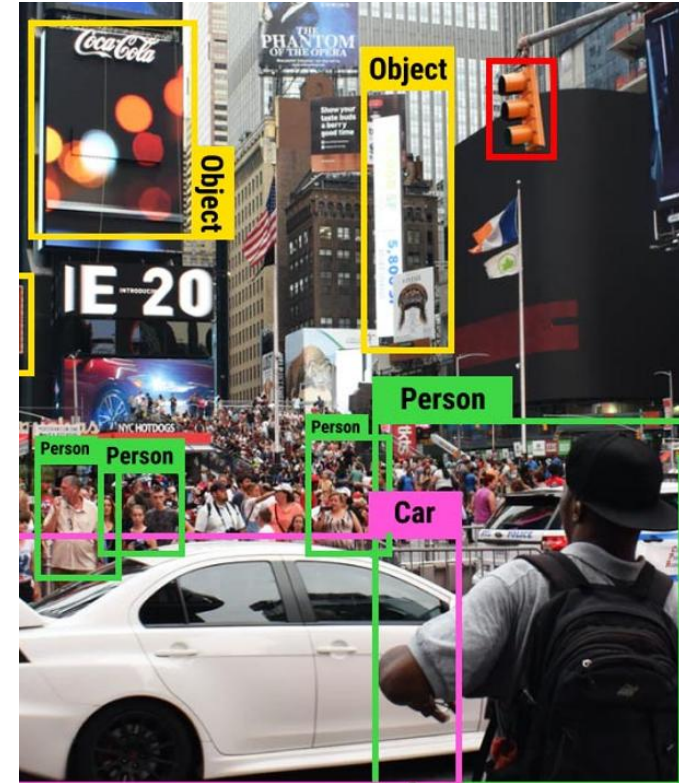
Security systems, **quality control**, **healthcare diagnostics**, and **retail**

Current models are **multimodal**, they can **accept** a **variety of inputs**, such as **text**, **images**, and **audio** and **generate** a **variety of outputs**

Ask **natural language questions** about **images**, bridging the gap between **computer vision** and **language understanding**

For example: How many people are present and is it raining?

MEAI and **GitHub Models**, so instantiate the **IChatClient** as we have been. Also start to create a chat history.



Basic Flow for Image Recognition

Obtain an Image

Could come from file uploads, camera capture or downloading from a URL

Send Image to the Model

Encode the image in a base64 format or pass it as binary data through an API

Ask Questions or Request Descriptions

Chat-based approach to prompt the model about the image

“Describe what you see in this image. Are there any people? Is it sunny?”

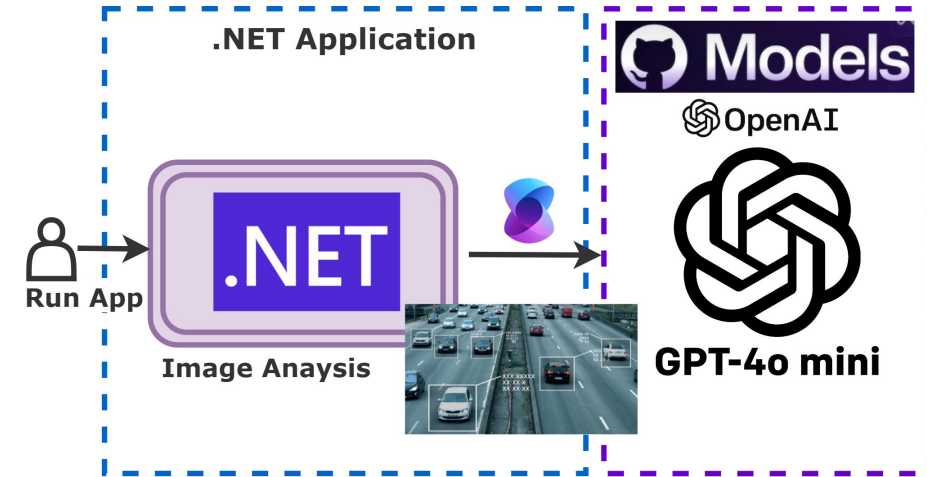
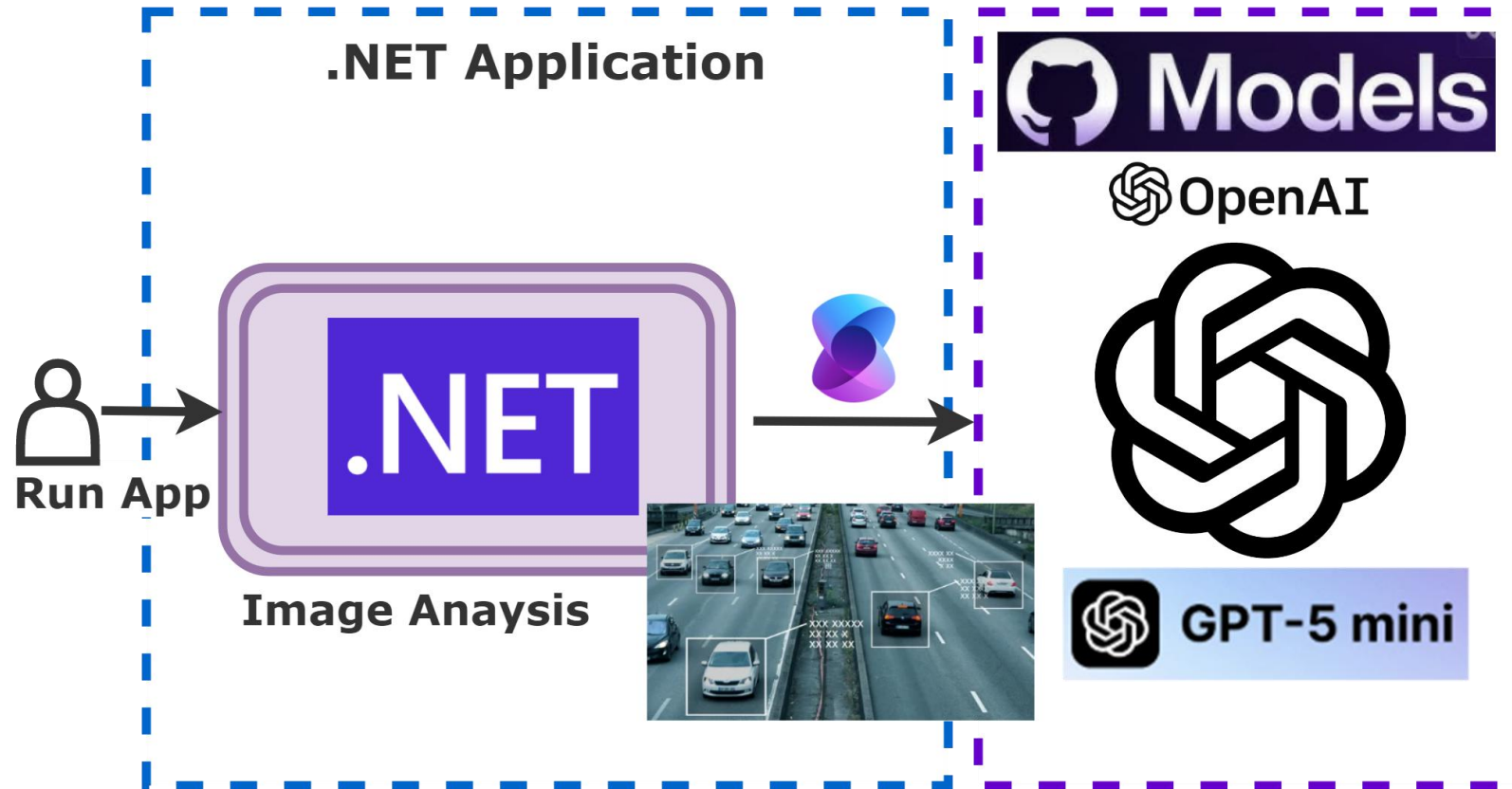


Image Analysis Apps with .NET AI



Traffic Cams Analysis App with Structured Output

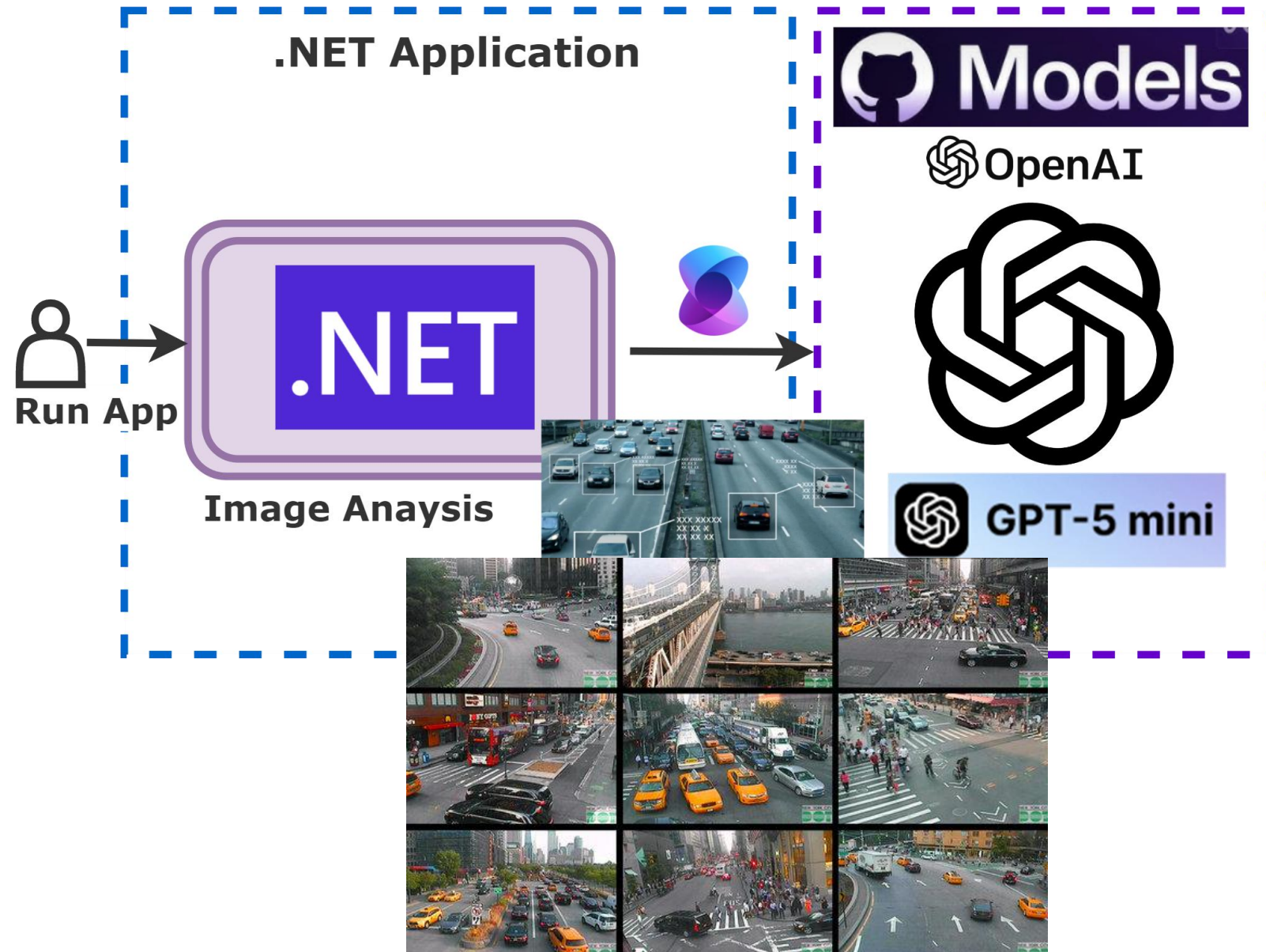
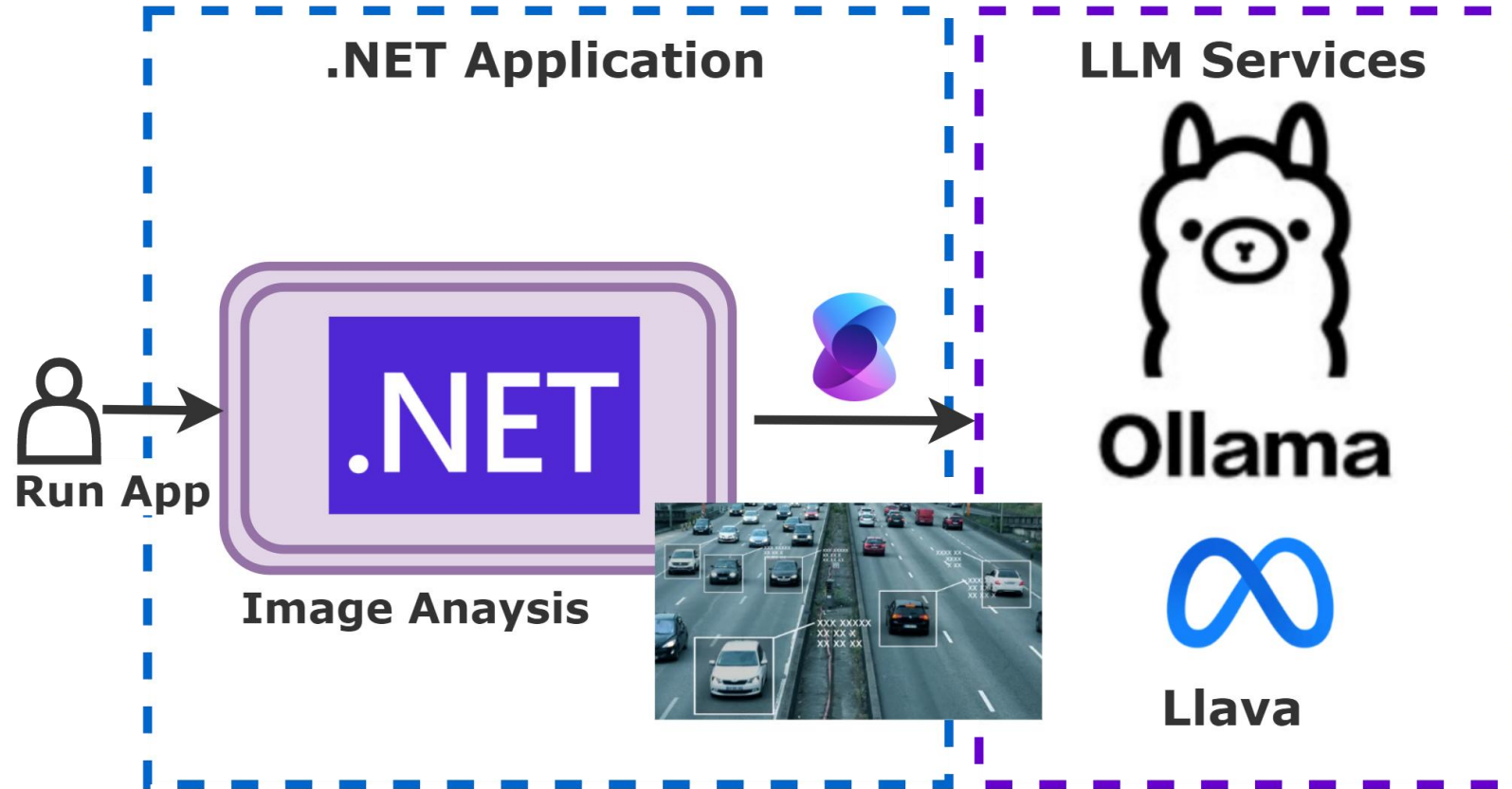


Image Analysis Apps with .NET AI - Ollama



Introduction to This Section

Starting Point: EShop Distributed Microservices Architecture with .NET Aspire

Add Chat Capability in EShop integrate GitHub Models OpenAI gpt-5-mini model

Add Qdrant Vector Database into .NET Aspire Distributed Environment

Client Integration Packages for Qdrant and Extensions.VectorData

Register Qdrant Client Services into .Net DI services in Catalog/Program.cs

Register EmbeddingGenerator services in Catalog/Program.cs

Develop ProductVector Domain Entity for Storing Vector Data

Develop ProductAIService.cs for Semantic Search Implementation

Develop Search Endpoints in ProductEndpoints.cs for Semantic Search

Blazor FrontEnd Search Page Development

Build Eshop Vector Search App w/ .NET Aspire, gpt-5-mini and Qdrant Vector DB

EShop Distributed Microservices Architecture with .NET Aspire

Add Chat Capability integrate GitHub Models OpenAI gpt-5-mini

Develop ProductAIService.cs Class for Business Layer

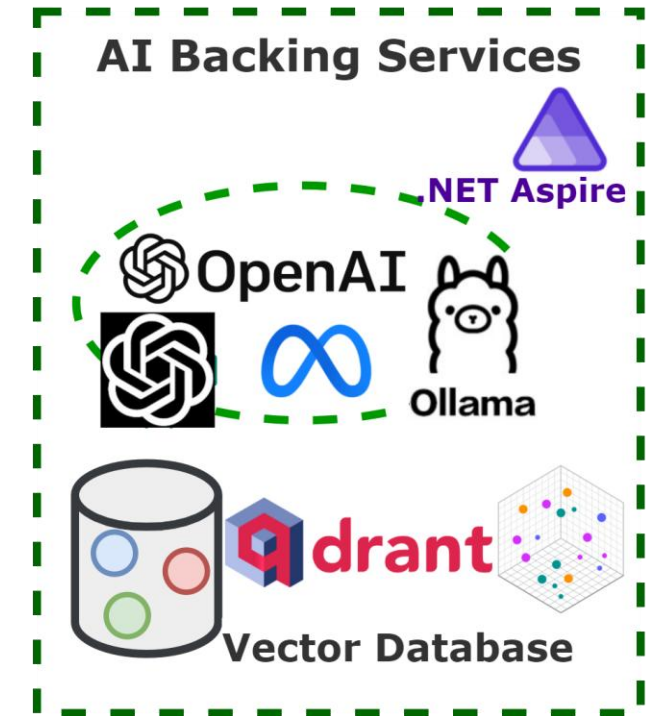
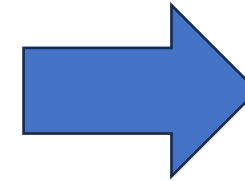
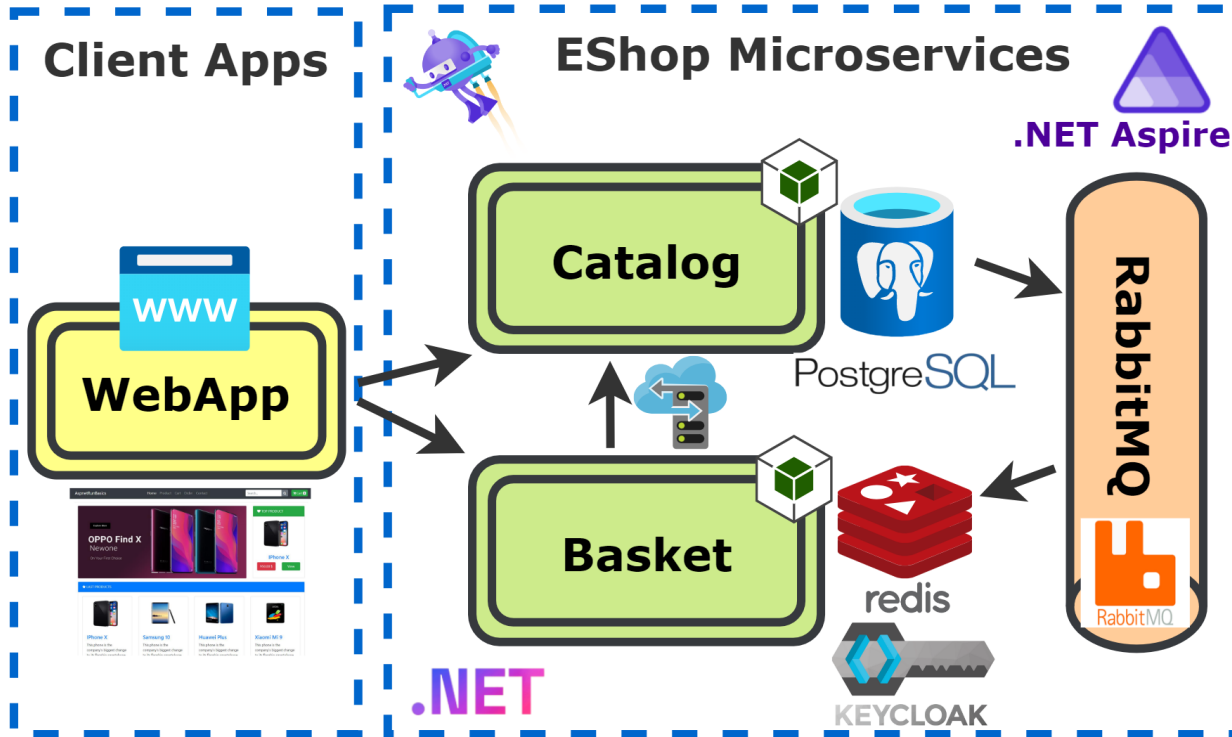
Develop ProductEndpoints Search Query Endpoint for Chat Q&A

Add Qdrant Vector Database into .NET Aspire

Mehmet Ozkaya



Starting Point: EShop Distributed Microservices Architecture with .NET Aspire



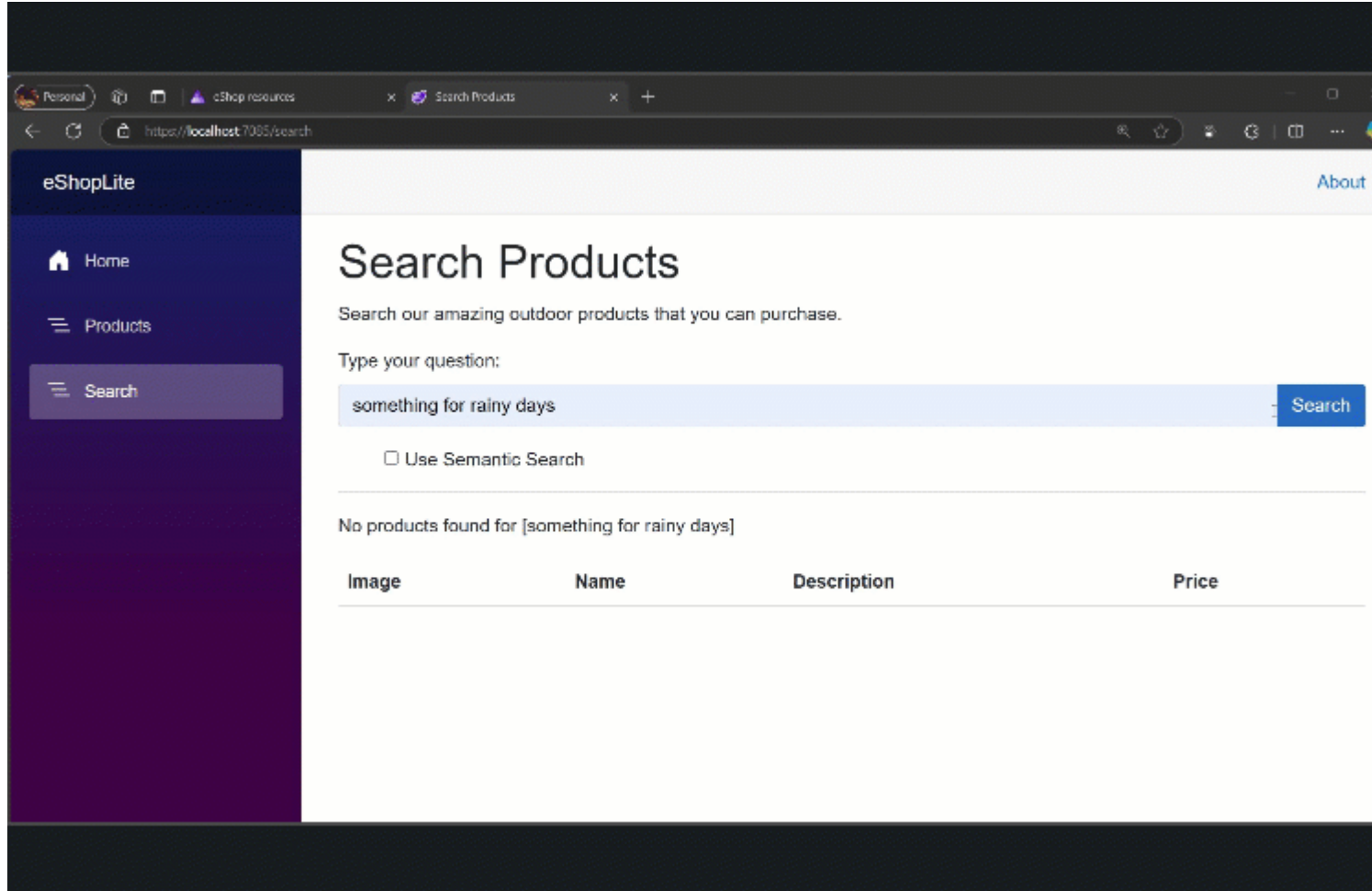
Develop Blazor Web Application

Products

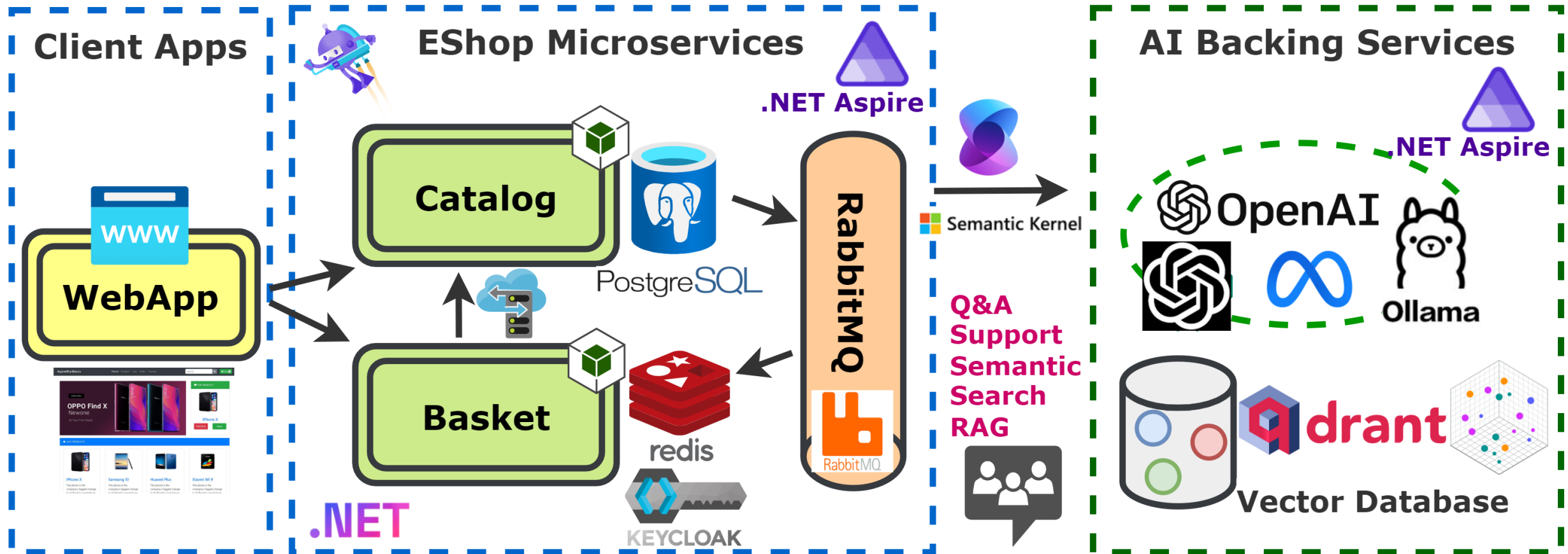
Here are some of our amazing outdoor products that you can purchase.

Image	Name	Description
	Hiking Poles	Ideal for camping and hiking trips
	Outdoor Rain Jacket	This product will keep you warm and dry in all weathers
	Survival Kit	A must-have for any outdoor adventurer
	Outdoor Backpack	This backpack is perfect for carrying all your outdoor essentials

Develop Blazor Web Application



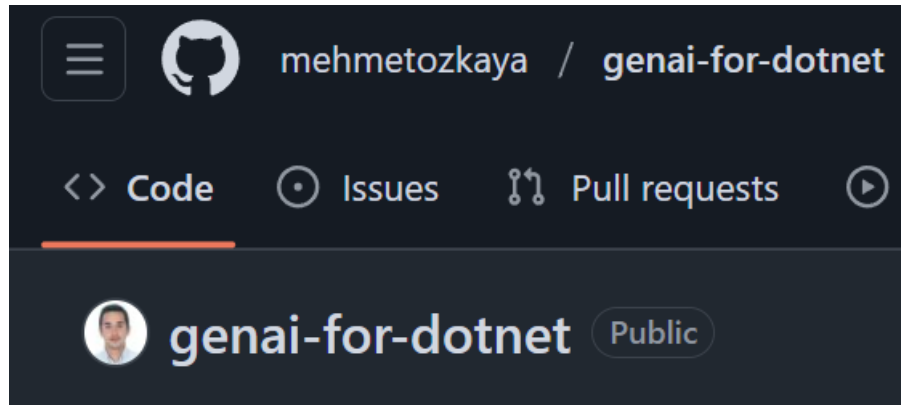
Course Project: Eshop Vector Search App w/ .NET Aspire, gpt-5-mini and Qdrant Vector DB



Slide Source Code on GitHub

All the code written and demonstrated in this course is available on GitHub

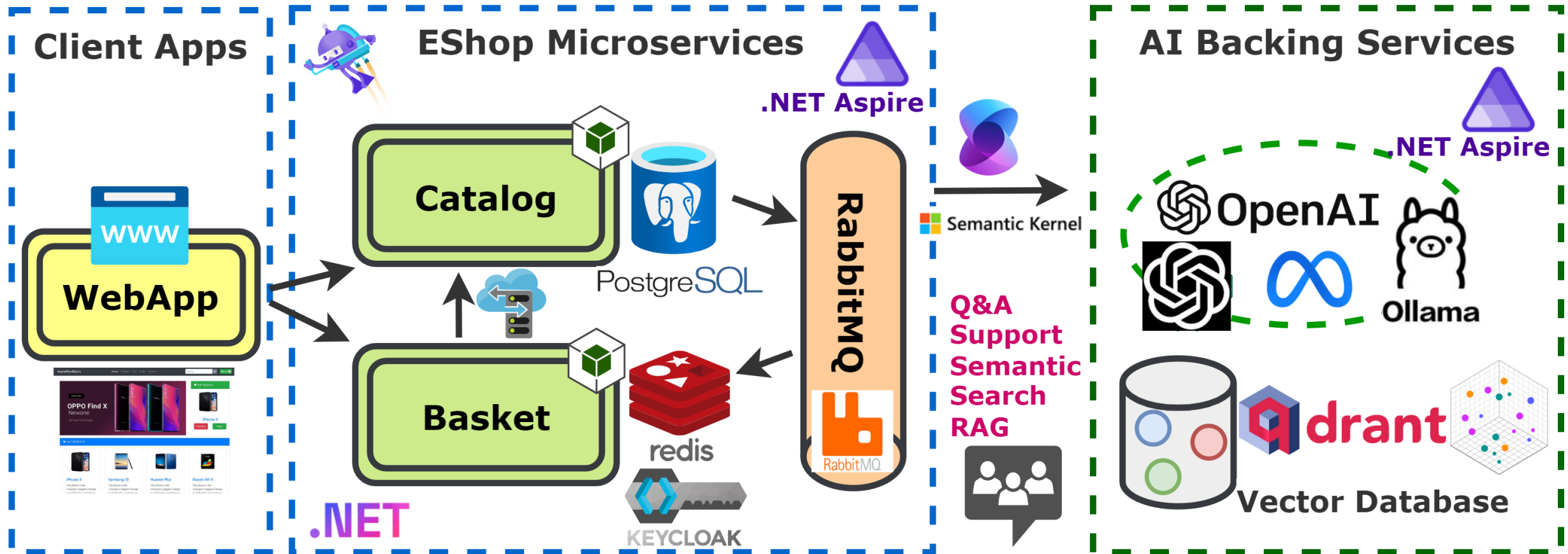
- <https://github.com/mehmetozkaya/genai-for-dotnet>



Starting Point: EShop Distributed Microservices Architecture with .NET Aspire

- <https://github.com/mehmetozkaya/genai-for-dotnet/tree/main/final-project/start/eshop-microservices>

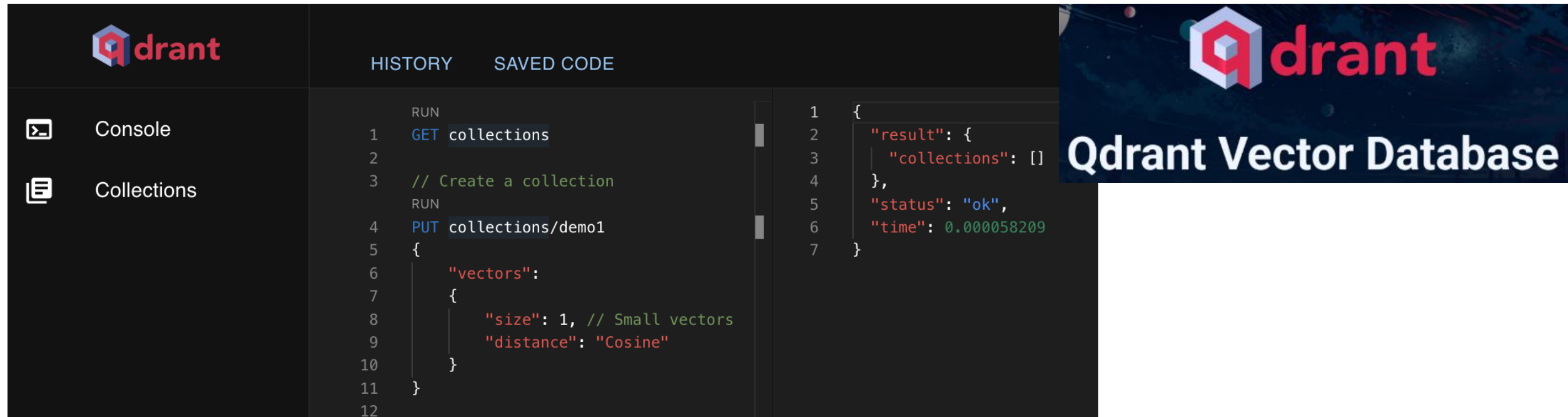
Course Project: Eshop Vector Search App w/ .NET Aspire, gpt-5-mini and Qdrant Vector DB



Why Qdrant ?

Vector search engine, perfect for storing embeddings from text, images, or other high-dimensional data

Enables semantic searches and is straightforward to set up in .NET Aspire with the Aspire.Hosting.Qdrant package



The screenshot displays the Qdrant Vector Database interface. On the left, there is a sidebar with the Qdrant logo and two menu items: 'Console' and 'Collections'. The main area is divided into two panels. The left panel, titled 'HISTORY', shows a REST client console with the following code:

```
1 RUN
2 GET collections
3 // Create a collection
4 RUN
5 PUT collections/demo1
6 {
7   "vectors":
8   {
9     "size": 1, // Small vectors
10    "distance": "Cosine"
11  }
12 }
```

The right panel, titled 'SAVED CODE', shows the JSON response from the PUT request:

```
1 {
2   "result": {
3     "collections": []
4   },
5   "status": "ok",
6   "time": 0.000058209
7 }
```

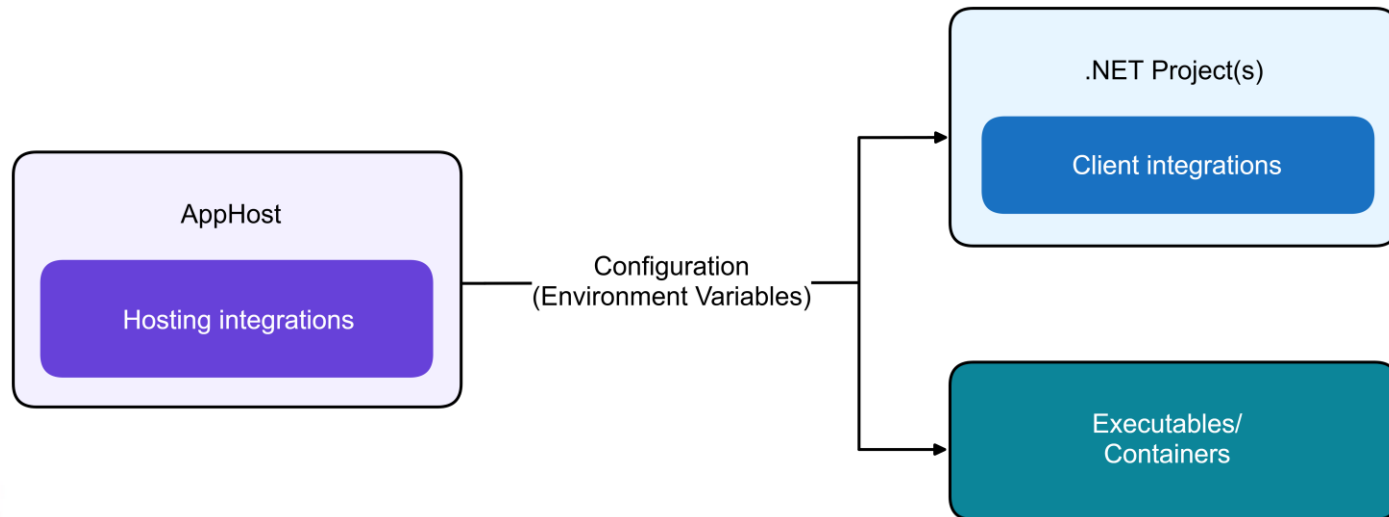
On the far right, there is a banner for the Qdrant Vector Database with the Qdrant logo and the text 'Qdrant Vector Database'.

.NET Aspire + Qdrant ?

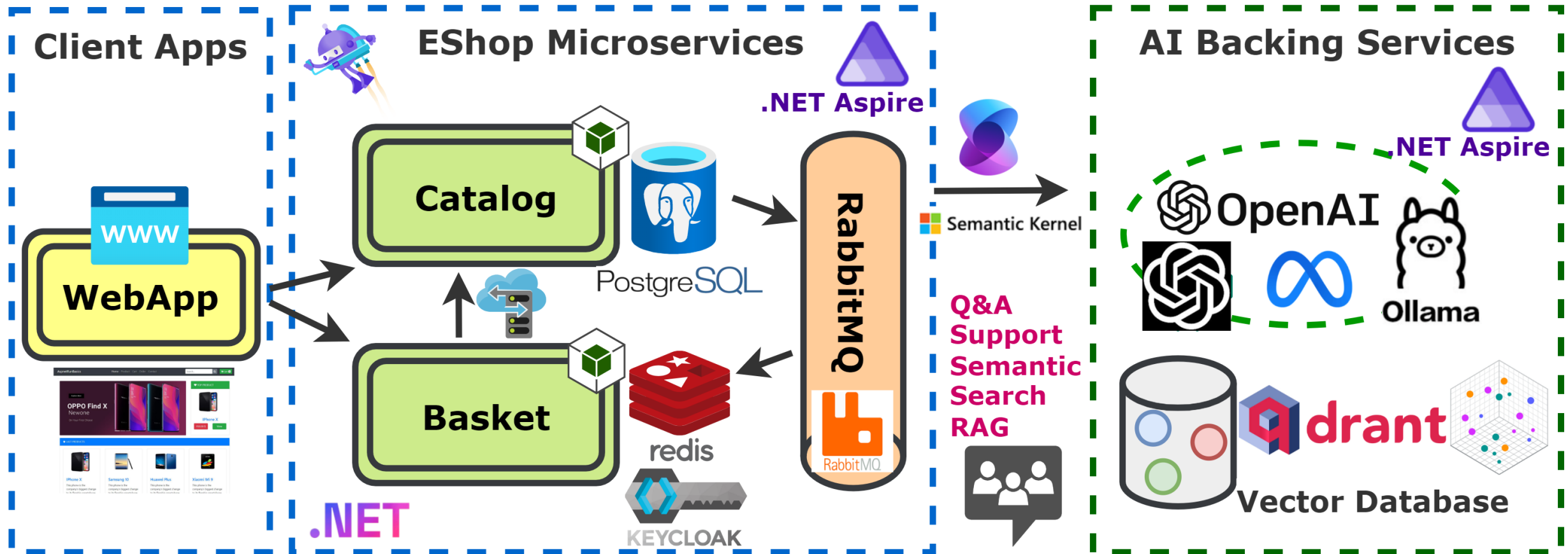
How you host and manage microservices—particularly around environment variables, connection strings, and container orchestration

By referencing Qdrant in Aspire, we can share environment variables and ensure the vector DB is fully up before a dependent microservice starts

Aspire Hosting Integration: <https://learn.microsoft.com/en-us/dotnet/aspire/database/qdrant-integration?tabs=dotnet-cli>



Course Project: Eshop Vector Search App w/ .NET Aspire, gpt-5-mini and Qdrant Vector DB



Microsoft.Extensions.VectorData.Abstractions

Provides Common Interfaces

Abstract CRUD on vector data, standardized to connect VectorDBs (Chroma, Qdrant, Pinecone, Weaviate)

Supports Vector and Text Search

Unified methods for inserting embeddings, High-level search functions

Decouples Your App from Vendor SDKs

App code remains flexible, Swap underlying vector store implementations (local or cloud-based) with minimal changes

Understanding the Library Stack

The Foundation - Microsoft.Extensions.VectorData

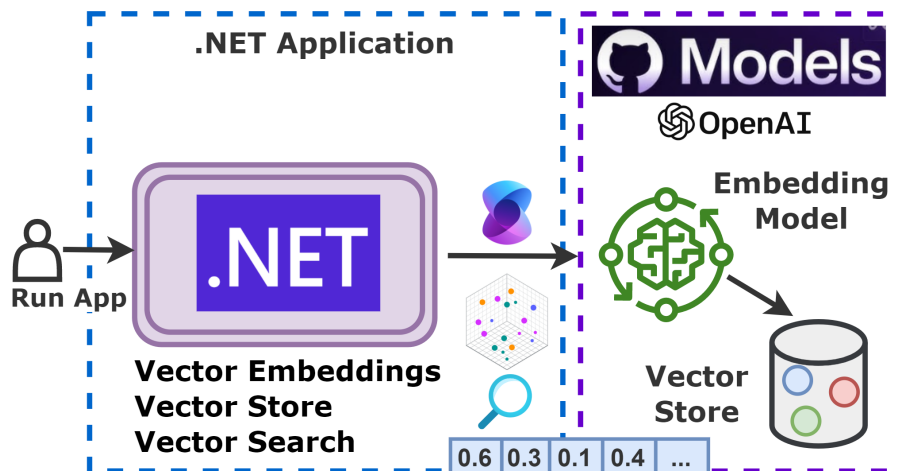
This library is like Entity Framework for vectors. Unified way to perform create, read, update, and search operations

The Implementation - Semantic Kernel Connectors

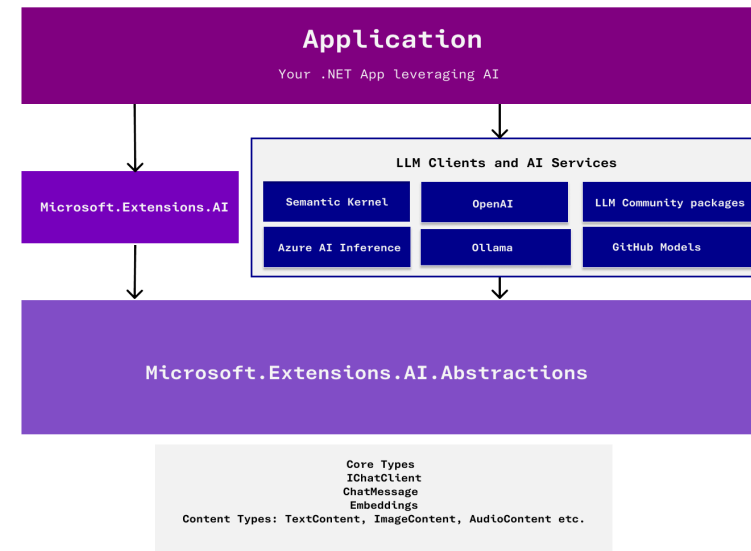
SK team provides a connector for Qdrant, another for Chroma, another for Pinecone..

The Integration - Aspire Client Libraries

Aspire.Qdrant.Client, make dependency injection and configuration effortless.



Support for
DI, Pipelines and
Middleware
(Caching, Tool Calling,
Telemetry)



Provides an
IChatClient that
connects to LLM
providers

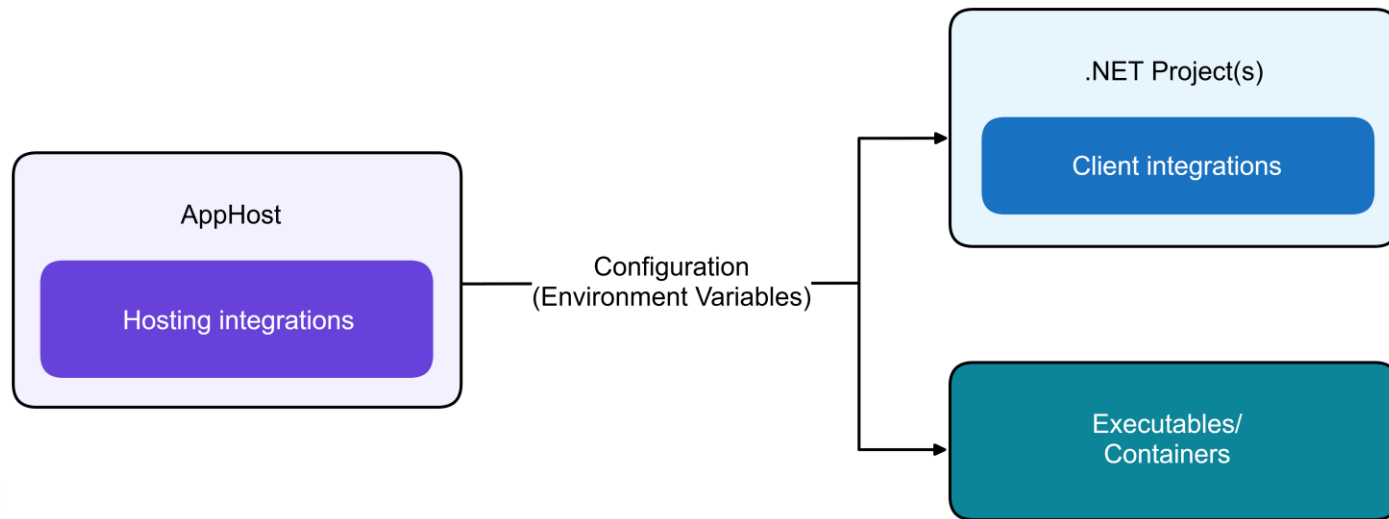
Required Package sfor .NET Aspire & Qdrant

Aspire Hosting Integration: Aspire.Qdrant.Client: <https://learn.microsoft.com/en-us/dotnet/aspire/database/qdrant-integration?tabs=dotnet-cli>

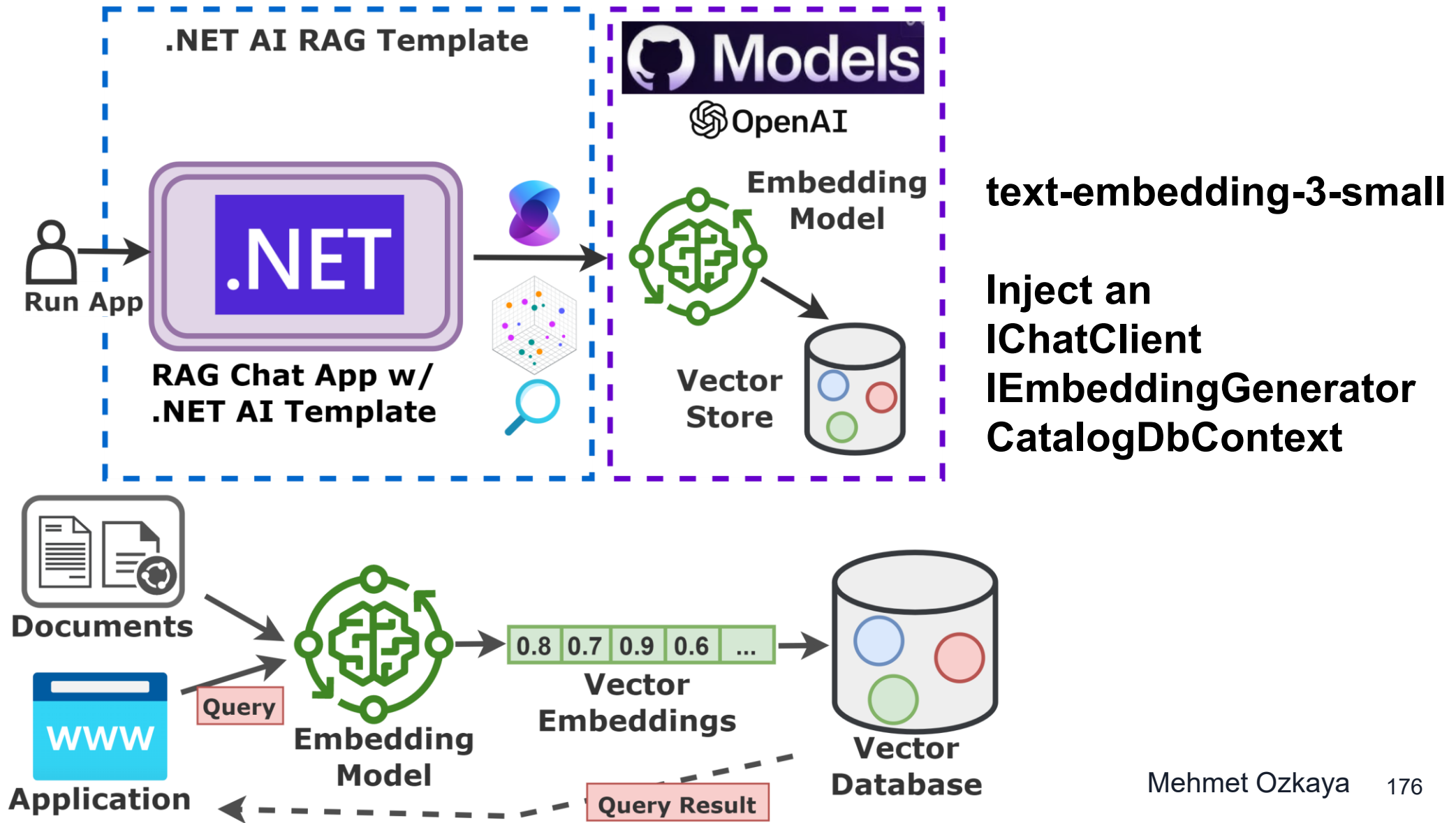
Microsoft.SemanticKernel.Connectors.Qdrant : Actual Qdrant driver from the Semantic Kernel team that works with the VectorData abstractions



.NET Aspire



Retrieval Augmented Generation (RAG) Application w/ .NET



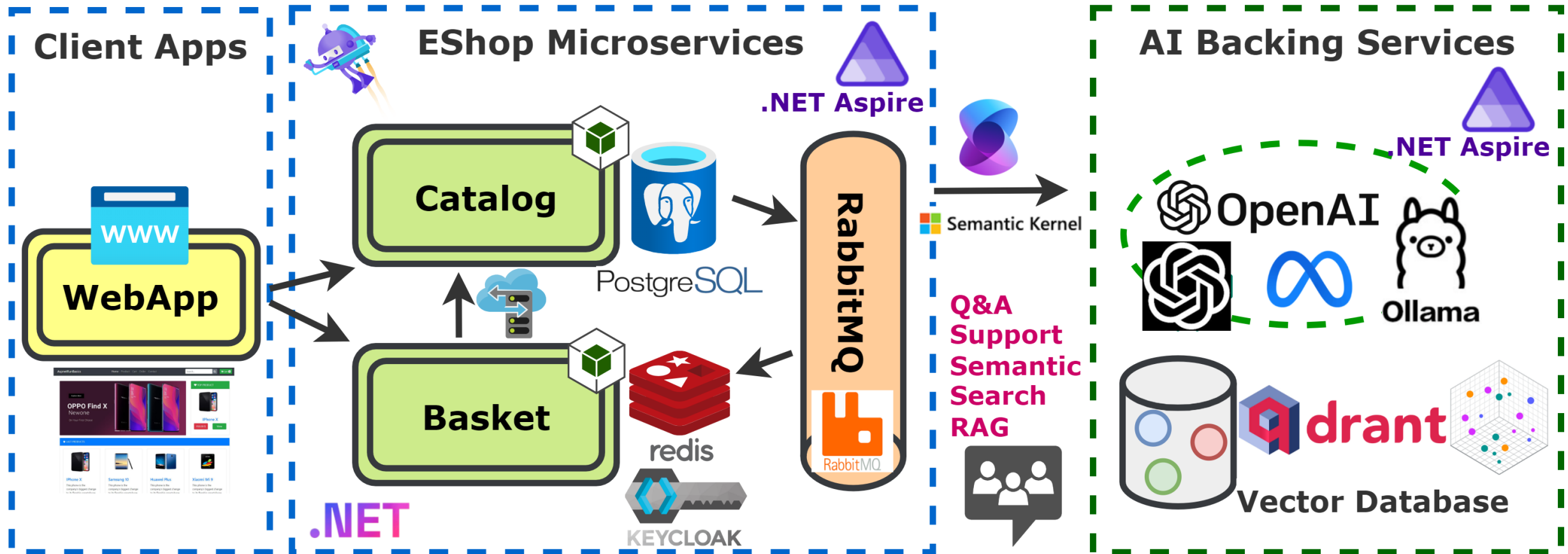
Develop Blazor Web Application

Products

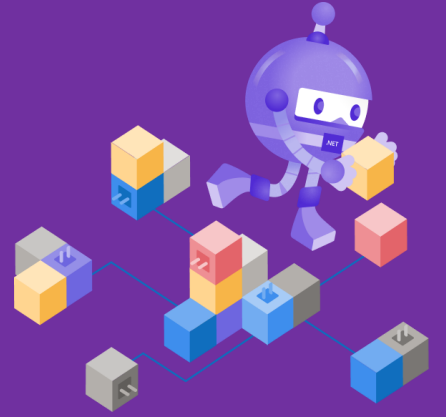
Here are some of our amazing outdoor products that you can purchase.

Image	Name	Description
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	Outdoor Backpack	This backpack is perfect for carrying all your outdoor essentials

Course Project: Eshop Vector Search App w/ .NET Aspire, gpt-5-mini and Qdrant Vector DB



Thanks



Thank you so much for being with me on this journey.

Reviews and feedback is really encourage to me for pushing forward to create new courses like this.

Mehmet Ozkaya

